Paulus Gerdes & Ahmed Djebbar

MATHEMATICS IN AFRICAN HISTORY AND CULTURES:

AN ANNOTATED BIBLIOGRAPHY

African Mathematical Union
Commission on the History of Mathematics in Africa
(AMUCHMA)

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Cover design inspired by a pattern on a mat woven in the 19th century by a Yombe woman from the Lower Congo area (Cf. GER-04b, p. 96).

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Preface

One cannot but welcome this very important annotated bibliography on *Mathematics in African History and Culture*. We are already at the beginning of the third millennium and, yet, one is often struck by attitudes, largely based on ignorance, towards the mathematical contributions from Africa and by Africans. I am sure that it is this phenomenon and the collective experience and knowledge of Gerdes and Djebbar that have led to the conceptualisation of this publication. It is, indeed, long overdue. This publication informs us about both the history of mathematics in Africa and the mathematics in the history of Africa. It is also appropriate that the contributions of Africans outside Africa, or as is commonly referred to, the African Diaspora, are included. For, often the involvement and impact of Africans on life and developments outside Africa, especially in developed countries, are knowingly and unknowingly underplayed or even ignored.

Mathematics in African History and Culture: An Annotated Bibliography, is bound to have a major impact on the curricula of courses on (the history of) mathematics in Africa. The role of African mathematicians in astronomy, time-reckoning and calendars can now be researched and appreciated more fully. By including the mathematics in African culture, the authors have attached meaningful value to the systematic, analytical and structured nature of African cultures. Thus string figure and board games emerge as meaningful mathematical activities in addition to being enjoyed as forms of relaxation. As far as it is known, this bibliography, of over a thousand references, is not just the most comprehensive ever produced, but also covers the whole African continent over many centuries as well as recognising "the historical links across the Mediterranean and the oceans". This latter aspect is important because it puts the achievements in so-called western mathematics into perspective.

I believe that there is a need for African students and researchers, especially the younger generation, to realise that Africans have made meaningful contributions to science and mathematics. This realisation should serve as inspiration to them. The technological and economic development of Africa in this modern age depends on various applications of mathematical sciences. Getting to grips with what has been produced by our forebears is potentially important for the generation of new knowledge, particularly in this era of knowledge-based economies.

On behalf of the African Mathematical Union, I sincerely wish to thank Professors Paulus Gerdes and Ahmed Djebbar for their contribution. They are, indeed, two stalwarts in our quest for unearthing and highlighting contributions by Africans to mathematical research and teaching.

Jan Persens, Ph.D

President of the African Mathematical Union (2000-2004)

Bellville, South Africa

June 2004

Introduction

One of the first measures taken by the Executive Committee of African Mathematical Union (AMU), elected at the 2nd Pan-African Congress of Mathematicians (April 1986, Jos, Nigeria), under the chairmanship of Professor Aderemi Kuku, was to create an AMU Commission on the History of Mathematics in Africa (AMUCHMA). The two authors were appointed chairman and secretary. At the subsequent congresses in 1991 (Nairobi, Kenya), in 1995 (Ifran, Morocco) and 2000 (Cape Town, South Africa), the authors were re-elected. As co-ordinators of the commission, we have tried to stimulate research, and to collect and disseminate as much information as possible about the history of mathematics in Africa. Along with the many papers delivered at conferences and seminars organised during the years, we published so far 28 issues of the AMUCHMA Newsletter. To the delegates of the 6th Pan-African Congress of Mathematicians (September 2004, Tunis, Tunisia), we would like to present the following bibliography on mathematics in African history and cultures.

The bibliography contains over one thousand five hundred references. It is the result both of the information we collected in the context of AMUCHMA and of our personal research. The first author used also the information he gathered as secretary (1991-1995) of the Southern Africa Mathematical Sciences Association (SAMSA) for the *Who is Who in Mathematics and Mathematics Education in Southern Africa* (5-GER-92, 93, 95).

Our bibliography attempts to encompass the African continent as a whole, from immortal times to the present, without forgetting the historical links across the Mediterranean and the oceans. For instance, several references included in the bibliography highlight the circulation of mathematicians and of mathematical ideas between the Maghreb (Northwest Africa) and Andalusia (Iberian Peninsula) during the 'Middle Ages.'

The present bibliography is as embracing, complete and update as was possible. Mainly references in Arabic, French, English, Portuguese, official languages of the continent, were selected. For studies from and about the mathematicians of Alexandria (Egypt) we tried to include the most relevant references from 1980 onwards. Partial bibliographies on Africa South of the Sahara and the Maghreb were published earlier in the *AMUCHMA Newsletter* (GER-92b, 92d; DJE-95a, 95b), in the international journal *Historia Mathematica* (GER-94f) and in the Spanish journal for the history of science and technology *LLULL* (GER-04e).

Organisation of the bibliography

Bibliographic references directly referring to mathematical ideas in African history and cultures are included in the main body of the bibliography. Several appendices present complementary bibliographies on themes related to the main theme. For instance, as several entries in the main body refer to mathematicians of African descent, Appendix 1 presents additional bibliographic information on mathematicians of the Diaspora. As during history African mathematicians were often involved in astronomy, Appendix 3 presents an additional bibliography on time-reckoning, calendars and astronomy in African cultures. As several authors discuss the use of string figure games in mathematics education, Appendix 4 presents an

additional bibliography on string figures in Africa. As various studies referred to in the main body analyse mathematical ideas of players of African board games, Appendix 6 presents an additional bibliography on board games in Africa.

To complete the image of what research is done by African scholars in the field of the history of mathematics, Appendix 2 lists publications of African scholars on the history of mathematics outside Africa. This research may be related, for instance, to mathematics in Islamic or Arab cultures or to the application of research methodologies developed in Africa to other cultural contexts, like the analysis of mathematical ideas of basket weavers in the Amazon.

As instances of AMUCHMA projects to be continued, Appendix 5 presents examples of books and booklets published by African mathematicians. Appendix 7 lists some examples of African mathematical pioneers in the 20th century. Appendix 8 presents an example of a mathematical research field inspired by the historical study of 'sona' ideograms from Angola.

To try to make the bibliography as useful as possible, several indexes have been included, making it possible to retrieve information by subject, country, region, ethnic or linguistic group, author, journal and mathematician.

Bibliographic entries

The references are as complete as possible as we were able to collect. The entries in the bibliography are presented in alphabetic order of the authors. The bibliographic information comes in the following sequence: year of publication, author's surname, author's first name(s), co-authors or co-editors, title (in original language), translated title, publisher or journal, place and country of publication, volume, issue, page numbers or total number of pages.

A bibliographic reference is followed by a brief annotation describing the contents of the publication. No annotation means either that the title of the reference presents already a reasonable description of its contents or that we were so far not able to see ourselves the publication. An annotation between "…" means a quotation from the author or editor. We welcome any complementary information.

Invitation

If a bibliographic reference is given incompletely, it means that we have not been able to establish the complete reference, and any reader that has the missing information is invited to send it to us. Similarly we extend this invitation to any reader who knows of references that are missing in the bibliography, or is able to present additional information about the contents of a book or article in the bibliography.

Reference codes

Each entry is referred to in the indexes by a code composed of the first letters of the surname of the (first) author followed by the last two digits of the year of publication, like AUT-04. If more than one publication of the same author appears in the bibliography, a letter is added to indicate its place in the order of publications: AUT-04a, AUT-04b, etc.

A reference in one of the appendices is coded like 3-AUT-04, indicating that it appears in the third appendix.

The bibliography we present to the 6th Pan-African Congress of Mathematicians is a first attempt, and as such necessarily incomplete. We hope it may be updated regularly, and that it may become available in various languages, and in several forms (book, CD, web). We hope to include in update further an overview of mathematics in African history and cultures and further appendices, in particular, an appendix on the development of mathematics in Egypt from the Middle Ages to the 19th century.

Acknowledgements

We should like to thank wholeheartedly all members of AMUCHMA and colleagues who gave us information over the years. In particular, we should like to thank those colleagues who in the final stage of the preparation verified or complemented some of the entries:

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We should finally like to thank the successive presidents of the African Mathematical Union, Professors Aderemi Kuku (1986-1995, Nigeria), Mohamed Kerkour (1995-2000, Morocco) and Jan Persens (2000-2004, South Africa) for their encouragement of the activities of AMUCHMA.

Paulus Gerdes & Ahmed Djebbar

June 2004

Introduction to the new edition

The reactions to the first edition of this bibliography have been very positive and encouraging. For instance, the African Studies Association attributed it a 'special mention' in the 2006 Conover-Porter Award competition.

In the second edition 170 new entries are introduced. Appendix 7 about mathematical pioneers in the 20th century has been withdrawn, as this theme is analysed in detail in the book *African Doctorates in Mathematics: A catalogue* (GER-07). By consequence, Appendix 8 of the first edition becomes the new Appendix 7. The catalogue of doctorates includes also a list of over 300 doctorates in mathematics education, of which only a few are referred to in the present bibliography as examples.

Illustrations are included in the new edition. They may give an image of some mathematical ideas in African history and cultures.

Paulus Gerdes & Ahmed Djebbar

March 2007

Bibliography

A

AAB-64
1964 Aaboe, Asger: *Episodes from the early history of mathematics*, Mathematical Association of America, Washington DC (USA), 133 p.

Chapter 2 is on Euclid's construction of the regular pentagon (35-72) and chapter 4 on Ptolemy's construction of a trigonometric table (101-127).

Translation: AAB-84.

AAB-84
1984 Aaboe, Asger: *Episódios da história antiga da matemática*, Sociedade Brasileira de Matemática, Brasília (Brazil), 170 p. (in Portuguese).

Translation of: AAB-64.

ABA-86

1986 Aballagh, Mohamed & Ahmed Djebbar: *Decouverte d'un écrit mathématique d'al-Hassâr (XIIe S.): Le livre I du Kâmil* [Discovery of a mathematical text of al-Hassâr (12th century): Book I of the *Kâmil*], Pré-publications d'Orsay, Vol. 86-14, Paris (France), 20 p. (in French).

Informs about the recent discovery in Marrakech (Morocco) of the first book of *Kitâb al-kâmil* (Complete Treatise on the Art of Number), a manual written by Abû Bakr (or: Abû Zakariyâ') al-Hassâr (12th century, Maghreb). This treatise together with the little book *Kitâb al-bayân wa t-tadhkâr* of the same author played an important role in mathematics education in the Maghreb from the 12th century until the beginnings of the 16th century. Probably they constitute the oldest written proofs of mathematical activity in this region of North Africa.

ABA-87

1987 Aballagh, Mohamed & Djebbar, Ahmed: Découverte d'un écrit mathématique d'al-Hassâr (XIIe S.): Le livre I du Kâmil, *Historia Mathematica*, New York (USA), Vol. 14, 147-158 (in French).

See ABA-86.

ABA-88

1988 Aballagh, Mohamed: *Raf' al-Hijâb d'Ibn al-Bannâ (édition critique, traduction française, étude philosophique et analyse mathématique)* [Critical edition, French translation, philosophical study and mathematical analysis], doctoral thesis ('Thèse de Nouveau Doctorat'), Université Paris I - Pantheon-Sorbonne, 2 volumes, 747 p.

This thesis includes a critical edition (based on 8 manuscripts), a translation into French and an analysis of the most important mathematical treatise of the Maghrebian scientist Ibn al-Bannâ (1256-1321), born in Marrakech (Morocco). In this treatise, on the basis of philosophical or mathematical arguments the author justifies certain definitions of the 'Science of Arithmetic', like those that relate to the concepts of unity, number and base, definitions that he had given in his famous work on arithmetic, *Talkhîs*, and that had been criticized by his contemporaries. In this sense this treatise is a commentary of *Talkhîs*. But at the same time 'Science of Arithmetic' is a complement of

Talkhîs as it contains some original contributions, like the demonstration of the famous rule of signs, the justification of the algorithm for the square and cubic root of arbitrary whole numbers, the demonstration of the existence of solutions of quadratic equations by a procedure that had been completely freed from geometry and finally the deduction of propositions, like the one that permits it to express the number of combinations of n objects taken p at a time, with the help of an arithmetical formula.

ABA-89

1989 Aballagh, Mohamed & Djebbar, Ahmed: Discovery of the first part of the *Complete Book on the Art of Number* of al-Hassâr, *Revue de la Faculté des Lettres et des Sciences Humaines*, Fez (Morocco), No. 10, 189-203 (in Arabic).

ABA-92

1992 Aballagh, Mohamed: Les fractions entre la théorie et la pratique chez Ibn al-Bannâ al-Murrâkushî (1256-1321) [Fractions between theory and practice in the work of Ibn al-Bannâ al-Murrâkushî (1256-1321)], in: BEN-92, 247-259 (in French).

The article presents certain aspects of the intervention of fractions in Ibn al-Bannâ's mathematical papers, in particular as tools allowing to express and to resolve problems of inheritances and as objects of a theoretical study within the framework of the reflection of this author on the notion of number.

ABA-94

1994 Aballagh, Mohamed: *To take the veil of the methods of calculation of Ibn al-Bannâ al-Murrâkushî (d. 721/1321)*, Publications de la Faculté des Lettres et Sciences Humaines, No. 5, Université Sidi Mohamed Ben Abdallah, Dhar el-Mehrez, Fez (Morocco), 360 p. (in Arabic).

Translation into Arabic (preceded by a new Introduction) of the doctoral thesis that Mohamed Aballagh defended on May 5, 1988, at the University of Paris I-Panthéon-Sorbonne (ABA-88).

ABA-00

2000 Aballagh, Mohamed: Introduction à l'étude de l'influence d'Ibn al-Bannâ sur les mathématiques en Egypte à l'époque ottomane [Introduction to the study of the influence of Ibn al-Bannâ on mathematics in Egypt during the Ottoman epoch], in: IHS-00, 75-80 (in French).

The author presents information concerning the circulation of mathematics in the north of Africa through the example of three works of Ibn al-Bannâ.

ABAS-95

1995 Abas, Syed Jan & Salman, Amer Shaker: *Symmetries of Islamic geometrical patterns*, World Scientific, Singapore, 396 p.

Contains five chapters: 1. Islamic patterns and their geometrical construction (1-28), 2. In praise of pattern, symmetry, unity and Islamic art (29-44), 3. The gateway from Islamic patterns to invariance and groups (45-72), 4. Classification, identification and construction of the seventeen types of two-dimensional periodic patterns (73-134), 5. Islamic patterns and their symmetries (135-139), Examples (140-388). Includes various examples from North Africa.

ABD-81

1981 Abdeljaouad, Mahdi: Vers une épistémologie des décimaux [Towards an epistemology of decimals], in: *Fragments d'histoire des mathématiques,* Brochure APMEP, Paris (France), No. 41, 69-97.

ABD-86

1986 Abdeljaouad, Mahdi: L'enseignement des mathématiques en Tunisie au XIXe siècle [Mathematics education in Tunisia in the 19th century], *Cahiers de Tunisie*, Tunis (Tunisia), Vol. 41-42, No. 151-154, 247-263 (in French).

"This is the first part of a study on the teaching of mathematics in 19th century Tunisia. The paper starts by introducing the historical context, in particular the reforms promoted by Mehemet Ali in Egypt and by Chekir Sahab at-Tabaa and Mustapha Khaznadar in Tunisia. Then we describe the teaching of mathematics in the traditional school system at the Zitouna and the parallel development of a modern educational

system embodied by the Military School of Bardo (1840-1864) and by the Sadiki College (1875)."

ABD-02

2002 Abdeljaouad, Mahdi: *Introduction à l'arithmétique* [Introduction to arithmetic], Centre des Publications Universitaires, Tunis (Tunisia), 270 p. (in French).

In this handbook for first year university students the author presents a chronology of arithmetic that gives its place back to the contribution of the Arabs. Each chapter concludes with an historical appendix that shows how each civilization contributed to the development of the concerned concepts.

ABD-03

2003 Abdeljaouad, Mahdi: *Ibn al-Hâ'im, Sharh al-Uujûza al-Yâsamîniyya*, Association Tunisienne des Sciences Mathématiques, Tunis (Tunisia), 427 p. (in Arabic and French).

Edition accompanied by commentaries in Arabic and in French, of a work by the Egyptian mathematician Ibn al-Hâ'im (1352-1412). This work is entirely dedicated to a detailed commentary of the algebraic poem *al-Yâsamîniyya* of the Maghrebian mathematician Ibn al-Yâsamîn (d. 1204).

ABD-04a

2004 Abdeljaouad, Mahdi: Le manuscrit mathématique de Jerba: Une pratique des symboles algébriques maghrébins en pleine maturité [The mathematical manuscript of Jerba: A praxis of Maghrebian algebraic symbols in complete maturity], in: *Actes du Septième Colloque Maghrébin sur l'histoire des mathématiques arabes (30-31 mai 2002),* Marrakech (Morocco) (in French) (in press).

ABD-04b

2004 Abdeljaouad, Mahdi: La bilatérallité dans le discours mathématique: une contrainte institutionnelle [Bilateralism in mathematical discours: an institutional constraint], *Revue de didactique des mathématiques 'petit x'*, Grenoble (France), 20 p. (in French).

The author dedicates an important section to bilateralism in certain periods of the history of Arab mathematics.

ABDUL-95
1995 Abdullah, Ustaz Yoonus: *Yizarika: Ebira counting*, Shebiotimo, Ijebu-Ode (Nigeria), 39 p.

ABDU-93
1993 Abdullatif, Ali I.: *Ibn al-Haytham, scholar of geometry*, University of Jordan, Amman (Jordan), 626 p. (in Arabic).

The work contains 15 chapters that deal with the life of the mathematician Ibn al-Haytham and his contributions to different mathematical fields, like the conics, the calculation of areas and volumes, the regular heptagon, the lunes; and to geometrical optics.

ABE-52
1952 Abel, H.: Déchiffrement des poids à peser l'or en Côte d'Ivoire [Deciphering the gold weights in Côte d'Ivoire], *Journal de la Société des Africanistes*, Paris (France), Vol. 22, 95-114.

ABU-73
1973 Abû, Fâris: A new proof of the Arabicity of the ciphers used in the Arab Maghreb, *al-Lisân al-ᶜarabî*, Cairo (Egypt), Vol. 10, Part 1, 231-233 (in Arabic).

ACT-88
1988 *Actes du Premier Colloque International d'Alger sur l'Histoire des Mathématiques Arabes* [Proceedings of the first International Colloquium on the History of Arabic Mathematics], La Maison des Livres, Algiers (Algeria), 205 p. (in French).

The proceedings of the first International Colloquium on the History of Arabic Mathematics, held in Algiers, Algeria (1986), include the following contributions:

* Souissi, M.: The Maghrebian mathematical school: some examples of its works and certain of its particularities (9-24)
* Sadallah, A.: Some scientific practices in Algeria during the period of scientific retardation (15th –18th centuries) (25-36)

* Jaouiche, K.: Analysis and synthesis in the Arabic-Islamic mathematics: the book of Ibn al-Haytham (37-50)
* Hogendijk, J.: The king-geometer al-Mu'taman Ibn Hûd and his book of perfection (Kitab al-istikmal) (53-66)
* Sesiano, J.: The *Liber Mahamaleth*, a Latin mathematical treatise composed in the 12th century in Spain (67-98)
* Djebbar, A.: Some aspects of algebra in the mathematical tradition of the Mussulman West (99-124)
* Bebbouchi, R. & Taleb, K.: The infinitely great quantities of Thâbit Ibn Qurra (125-132)
* Aballagh, M.: The foundations of mathematics in 'Raf' al-hijâb' of Ibn al-Bannâ (1256-1321) (133-154)
* Abdeljaouad, M. & Hedfi, H.: Towards a study of the historical and mathematical aspects of the open problems of Ibn al-Khawwâm (13th century) (155-178)
* Guergour, Y.: A Maghrebian mathematician: Ibn Qunfudh al-Qasantînî (740-809 / 1339-1406) (179-190)
* Zemouli, T.: The poem of Ibn al-Yâsamîn on irrational quadratic numbers (191-203).

ACT-91

1991 *Actes du Deuxième Colloque Maghrebin sur l'Histoire des Mathématiques Arabes* [Proceedings of the second Maghrebian Colloquium on the History of Arabic Mathematics], Maghreb-Éditions, Tunis (Tunisia), 206 p. (most papers in French).

The proceedings of the second Maghrebian Colloquium on the History of Arabic Mathematics, held in Tunis, Tunisia (1988), include the following contributions:

* Abdullatif, A.: The lunes of Ibn Al-Haitham (in Arabic, 40-67; French summary p.195)
* Atik, Y: The algebraic epistle of Sinân Ibn al-Fath (10th century) (5-19)
* Benrebia, Y.: Mechanical geometry in the Arab mathematical tradition (in Arabic, 143-152)
* Bebbouchi, R.: The infinite and the Arab mathematicians (20-26)
* Borowczyk, J.: Proof and complexity of the algorithms for the solution of polynomial equations by al-Tûsi and Viète (27-52)
* Bruins, E.: Mathematics before and after the so-called Islamic period (in Arabic; summary in French, 196)

* Chelhoub, S.: Arithmetic and Algebra of Abû Kâmil Shujâ' ibn Aslam and his influence on the works of al-Karajî and of Leonardo Fibonacci (in Arabic, 23-39)
* Djebbar, A.: Some new elements on the Arabic mathematical activities in the East Maghreb (9th – 16th century) (53-73)
* Dold-Samplonius, Y: Al-Kâshî's measurement of Muqarnas (in English, 74-84)
* Folkerts, M.: The Arabic Euclid in the Latin West (85-94)
* Guillemot, M.: From Egyptian arithmetic to Arabic-Islamic arithmetic (95-105)
* Hadfi, H.: The book of Data (al-mafrûdât) of Thâbit Ibn Qurra (in Arabic; summary in French, 197-198)
* Hamzaoui, R.: About unification and normalization of Arabic scientific terminology (in Arabic; summary in French, 199)
* Jaouchi, K.: Some aspects of the evolution of the role of geometry and algebra from the 9th to the 13th century (106-124)
* Kane, A.: Arabic alphabetic numeration and decimalization of the Mandé numeration systems (West-Africa) (in Arabic; French summary p.200)
* King, D.: An overview of the sources for the history of astronomy in the medieval Maghreb (in English, 125-157)
* Laïb, A.: Infinitesimal determination through the epistle of Ibn al-Haytham on the volume of the sphere (in Arabic; French summary p.202)
* Lorch, R.: Remarks on Greek mathematical texts in Arabic (in English, 158-163)
* Martzloff, J.: The contacts between Arabic and Chinese astronomy and mathematics principally seen from Chinese sources (164-182)
* Saïdan, A.: Mathematics between the Islamic West and East (in Arabic; French summary p.203)
* Sesiano, J.: The place of geometry in establishing the foundations of Islamic algebra (183-194)
* Souissi, M.: Some problems and their Arabic solutions (in Arabic; French summary p.205)
* Zemouli, M.: Birth and evolution in the Arabic algebraic terminology (in Arabic; French summary p.206)
* Zerrouki, M.: Fractions in the Maghrebian mathematical tradition between the 12th and the 15th century through an anonymous manuscript (in Arabic, 97-109)

ACT-98a

1998a *Actes du Troisième Colloque Maghrebin sur l'Histoire des Mathématiques Arabes* [Proceedings of the 3rd Maghrebian Colloquium on the History of Arabic Mathematics], Office des Publications universitaires, Algiers (Algeria), Vol. 1, 280 p. (mostly in French and some papers in English), Vol. 2, 111 p. (in Arabic).

The proceedings of the 3rd Maghrebian Colloquium on the History of Arabic Mathematics, held in Tipaza (Algeria, 1990), include the following contributions:

* Aballagh, M.: The mathematical thinking of Ibn Haydûr (Vol. 2, 5-22)
* Al Daffa, A.: The role of al-Khwârizmî in algebra (Vol. 2, 103-104, (abstract)
* Alaoui, J.: The problematics of the links between mathematics and metaphysics or from mathematics to the first philosophy in the work of Ibn Rushd (Vol. 2, 105-107, abstract)
* Bebbouchi, R.: Arabic heritage in the redaction and the explication of mathematical texts (5-11)
* Berggren, J. L.: Geometric Methods in Medieval Islam: The case of the Azimuth Circles (13-21, in English)
* Brentjes, S.: The Arab transmission of the Introduction to Arithmetic in non-mathematical works during the 9th century (23-29)
* Calvo, E.: The graphic resolution astrologic questions in Andalusia (31-44)
* Cassinet, J.: The fund of ancient Arab mathematical manuscripts in the Laurentian library in Florence (45-59)
* Comes, M.: The deferent of Mercury in the al-Andalus' Equatoria, (61-71, in English)
* Dhombres, J.: The theory of proportions in the 17th century: Variety of Arab or Latin influences based on Greek foundations, and new developments (277, abstract)
* Djebbar, A.: Mathematical activities in the cities of the Central Maghreb (9th - 14th century) (73-115)
* Folkerts, M.: Remarks on al-Khwârizmî's *Arithmetic* (117-123, in English)
* Gari, L.: The unity of measurement in Islamic architecture (Vol. 2, 39-63)

* Guergour, Y.: Introduction to the art of geometry of Qustâ Ibn Lûqâ (m. 910) (Vol. 2, 65-71)
* Guillemot, M.: The methods of simple false position in Egyptian and Arab mathematics (125-145)
* Hadfi, H.: The contribution of the Island of Jerba to mathematical activities (Vol. 2, 107-108, abstract)
* Hogendijk, J. P.: The study of conic sections in the Arab tradition (147-158)
* Hoyrup, J.: "Oxford" and "Gherardo da Cremona" on the relation between two versions of al-Khwârizmî Algebra (159-178, in English)
* Jaouiche, K.: Overview of the problem of tangent circles in the works of Ibrâhîm Ibn Sinân, Ibn al-Haytham and Viète (179-193)
* King, D. A.: Maghrebi Astronomical Instruments (278, abstract, in English)
* Koelblen, S.: Ahmad Ibn Yûsuf and his treaty on the theory of proportions (195-206)
* Laabid, E.: The donations in medieval mathematics, the example of al-Hubûbî (207-220)
* Lapousterle, P.: Description of three mathematical manuscripts in the Ahmed Baba library of Timbuktu (Mali) (277-278, abstract)
* Lorch, R.: Graphical Methods in Spherical Astronomy in treatises by Habash al-hâsib and al-Mâhânî (221-226)
* Martzloff, J. C.: The Qi Zheng Tuibu of Bei Lin (vers 1477) (227-237)

  Mawaldi, M.: Kamâl ad-Dîn al-Fârisî and his book *The essential rules for the foundations of useful things* (Vol. 2, 93-101)
* Mesbahi, M.: Unity between accident and essence: Ibn Sînâ and Ibn Rushd (Vol. 2, 73-91)
* Mili, A.: Breathe life into arithmetic algorithms of the past millennium (Vol. 2, 107, abstract)
* Rebstock, U.: If Numbers are right: on the Use of Reckoning in the Islamic Middle Age (239-249, in English)
* Sadallah, A.: The epistle on the celestial sphere of Ibn Hamadush (12th -13th century) (Vol. 2, 23-30)
* Sesiano, J.: Some constructions of simple magic squares in Arab texts (251-262)

* Shawqi, J.: The science of magic squares in Islamic civilization (Vol. 2, 104, abstract)
* Souissi, M.: The teaching of mathematics in Arabic in the Maghreb, particularly in Tunisia during the 13th century and in the first half of the 14th century of the hegira (Vol. 2, 31-37)
* Taha, A.: The Arabic version of the lemmas of Archimedes (263-275).

ACT-98b

1998b *Actes du 5e Colloque Maghrebin sur l'Histoire des Mathematiques Arabes* [Proceedings of the 5th Maghrebian Colloquium on the History of Arabic Mathematics], Imprimerie Impak, Tunis (Tunisia), 257 p. (most papers are in French)

The proceedings of the 5th Maghrebian Colloquium on the History of Arabic Mathematics held in Tunis, Tunisia (December 1-3,1994), include the following contributions:

* Ben Miled, M.: Undecidability in the work of as-Samaw'al (7-11)
* Bebbouchi, R.: The memory of symbols from Antiquity to our days (12-20)
* Berggren, J.L.: Abû Sahl on a Lacuna in Archimedes (in English) (21-26)
* Calvo, E.: Analysis of six geometrical models to calculate the length of the solar year in Ibn al-Hâ'im' "al-Zîj al-Kâmil fî l-Ta'âlîm" (in English) (27-39)
* Cassinet, J.: The treatise concerning the methods of numerical problems by al-Husayn as-Samarqandi (d. 1235) (40-48)
* Comes, M.: The unknown 'Equatoria' of Abûl-Hasan al-Marrâkushî (49-61)
* Djebbar, A.: The Euclidian arithmetic tradition in the Kitâb al-istikmâl of al-Mu'taman and its continuation in Andalusia and in the Maghreb (62-84)
* Dold-Samplonius, Y.: Al-Kâshî's Constructions of Arches, Vaults and Domes (in English) (85-100)
* Folkerts, M. & Lorch, R.: The Mathematical and Astronomical Writings of al-Khwârizmî (in English) (109-119)
* Guergour, Y.: Comparative study of the species 2 and 3 of the book *Istikmâl* of al-Mu'taman ibn Hûd (d. 1085) (in Arabic, 31-46)

* Guillemot, M.: Is it possible to speak of methods of false position in the case of Egyptian mathematics? (120-147)
* Hoyrup, J.: On the Mensuration of the *Liber Mensurationum* (in English) (148-183)
* Koelblen, S.: The debates around the definition of proportionality in book V of Euclid's *Elements* in the Arab and Latin traditions from the 10th to the 17th century (184-196)
* Lamrabet, D.: The reasons for the study of mathematics according to some Maghrebian scholars (101-108)
* Mawaldi, M.: Edition and study of the epistle *The redaction of Taqiy ad-Dîn Ibn Ma'rûf of the two proofs of the Banû Mûsâ brothers of Heron's formula* (47-58)
* Pinel, P. & Taha, A.: On a premature, anonymous Arabic version of Menelaus' *Sphaerica* conserved in the Latin translation of Gérard de Crémone (198-225)
* Puig, R.: The *Risâla fî l-'amal bi l-shabîha* of Ibn al-Bannâ al-Marrâkushî (1256-1321) (226-235)
* Rebstock, U.: The *al-Mu'âmalât* of Ibn al-Haytham (236-262)
* Samso, J.: The tables of planetary equations in the *Minhâj* of Ibn al-Bannâ (263-272)
* Schubring, G.: Actual tendencies in the research on the institutional history of the sciences and its application to the Islamic culture (273-283)
* Souissi, M.: Epîstle of al-Kindî on the determination of the dimensions of an optical instrument (in Arabic, 9-30)
* Taha, A.: Note on Menelaus' *Sphaerica* of in the version of at-Tûsî (284-296)

ACT-98c

1998c *Actes du Colloque de Marrakech sur "Le raisonnement géométrique, enseignement et apprentissage"*, Imprimerie Walili, Marrakech (Morocco), 214 p.

These proceedings of an international colloquium held in Marrakech on "Geometrical Reasoning, Teaching and Learning" (May 28-31, 1997) include the following papers related to the history of mathematics in Africa:

* Ahmed Djebbar: Geometrical reasoning in the Arab mathematical tradition (9th –15th centuries) (89-121)
* A. El-Idrissi: The instruments used in geometrical reasoning: history and didactics (134-144).

ADA-82
1982 Adaaku, J.: *The mathematical heritage of the Tiv people*, M.Ed. project, Ahmadu Bello University, Zaria (Nigeria).

ADD-66
1966 Addy, L.: The Entebbe Maths, *Ghana Teachers' Journal*, Vol. 52, No. 4, 1-12.

ADJ-95
1995 Adjamagbo, Pascal Kossivi & Diop, Cheikh M'Backé: Sur la mesure du cercle et de la sphère en Égypte ancienne [On measure of the circle and the sphere in Ancient Egypt], *Ankh, Revue d'Égyptologie et des Civilisations africaines*, Paris (France), No. 4/5, 1995/1996, 222 – 245 (in French).

Discusses the calculation of the area of a circle (Papyrus Rhind) and of the surface of a hemisphere (Moscow Papyrus).

ADL-88
1988 Adler, Jill B.: Newspaper-based mathematics for adults in South Africa, *Educational Studies in Mathematics*, Dordrecht (Netherlands), Vol. 19, No. 1, 59-78.

ADL-91
1991 Adler, Jill B.: How to do it? Politics and practice in mathematics education in South Africa, *Perspectives in education*, Johannesburg (South Africa), Vol. 13, No. 2, 21-31.

ADL-95
1995 Adler, Jill B.: Insights from mathematics education developments in South Africa in transition, *Mathematics Education Research Journal*, Vol. 6, No. 3, 101-112

ADL-96
1996 Adler, Jill B.: Secondary School Teachers' Knowledge of the Dynamics of Teaching and Learning Mathematics in Multilingual Classrooms, doctoral thesis, University of The Witwatersrand, Johannesburg (South Africa).

ADL-01
2001 Adler, Jill: *Teaching mathematics in multilingual classrooms*, Kluwer, Dordrecht (Netherlands), 169 p.

AFO-90
1990 Afolayan, Adebisi (Ed.): *A vocabulary of primary science and mathematics in nine Nigerian languages*, Nigerian Educational Research and Development Council, Enugu (Nigeria), 3 volumes.

AGB-69
1969 Agbo, Casimir: La numération au Dahomey, *Études Dahomeennes* (Nouvelle Série), Porto Novo (Benin), Nos. 14-15, 59-110; 1970, No. 16, 5-112.

Study concluded in 1942. It presents the numerals in several languages spoken in the Republic of Benin: Fon or Fongbé, Mina or Ghen, Ghin or Ghinbe, Nagot or Yoruba.

AGW-98
1998 Agwu, Nkechi: *Mathematical teaching techniques inherent in Nigerian cultures*, paper presented at the Conference in Honor of the 65th Birthday of Ubiratan D'Ambrosio, January 6, Baltimore (USA).

AHMA-92
1992 Ahmadi, M. H.: On Egyptian fractions, in: *Proceedings of the 21st Annual Iranian Mathematics Conference*, Isfahan (Iran), 1-20.

AHR-22
1922 Ahrens, W.: Die magischen Quadrate al-Bunis [The magic squares of al-Buni], *Der Islam*, Vol. 12, 157-177 (in German).

AIS-92a
1992a Aïssani, Djamil: Exhumation des témoignages sur les activités mathématiques à Béjaïa au moyen âge , in: J. Cassinet (Ed.), Mathématiques Arabes et Occident, Actes des Journées de l'AFEMAM, Toulouse (France), 10 p. (in French).

Several papers confirm the existence of an important mathematical school in Béjaïa (Algeria) during the Middle Ages. The objective of this article is to clarify the strategy of the GEHIMAB association concerning the exhumation of testimonies: history of its education, methods and disciplines (name, objects, tools, algorithms, proofs and domains), of the outstanding personalities of the time.

AIS-92b

1992b Aïssani, Djamil: Impact de l'exhumation des témoignages concernant les activités mathématiques à Béjaïa à l'époque médiévale dans l'enseignement actuel [Impact of the exhumation of testimonies concerning mathematical activities in medieval Béjaïa for education today], in: *Printemps de la Didactique des Mathématiques à Fès*, *Actes du Premier Séminaire Franco-Maghrébin sur la Didactique des Mathématiques*, Fez (Morocco), 1-7 (in French).

Indicates the possible educational, didactic and cultural impact that the exhumation of testimonies of mathematical activities in medieval Béjaïa may have on the current education.

AIS-93

1993 Aïssani, Djamil: *Bougie á l'époque médièvale: les mathématiques au sein du mouvement intellectuel* [Béjaïa during the Middle Ages: Mathematics within the intellectual movement], IREM de Rouen, Rouen (France), 112 p. (in French).

Analyses the bio-bibliographical sources and presents a synthesis of testimonies known on mathematical activities about Béjaïa in the Middle Ages. In particular, the author indicates names on which the attention of the specialists of the history of sciences did not yet focus, and he proposes a certain number of tracks of reflection and study that would allow to analyse better the contents of studied disciplines.

AIS-94

1994 Aïssani, Djamil: Les mathématiques dans la Bougie médiévale et Fibonacci [Mathematics in the medieval Béjaïa and Fibonacci], in: *Leonardo Fibonacci, Il tempo, le opere, l'eredità scientifica*, Pacini Editore, Pisa (Italy), 67-82 (in French).

The article presents the political, cultural and economic context in which the scientific activities in Béjaïa were developed during the Middle Ages. It describes some aspects of mathematical production in this city (science of calculation and algebra) and it concludes with a remark about Fibonacci, who had studied mathematics in Béjaïa.

AIS-95a

1995a Aïssani, Djamil: *Bougie médiévale – Centre de transmission méditerranéen* [Medieval Béjaïa – Centre of Mediterranean transmission], in IRE-95, 499 – 506 (in French).

The first part of the paper analyses the structure of the scientific environment in medieval Béjaïa, as well as the Mediterranean peculiarities that have played a role in the development of mathematical activities. The second part concerns the role of the city as a centre of influence and exchange with Christianity. The process of transmission is evoked through a dozen of scholars, who were native of various regions of the Mediterranean, and who were specialized in different mathematical disciplines.

AIS-95b

1995b Aïssani, Djamil: Les mathématiques à Bougie médièvale et Fibonacci [Mathematics in medieval Béjaïa and Fibonacci], *Revue Algérienne de l'Éducation*, Algiers (Algeria), No. 2, 5-19 (in French).

Overview of research realised during the last decades about the role of Béjaïa as a scientific centre in the 12th and 13th century.

AIS-96a

1996a Aïssani, Djamil: Le mathématicien Eugène Dewulf (1831-1896) et les manuscrits médiévaux du Maghreb [The mathematician Eugène Dewulf (1831-1896) and the medieval manuscripts in the Maghreb], *Historia Mathematica*, New York (USA), Vol. 23, No. 3, 257-268 (in French).

Presents some aspects of the investigations of the French geometer Eugène Dewulf, founding member of the Mathematical Society of France, concerning Maghrebian medieval manuscripts.

AIS-96b

1996b Aïssani, Djamil & Mechehed, Djamel Eddine: *Catalogue de la Collection de Manuscrits Ulahbib (Béjaïa)*[Catalog of the Ulahbib Collection of Manuscripts (Béjaïa)], GEHIMAB, Béjaïa (Algeria), 189 p. (in French).

The Ulahbib Collection regroups the manuscripts found in the *Khizana* (learned library of manuscripts) of the Sheik Lmuhub. This library was established in the middle of 19th century in the mountain of Beni

Ourtilane in the Southeast of Kabylia (Algeria). The catalog is the first of this kind on the manuscripts of Kabylia. The manuscripts are classified by discipline, among which: Science of calculation (15 - 19), Algebra and Geometry (20 - 21), Science of Inheritance (22 - 27), Astronomy (28 - 33), Astrology (34 - 40), Logic (106 - 111).

AIS-98a

1998a Aïssani, Djamil: *Mathématiques et Mathématiciens en Algérie (de l'époque médiévale au XIX-ème siècle)* [Mathematics and mathematicians in Algeria from the Middle Ages until the 19th century], in: *Alger fête la Science*, Bibliothèque Nationale d'El Hamma, Algiers (Algeria), 5-12 (in French).

Presents an overview of 900 years of mathematical activities in Algeria. Three aspects are discussed: (1) The contribution of the Algerian centers of knowledge to the creation of the medieval mathematical tradition of the Maghreb; (2) The mathematical knowledge of the local Algerian scholars in the 18th and 19th century; (3) The contribution of several French mathematicians in Algeria (François Arago, Eugène Dewulf, Albert Ribaucour, ...).

AIS-98b

1998b Aïssani, Djamil & Mechehed, Djamel Eddine: *La Khizana (bibliothèque savante) de Cheikh Lmuhub: lettrés locaux et culture écrite en Kabylie au milieu du XIXe siècle* [The Khizana (learned library) of Sheik Lmuhub: local scholars and written culture in Kabylia in the middle of the 19th century], Association Gehimab Ed., Béjaïa (Algeria), 170 p. (in French).

Regroups the information collected during the investigation that led to the elaboration of the Catalog of the Ulahbib Collection of manuscripts (Béjaïa). In particular, it tries to indicate the knowledge available to the local scholars in Kabylia in the middle of the 19th century. Chapters 8 and 9 are dedicated to the mathematical disciplines (55 - 80).

AIS-99a

1999 Aïssani, Djamil: Centri del Sapere Maghrebino ed i loro Rapporti con l'Occidente Cristiano [The Maghrebian Centres of knowledge and their relationship with the Christian West], in: *Proceedings of the International Seminar "Natura, Scienza*

e Società nel Mediterraneo", Unesco, Cosenza (Italy), 121 – 127 (in Italian).

Based on subjects already analysed by the Italian National Commission of UNESCO (Elaboration of knowledge, circulation of knowledge, history of journeys and travelers, borders and contact zones of in the Mediterranean), this article tries to describe the contribution of the Maghrebian centres of knowledge in the process of the development of scientific knowledge, since the period with translations (in the East), until the fixation of the medieval mathematical tradition of the Maghreb.

AIS-00a

2000a Aïssani, Djamil: Qal`at Beni Hammad à l'époque médiévale: les mathématiques au sein du mouvement intellectuel [Qal`at Beni Hammad in the medieval epoch: mathematics within the context of the intellectual movement], *Actes du RAMAII (Rencontre d'Analyse Mathématique et ses Applications – dans le cadre du WMY 2000)*, Msila (Algeria), 1 – 15 (in French).

The author proposes a synthesis of testimonies known (of bio-bibliographical or scientific sources) on mathematical activities in Qal`at Beni Hammad from the 11th to the 13th century, giving information on the connections with Ifrikiya. He analyses the influence of the educational tradition of Qal`at on the development of mathematical activities in Béjaïa.

AIS-00b

2000b Aïssani, Djamil: Le séjour Algérien du célèbre mathématicien François Arago (1808-1809) [The stay in Algeria of the famous mathematician François Arago (1808-1809), in: *Actes de RMA'2000 (Rencontre des Mathématiciens Algériens - dans le cadre du WMY 2000)*, Algiers (Algeria), 1-9 (in French).

After having continued up to Barcelona the first measurement of the terrestrial meridian, the famous French mathematician François Arago (1786-1853) made a spectacular visit of Kabylia (1808). He lectured afterwards the first course of both theoretical and applied probability theory in France, entitled "social arithmetic". Besides the presentation of Arago's "Algerian observations", the purpose of this article is to stimulate a reflection on the teaching of probability theory in Algeria during the last twenty-five years.

AIS-02a

2002a Aïssani, Djamil: La Zawiyya de Chellata: un Institut Supérieur au Fin Fond de la Kabylie, in: *Actes de la Journée d'études "Les Manuscrits Berbères au Maghreb et dans les Collections Européennes ",* CCL Arles & IREMAM, Aix-en-Provence (France), 22 p. (in French) (in press).

This article presents one of the most renowned religious and scientific centres of Northern Africa. Founded at the beginning of 18th century, the Zawiyya of Chellata was the centre of activity of the renowned astronomer Mohammed Ben Ali Cherif ash Shellati, commentator of the as-Susi. A presentation and an analysis of the work *Ma`alim al-Istibsar bi Tafdhil al-Azman wa Manafi` al-Bawadi wa l-Amsar* (commented overview of the times and benefactions of the regions and the countries), more known in Kabylia under the name of *Hashiyat Ibn `Ali Sherif `ala `Ilm al-Falak Susi* , are included.

AIS-02b

2002b Aïssani, Djamil & Valerian, Dominique: Mathématiques, Commerce et Société à Béjaïa (Bugia) au moment du séjour de Leonardo Fibonacci [Mathematics, commerce and society in Béjaïa (Bugia) at the time of the stay of de Leonardo Fibonacci], in: Enrico Giusti & Marco Tangheroni (Eds.), *Leonardo Fibonacci. Mathematica e Società nel Secolo XIII*, Pisa (Italia), 19 p. (in French) (in press).

This article analyses the relationship between the environment in which the Italian mathematician Léonardo Fibonacci (1170-1240) lived, notably the environment of traders from Pisa in Béjaïa (Algeria), and the formulation of his mathematical knowledge. Indeed, in the first part of the *Liber Abaci*, the explanations and demonstrations of Fibonacci are constantly based on examples and problems which come from the daily activities of these traders and sailors: problems of exchange, weights and measures, of loads of vessels, of price calculations. Also, the products, which appear are mostly those that one finds on the market of Béjaïa.

AISS-83
1983 Aïssata, Moumouni Kane: *Étude de quelques problèmes pédagogiques et linguistiques concernant l'enseignement des mathématiques au Niger* [Study of some pedagogical and linguistic problems concerning the teaching of mathematics in Niger], doctoral thesis, Université Paris VII (France) (in French).

AJO-78
1978 Ajose, Sunday A.: *Research on mathematics education in sub-Saharan Africa*, paper presented at the 21st Annual Meeting of the African Studies Association, Baltimore MD (USA), 16 p.

Examines the research that has been done in mathematics education in Africa. It discusses the scope and the significance of these studies and concludes with an outline of needed research

AKI-85
1985 Akin, F. & Fapenle, I.: *Indigenous mathematics: a case study of the Aweri community of Ogun state, Nigeria*, B.Sc. project, Ahmadu Bello University, Zaria (Nigeria).

AKIN-92
1992 Akinyele, O.: Adegoke Olubummo [1923-1992]: The man, the teacher, the mathematician, in MEM-92, i-ii.

AKK-02
2002 Akkar, Mohamed: L'enseignement des mathématiques dans l'enseignement secondaire maghrébin [Mathematics teaching in secondary schools of the Maghreb region], *Zentralblatt für Didaktik der Mathematik - International Reviews on Mathematical Education*, Karlsruhe (Germany), Vol. 34, No. 4, 179-185 (in French).

Analyses "the following questions. Does the mathematics teaching in the secondary schools in the Maghreb prepare to University studies and more specifically does it initiate students to modern science and technology? Is anyone able to understand mathematics or is mathematics only accessible to the happy few. Is it a means of selection? Is mathematics omnipresent in our modern society? What relationships can one hope to find between mathematics and other disciplines? Has mathematics evolved to such an abstract and formal

state that it seems difficult to relate it to any other topic? All these questions are discussed in relationship with the particular problems in the Maghreb, namely the mathematics program as taught today in these countries."

AKO-88
1988 Akonambi, Ngilambi tè: Opérativité sémantique, schématique et algorithmique dans l'apprentissage de la notion d'application et de bijection: étude des relations avec la réussite ou l'échec chez les élèves zaïrois, 13-15 ans, doctoral thesis, Université de Bordeaux 2 (France) (in French).

ALB-90
1990 Alberich, Julio Cola: Números simbólicos y rituales en el Africa subsahariana [Symbolic numbers and rituals in sub-Saharan Africa], in: *Homenagem a J. R. dos Santos Júnior*, Instituto de Investigação Científica Tropical, Lisbon, 99-104 (in Spanish).

Describes the symbolical significance or use of the numbers two, three, four, five, six and seven in various African cultures.

ALA-01
2001 Alausa, Yesir Adeleke: Gender differences in the Namibian students' perception of their mathematics classroom environment, *Zimbabwe Journal of Educational Research*, Harare (Zimbabwe), Vol. 13, No. 1, 22-37

Investigates how Namibian secondary school students perceive their mathematics classroom environments, particularly determining differences in perception that can be attributed to students' and teachers' gender and the interaction between the two. The study was carried out in the Khorixas educational region of Namibia.

ALBE-91
1991 Albertini, Tamara: La quadrature du cercle d'ibn al-Haytham : solution philosophique ou mathématique? [The quadrature of the circle by Ibn Al-Haytham: a philosophical or a mathematical solution], *Journal for the History of Arabic Science*, Aleppo (Syria), Vol. 9, Nos. 1-2, 5-21, 132 (in French).

ALE-89
1989 Ale, Sam O.: Mathematics in rural societies, in: C. Keitel, P. Damerow, A. Bishop & P. Gerdes (Eds.), *Mathematics, Education, and Society*, UNESCO, Paris (France), 35-38.

Gives examples of the oral mathematics used by the nomadic Fulani (Nigeria): elements of statistics, inequality, probability, geometry, and basic algebra, and suggests that a relevant curriculum for rural communities must build upon the mathematics existing in these communities. The paper shows also how the Fulani use symbols to represent the number of cows or goats they possess: 100 is represented by two short sticks in the form V, 50 by two sticks in the form X, 10 by one stick _ , 3 by three sticks | | |, etc.

ALM-47
1947 Almeida, António de: Sobre a matemática dos indígenas da Guiné Portuguesa [About the mathematics of the natives of Portuguese Guinea], *Boletim Cultural da Guiné Portuguesa*, Lisbon (Portugal), Vol. 6, 389-440 (in Portuguese).

Deals with numerals, arithmetic operations, measurement, monetary system and time reckoning in Guinea-Bissau.

AMA-00
2000 Amazigo, John C.: *Mathematics phobia: diagnosis & prescription*, National Mathematical Centre, Abuja (Nigeria), 31 p.

ANB-63
1963 Anbouba, Adel: Un algébriste arabe, Abû Kâmil Shuga' Ibn Aslam [An Arab algebraist: Abû Kâmil Shuga' Ibn Aslam], *Horizons Techniques du Moyen Orient*, Beyrouth (Lebanon), No. 3, 6-15.

AND-80
1980 Andrzejeweskis, B. W.: The use of Somali in Mathematics and Science, *Afrika und Übersee*, Berlin (Germany), Vol. 63, 103-117.

Discusses the ways in which Somali language has been used in mathematics and science teaching in Somalia since 1972, substituting the use of the foreign languages, Italian, English and Arabic. In

particular, it analyses the formation of new scientific terms by composition, semantic shift and borrowing.

ANG-97
1997 Angoué Ndoutoume, Robert: *Genèse du nombre et conservation numérique chez l'enfant Fang du Nord-Gabon* [Genesis of number and number conservation among the Fang children of north Gabon], doctoral thesis, Université de Montpellier 3 (France), 278 p. (in French).

ANI-92
1992 Animalu, A. O. E.: Professor Chike Obi [b. 1921], *Journal of the Nigerian Mathematical Society*, Ibadan (Nigeria), Vol. 11, No. 1, i-iv.

Introduction to special issue in honour of Professor Chike Obi, 114 p.

ANS-96
1996 Anselin, Alain: Les gestes du nombre, in: Alain Anselin, *La Cruche et le Tilapia, Une lecture africaine de l'Egypte nagadéenne*, Éditions de l'UNIRAG, Abymes (Guadeloupe), 103-115.

Presents a comparative analysis of number words in Ancient Egypt and several African languages. The chapter analyses different ways of counting and presents a 'human ecology' of the numbers in Ancient Egyptian.

ANT-98
1998 Antoine, Yves: *Inventeurs et savants noirs* [Black inventors and scientists], L'Harmattan, Paris (France), 142 p.

ANZ-88
1988 Anzenge, Hirazaan H., Bako, Danladi W., Ezenduka, Patricia N., Nyomo, Daniel J. & Sambo, Madaki H.: *Indigenous mathematical algorithms*, B. Ed. project, Ahmadu Bello University, Zaria (Nigeria).

Reports on field work on mathematical algorithms used by unschooled, illiterate of the Igbo, Tiv and other home communities of the students in the southern part of Kaduna State (Nigeria).

ARC-27

1927 Archibald, Raymond Clare: Bibliography of Egyptian mathematics with special references to the Rhind mathematical papyrus and sources of interest in its study, Mathematical Association of America, Oberlin O. (USA), 84 p.

ARC-50

1950 Archibald, Raymond C.: The first translation of Euclid's elements into English and its source, *American Mathematical Monthly*, Washington DC (USA), Vol. 57, 443-452.

ARG-94

1994 Argoud, Gilbert (Ed.): *Science et vie intellectuelle à Alexandrie (I^e^-IIIe siècles ap. J.C.* [Science and intellectual life in Alexandria (1st – 3rd Cent.)], Publications de l'Université de Saint Etienne, Saint Etienne (France), 225 p. (in French).

Collection of articles of which the majority are dedicated to Heron of Alexandria (1st century) and his contributions to mechanics and mathematics.

ARI-65

1965 Arif, Aida S. & Hakima, Ahmad M. Abu: Descriptive catalogue of Arabic manuscripts in Nigeria: Jos Museum and Lugard Hall Library, Kaduna, London (UK), 216 p.

This catalogue contains over one thousand manuscripts, including manuscripts about mathematics and astronomy.

ARM-62

1962 Armstrong, Robert G.: *Yoruba numerals*, Nigerian Institute of Social and Economic Research / Oxford University Press, Ibadan (Nigeria), 36 p.

"The traditional Yoruba numeral system is a fascinating chapter in the history of mathematics and of the development of human thought. It is a vigesimal system, which is to say that it reckons the higher numbers by twenties (*ogún*). Thus 'forty' is 'two twenties' (*ogójì*, from *ogún èjì*) and 'sixty' is 'three twenties' (*ogóta*). … it is based on finger-and-toe counting … " (p. 5). The author proposes a decimal number system "which uses Yoruba words throughout and in a regular way" (p. 21).

ARM-71
1971 Armstrong, Robert G. & Bamgbose, Ayo: *Mathematical concepts in Yoruba, a manual for Yoruba teachers*, University of Ibadan Institute of African Studies, Ibadan (Nigeria), 14 p. (mimeo).

Brief manual prepared for the use of Yoruba speaking primary teachers of Entebbe Mathematics (Africa Mathematics Program).

ARO-95
1995 Aronson, Lisa: Review of Gerdes' & Bulafo's *Sipatsi: Technology, Art and Geometry in Inhambane* (GER-94c), *African Arts*, Los Angeles CA (USA), Vol. 28, No. 2, 89-90.

ART-99
1999 Artmann, Benno: *Euclid — the creation of mathematics*, Springer-Verlag, New York (USA), 368 p.

ASA-88
1988 Asar, Reda Mosad El-Said: *A critical appraisal of mathematics education research carried out in Egypt, with special reference to techniques of research methodology and statistical analysis*, doctoral thesis, University of Cardiff (UK).

ASC-88
1988 Ascher, Marcia: Graphs in cultures (II): a study in ethno-mathematics, *Archive for History of Exact Sciences*, Berlin (Germany), Vol. 39, No. 1, 75-95.

This paper discusses and analyses interest in continuous tracing of figures as it is evidenced in Africa among the Bushoong and Cokwe (Angola / Congo / Zambia region). Included are figures, statements about the cultural context, and associated geometric and topological ideas. Emphasis is on the structure of the figures and also, where possible, processes of construction are elaborated.

ASC-90
1990 Ascher, Marcia: A River-Crossing Problem in Cross-Cultural Perspective, *Mathematics Magazine*, Washington DC (USA), Vol. 63, No. 1, 26-29.

Analyses the logical structure behind traditional story puzzles from Algeria, Cape Verde Islands, Ethiopia, Liberia, Tanzania, Zambia.

ASC-91

1991 Ascher, Marcia: *Ethnomathematics: A multicultural view of mathematical ideas*, Brooks / Cole, Pacific Grove CA (USA), 203 p. (Chapman & Hall / CRC, 1994).

Sections 2.3 and 2.4 deal with mathematical aspects of sand drawings among the Kuba (Congo / Zaire) and the Cokwe (Angola); section 4.8 deals with mathematical aspects of river-crossing puzzles.

Translation: ASC-98.

ASC-97

1997 Ascher, Marcia: Malagasy *Sikidy*: A Case in Ethnomathematics, *Historia Mathematica*, New York (USA), Vol. 24, 376-395.

"*Sikidy* is a system of divination that plays a significant role in the lives of the people of Madagascar. Here we focus on the mathematical ideas, which it embodies. Formal algebraic algorithms are applied to initial random data, and knowledge of the internal logic of the resulting array enables the diviner to check for and detect errors. *Sikidy* and the mathematical ideas within it are placed in their cultural and historical contexts."

ASC-98

1998 Ascher, Marcia: Mathématiques d'ailleurs, Nombres, forms et jeux dans les sociétés traditionelles, Editions du Seuil, Paris (France), 278 p. (in French).

French edition of the already classical study ASC-91. Translation and afterword by Karine Chemla and Serge Pahaut.

ASC-00

2000 Ascher, Marcia: Review of Gerdes' *Geometry from Africa* (GER-99a), *Mathematical Reviews*, Lancaster PA (USA), MR2000e:01009.

ASC-02

2002 Ascher, Marcia: *Mathematics Elsewhere: An Exploration of ideas Across Cultures*, Princeton University Press, Princeton (USA), 207 p.

Some sections are related to the African continent. The first chapter is about divination and includes detailed discussions of Sikidy, as practiced in Madagascar, and Ifa, as practiced by the Yoruba (Nigeria). In the third chapter, which is about calendars, there is a brief mention of the Akan calendar. The fifth chapter includes a detailed discussion of the Gada system (essentially a system of social organization) of the Borana. And the seventh chapter has brief mentions of the Cokwe *sona* (Angola) and designs of the Kuba (Congo).

ASC-03
2003 Ascher, Marcia: Review of Verran's *Science and an African Logic* (VERR-01), *Current Anthropology*, Chicago IL (USA) (in press).

ASH-00
2000 Ashbacher, Charles: Review of P. Gerdes' *Geometry from Africa* (GER-99a), *Journal of Recreational Mathematics*, Amityville NY (USA), Vol. 30, No. 1, 59.

ASS-00
2000 Assali, Sidi Amar: *The mathematical instruments of astronomy through the work of al-Hasan al-Murrakushi in the "Book of principles and objectives of the science of time"*, 'Magister' thesis in the History of Mathematics, École Normale Supérieure d'Alger, Algiers (Algeria), 210 p. (in Arabic).

Analysis of the arithmetical, algebraic, geometrical and trigonometric tools which intervene in the statement and the resolution of the problems of astronomy and more particularly those that are related to the various instruments described in the work of al-Murrâkushî.

ATK-61
1961 Atkins, Guy: Notes on the concords and classes of Bantu numerals, *African Language Studies*, London (UK), Vol. 2, 42-48.

Describes "the diversity of grammatical forms of the Bantu numerals as a whole."

AUJ-86
1986 Aujac, Germaine: Le rapport 'di isou' (Euclide V, définition 17): Définition, utilisation, transmission [Le relationship 'di

isou' (Euclid V, definition 17): Definition, utilisation, transmission], *Historia Mathematica*, New York (USA), Vol. 13, No. 4, 370-386.

AUJ-93

1993 Aujac, Germaine: *La Sphère, instrument au service de la découverte du monde: D'Autolycos de Pitane à Jean de Sacroboso* [The sphere, instrument in the service of th discovery of the world: From Autolycos of Pitane to John of Sacroboso], Paradigme, Caen (France), 380 p.

The articles in this collection are grouped under three headings: 1. Spherics and geocentrics; 2. Spherics; 3. Practical applications. The following contributions concern the history of mathematics in Africa: Euclid and Spherics (151-156); Greek geography in Alexandria in the 2nd century (347-368).

Review: VIT-95.

B

BAB-02

2002 Babunguru, A.: *Infusing ethnomathematics and ethnoscience in the curriculum*, paper presented at the symposium 'African Universities in the 21st Century', University of Illinois, Chicago (USA), 25th 27 April, 2002.

Paper based on the author's experience at the University of Botswana.

BAD-97

1997 Badmus, Gani Ademola & Ocho, Lawrence Offie: *Science, mathematics, and technology education in Nigeria*, Everlead, Lagos (Nigeria), 322 p.

BALL-97

1997 Ballieu, Michel & Aïssani, Djamil: Le savoir Mathématique disponible en Petite Kabylie au XIX-ème siècle [Mathematical knowledge available in the Small Kabylia in the 19th century], *Actes de la Conference Internationale "Béjaïa et sa région à travers les âges: Histoire, Société, Sciences, Culture"* [Proceedings of the International Conference "Béjaïa and its region through the ages: history, society, sciences, culture"], GEHIMAB, Béjaïa (Algeria), 252 – 260 (in French).

"This paper stresses the principal sources of interest in and the level of knowledge of mathematics in the 19th century in Small Kabylia (Algeria). The recent discovery in the Ath Urtilan area of a scholarly library of manuscripts (the Ulahbib Collection) allows some conclusions through a method of mathematical analysis of social facts."

BAN-66a

1966a Bantu Education Department: Proposed contracted numerals for northern Sotho, *Bantu Education* (South Africa), February, 12-16.

Proposal by Radio Bantu of contracted numerals for northern Sotho (South Africa).

BAN-66b
1966b Bantu Education Department: Proposed improvement on contracted numerals for northern Sotho, *Bantu Education* (South Africa), September, 17-21.

BAN-69
1969 Bantu Education Department: Northern Sotho numerals, *Bantu Education* (South Africa), August, 5-6.

Presents the contracted numerals for northern Sotho officially recognised by the Lebowa Territorial Authority (South Africa).

BARN-87
1987 Barnard, Anna: *'n Histories-pedagogiese ondersoek na die opleiding van wiskunde-onderwysers vir die primêre skool* [A historic-pedagogical investigation of the training of mathematics teachers for primary schools], doctoral thesis, University of Pretoria (South Africa) (in Afrikaans).

BAR-71
1971 Barreto, Manuel Cabrera: Die Zahlwörter der Altkanarier [The number words of the ancient Canarians], *Almogaren*, Hallein (Austria), Vol. II, 151-167 (in German).

"The author examines critically the information available regarding the numerals in the language of the natives of the Canary Islands. Their basis of counting is the decimal system, which is clearly proven by all recent critical and historical studies. Ancient Canarian and Berber numerals are closely akin as regards language and counting, which shows the North African origin of the ancient Canary islanders also in this domain. Apparently Semitic traits can be explained by the presence of Negro and Berber slaves in the Canary Islands, as stated by Bosch Millares, an assumption which is better established than that of a linguistic hybridization of the Canarian natives" (p.167).

BARR-93a
1993a Barrios García, José: Notas sobre los conocimientos matemáticos y astronómicos de los Benahoaritas, según las fuentes escritas anteriores al siglo XVII [Remarks about the mathematical and astronomical knowledge of the Benahoaritas according to written sources before the 17th century] (paper

presented at the 1st Las Palmas Congress on History, Art and Geography, Las Palmas (Spain), 15-19 March 1993), 6 p. (mimeo) (in Spanish).

BARR-93b
1993b Barrios García, José: *Matemáticas tribales y cultura. El caso de las canarias Bereberes durante los siglos XIV y XV* [Tribal mathematics and culture. The case of the Canarian Berbers during the 14th and 15th century] (paper presented at the 6th Congress of the Federation of Spanish Anthropology Associations, La Laguna (Canary Islands, Spain), 6-11 September 1993), 9 p. (mimeo) (in Spanish).

BARR-94a
1994 Barrios García, José: Notas sobre los conocimientos matemáticos y astronómicos de los antiguos palmeros según fuentes escritas [Notes about the mathematical and astronomical knowledge of the ancient inhabitants of Las Palmas according to written sources], in: *I Encuentro de geografía, Historia y Arte de la Ciudad de Santa Cruz de La Palma*, Santa Cruz de la Palma (Canary Islands, Spain), Vol. 1, 112-118 (in Spanish).

"Drawing evidence from archaeological and ethnographical literature, this paper summarizes a number of mathematical practices of the Berber populations of the Canary Islands in the 14th – 15th centuries, in relation with their economical and socio-cultural context."

BARR-94b
1994 Barrios García, José: La lista de numerales canarios atribuida a Antonio Cedeño [The list of Canarian numerals attributed to Antonio Cedeño], in: *X Coloquio de Historia Canario-Americana, Las Palmas de Gran Canaria*, Cabildo Insular (Canary Islands, Spain), Vol. 2, 859-878 (in Spanish).

Analyses Cedeño's list (about 1687) of Berber numerals from the Canary Islands and compares it to other reported lists of these numerals.

BARR-96a

1996 Barrios García, José: The Guanche lunar calendar and the Virgin of Candelaria (Tenerife, 14th -15th centuries), in: Schlosser, W. (Ed.), *Proceedings of the Second SEAC Conference (Bochum 1994),* Astronomisches Institut der Ruhr-Universität, Bochum (Germany), 151-162.

"Study of the lunar calendar of the Berber populations of Tenerife island in the 14th – 15th centuries, the so called *Guanches*, mainly based on the ethnographic written sources arisen from the Spanish conquest in late 15th century. It is argued that the first moon of the Guanche lunar calendar was fixed by the heliacal rise of *Canopus* about middle August. The Guanche cult to this star was later transferred to the main catholic cult of the island after the conquest: the *Virgin of Candelaria*."

BARR-96b

1996 Barrios García, José: Cuentas que pasaron ante Juan de Anchieta, escribano público. Un estudio sobre los sistemas de numeración y algoritmos de cálculo utilizados en Tenerife a mediados del siglo XVI [Reckoning in front of Juan de Anchieta, the public scribe. A study of the numeration systems and algorithms used in Tenerife in the middle of the 16th century], in: Morales Padrón, F. (Ed.), *XI Coloquio de Historia Canario-Americana (Las Palmas 1994)*, Cabildo, Las Palmas de Gran Canaria (Canary Islands, Spain), Vol. 1, 409-426 (in Spanish).

"An arithmetical study of the protocols of the notary Juan de Anchieta (father of José de Anchieta, apostle of Brazil), preserved at the 'Archivo Histórico Provincial de Santa Cruz de Tenerife', and dating from 1541."

BARR-97a

1997a Barrios García, José: *Sistemas de Numeración y Calendarios de las Poblaciones Bereberes de Gran Canaria y Tenerife en los Siglos XIV-XV* [Numeration systems and calendars of the berber populations of Grand Canary and Tenerife in the 14th and 15th century], doctoral thesis, Universidad De La Laguna, Tenerife (Canary Islands, Spain), 244 p. (in Spanish).

"Doctoral thesis on the number systems and calendars used by the Berber populations of Grand Canary (Canarians) and Tenerife (Guanches) in the 14th and 15th centuries. It is established for both islands the use of a pure 10-based system (deeply related with both proto-Berber and ancient Egyptian numeral systems), the existence of census of their inhabitants, as well as the existence of systematic records of lunar, solar and sidereal counts. Not withstanding several basic similarities, the calendrical practices of both islands show several conceptual differences. On the one hand, the Canarians recorded numerical, astronomical and calendar data by means of geometrical figures (squares, triangles, circles, etc.), painted in black, red and white, using the *acano* (a chessboard of 3 x 4 squares representing 12 moons), to record data, as well as to perform 'lunisolar' and eclipse counts; nothing of which can be documented for the Guanches. On the other hand, in contrast with some (weak) notices supporting the existence of a Sirius calendar in Grand Canary, the main result with respect to the Guanche calendar is the fundamental role played by the phases of the star Canopus. Additional evidence drawn from continental Berbers, supports the antiquity and widespread of a Canopus cosmological system in Northwest Africa."

BARR-97b

1997b Barrios García, José: Tara: A study of the Canarian astronomical pictures. Part II: The acano chess board, in: Jaschek, C. & Atrio Barandela, F. (Eds.), *Proceedings of the IVth SEAC meeting "Astronomy and Culture*," Universidad de Salamanca, Salamanca (Spain), 47-54.

Paper presented at a meeting of the SEAC (Société Européenne pour l'Astronomie dans la Culture; European Society for Astronomy in Culture). The paper presents the 'acano' as a Berber lunar calendar and shows "how to number its squares to force the solisticial, equinoctial and eclipse moons to move across the board with very simple and stable patterns. These patterns provide a safe and clear mnemonic guide for performing on the acano an easy calculus of seasonal and eclipse moons over extended periods of time, just using the difference in days of the lunar year with either the solar year or the eclipse year to perform an elementary saw function on the squares. This calculus establishes the octaeteris, the metonic cycle and the 135-moon eclipse cycle as basic periods of the 'acano'. … The proposed calculus on the acano would reveal an unsuspected high level of Canarian

mathematical astronomy [in the 14th and 15th century] and pose the question of the origin of this set of techniques."

BARR-98

1998 Barrios García, José: The Canary Islands also count: on the ancient number systems of Northwest Africa, in: Oliveras, M. L.; Fernández, J. & Fuentes, J. (Eds.), *Ethnomathematics and Mathematic Education. Building an Equitable Future. Proceedings of the First International Conference on Ethnomathematics (Granada 1998)*, Universidad de Granada, Granada (Spain), 12 p. (CD-ROM).

"In spite of its title, Claudia Zaslavsky's book *Africa Counts* is admittedly devoted to sub-Saharan cultures, as clearly shows the great blank recovering all the north-western part of the different distribution maps depicting the entire continent … The purpose of the presentation is, precisely, to contribute to fill this notable void in Zaslavsky's book, presenting sound evidence about the number system used by the Berber populations of Grand Canary Island in the 14th – 15th centuries, considered against its wider North African context. While briefly summarizing the role of mathematics in the socio-economical organization of the Island, the paper stresses the importance of the Canarian studies in relation to modern research on ancient North African mathematics."

BARR-99

1999 Barrios García, José: Tara: A study of the Canarian astronomical pictures. Part I: Towards an interpretation of the Gáldar Painted Cave, in: Florin Stanescu (Ed.), *Proceedings of the Third SEAC Meeting*, Lucian Blaga University, Sibiu (Romania), 24-36.

Paper presented at a meeting of the SEAC (Société Européenne pour l'Astronomie dans la Culture; European Society for Astronomy in Culture). "The first part of the paper analyses the archaeological, ethnohistorical and linguistic evidence that led the author to propose that in the 14th and 15th century the Berber populations of Grand Canary Island systematically recorded numerical, astronomical and calendar data by means of certain geometrical figures named 'tara', painted in white, red and black on wooden planks and on the walls of certain caves. One main conclusion of the study was the discovery of

the use of a type of chess board of 3 vertical x 4 horizontal squares, named 'acano' by the author, to represent 12 moons."

BARR-00

2000 Barrios García, José: Sobre la existencia de censos de población entre los antiguos Canarios (Gran Canaria, Siglos XIV-XV) [On the existence of population census among the ancient Canarians (Grand canary island, 14th – 15th centuries)], in: Morales Padrón, F. (Ed.), *XIII Coloquio de Historia Canario-Americana*, Ediciones del Cabildo de Gran Canaria, Las Palmas de Gran Canaria (Canary Islands, Spain), 1697-1704 (in Spanish).

"Paper gathering the ethnographical written sources supporting the existence of systematic census of inhabitants carried out by the Berber populations of Grand Canary Islands in the 14th - 15th centuries, just those preceding the Spanish conquest of the Island."

BARR-02a

2002 Barrios García, José: Investigaciones sobre matemáticas y astronomía Guanche. Parte I: Señales para recuerdo [Research on Guanche mathematics and astronomy. Part 1: Signs for recording], in: Morales Padrón, F. (Ed.), *XIV Coloquio de Historia Canario-Americana* (Las Palmas 2000), Cabildo, Las Palmas de Gran Canaria (Canary Islands, Spain), 508-517 (in Spanish) (CD-ROM).

"The first part of this paper opens with a very brief summary of what is known about the Guanches (ancient Berber inhabitants of Tenerife island) in the 14th - 15th centuries, and goes on analyzing the archaeological and ethnographical evidence documenting their use of marks on the mummies, strings of clay beads, as well as marks and pictures on wood planks and stones for recording several kind of data, mainly calendar and numerical ones."

BARR-02b

2002 Barrios García, José: Investigaciones sobre matemáticas y astronomía Guanche. Parte II: Sistemas de numeración [Research on Guanche mathematics and astronomy. Part 2: Numeration systems], in: Morales Padrón, F. (Ed.), *XV Coloquio de Historia Canario Americana (Las Palmas 2002)*,

Cabildo, Las Palmas de Gran Canaria (Canary Islands, Spain) (in Spanish) (CD-ROM).

"The second part of this paper studies the numeral systems used by the ancient inhabitants of Tenerife Island to perform several economical and socio-cultural practices as mentioned in the written ethnographic sources. On this ground an hypothesis is formulated about the numeral systems used, the name of the numbers and the scope of the system."

BARRO-01
2001 Barrow, John D.: Review of Gerdes' *Geometry from Africa* (GER/99a), *PLUS Magazine*, Cambridge University, Cambridge (UK) [available online at: http://plus.maths.org/issue15/reviews/book2/].

BAS-75
1975 Bascom, William R.: *African dilemma tales*, Mouton Publishers, The Hague (Netherlands), 162 p.

BASH-97
1997 Bashmakova, Izabella G.: *Diophantus and Diophantine Equations*, The Mathematical Association of America, Washington DC (USA), 90 p.

Presents the works of Diophantus of Alexandria, focusing on Diophantus' general methods of obtaining rational solutions of indeterminate equations of the second and third order. The second part of the book considers the evolution of the theory of Diophantine equations from the Renaissance to the middle of the 20th century.

BAU-95
1995 Bauval, R. G.: Logistics of the Shafts in Cheops' Pyramid. A Religious "Function" Expressed with Geometrical Astronomy and Built in Architecture, *Discussions in Egyptology*, Oxford (UK), No. 31, 5-13.

BAZ-95
1995 Bazin, Maurice & Modesto Tamez: *Math across cultures*, Exploratorium Teacher Activity Series, San Francisco CA (USA), 48 p.

Booklet with suggestions for teachers on how to use a multicultural approach in the maths classroom. Chapter 3 is on mathematics in Africa: *Counting like an Egyptian: Egyptian math* (23-32).

BAZ-02
2002 Bazin, Maurice & Modesto Tamez: Math and Science Across Cultures, Activities and Investigations from the Exploratorium, The New Press, New York (USA), 176 p.

Papers with activities related to mathematics from Africa are:
* Paulus Gerdes: Sona: Sand drawings from Africa (3-15)
* Robert Lange: Madagascar Solitaire: Playing Games (25-31)
* Maurice Bazin & Modesto Tamez: Counting like an Egyptian: Egyptian math (47-59).

BEA-55
1955 Beart, Charles: *Jeux et jouets de l'ouest africain* [Games and toys of West Africa], Institut Français d'Afrique Noire (today: Institut Fondamental d'Afrique Noire), Dakar (Senegal), 2 volumes, 800 p. (in French).

Presents a descriptive inventory of the West-African games collected by the author. Among the games with mathematical aspects are: Order games (arrange and collect), Combination games like checker-board games as 'awalé', hop-scotch playing, 'tiouk-tiouk', 'dili', magic squares and games that resemble the game of draughts, and Gambling games (cowry games).

Review: DOU-89.

BEC-57
1957 Becker, Oskar: *Das mathematische Denken der Antike* [Mathematical thinking of the Ancients], Vandenhoeck & Ruprecht, Göttingen (Germany), 128 p. (New edition: 1966, 131 p.) (in German).

Includes chapters on mathematics in Egypt.

BEC-61
1961 Becker, Oskar: Über die Proportionen der ägyptischen Pyramiden I. Die klassischen Pyramiden des Alten Reiches [On the proportions of the Egyptian pyramids I. The classic

pyramids of the Old Kingdom], *Praxis der Mathematik*, Köln (Germany), Vol. 3, 260-266 (in German).

BEL-02
2002 Bello, Muhammad Yahuza: Indigenous Hausa Number System, *Afrikanische Arbeitspapiere* (African Working Papers), Institut für Afrikanistik, Universität Köln, Köln (Germany), No. AAP-70, 191-198.

Analyses the structure of the Arabic- and English-free number words in the Hausa language (Nigeria). The indigenous Hausa number system has base ten, with numerals for the first four powers of ten. Both the additive and the subtractive principles are used. For instance, 98 is expressed as 'xari ba goma' (100 less 2).

BELL-95
1995 Belluccio, A.: Le nombre caché dans l'Oeil d'Horus [The hidden number in the Horus Eye], *Discussions in Egyptology*, Oxford (UK), No. 32, 7-8.

BENC-74
1974 Benchekroun, Ridwan: The works of Ibn al-Bannâ and his method of notation, *Al-Manâhil*, Rabat (Morocco), Vol. 33, 207-229 (in Arabic).

BEN-92
1992 Benoit, Paul; Chemla, Karine & Ritter, Jim (Eds.): *Histoire de fractions, fractions d'histoire* [History of fractions, fractions of history], Birkhäuser Verlag, Basel (Switzerland), 436 p. (in French).

Proceedings of the international colloquium on the History of Fractions held in Paris, France (1987). The following chapters concern the history of mathematics in Africa:

* J. Ritter: Metrology and the prehistory of fractions (19-35);
* M. Caveing: The arithmetic status of the Egyptian 'quantième' (39-52);
* M. Guillemot: Do notational and operational practices allow us to speak of Egyptian fractions? (53-70);
* A. Djebbar: The treatment of fractions in the Arab mathematical tradition of the Maghreb (223-246);

* M. Aballagh: Fractions between theory and practice in the work of Ibn al-Banna al-Marrakusi (1256-1321) (247-259).

BENTA-99
1999 Bentaleb, Farès: *Al-Qalasâdî, Commentary on the "Summary of arithmetic operations"*, Dâr al-Gharb al-islami, Beyrouth (Lebanon), 614 p. (in Arabic).

Critical edition and translation into French of the commentary written by the mathematician al-Qalasâdî (d. 1486) on the famous handbook "Summary of arithmetic operations" of the Maghrebian mathematicians Ibn al-Bannâ (d. 1321).

BENT-77
1977 Bentley, A.: Symmetry in pattern reproduction by Scottish and Kenyan children, *Journal of Cross-Cultural Psychology*, Beverly Hills CA (USA), Vol. 8, No. 4, 415-424.

BERGD-76
1976 Berg, Daniel J. Van Den: *The training of mathematics teachers in the Republic of South Africa and in some western countries*, South African Human Sciences Research Council, Pretoria (South Africa), 353 p.

BERI-00
2000 Berisso, Taddesse: The riddles of number nine in Guji-Oromo culture, *Journal of Ethiopian Studies*, Addis Ababa (Ethiopia), 2000, vol. 33, no. 1, p. 49-66

In Guij-Oromo culture (southern Ethiopia), the number nine is associated with critical times, with ghosts, and with illness and death. This is evident in proverbs, in children's games, and when a woman is pregnant with and gives birth to her ninth child.

BER-87
1987 Bernal, Martin: *Black Athena, the Afroasiatic roots of classical civilization*. Vol. 1: *The fabrication of Ancient Greece 1785-1985*, Free Association Books, London (UK), 576 p.

BER-91
1991 Bernal, Martin: *Black Athena, the Afroasiatic roots of classical civilization*. Vol. 2: *The archaeological and documentary evidence*, Rutgers University, New Brunswick NJ (USA), 736 p.

BERTE-92
1992 Berté, Zakaria: *L'apprentissage de contenus logico-mathématiques opératoires formels chez des élèves du secondaire de Côte d'Ivoire*, doctoral thesis, Université de Montréal (Canada) (in French).

BERT-02
2002 Bertolini, Marina: *Arte e Geometria nelle Culture Africane* [Art and Geometry in African Cultures], Dipartimento di Matematica, Universitá degli Studi di Milano, Milan (Italy), 60 p. (in Italian).

Presents an introduction to Gerdes' studies on geometrical ideas embedded in African cultural activities.

BHA-71
1971 Bhagat, H.: Socialist transformation of Tanzania and the teaching of mathematics, in: *Mathematics Institute 1971*, University of Dar es Salaam, (Tanzania).

Presents examples of the mismatch between the socialist aims, and both the construction of, and examples used in, the mathematics texts in use in secondary schools in Tanzania.

BIN-96
1996 Binsbergen, Wim van: Regional and historical connections of four-tablet divination in Southern Africa, *Journal of Religion in Africa*, Leiden (Netherlands), Vol. XXVI, 1, 3-29

Presents regional and historical connections of four-tablet divination.

BIS-01
2001 Bisher, Hisham Barakat: *The effect of using Bedouin ethnomathematics in teaching primary stage mathematics courses on the achievement and behaviour in daily life*, Masters

thesis in ethnomathematics, Ain Shams University, Cairo (Egypt) (in Arabic).

Analyses mathematical ideas in nomad Bedouin culture and possibilities to embed them into mathematics teaching.

BLE-00

2000 Bleicher, Michael N.: Egyptian fractions, in: Anatole Beck, Michael N. Bleicher & Donald W. Crowe, *Excursions into Mathematics. The Millennium Edition*, A. K. Peters, Natick MA (USA), 421-434.

Presents a "number of solved and unsolved problems" related to Egyptians fractions. The problems which "arise from the oldest known mathematical manuscripts" are "easily accessible to the mathematical novice" (p. 421).

BOC-88

1988 Bockaire, A.: *Mathematics used and needed by Mende farmers of Moyamba and Kailahun districts in Sierra Leone*, International Development Research Centre, Dakar (Senegal).

BOG-87

1987 Bogoshi, Jonas; Naidoo, Kevin & Webb, John: The oldest mathematical artifact, *The Mathematical Gazette*, London (UK), Vol. 71, 294.

"A small piece of the fibula of a baboon, marked with 29 clearly defined notches, may rank as the oldest mathematical artifact known. Discovered in the early seventies during an excavation of Border Cave in the Lebombo Mountains between South Africa and Swaziland, the bone has been dated to approximately 35 000 BC." It has been noted that the bone resembles calendar sticks still in use in Namibia.

BON-89

1989 Bonini, Nathalie: *Numération et évaluation du temps dans trois sociétés d'Afrique orientale. L'exemple des Borana, des Chaga et des Maasai*, Mémoire présenté en vue de la maitrise d'ethnologie, Université de Paris-X Nanterre, Laboratoire d'Ethnologie et de Sociologie comparative, Paris (France), 94 p. (in French).

'Maitrise' thesis in ethnology on numeration and time measurement in three societies of Eastern Africa, discussing the examples of the Borana, the Chaga and the Maasai.

BOP-98

1998 Bopape, Mathume: *South African new mathematics curriculum: people's mathematics for people's power?* (on-line available at: www.nottingham.ac.uk/csme/meas/papers/bopape .html)

Focuses "on the place of People's Mathematics for People's Power in the new South African mathematics Curriculum. Particular attention is given to one aspect of the People's way of life, *botho*, that enable blacks to sustain togetherness among the people, through serious economic hardships, leading to the people's regaining of political strength. Questions are raised as to what extent the framework of the new curriculum provides room for the previously disenfranchised and whether they will be able to gain access to economic and political power through engaging the strength of *botho*."

BOU-95

1995 Bouazzi, Marie: *Mathématiques et compositions décoratives régulières: Faiences murales tunisoises du XVIIIe siècle* [Mathematics and regular decorative compositions: Tunisian mural faience from the 18th century], Institut Technologique d'Art, d'Architecture et d'Urbanisme de Tunis, Tunis (Tunisia), 28 p. (mimeo) (in French).

BOUQ-62

1962 Bouquiaux, Luc: A propos de numération: L'emploi du système décimal et du systéme duodécimal dans la langue birom (Nigéria septentrional), *Africana Linguistica*, Tervuren (Belgium), 7-10 (in French).

Describes the traditional duodecimal numeration system of the Birom (Plateau Province, central Nigeria) and its interaction with decimal systems.

BOUZ-99

1999 Bouzari, Abdelmalek: *Les sections coniques dans la tradition mathématique arabe à travers le traité attribué à al-Khâzin* [The conic sections in the Arab mathematical tradition through

a treatise attributed to al-Khâzin (10th century)], 'Magister' thesis, E.N.S., Algiers (Algeria), 271 p. (in French).

The thesis contains a historical presentation of the conic sections in the Greek and Arab traditions, a critical edition – on the basis of two existing manuscripts – of a text from the 10th century, attributed to the mathematician al-Khâzin, and a mathematical analysis of the contents of this text.

BOUZ-03

2003 Bouzari, Abdelmalek: Procédures des nombres pensés en Occident musulman [Procedures of thought numbers in the Islamic West], *Actes du Colloque International "De la Chine à l'Occitanie, chemins entre arithmétique et algèbre" (Toulouse, 22-24 septembre 2000)*, Éditions CIHSO, Toulouse (France), 15-27.

BOW-91

1991 Bowen, Alan C.: Euclid's 'Sectio canonis' and the history of Pythagoreanism, in: Bowen, Alan C. (Ed.), *Science and philosophy in classical Greece*, Garland, New York (USA), 167-187.

BRA-94

1994 Brading, Mary: *Mathematics from History: The Egyptians*, Educational Television Company, London (UK), 48 p.

The topics covered in the booklet include: number system, arithmetic and fractions calculations, calendar, measurement of land area and boundaries, standardized weights, construction of pyramids and temples, logical and strategic mathematical games and puzzles. The explanations are followed by activity, resource information sheets for children and notes for teachers.

BREE-03

2003 Breen, Chris; Vithal, Renuka; Mtetwa, David & Setati, M.: Joining and re-forming: Towards a strategy for optimising SAARMSTE influence in the broader mathematics education community, *Pythagoras*, Johannesburg, (South Africa), Vol. 57, 19-26.

"This was the title of an energetic Round Table Discussion, which took place during the SAARMSTE Conference in Swaziland in

January 2003. In the belief that the issues raised are extremely important and need to be debated by the wider community, the main presenters (authors of this article) have attempted to capture the essence of their arguments in this article."

BRE-97

1997 Brenner, Klaus-Peter: *Chipendani und Mbira: Musikinstrumente, nicht-begriffliche mathematik und die Evolution der harmonischen Progressionen in der Musik der Shona in Zimbabwe* [Chipendani and Mbira: Musical instruments, non-lexical mathematics and the evolution of the harmonic progressions in the music of the Shona in Zimbabwe], Vandenhoeck & Ruprecht, Göttingen (Germany), 559 p. (plus 2 CDs) (in German).

The author presents an English language summary (367-374), entitled: "Hypotheses on the role of the 'chipendani' mouth bow, of the non-lexical mathematics and of the 'mbira' lamellophone in the evolution of the harmonic progressions of Shona music."

BRE-04

2004 Brenner, Klaus-Peter: *Die kombinatorisch strukturierten Harfen- und Xylophonpattern der Nzakara (Zentralafrikanische Republik) als klingende Geometrie – eine Alternative zu Marc Chemilliers Kanonhypothese* [The Combinatorically Structured Harp and Xylophone Patterns of the Nzakara (Central African Republic) as Sounding Geometry – an Alternative to Marc Chemillier's Canon-Hypothesis], Holos-Verlag, Bonn (Germany), 209 p. (plus 1 Audio-CD) (in German with English summary).

BRI-79

1979 Bril, Blandine: Analyse des nombres associés à l'homme et à la femme en Afrique de l'Ouest [Analysis of the numbers associated with male and female in West Africa], *Africa: Journal of the International African Institute*, London (UK), Vol. 49, No. 4, 367-376 (in French).

Presents an analysis of the numbers associated with male and female in West Africa. "The opposition and the complementarity of male and female have been brought out in different societies with the aid of pairs of symbols based on left-right, points of the compass, color, etc.

Number also appears to be an apt means of expressing this idea. By studying the rituals of birth and death in West Africa it has been possible to distinguish four pairs of numbers widely associated with male and female: (3,4), (4,3), (9,7) and (5,4). The geographical distribution of these pairs of numbers shows marked grouping. Explanations of the use of the different numbers are generally based on myths or on physiological differences between the sexes and are not very convincing. However, the pairs of numbers are widely used in the 'numerical system' of a society that determines the ritual calendar. These systems also make great use of the number 7 and the author contends that not only is this widely seen as the sum of 4 and 3 in areas using that pair but that some evidence can be found that the area using the pair (5,4) tends similarly to use the sum, 9, in its numerical system for the ritual calendar" (p. 376).

BRIT-79

1979 British Council (Ed.), *The development of teaching materials for school mathematics*, British Council, London (UK).

Contains the following papers concerning African countries:

* Kenya: Mathematics in society (30-36)
* Swaziland: Language problems in mathematics education (43-46)
* Swaziland: Mathematics and language (47-51)
* Tanzania: Language problems in teaching mathematics (52-57).

BRO-88

1988 Bronshtehn, V. A.: *Claudius Ptolemy. Second Century AD*, Nauka, Leningrad (Petersburg, Russia), 24 p. (in Russian).

Edited with a preface and an afterword by A. Gurshtein. An overview that besides the contributions of Ptolemy to astronomy includes discussions of his work in optics, music, geography, and astrology.

BROW-81

1981 Brownson, C. D.: Euclid's 'Optics' and its compatibility with linear perspective, *Archive for History of Exact Sciences*, Berlin (Germany), Vol. 24, No. 3, 165-194.

BRU-45

1945 Bruins, Evert: Over de benadering van pi/4 in de Aegyptische meetkunde [On the approximation of pi/4 in Egyptian

geometry], *Indagationes Mathematica*, Amsterdam (Netherlands), Vol. 7, 11-15 (in Dutch).

BRU-52
1952 Bruins, Evert: Ancient Egyptian arithmetic: 2/n, *Indagationes Mathematica*, Amsterdam (Netherlands), Vol. 14, 81-91.

BRU-57a
1957a Bruins, Evert: The icosahedron from Heron to Pappus, *Janus, the International Journal for History of Science, Technology, Medicine and Pharmacy*, Amsterdam (Netherlands), Vol. 46, 173-182.

BRU-57b
1957b Bruins, Evert: Platon et la table 2/n égyptiennes [Plato and the Egyptian table 2/n], *Janus*, Amsterdam (Netherlands), Vol. 46, 253-263 (in French).

BRU-62
1962 Bruins, Evert: Rationalitätsfragen bei Pyramiden [Questions of rationality in pyramids], *Praxis der Mathematik*, Köln (Germany), Vol. 4, 281-284 (in German).

BRU-64
1964 Bruins, Evert: Babylone et Héron versus Euclide [Babylon and Heron versus Euclid], *Revue d'Assyriologie et d'Archéologie Orientale*, Vol. 58, 173-181 (in French).

BRU-65
1965 Bruins, Evert: The Egyptian shadow clock, *Janus*, Amsterdam (Netherlands), Vol. 52, 127-137.

BRU-75a
1975a Bruins, Evert: The part in ancient Egyptian mathematics, *Centaurus*, Copenhagen (Denmark), Vol. 19, 241-251.

BRU-75b
1975b Bruins, Evert: Contribution to the interpretation of Egyptian mathematics, *Actes du XXIXe Congrès International des Orientalistes, Section Égyptologie*, L'Asiathèque, Paris (France), Vol. 1, 25-28.

BRU-77
1977 Bruins, Evert; P. Sijpesteijn & K. Worp: Fragments of mathematics on papyrus, *Chronique d'Egypte*, Brussels (Belgium), Vol. 52, 105-111.

BRU-81a
1981a Bruins, Evert: Egyptian Arithmetic, *Janus*, Amsterdam (Netherlands), Vol. 68, 33-52.

BRU-81b
1981b Bruins, Evert: Reducible and trivial decompositions concerning Egyptian arithmetic, *Janus*, Amsterdam (Netherlands), Vol. 68, 281-297.

BRU-83
1983 Bruins, Evert: On some hau-problems: a revision, *Janus*, Amsterdam (Netherlands), Vol. 70, 229-262.

BRU-88
1988 Bruins, Evert; W. Liesker & P. Sijpesteijn: A Ptolemaic papyrus from the Michigan collection, *Zeitschrift für Papyrologie und Epigraphik*, Bonn (Germany), Vol. 74, 23-28.

BRU-90a
1990a Bruins, Evert: Ptolemaic and Islamic trigonometry: the problem of the qibla, *Janus*, Amsterdam (Netherlands), Vol. 73, 125-148.

BRU-90b
1990b Bruins, Evert: Review of Robins' & Shute's The Rhind mathematical papyrus (ROB-87), *Mededelingen van het Wiskundig Genootschap*, Utrecht (Netherlands), Vol. 33, No. 3.

Reproduced in: AMUCHMA Newsletter, No. 7, 1990.

BRUM-93a
1993a Brummelen, Glen Robert van: *Mathematical Tables in Ptolemy's 'Almagest'*, doctoral thesis, Simon Fraser University (Canada), 428 p.

Attempts to understand the methods used to construct the tables in the *Almagest*.

BRUM-93b

1993b Brummelen, Glen Robert van: Ptolemaic Interpolation: Method, Application and Tabulation in the *Almagest*, in: Tattersall, James (Ed.), *Proceedings of the CSHPM/SCHPM 19th Annual Meeting*, Carleton University, Ottawa (Canada), Vol. 6, 71-80.

An explanation for the errors that appeared in the interpolation tables in Ptolemy's *Almagest*, and a reconstruction of the tables that lends insight into Ptolemy's numerical methods.

BRUM-94

1994 Brummelen, Glen Robert van: Lunar and Planetary Interpolation: Tables in Ptolemy's *Almagest*, *Journal for the History of Astronomy*, Chalfont St. Guiles (UK), Vol. 25, 297-311.

Errors in the numerical tables in Ptolemy's *Almagest* are usually quite minor. Several auxiliary tables, however, contain some more serious errors. These errors are analyzed and explained.

BUS-67

1967 Busard, Hubertus L. L.: The translation of the 'Elements' of Euclid from the Arabic into Latin by Hermann of Carinthia (?), *Janus*, Amsterdam (Netherlands), No. 54, 1-140.

BUS-68

1968 Busard, Hubertus L. L.: *The translation of the Elements of Euclid from the Arabic into Latin by Hermann of Carinthia (?)*, Brill, Leiden (Netherlands), 142 p.

BUS-77

1977 Busard, Hubertus L. L.: *The translation of the Elements of Euclid from the Arabic into Latin by Hermann of Carinthia (?), Books VII-XII*, Mathematisch Centrum, Amsterdam (Netherlands), 198 p.

BUS-83

1983 Busard, Hubertus L. L.: *The Latin translation of the Arabic version of Euclid's 'Elements' commonly ascribed to Adelard of Bath: books I-VIII and books X.36-XV.2*, Pontifical Institute of Medieval Studies, Toronto (Canada), 425 p.

BUS-87
1987 Busard, Hubertus L. L. (Ed.), *The Medieval Latin translation of Euclid's 'Elements': Made directly from the Greek*, Steiner Verlag, Wiesbaden (Germany), 411 p.

BUS-92
1992 Busard, Hubertus L. L. & Folkerts, Menso (Eds.): *Robert of Chester's (?) redaction of Euclid's Elements, the so-called Adelard II version*, Birkhauser, Basel (Switserland), 2 Vol., 959 p.

BUS-01
2001 Busard, Hubertus L. L. (Ed.): *Johannes de Tinemme's redaction of Euclid's Elements, the so-called Adelard III version*, Steiner, Stuttgart (Germany), 2 vol., 632 p.

BUI-96
1996 Buikema-Draisma, Frouke: *Concepts of multiplication of children and teachers in Mozambique*, in: Luis Puig & Angel Gutiérrez: *Proceedings of the 20th Conference of the International Group for the Psychology of Mathematics Education,* University of Valência, Valência (Spain), Vol. 2, 169–176.

BUI-99
1999 Buikema-Draisma, Frouke: O que significa 3 x 4? O uso de duas definições de multiplicação no ensino em Moçambique' [What is the meaning of 3 x 4? The use of two definitions of multiplication in Mozambican schools], in, *Actas do ProfMat 99*, Associação de Professores de Matemática, Lisbon (Portugal), 195–213.

BUL-84
1984 Bulmer-Thomas, I.: Guldin's theorem - or Pappus's?, *Isis*, Madison WI (USA), Vol. 75, No. 277, 348-352.

BUR-52
1952 Burssens, Amaat: Les numéraux en Amashi (Kivu) [The numerals in Amashi], *Kongo-Overzee*, Antwerpen (Belgium), Vol. 43, No. 1, 66-76 (in French).

Lists the numerals in Amashi, the language of the Abashi (Kivu, Congo / Zaire) and discusses grammatical aspects.

BUR-54
1954 Burssens, Amaat: La numération [Numeration], in: Amaat Burssens, *Introduction à l'étude des langues bantoues du Congo belge* [Introduction to the study of Bantu languages of Belgian Congo], Kongo-Overzee Bibliotheek, Vol. VIII, Antwerpen (Belgium), Chapter 13 (in French).

BURS-58
1958 Burssens, Herman: Arithmétique [Arithmetic], in: H. Burssens, *Les peuplades de l'Entre Congo-Ubangi (Ngbandi, Ngbaka, Mbandja, Ngombe et Gens d'Eau)*, International African Institute, London (UK), 171-172 (in French).

Presents brief information on the numeration systems among the Ngbandi, Ngbaka ['7'='6+1'; '9'='5+4'], Mbandja ['7'='6+1'; '9'='8+1'] and Ngombe (Congo / Zaire).

BURT-45
1945 Burton, H. E.: The optics of Euclid, *Journal of the Optical Society of America*, Washington DC (USA), Vol. 35, 357-372.

BYN-67
1967 Bynon-Polak, L.: L'expression des ordinaux dans les langues bantoues [The expression of ordinal numbers in Bantu languages], *Africana Linguistica II, Annales du Musée Royal de l'Afrique Centrale, Sciences Humaines*, Tervuren (Belgium), No. 55, 127-160 (in French).

Presents a comparative linguistic study of the construction of the words for ordinal numbers in Bantu languages. Includes maps on the geographical distribution of the four basic methods of construction analyzed by the author.

C

CAM-76
1976 Campbell, Paul: An experimental course on African mathematics, *Historia Mathematica*, New York (USA), Vol. 3, 477-478.

Describes an experimental liberal arts mathematics course (St. Olaf College, Northfield, USA) on African mathematics: consideration of numeration systems, geometry in art and architecture, and mathematical games; together with an analysis of important concepts of 'western' mathematics they suggest.

CAP-83
1983 Caprile, Jean-Pierre; Adoum Khamis & Ndjerassem Ngabot: Pour une terminologie de l'enseignement du calcul dans les langues africaines: la structure d'expression des nombres et des techniques opératoires dans deux langues "sara" du sud du Tchad, le "ngambay" et le "mango", *Bulletin de l'AELIA (Association d'études linguistiques interculturelles africaines)*, Vol. 6, 273-287 (in French).

Contributes towards a terminology for the teaching of arithmetic in African languages. Discusses the expressions used for numbers and operations in two Sara languages from Chad: Ngambay and Mango.

CAP-86
1986 Caprile, Jean-Pierre & Irumu, Agozia-Kario: La numération orale et les systèmes de mesure en Logoti (Nord-est du Zaire) [Oral numeration and measurement systems in Logoti (Northeast of Congo / Zaire)], *Cahiers du LACITO*, Paris (France), No. 1 (in French).

CAP-87
1987 Jean-Pierre Caprile: Numérations orales et enseignement des mathématiques en Afrique [Oral numeration and the teaching of mathematics in Africa], *LENGAS, revue de sociolinguistique*, Université Paul Valéry, Montpellier (France), No. 21, 143-162 (in French).

Paper presented at a session organized by the African Bureau of Educational Sciences in Kisangani (Congo / Zaire) in December 1984. It gives some information on systems of numeration in Africa (Sara-Ngambay in Chad; Birom in Nigeria; Banda in Central-Africa) and outside Africa.

CAR-70

1970 Careccio, John: Mathematical heritage of Zambia, *The Arithmetic Teacher*, Reston VA (USA), 391-395.

The author compiled information on traditional ways of measuring time, distance, weight, and volume in Zambia. The information was collected by "using University students who sought out the oldest people of their villages to find out how these things were done before the European types of measurement replaced the African methods."

CARR-48

1948 Carra de Vaux, Bernard: Une solution arabe du problème des carrés magiques [An Arabic solution of the problem of magic squares], *Revue d'Histoire des Sciences*, Paris (France), No. 1, 206-212 (in French).

CAS-70

1970 Case, John H.: *Annotated bibliography on science and mathematics education in sub-Saharan Africa*, UNESCO, Paris (France), 234 p.

Lists references up to 1967 on science and mathematics education in the English-speaking countries of East, West, Central and Southern Africa.

CASM-75

1975 CASME: *Languages and the teaching of science and mathematics with special reference to Africa*, The Commonwealth Secretariat, London (UK).

Report of a seminar organized by the Commonwealth Association for Science and Mathematics Education (CASME), held in Accra (Ghana). It includes among other papers TAI-75 and a reproduction of MMA-74.

CASS-03
2003 Cassy, Bhangy: *Effect of classroom interaction and gender on mathematics performance and attitudes toward mathematics of secondary pupils in Mozambique*, doctoral thesis, University of the Witwatersrand, Johannesburg (South Africa).

CAV-92
1992 Caveing, Maurice: Le statut arithmétique du quantième égyptien [The arithmetic status of the Egyptian 'quantième'], in: BEN-92, 39-52 (in French).

The author analyses Egyptian calculations, both as concrete calculations concerning daily life and as an art of the abstract calculation. He concludes from this philological and mathematical analysis that, contrary to what assert other historians of science, that the calculators of this civilization do not consider 'quantième' as 'part of the unity' but rather as 'part of a collection.'

CAV-94
1994 Caveing, Maurice: *Essai sur le savoir mathématique dans la Mésopotamie et l'Egypte anciennes* [Essay on mathematical knowledge in ancient Mesopotamia and Egypt], Presses Universitaires de Lille, Lille (France), 417 p. (in French).

This book is the up-dated version of the first volume of the thesis defended by Caveing in 1977, entitled 'The constitution of the mathematical type of the ideality in Greek thinking'. The first part of Volume 1 is dedicated to the study of Babylonian mathematical texts (19-236). The second part deals with the art of calculation of the ancient Egyptians (237-404).

Review: VIT-99b

CEN-63
1963 Centner, Th.: *L'Enfant Africain et ses jeux dans le cadre de la vie traditionnelle au Katanga* [The African child and its games in the context of traditional life in Katanga], CEPSI, Elisabethville / Lubumbashi (Congo / Zaire), 412 p. (in French).

Includes games of Katanga (Shaba): sand drawings, games of chance, counting chants, *kisolo* (mancala) game, string figures, memory games.

CHA-94

1994 Chabert, Jean-Luc et al.: *Histoire d'algorithmes du caillou à la puce* [History of algorithms from the pebble to the flea], Éditions Belin, Paris (France), 591 p. (in French).

Collective work of seven authors (among them Ahmed Djebbar) on the history of algorithms, including analyses, comments and (translations of) original texts. The chapters and those sections directly related to the history of mathematics in Africa are the following:

1. Algorithms of arithmetical operations (11-58)
* Egyptian arithmetical algorithms: Rhind Papyrus (1650 BC) (20-25)
* Optimization of calculations: *Hawî l-lubâb* (1437) of Ibn al-Majdî (Egyptian mathematician and astronomer) (34-36)
2. Magical squares (59-94)
* A plotting procedure: Ibn Qunfudh (Maghreb, 14th century), *The unveiling of the operations of calculation* (69-74)
3. Around the methods of false position (95-128)
* Egypt: Problem 26 of the Rhind Papyrus (101-104)
* The *Talkhîs* of Ibn al-Bannâ (Maghreb, 13th century) (116-118)
4. Around Euclid's algorithm (129-158)
* Euclid, *Elements, Book VII* (3rd century BC) (129-134)
5. From circle measurement to π (159-192)
6. Newton's methods (193-226)
7. Solution of equations by successive approximations (227-262)
* Heron of Alexandria, *Metrica* (1st century) (231-232)
* Theon of Alexandria, Comments on the *Almagest* (4th century) (232-234)
* Medieval binomial algorithms, Ibn al-Banna, *Talkhis* (234-237)
8. Algorithms of number theory (271-318)
* The sieve of Eratosthenes: Nicomachus of Gerasa, Introduction to Arithmetic (2nd century) (274-77)
* Diophantus of Alexandria, The six arithmetic books (about 250) (309-311)
9. Solution of systems of linear equations (319-354)
10. Tables and interpolation (355-392)
* Ptolemy of Alexandria, *Mathematical composition* (about 150) (358-364)
11. Approximate quadratures (393-414)
12. Approximate solutions of differential equations (415-448)
13. Approximation of functions (449-476)

14. Acceleration of convergence (477-536)
Biographical notes (540-576), with information on the following mathematicians who were Africans or worked (some time) in Africa: Abû Kâmil, Archimedes, Diophantus, Eratosthenes, Euclid, Fibonacci, Heron, Hypathia, Ibn al-Bannâ, Ibn al-Haytham, Ibn al-Majdî, Ibn Qunfudh, Ptolemy, Theon.

CHAC-27
1927 Chace, Arnold B.: *The Rhind mathematical papyrus: British museum 10057 and 10058*, Mathematical Association of America, Oberlin, Ohio (USA), 109 pl. [Reprint: National Council of Teachers of Mathematics, Reston VA (USA), 1979, 140 p.]

CHAK-94
1994 Chakalisa, Paul: *Relationships of student gender, teacher experience and setting to students achievement and attitudes toward mathematics in Botswana junior secondary schools*, doctoral thesis, Ohio University, Athens (USA).

CHAM-02
2002 Chamdimba, Catherine Panji: *Co-operative learning and gender in mathematics education: A case study in a Malawian secondary school*, doctoral thesis, University of Waikato (New Zealand).

CHES-05
2005 Che, Stacy Megan: *Cameroonian teachers' perceptions of culture, education, and mathematics*, doctoral thesis, University of Oklahoma (USA).

CHEM-02
2002 Chemillier, Marc: *Polyrythmies de l'Afrique centrale* [Polyrhythms of Central Africa] (online available at: www.users.info.unicaen.fr/~marc/publi/diderot/pygmees.html)

Analyses "certain asymmetrical rhythmic structures appearing in the music of the culture of the Aka Pygmies of Central Africa. Complex mathematical patterns have been carefully woven into this music. Since these patterns are imperceptible to the listener, the author concludes that they must have been incorporated into the music as a result of mathematical rather than aesthetic concerns."

CHE-99
1999 Cherinda, Marcos: *Weaving geometric shapes: exploring the weaving board*, Universidade Pedagógica, Maputo (Mozambique), 30 p.

Booklet for children on exploring geometrical designs, using a weaving board, inspired by African basket and mat weaving.

CHE-02
2002 Cherinda, Marcos: *The use of a cultural activity in the teaching and learning of mathematics: The exploration of twill weaving in Mozambican classrooms*, doctoral thesis, Witwatersrand University, Johannesburg (South Africa), 270 p.

CHET-91
1991 Chetty, Devanathan: *Mathematics anxiety among Indian primary school children*, doctoral thesis, University of South Africa, Pretoria (South Africa).

CHI-74
1974 Chimuka, S. S. & Zulu, R. S.: The medium of instruction in primary schools in Zambia, UNESCO (ED-74/CONF.808/14), Paris (France).

Paper presented at the UNESCO Symposium on 'Interactions between Linguistics and Mathematical Education' (Nairobi, Kenya, 1-11 September 1974). Discusses the reasons for Zambia's language policies in school education, and the resultant implications for the teaching and learning of mathematics.

CHIO-95
1995 Chiocca, Catherine-Marie: *Analyse du discours de l'enseignant de mathématiques en classe de mathématiques - représentations des lycéens sénégalais*, doctoral thesis, Université Paris VII (France) (in French).

CHRI-03
2003 Chrisomalis, Stephen: The Egyptian origin of the Greek alphabetic numerals, *Antiquity*, Vol. 77, No. 297, 485-496.

CHR-91

1991 Christianidis, Jean: *Aristhmetikè Stoicheíosis*: Un traité perdu de Diophante d'Alexandrie? [A lost treatise of Diophantus of Alexandria?], *Historia Mathematica*, New York (USA), Vol. 18. No. 3, 239-246 (in French).

"The author suggests a conjecture about the existence of a lost theoretical treatise of Diophantus, entitled *Teaching of the Elements of Arithmetic*. His claims are based on a scholium of an anonymous Byzantine commentator."

CLA-89

1989 Clagett, Marshall: *Ancient Egyptian Knowledge. A source book*, American Philosophical Society, Philadelphia (USA), 2 vols., 863 p.

CLE-98

1998 Cleghorn, A., Mtetwa, D., Dube, R., & Munetsi, C.: Classroom language use in multilingual settings: mathematics lessons from Quebec and Zimbabwe, *Journal of Qualitative Studies in Education*, London (UK), Vol. 11, No. 3, 463-477.

"This article is concerned with language use in mathematics lessons in settings where the language of instruction is a second language for all or most of the learners. Four lessons taken from primary schools in Montreal and in Zimbabwe are compared, illustrating ways in which the teachers in each setting couple development of the second language with teaching of the subject content. By doing so, we believe that instruction is effective in helping children to make the shift from the primary school emphasis on computing numbers to the secondary level emphasis on solving problems; in the long term children are also better prepared for the language-related demands of higher education."

COLE-74

1974 Cole, Michael; Gay, John & Glick, J.: Some experimental studies of Kpelle quantitative behaviour, in: John W. Berry & Pierre R. Dasen (Eds.), *Culture and cognition: Readings in cross-cultural psychology*, Methuen, London (UK), 159-195.

Examines cognitive behavior involved in making quantitative judgments among the Kpelle people of Liberia. Deals with subjects such as geometric concepts, disjunction and conjunction, and estimates of volume, length, time and number.

COLES-59
1959 Coles, W. D.: Unified mathematics, *West African Journal of Education*, Vol. 3, No. 1, 32-36.

The gradual merging of algebra, arithmetic and geometry into a unified subject in schools in the UK is taken as the basis for recommending a similar change to take place in the British colonies of West Africa.

COL-73
1973 Collard, Chantal: Les "noms-numéros" chez les Guidar [The "names-numbers" among the Guidar], *L'Homme, revue française d'anthropologie*, Paris (France), Vol. 13, No. 3, 45-59 (in French).

Analyses the way the Guidar in North-Cameroon give names to their children. The first name indicates the order in which the mother gave birth (and also the sex in the case of the first four children); the second name is the name-number of the father of the child. E.g. the first of an individual called Tizi Dawaï expresses that he is a boy and the first child of his mother; his surname indicates that his father is the seventh child of his respective mother.

COLL-74
1974 Collison, G. O.: *Language and mathematical concept development in Ghanaian elementary school children*, UNESCO (ED-74/CONF.808/20), Paris (France).

Paper presented at the UNESCO Symposium on 'Interactions between Linguistics and Mathematical Education' (Nairobi, Kenya, 1-11 September 1974). Discusses the relation between language and the development of mathematical concepts at the primary school level in Ghana.

COM-05
2005 Communay, Pierre Henri: *Les pyramides d'Egypte: une histoire simple et analyse mathématique*, Groupe de recherche et d'édition, Saubens (France), 587 p. (in French).

COU-83
1983 Couchoud, Sylvia: *Recherche sur les connaissances mathématiques de l'Egypte pharaonique* [Research on mathematical knowledge in pharaonic Egypt], doctoral thesis,

Institute d'Egyptologie, Université de Lyon II, Lyon (France), 420 p. (in French).

Couchoud's thesis on mathematical knowledge in Pharaonic Egypt, deals with 1) arithmetical operations and the notion of fraction, including a study of 'red auxilaries' (14-39); 2) geometry (metrology, plane figures and solids, *nbt*-notion) (40-188); 3) procedures which are equivalent to equations and series (189-330); 4) solutions of concrete problems (distribution of daily food rations, production of sandals, delivery of wood, etc.) (331-371).

COU-86
1986 Couchoud, Sylvie: Essai d'une nouvelle interpretation du premier problème du Papyrus mathématique démotique 10520 du British Museum [Attempt at a new interpretation of the first problem of the Demotic mathematical papyrus No. 10520 of the British Museum], *Centaurus*, Copenhagen (Denmark), Vol. 29, 1-4 (in French).

In the problem, the scribe calculates two sums of natural numbers. The first is the sum of the first ten natural numbers. According to R. A. Parker, the second should be the sum of the first ten square numbers. But the sum given by the scribe is 220 and not 385. The author of the paper thinks that in fact the scribe wanted to calculate the following sum: $S_1 + S_2 + ... + S_{10}$, with: $S_1 = 1$, $S_2 = 1+2$, $S_3 = 1+2+3$, ... , $S_{10} = 1+2+...10$, that is indeed equal to 220. If this interpretation is exact, then "only Egypt [among the peoples of Antiquity] could give evidence, by means of the formulation of problem number 53 of the Demotic papyrus B.M.10520 and the solution that may have obtained, of this very advanced knowledge."

CRO-71
1971 Crowe, Donald W.: The geometry of African art I. Bakuba art, *Journal of Geometry*, München (Germany), Vol. 1, 169-182.

Uses a group theoretic analysis of repeated patterns to study strip and plane patterns on Bakuba raffia cloths and carved wooden boxes and cups (Congo / Zaire). All seven possible strip patterns and (at least) 12 plane patterns occur in Bakuba art.

CRO-73
1973 Crowe, Donald: Geometric symmetries in African art, in: ZAS-73a, 190-196.

Presents examples of the geometric analysis of the symmetries of repeated patterns as appearing in Bakuba art (Congo / Zaire), Benin bronzes (Benin, Nigeria), Yoruba *adire* cloth (Nigeria).

CRO-75a

1975a Crowe, Donald W.: The geometry of African art II. A catalog of Benin patterns, *Historia Mathematica*, New York (USA), Vol. 2, 253-271.

Investigates the repeated patterns found in the art from Benin, classifying them on the basis of the 24 plane crystallographic group. All 7 possible strip patterns and 12 of the 17 frieze patterns occur. A catalog is given with most of the strip patterns the author has found in Benin art, along with one example of each of the 12 plan patterns that occur.

CRO-75b

1975b Crowe, Donald W.: Erratum to *The geometry of African art. I, II* (CRO-71, CRO-75a), *Historia Mathematica*, New York (USA), Vol. 2, No. 4, 617.

CRO-82a

1982a Crowe, Donald: The geometry of African art III. The smoking pipes of Begho, in: C. Davis, B. Grünbaum, F. Sherk (Eds.), *The geometric vein, the Coxeter Festschrift*, Springer Verlag, New York (USA), 177-189.

Applies the symmetry classification scheme for repeated patterns to the analysis of the decorated pipes excavated from the K2 site of he Kramo quarter of Begho (Ghana). All seven one-dimensional types appear, and seven of the seventeen possible two-dimensional patterns were found on Begho K2 pipes.

CRO-82b

1982b Crowe, Donald: Symmetry in African art, *Ba Shiru, Journal of African Languages and Literature*, University of Wisconsin, Madison (USA), Vol. 11, No. 1, 57-71.

Investigates the repeated patterns found in African art, classifying them on the basis of the 24 plane crystallographic groups. Of these, seven admit translations in only one direction (the corresponding patterns are called strip patterns), while the remaining 17 admit two

independent translations (so-called plane patterns). Presents examples from Cameroon, Congo / Zaire, Ghana, and Nigeria.

CRO-01
2001 Crowe, Donald: Review of Gerdes' *Geometry from Africa* (GER-99a) and *Le cercle et le carré* (GER-00b), *The Mathematical Intelligencer*, New York (USA), Vol. 23, No. 2, 65-68.

CRO-05
2005 Crowe, Donald W. & Dorothy K. Washburn: Geometrical, Perceptual, and Cultural Perspectives on Figure / Ground Differences in Bakuba Pattern, *Visual Mathematics*, Beograd (Serbia), Vol. 7, No. 3 [online available at: www.mi.sanu.ac.yu/vismath/bridges2005/crowe/ index.html]

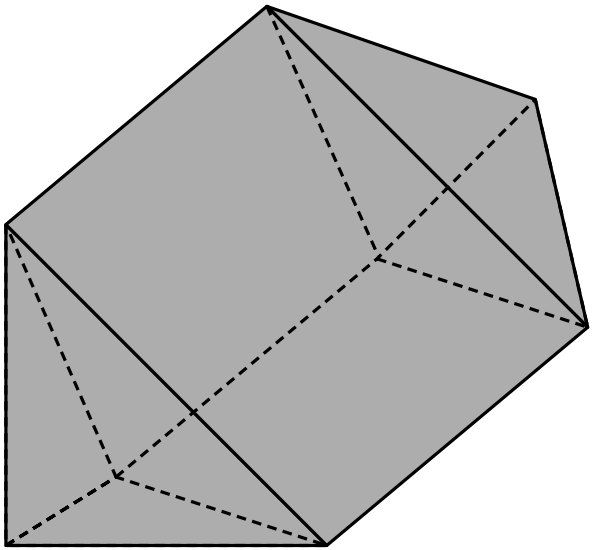
"Two tabletops carved by a Bakuba wood-carver reveal a surprising duality. Although the carvings at first glance appear completely different, closer attention shows that the carved portion of each is exactly the uncarved portion of the other. Hence, in a certain sense, they have exactly the same symmetries. We discuss the cultural insights suggested and supported by this observation." [DR Congo]

CROZ-96
1996 Crozet, Pascal: *Eléments pour une histoire de la modernisation des sciences exactes en Egypte au XIX ème siècle*, (1805-1902) [Elements for a history of the modernization of the exact sciences in Egypt in the 19th century], doctoral thesis, Université Paris 7 (France) (in French).

CUO-00
2000 Cuomo, Serafina: *Pappos of Alexandria and the Mathematics of Late Antiquity*, Cambridge University Press, Cambridge (UK), 234 p.

Chapter 1 examines the place that have mathematicians or those that have to do with mathematics (artisans, jurists, astrologers, ...). The following three chapters are dedicated to the Books III, IV, V, VIII or to the specific questions posed in Pappus' *Collection* (classifications of problems, regular polyhedra, geometry of curves, isoperimetric figures, mechanical questions...). The fifth and last chapter tries to determine Pappus' motivations and intentions, his use of the tradition

and his relationship with history and his predecessors. This approach of Pappus is innovative as such and constitutes one of the strong points of the work.



Shape of a plaited nonahedron (Mozambique) (cf. GER-05c)

D

DAM-81

1981 Damerow, Peter: Die Entstehung des arithmetischen Denkens, in: P. Damerow & W. Lefèvre, *Rechenstein, Experiment, Sprache: Historische Fallstudien zur Entstehung der exakten Wissenschaften*, Klett-Cotta, Stuttgart (Germany), 11-113 (in German).

Original version of DAM-96.

DAM-96

1996 Damerow, Peter: The development of arithmetical thinking: on the role of calculating aids in Ancient Egyptian and Babylonian arithmetic, in: P. Damerow, *Abstraction and Representation. Essays on the Cultural Evolution of Thinking* [Boston Studies in the Philosophy of Science, Vol. 175], Kluwer, Dordrecht (Netherlands), 173-273.

Translation of DAM-81. Includes sections on the 'Structural characteristics of Ancient Egyptian Arithmetic' (176-188) and 'The means of calculation in Ancient Egypt' (188-199).

DAMB-98

1998 D'Ambrosio, Ubiratan: Review of Gerdes' *Femmes et Géométrie en Afrique Australe* (GER-96b) / *Women, Art and Geometry in Southern Africa* (GER-98d) [available online at: www.mox.uniandes.edu.co/voc/Paulus_Gerdes.htm].

DAR-03

2003 Darvas, György: Review of Gerdes' *Awakening of Geometrical Thought in Early Culture* (GER-03a), *Symmetry: Culture and Science*, Vol. 12, No. 1-2, Budapest (Hungary), 229-230.

DAV-88

1988 Davies, Richard: *An introduction to shax: a Somali game* (updated in 1996) [available online at: www.swan.ac.uk/cds/shax.htm].

Describes the three-in-a-row game from Somalia, called *shax*, indicating the differences with the *morabaraba* game from Lesotho.

DEA-92
1992 Deakin, Michael: Hypathia of Alexandria, *Mathematics Education*, No. 8, 187-191.

Describes the life, times, and work of Hypatia of Alexandria (370-415 AD).

DEA-94
1994 Deakin, Michael: Hypathia and her mathematics, *American Mathematical Monthly*, Washington DC (USA), No. 101, 234-243.

Evaluates the sources of knowledge about Hypatia of Alexandria (around 370-415 AD), and describes what is known of her mathematical activities.

DEA-95
1995 Deakin, Michael: *The Primary Sources for the Life and Work of Hypathia of Alexandria*, History of Mathematics Paper No. 63, Department of Mathematics, Monash University, Clayton (Australia), 16 p.

Describes the primary sources for the life, times, and work of Hypatia of Alexandria.

DEA-96
1996 Deakin, Michael: Review of Dzielska's *Hypatia of Alexandria* (DZI-95), *The American Mathematical Monthly*, Washington DC (USA), Vol. 103, No. 1, 83-87.

This review of DZI-95 welcomes Dzielska's book: "We have waited over two centuries since the last book-length biography of Hypatia of Alexandria was published in English" (p. 83), analyses Dzielska's reconstruction of Hypatia's philosophical ideas, and criticizes the treatment of Hypatia's mathematics.

DEL-28
1928 Delafosse, Maurice: La numération chez les Nègres [Numeration among the Black], *Africa, Journal of the International Institute of African Languages*, London (UK), Vol. 1, No. 3, 387-390 (in French).

Analyses the most frequent structures of number words in African languages: ‘6’ = ‘5+1’ = ‘+1’ = ‘2x3’; ‘7’ = ‘5+2’ = ‘2nd 6’; ‘8’ = ‘5+3’ = ‘+3’ = ‘2x4’ = ‘4+4’; ‘9’ = ‘5+4’ = ’10 – 1’ = ‘missing 1’

DELE-81
1981 Deledicq, André: Numération et langues africaines [Numeration and African languages], *Bulletin de liaison des professeurs de mathématiques*, No. 27, 3-9.

DER-72
1972 Deregowski, Jan: The role of symmetry in pattern reproduction by Zambian children, *Journal of Cross-Cultural Psychology*, Beverly Hills CA (USA), Vol. 3, No. 3, 303-307.

DER-76
1976 Deregowski, Jan: Coding and drawing of simple geometric stimuli by Bukusu school-children in Kenya, *Journal of Cross-Cultural Psychology*, Beverly Hills CA (USA), Vol. 7, No. 2, 195-208.

DEY-84
1984 DeYoung, Gregg: The Arabic textual traditions of Euclid’s ‘Elements’, *Historia Mathematica*, New York (USA), Vol. 11, No. 2, 147-160.

DEY-94
1994 De Young, G.: Ibn al-Sari on ex aequali ratios : his critique of ibn al-Haytham and his attempt to improve the parallelism between books V and VII of Euclid’s ‘Elements’, *Zeitschrift für Geschichte der Arabisch-Islamischen Wissenschaften*, Frankfurt (Germany), Vol. 9, 99-152.

DHO-87
1987 Dhombres, Jean & A. Dahan-Dalmedico, R. Bkouche, C. Houzel and M. Guillemot (Eds.): *Mathématiques au fil des âges* [Mathematics during the ages], Gauthier-Villars, Paris (France), 327 p.

This book is addressed to high school pupils and their teachers. It contains extracts of texts by mathematicians throughout history. The extracts are accompanied by commentaries, and are grouped into six chapters: Object and Utility of Mathematics, Arithmetic and Number

Theory, Algebra, Calculus, Probability, Geometry. University lecturers and high school teachers have worked together on the conception of this book. Ahmed Djebbar has contributed with the topics on Arabic mathematics.

DIAG-80
1980 Dia, Galaye: Le raisonnement mathématiques dans le milieu culturel, in: Haberland, Elke (Ed.), *Symposium Leo Frobenius II; le rôle des traditions dans le développement de l'Afrique*, Deutsche UNESCO-Kommission, Bonn (Germany), 394-398 (in French).

Argues that a teacher of mathematics should start with concrete situations drawn from the socio-cultural context of the child to give it the opportunity to discover structures. The mother tongue is the language to think of these structures (Senegal).

DIA-82
1982 Diagne, Bachir S.: Note sur la question: faire des mathematiques en Ouolof, *Langues africaines et échange des connaissances*, UNESCO & Conseil Interafricain de Philosophie, Cotonou (Benin) (in French).

Paper on doing mathematics in the Wolof language (Senegal).

DIAL-79
1979 Diallo, Fatoumata Câmara: *Recherche des conditions de possibilité d'une didactique mathématique au Mali*, doctoral thesis, Université de Bordeaux 2 (France) (in French).

DIO-59
1959 Diophantus of Alexandria: *Les six livres arithmétiques et le livre des nombres polygones* [The six arithmetical books and the book on polygonal numbers], Blanchard Paris, (France), 390 p. (in French).

Translated for the first time from Greek into French by Paul Ver Eecke.

DIO-74
1974 Diophantus of Alexandria: *Diophanti Alexandrini Opera omnia cum graecis commentariis* [The complete works of Diophantus of Alexandria] (reprint of the 1893-1895 edition by Paul

Tannery), Teubner, Stuttgart (Germany), 298 p. (in Latin and Greek).

DIO-82
1982 Diophantus of Alexandria: *Arithmetica, Arabic translation attributed to Qusta ibn Luqa* (Edition by Jacques Sesiano), Springer Verlag, New York (USA), 502 p.

DIO-84a
1984a Diophantus of Alexandria: *Les arithmétiques, Tome III: Livre IV* [*Arithmetica* Vol. III: Book IV] (Translation by Roshdi Rashed), Les Belles Lettres, Paris (France), 216 p. (in French).

DIO-84b
1984b Diophantus of Alexandria: *Les arithmétiques, Tome IV: Livres V-VII* [*Arithmetica* Vol. IV: Books V-VII] (Translation by Roshdi Rashed), Les Belles Lettres, Paris (France), 319 p. (in French).

DJE-81
1981 Djebbar, Ahmed: *Enseignement et recherche mathématiques dans le Maghreb des XIIIème - XIVème siècles* [Mathematical education and research in the Maghreb from the 8th to the 14th century], Publications Mathématiques d'Orsay, Paris (France), No. 81-02, 146 p. (in French).

This study is based on a series of unpublished manuscripts and has three chapters. The first deals with different classifications of equations of degree inferior or equal to two, the role of geometry in the study of these equations and the contribution of the Maghrebians in this domain. The second chapter deals with the arithmetical and algebraic symbolism that was used in the Maghreb from the 12th century on and that would be brought to Egypt from the 14th century on. The third chapter reveals and gives an exposition – for the first time – of certain aspects of the contribution to combinatorics by mathematicians from the Maghreb.

DJE-84a
1984a Djebbar, Ahmed: Quelques remarques sur les rapports entre Philosophie et Mathématiques arabes [Some remarks on the relationships between Arab philosophy and mathematics],

Revue Tunisienne des Etudes Philosophiques, Tunis (Tunisia), No. 2, 3-21 (in French).

DJE-84b
1984b Djebbar, Ahmed: Les scientifiques arabes face à leur patrimoine [Arab scientists facing their heritage], *Revue de la Documentation française, Maghreb-Machrek*, Paris (France), No. 105, 48-64 (in French).

DJE-85a
1985a Djebbar, Ahmed: *L'analyse combinatoire au Maghreb: l'exemple d'Ibn Mun^{c}im (XIIe-XIIIe s.)* [Combinatorics in the Maghreb: the example of Ibn Mun^{c}im (12th – 13th century)], Publications Mathématiques d'Orsay, Paris (France), Vol. 85-01, 124 p. (in French).

Contains a commentary and a translation of section XI of 'Fiqh al-Hisab', a manual written by Ibn Mun^{c}im (Maghreb) between 1207 and 1212. On the basis of some linguistic problems (number of Arabic words of given length etc.), Ibn Mun^{c}im develops his combinatorics. He presents an arithmetic triangle (the so-called Pascal's triangle) and deduces the equivalents of formulas like

$$C_{p-1}^{p-1} C_{n}^{p} = C_{n-1}^{p-1} + C_{n-2}^{p-1} + \ldots + C_{p-1}^{p-1},$$

$$P_n = n!,$$

$$P^{k_1,\ldots k_r} = \frac{P_n}{P^{k_1} \ldots P^{k_r}},$$

centuries before Cardano, Tartaglia, Mersenne, Frenicle, etc., in Europe.

Detail of Ibn Munᶜim's manuscript with his arithmetic triangle
four centuries before Pascal (1623-1662)

DJE-85b

1985b Djebbar, Ahmed: *Les nombres figurés dans la tradition mathématique de l'Andalousie et du Maghreb* [Figurate numbers in the mathematical tradition of Andalusia and the Maghreb], Prépublications Mathématiques d'Orsay, Paris (France), Vol. 85 T 44, 29 p. (in French).

DJE-86a

1986a Djebbar, Ahmed: Les Mathématiques arabes et leur environnement [Arabic mathematics and its environment], in: *Actes de l'université d'été sur l'histoire des mathématiques (6-13 Juillet 1984)*, Université du Maine, Le Mans (France), 36-70 (in French).

This paper presents the economic, political, cultural and ideological context in which the mathematical activities in the Arab-Islamic civilization were born and developed. It deals equally with the internal and external factors that could explain the retardation, from the 14th century onwards, of the scientific activities of this civilization.

DJE-86b

1986b Djebbar, Ahmed: L'Algèbre arabe [Arab algebra], *L'Ouvert*, IREM de Strasbourg, Strasbourg (France), Vol. 44, 26-30 (in French).

Text of a public lecture given in Strasbourg in 1986 on the main directions in algebra in the Arabic mathematical tradition from the 9th to the 15th century.

DJE-87a

1987a Djebbar, Ahmed: Les Mathématiques au Maghreb à l'époque d'Ibn al-Bannâ [Mathematics in the Maghreb at the time of Ibn al-Bannâ], in: *Actes du Colloque de la Société de Philosophie au Maroc*, L'Harmattan, Paris (France) & Okad., Rabat (Morocco), 31-46 (in French).

DJE-87b

1987b Djebbar, Ahmed: L'analyse combinatoire au Maghreb entre le XIIe et le XIVe siècle [Combinatorics in the maghreb between the 12th and the 14th century], in: Dhombres, Jean (Ed.), *Cahiers d'Histoire et de Philosophie des Sciences*, Nouvelle série, Paris (France), No. 20, 232-239 (in French).

DJE-87c

1987c Djebbar, Ahmed: Algorithmes et optimisation dans les mathématiques arabes [Algorithms and optimization in Arab mathematics], in: M. Amara et al. (Eds.), *Proceedings of the First International Symposium of ICOMIDC on "Informatics and the teaching of mathematics in developing countries*," ICOMIDC, Tunis (Tunisia), 10 p. (in French).

DJE-88a

1988a Djebbar, Ahmed: Quelques aspects de l'Algèbre dans la tradition mathématique arabe, in: *Actes de l'université d'été sur*

l'histoire des mathématiques (6-12 Juillet 1986), IREM de Toulouse, Toulouse (France), 257-286 (in French).

This paper discusses the origin, the beginnings and the development of algebra in the Moslem East from the 8th century on. Information is given on the contributions of al-Khwârizmî (d. 850), Abû Kâmil (d. 930), al-Karajî (d. 1029), as-Samaw'al (d. 1175), al-Khayyâm (d. 1131) and Sharaf ad-Dîn at-Tûsî (d.1213) as well as on the contribution of lesser known researchers like Sinân Ibn al-Fath (10th C.), who have participated in the development of this discipline.

DJE-88b

1988b Djebbar, Ahmed: Mathématiques et linguistique dans le Moyen Age arabe [Mathematics and linguistics in the Arab Middle ages], in: *Résumé des communications du Colloque Sciences au Moyen Age (22-23 Avril 1988)*, Université d'Orléans, Orléans (France), 21-24 (in French).

This communication is a summary of various studies by the author (published between 1981 and 1985), that concern the combinatorial practice in Arabic linguistics, music and lexicography and also the history of the progressive mathematisation of this combinatorial practice in the East and in the Maghreb (between the 9th and the 14th century).

DJE-88c

1988c Djebbar, Ahmed: *Le contenu de l'enseignement mathématique dans le nord de l'Afrique, au moyen âge et son rôle dans l'enseignement actuel* [The contents of mathematics teaching in North Africa during the Middle Ages and its role in present day teaching], École Normale Supérieure, Algiers (Algeria), 16 p. (in French).

After a short description of mathematical activity in North Africa during the Middle Ages, the author describes the mathematical contents taught at that time (decimal system, six arithmetical operations, polynomials and the algebraic and geometric solution of polynomial equations). In the last part he underlines the cultural value of this rich heritage of medieval mathematics for education today in North Africa.

DJE-89a

1989a Djebbar, Ahmed: Review of Kane's *Les systèmes de numération parlée des groupes ouest-atlantiques et Mandé* (KAN-87), *AMUCHMA Newsletter*, Maputo (Mozambique), No. 3, 8-11.

DJE-89b

1989b Djebbar, Ahmed: The contents of mathematics teaching in North Africa in the Middle Ages and its role in present day teaching, in: C. Keitel, P. Damerow, A. Bishop, P. Gerdes (Eds.), *Mathematics, Education, and Society*, UNESCO, Paris (France), 3-4.

Gives a brief overview of the contents of mathematics teaching in North Africa in the Middle ages and demonstrates, using as examples the Arabic-speaking regions of Africa, the possibilities which are offered by the mathematical heritage of these regions to enrich the pedagogy of teaching mathematics today.

DJE-90a

1990a Djebbar, Ahmed: Arab Mathematics and Linguistics in the medieval Maghreb: the example of combinatorics, *Revue Arabe des Technologies*, Paris (France), No. 3, 43-50 (in Arabic).

In the first part of this paper the author presents the different known aspects of combinatorial practices in various domains of medieval Arabic culture and science (linguistics, lexicography, grammar, poetry, astronomy, algebra). The second part is dedicated to the mathematization of these combinatorial practices and to the contribution of two Maghrebian scientists – Ibn Mun^{c}im (d. 1228) and Ibn al-Bannâ (d. 1321) – to this mathematisation: elaboration of definitions, propositions and demonstrations of combinatorial nature and the introduction of combinatorial techniques in different domains, both mathematical and non-mathematical.

DJE-90b

1990b Djebbar, Ahmed: Al-Qalasâdî, an Andalusian-Maghrebian scholar of the 15th century, *Revue Arabe des Technologies*, Paris (France), No. 9, 12-13 (in Arabic).

This article is dedicated to the Maghrebian scientist (of Andalusian origin) 'Ali al-Qalasâdî (1412-1486) who has been the most important

mathematician in the Maghreb during the 15th century. The paper contains a detailed biography of this scientist and an exposition of the contents of his mathematical works on arithmetic, algebra and also on the use of arithmetical techniques in the solution of heritage problems.

DJE-90c

1990c Djebbar, Ahmed: *Le traitement des fractions dans la tradition mathématique médiévale du Maghreb* [The treatment of fractions in the medieval mathematical tradition of the Maghreb], Université de Paris-Sud, Pré-publications Mathématiques d'Orsay, Paris (France), No. 90-04, 30 p. (in French).

The author exposes first the essential aspects of the theory of fractions in the Arabic mathematical tradition of the East and then, on the basis of a study of manuscripts from the 12th to the 16th century, he analyses the transmission of the concepts and techniques of the fractions from the East to the West and he exposes new elements that concern the practices of calculation with fractions as encountered in the mathematical works of the Maghreb.

DJE-90d

1990d Djebbar, Ahmed: *Mathématiques et Mathématiciens du Maghreb médiéval (IXe-XVIe siècles): Contribution à l'étude des activités scientifiques de l'Occident musulman* [Mathematics and Mathematicians in the medieval Maghreb (9th –16th century): contribution to the study of scientific activities in the Islamic West], higher doctorate, Université de Nantes, Nantes (France), 850 p. (in French and Arabic).

The dissertation has the following chapters:

1. General Introduction
2. Historical introduction: the context of the arrival and development of mathematical activities in the Maghreb
3. Mathematical education and research in the Maghreb during the 13th and 14th centuries
4. Combinatorics in the Maghreb: the example of Ibn Munᶜim (12th –13th centuries)
5. Some remarks on the relationship between Arabic philosophy and mathematics

6. Two little known mathematicians from 11th century Spain: al-Mu'taman and Ibn Sayyid
7. The mathematical contribution of al-Mu'taman and his influence on the Maghreb
8. The treatment of fractions in the Arabic mathematical tradition of the Maghreb
9. Abû Bakr Ibn Bâjja and the mathematics of his time
10. Discovery of a mathematical manuscript of al-Hassâr (12th century): Book I of al-Kâmil
11. Figurate numbers in the mathematical tradition in Andalusia and in the Maghreb
12. Some new elements on Arabic mathematical activity in the oriental Maghreb (9th –16th c.)
13. Some aspects of algebra in the Arabic mathematical tradition
14. The algebra book of Ibn al-Bannâ

* Introduction and mathematical analysis
* Translation into French
* Arabic text

DJE-90e
1990e Djebbar, Ahmed: The mathematical contribution of al-Mu'taman and his influence on the Maghreb, in: *The History of Science among the Arabs*, Bayt al-Hikma, Carthage (Tunisia), 21-42.

DJE-91a
1991a Djebbar, Ahmed: Mathématique et linguistique dans le Moyen-âge arabe. L'exemple de l'analyse combinatoire au Maghreb [Mathematics and linguistics in the Arab Middle Ages. The example of combinatorics in the Maghreb] in: *Actes du Colloque "Le Moyen-âge et Sciences" (Orléans, 22-23 April 1988)* [Colloquium on The Middle Ages and Science], Kincksieck, Paris (France), 15-29 (in French).

DJE-91b
1991b Djebbar, Ahmed: Scientific activities in Marrakech during the 12th – 13th century, *Revue Arabe des Technologies*, Paris (France), No. 15, 13-25 (in Arabic).

DJE-92a

1992a Djebbar, Ahmed: Las matematicas en al-Andalus através de las actividades de tres sabios del siglo XI [Mathematics in Andalusia through the activities of three scholars of the 11th century], in: *El Legado Cientifico Andalusi* [The Andalusian scientific heritage], Museo Arqueologico Nacional, Madrid (Spain), 340 p. (in Spanish).

Two of the three mathematicians presented in this paper, al-Mu'taman (d. 1085) and Ibn Sayyid (11th - 12th century), have written mathematical texts that have been used in the Maghreb during the 12th and 13th century. The third scholar, Ibn Bâjja (d. 1138), has lived the last part of his life in the Maghreb.

DJE-92b

1992b Djebbar, Ahmed: Le traitement des fractions dans la tradition mathématique arabe du Maghreb [The treatment of fractions in the Arab mathematical tradition of the Maghreb], in: BEN-92, 223-246 (in French).

See the summary in: DJE-90c.

DJE-95a

1995a Djebbar, Ahmed: Sur les activités mathématiques dans le Nord d'Afrique à partir du IXe siècle. Première partie: Les Mathématiques dans le Maghreb médiéval, *Bulletin de l'AMUCHMA*, Paris (France), No. 15, 3-38 (in French).

Overview of mathematical activities in medieval Maghreb. See DJE-95b.

DJE-95b

1995b Djebbar, Ahmed: On mathematical activities in North Africa since the 9th century. First part: Mathematics in medieval Maghreb, *AMUCHMA Newsletter*, Maputo (Mozambique), No 15, 3-42.

Translation of DJE-95a.

DJE-96a

1996a Djebbar, Ahmed: On mathematical activities in North Africa since the 9th Century, *International Study Group on the*

Relations between History and Pedagogy of Mathematics Newsletter, Washington DC (USA), No. 37, 11-13.

Partial reproduction of DJE-95b.

DJE-96b

1996b Djebbar, Ahmed: Quelques Commentaires sur les Versions arabes des *Eléments* d'Euclide et sur leur Transmission à l'Occident Musulman [Some Comments on the Arabic versions of Euclid's *Elements* and on their Transmission to the Muslim West], in: M. Folkerts, *Mathematische probleme im Mittelalter. Der lateinische und arabische Sprachbereich* [Mathematical Problems in the Middle Ages. The Latin and Arabic Language Area], Harrassowitz Verlag, Wiesbaden (Germany), 91-114 (in French).

Euclid's *Elements* were probably the most studied and most commented text by Arab mathematicians in the period between the end of the 8th century and the beginning of the 19th century. Several Arabic versions of the *Elements* were used. The paper presents new information concerning terminology and certain variants found by the author and relative to the transmission of the *Elements* to and their use in the Muslim West.

DJE-97a

1997a Djebbar, Ahmed: La rédaction de *L'Istikmâl d'al-Mu'taman* (XIe s.) par Ibn Sartaq, un mathématicien des XIIIe–XIVe siècles, *Historia Mathematica*, New York (USA), Vol. 24, 185-192 (in French).

The author presents a "14th-century manuscript that has not been studied before. It contains a complete redaction of the *Kitâb al-istikmâl* by the Andalusian mathematician, al-Mu'taman ibn Hûd (11th century), and informs us about the missing pieces of al-Mu'taman's book and about the content of his initial project that had never been completed."

DJE-97b

1997b Djebbar, Ahmed: *Les activités mathématiques dans le Maghreb Central* [Mathematical activities in the Central Maghreb], Université de Paris-Sud, Paris (France), Preprint No. 97, 43 p. (in French).

The article describes the conditions under which, between the 9th and the 15th century, emerged and developed a series of mathematical activities in some cities in the Central Maghreb. The description includes the links that were woven between these cities and other scientific centers in the west Mediterranean that exercised mutual influence and stimulated the circulation of ideas and men. The study presents also some mathematicians from this region of the Maghreb, by specifying their various known contributions, both with respect to their publications as to their scientific teaching.

DJE-98

1998 Djebbar, Ahmed: Le raisonnement géométrique dans la tradition mathématique arabe [Geometrical reasoning in the Arab mathematical tradition (9th - 15th century)], in: ACT-98b, 89-121 (in French).

The article presents first the context of the development of geometrical reasoning in the Arabic scientific tradition, then it evokes the pre-Islamic sources of demonstration. In the third part, it explains the status of the various types of geometrical justification. The fourth part treats geometrical reasoning as object of study by mathematicians. The last part discusses various types of geometrical reasoning in Arabic philosophic and mathematical writings.

DJE-00a

2000a Djebbar, Ahmed: Figurate Numbers in the Mathematical Tradition of al-Andalus and the Maghreb, *Suhayl*, Barcelona (Spain), Vol. 1, 57-70.

The paper analyses certain contributions made in Andalusia and the Maghreb to the theme of figurate numbers. These numbers are a geometrical representation of numbers and had been created by the Pythagorean School. The oldest known study of these numbers is found in the *Introduction to Arithmetic* by Nicomachus. An Arabic translation of this work circulated in Andalusia and in the Maghreb from the 10th century onwards.

DJE-00b

2000b Djebbar, Ahmed: Les activités mathématiques au Maghreb à l'époque ottomane [Mathematical activities in the Maghreb during the Ottoman epoch], in: HIS-00, 49-66 (in French).

The paper presents some unpublished information on the mathematical activities in the region of the Maghreb that, from the 16th century onwards, was under the political authority of the Ottoman power of Istanbul. A comparison is made between these activities and those in the Western Maghreb, which were autonomous.

DJE-00c

2000c Djebbar, Ahmed: Les récréations dans les mathématiques du monde musulman [Mathematical recreations in the Islamic world], *La Recherche*, Special issue, May-June, 70-72 (in French).

The paper presents little known element about the recreational and game aspects of Arab mathematics from the East and from the Maghreb.

DJE-00d

2000d Djebbar, Ahmed: La production scientifique arabe, sa diffusion et sa réception au temps des croisades: l'exemple des mathématiques [Arab scientific production, its diffusion and reception at the time of the crusades: the example of mathematics], in: *Actes du Colloque International sur "Occident et Proche-Orient: Contacts scientifiques au temps des croisades" (Louvain-la-Neuve, 24-25 mars 1997)*, Brepols, Brussels (Belgium), 343-368 (in French).

Study of the different types of circulation of mathematical knowledge since the 12th century, inside the Muslim empire, between the East and the West, and outside this empire to Latin Europe.

DJE-00e

2000e Djebbar, Ahmed: La place et le rôle de l'imagination dans les activités mathématiques de la tradition arabe médiévale [The place and role of imagination in the mathematical activities of the medieval Arab tradition], in: A. Benmaïssa (Ed.), *Actes du Colloque International sur "Imagination and Sciences" (Rabat, 1998)*, Publications de la Faculté des Lettres et des Sciences Humaines, Rabat (Morocco), 153-176 (in French).

Study of the different interventions of the imagination among the mathematicians of the Islamic countries, both in their scientific practice and their discourse.

DJE-01a

2001a Djebbar, Ahmed; Rommevaux, Sabine & Vitrac, Bernard: Remarques sur l'histoire du texte des Eléments d'Euclide [Remarks on the history of the text of Euclid's Elements], *Archives for the History of Sciences*, Berlin (Germany), No. 55, 221-295.

A comparative study of certain aspects of the contents of the three great traditions of Euclid's *Elements*, those of ancient Greece, of the Arab translators and commentators and of the medieval Latin translators and commentators.

DJE-01b

2001b Djebbar, Ahmed: Les transactions dans les mathématiques arabes: classification, résolution et circulation [Transactions in Arab mathematics: classification, solution and circulation], in: *Actes du Colloque International "Commerce et mathématiques du Moyen Âge à la Renaissance, autour de la Méditerranée" (Beaumont de Lomagne, 13-16 mai 1999)*, Editions du C.I.H.S.O, Toulouse (France), 327-344 (in French).

An analysis of the different transaction problems and the solution procedures included in the known Arab mathematical manuals that were published between the 9th and the 14th century.

DJE-01c

2001c Djebbar, Ahmed: *La phase arabe de l'histoire de l'algèbre* [The Arab phase in the history of algebra], in: Actes de la Troisième Université d'Été Européenne sur "*Histoire et épistémologie dans l'éducation mathématique*" *(Louvain-la-Neuve, 15-18 juillet 1999)*, Université Catholique de Louvain, Louvain (Belgium), Vol. 2, 203-217 (in French).

Summary of the most significant developments in algebra during the Arab phase, that is between the 9th and the 15th century. An important place is given to algebraic activities in Andalusia and in the Maghreb.

DJE-01d

2001d Djebbar, Ahmed: *Las Matemáticas árabes y su papel en el desarrollo de la tradición científica europea* [Arab mathematics and its role in the development of the European scientific tradition], in: *Galileo y la gestación de la ciencia*

moderna, (La Laguna and Las Palmas de Gran Canaria, October 1999 - May 2000), Fundación Canaria Orotava de Historia de la Ciencia, Las Palmas (Canarian Islands, Spain), 23-34 (in Spanish).

Paper presented at the Universities of La Laguna and Las Palmas in which information is given about the role of Andalusia in the development of certain mathematical activities and their diffusion to medieval Europe.

DJE-01e

2001e Djebbar, Ahmed & Aballagh, Mohamed: *The life and work of Ibn al-Bannâ al-Murrâkûshî (1256-1321)*, Faculté des Lettres et Sciences Humaines - Université Mohamed V, Rabat (Morocco), 238 p. (in Arabic).

The book constitutes a bio-bibliographical essay on the most important mathematician from the Maghreb of the 14th century. It is based essentially on the handwritten sources from the Maghreb, which the two authors have studied during this last decade. The book contains a detailed biography of the mathematician, reconstituted from testimonies both from historians and from mathematicians who commented on some of his works. It also contains the complete list of Ibn al-Bannâ's writings, reconstituted from information supplied by his commentators, as well as with references from the libraries containing these manuscripts.

DJE-01f

2001f Djebbar, Ahmed: *Une histoire de la science arabe* [A History of Arab Science, Ahmed Djebbar interviewed by Jean Rosmorduc], Editions du Seuil, Paris (France), 384 p. (in French).

The contents of this book of popularization are presented in 8 chapters in the form of interviews. The first three treat the emergence and development of the Moslem Empire, the place of the science in the Arab–Moslem societies of the 9th to the 15th century and the role of the ancient heritage in the development of these sciences. The five remaining chapters are dedicated to the presentation of the most important scientific disciplines that were practiced in this civilization: astronomy, mathematics, physics, earth and life sciences, chemistry.

DJE-02a

2002a Djebbar, Ahmed: Pratiques savantes et savoirs traditionnels en pays d'Islam: l'exemple des sciences exactes [Scholarly practices and traditional knowledge in Islamic countries: the example of the exact sciences], in: *Actes du Colloque International sur "Science and Tradition: Roots and wings for Development"*, Académie Royale des Sciences d'Outre Mer, Brussels (Belgium), 62-86 (in French).

Paper presented at a colloquium organised by the 'Académie Royale des Sciences d'Outre Mer' & UNESCO (Brussels, 5-6 April 2001). Partial analysis and reflection about the relationships between two types of knowledge that are often separated, in the discourse on science, but that have known important interactions. The question is illustrated by the study of the complex relationships that existed between the oral and written transmission and the theoretic and practic aspects of scientific activity in the countries of the Islam.

DJE-02b

2002b Djebbar, Ahmed: L'épître d'al-Khayyâm sur "l'explication des prémisses problématiques du livre d'Euclide", *Farhang*, Teheran (Iran), Vol. 14, No. 39-40, 79-136 (in French).

This is the French translation of an important book of al-Khayam that includes three chapters: the first contains an attempt of demonstration of the postulate of parallels. The second presents new definitions of the equality and the inequality of two proportions considered better than those given by Euclid in Book V of the *Elements*. The third chapter deals with the composition of the proportions, which was an operation very useful for the astronomers.

DJE-02c

2002c Djebbar, Ahmed: La circulation des mathématiques entre l'Orient et l'Occident musulman: interrogations anciennes et éléments nouveaux [The circulation of mathematics between the islamic east and West: old questions and new elements], in Y. Dold-Samplonius, J. W. Dauben, M. Folkerts & B. Van Dalen (Eds.), *From China to Paris: 2000 Years Transmission of Mathematical Ideas*, Franz Steiner Verlag, Stuttgart (Germany), 213-236 (in French).

Paper included in the Proceedings of the Colloquium on "2000 Years Transmission of Mathematical Ideas: Exchange and Influence from Late Babylonian Mathematics to Early Renaissance Science" (Bellagio, Italy, May 8-12, 2000).

DJE-03a

2003a Djebbar, Ahmed: A Panorama of Research on the History of Mathematics in al-Andalus and the Maghreb between the Ninth and the Sixteenth Century, in: Jan P. Hogendijk & A. Sabra (Eds.), *The Enterprise of Science in Islam, New perspectives*, MIT Press, Cambridge MA (USA), 309-350.

Paper presented at the Dibner Institute Conference on "New Perspectives on Science in Medieval Islam" (Boston, November 6-8, 1998).

DJE-03b

2003b Djebbar, Ahmed: Les activités mathématiques au Maghreb à travers le témoignage d'Ibn Khaldûn [Mathematical activities in the Maghreb through the testimony of Ibn Khaldûn], *Actes des journées sur "Les sciences dans la phase de déclin" (Marrakech, 8-11 February 2001)*, Faculté des Lettres et Sciences Humaines, Rabat (Morocco), 7-22 (in French).

DJE-03c

2003c Djebbar, Ahmed: Quelques exemples de scholies dans la tradition arabe des Eléments d'Euclide [Some examples of scholies in the Arab tradition of Euclid's Elements], *Revue d'Histoire des Sciences*, Paris (France), Vol. 56, No. 2, 293-321 (in French).

DJE-03d

2003d Djebbar, Ahmed: Les activités mathématiques médiévales, un exemple d'échanges scientifiques et interculturels en Méditerranée [Medieval mathematical activities, an example of scientific and intercultural exchange in the Mediterranean], in: E. Gallo, L. Giacardi & O. Robutti (Eds.), *Conferenze e Seminari 2002-2003*, Publication du Seminario di Storia delle Matematiche "Tullio Viola", Turin (Italy), 287-308.

DJE-03e

2003e Djebbar, Ahmed: Mathématiques et société à travers un écrit maghrébin du XIVe siècle [Mathematics and society through a Maghrebian writing of the 14th century], *Actes du colloque international "De la Chine à l'Occitanie, chemins entre arithmétique et algèbre"* (Toulouse, 22-24 September 2000), Editions du C.I.H.S.O., Toulouse (France), 29-54 (in French).

DJE-03f

2003f Djebbar, Ahmed: Les activités mathématiques en Andalus et leur prolongement au Maghreb (IXe-XVe siècles) [Mathematical activities in Andalusia and their extension to the Maghreb (9th – 15th century)], *Journées de la Société Catalane d'Histoire des Sciences et des Techniques* (Barcelona, 15-17 November 2002), Societat Catalana d'Historia de la Ciencia i la Tecnica, Barcelona (Spain), 87-112 (in French).

DJE-03g

2003g Djebbar, Ahmed: Nasîr ad-Dîn at-Tûsî, un savant polygraphe du XIIIe siècle [Nasîr ad-Dîn at-Tûsî, a polygraphic scholar of the 13th century], *Revue Farhang*, Teheran (Iran), 159-181 (in French).

DJE-03h

2003h Djebbar, Ahmed: La science islamique: naissance et développement à travers l'exemple des mathématiques [Islamic science: Birth and development through the example of mathematics], *Ayene-ye Miras* [Mirror of Heritage], *Quarterly Journal of Book Review*, Teheran (Iran), New series, Vol. 5, No. 4 (No. 20), 27-50 (in French).

DJE-04a

2004a Djebbar, Ahmed: La phase arabe de l'histoire de la trigonométrie [The Arab phase in the history of trigonometry], *Actes du colloque "Les instruments scientifiques dans le patrimoine: quelles mathématiques ?"* (Rouen, 6-8 April 2001), Editions Ellipse Paris (France), 415-435 (in French).

DJE-04b

2004b Djebbar, Ahmed: Du nombre pensé à la pensée du nombre: quelques aspects de la pratique arithmétique arabe et de ses prolongements en Andalus et au Maghreb [From the thought number to the thinking of number: Some aspects of the Arab arithmetical practice and its continuation in Andalusia and in the Maghreb], in: C. Alvarez, J. Dhombres & J.-C. Pont (Eds.), *Actes de la "Rencontre Internationale de Peyresc sur la pensée numérique"* (Peyresc, 7-10 September 1999), *Sciences et Techniques en Perspective*, Brepols (Belgium), Second Series, Vol. 8, No. 1, 303-322 (in French).

The article presents what is known about the arithmetical practices from the Islamic East that have circulated in Andalusia and in the Maghreb and that were continued in both regions.

DJE-04c

2004 Djebbar, Ahmed: Les sciences autour de la Méditerranée jusqu'à la guerre de Cent ans [Sciences around the Mediterranean until the 100-year war], *Cahiers art et science*, Université de Bordeaux 1 (France), numéro spécial 8, 75-90 (in French).

DJE-05a

2005 Djebbar, Ahmed: *L'algèbre arabe, genèse d'un art* [Arab algebra, genesis of an art], Vuibert-Adapt, Paris, 211 p. (in French) (Preface: Bernard Maitte).

Presents an overview of the genesis of algebra in Arab culture. The introduction explains the context in which the Arab algebraic tradition emerged (p. 11-18). The following chapters constitute the first part entitled "Arab algebra in the Muslim East": "The first steps of algebra as a discipline" (p. 19-48), "The Arab algebraic practices in the 9th century" (p. 49-51), "The contributions of the 10th century" (p. 51-54), "The new orientations of algebra in the 11th and 12th centuries (p. 54-70), and "The algebraic practices in the east after the 12th century" (p. 70-72). The second part "Arab algebra in the Muslim West" is composed of two chapters: "The beginnings of algebra in the Muslim West (p. 74-78) and "The algebraic practices through existing works" (p. 78-104). The third part is about Arab algebra in Europe (p. 105-116). The first appendix (p. 123-145) presents short biographies of

mathematicians, including the following North Africans Abu Kamil (d. 930), Abu Bakr al-Hassar (12th century), Samaw'al (d. 1175), Ibn al-Yasamin (d. 1204), Ibn Rashiq (c. 1275), Ibn al-Banna (1256-1321), Uqbani (1320-1408), Ibn Qunfudh (1339-1407), Ibn al-Ha'im (1352-1412), Ibn Haydur (d. 1413), Ibn al-Majdi (1365-1447), Qatrawani (15th century), Sibt al-Maradini (1423-1506), Ibn Ghazi (1437-1513), and of mathematicians born outside Africa but who lived for many years in North Africa, like Ibn al-Haytham (965-1041), Al-Qurashi (d. 1184), and Al-Qalasadi (1412-1485). Appendix 2 (p. 147-180) presents some types of algebraic problems. Appendix 3 (p. 181-184) presents testimonies on problems not solved by mathematicians from the countries of the Islam. Appendix 4 (p. 185-190) presents a lexicon of technical terms, followed by the general bibliography in appendix 5 (p. 191-206) and an index.

DJE-05b

2005 Djebbar, Ahmed: *L'âge d'or des sciences arabes* [The golden age of the Arabic sciences], Éditions Le Pommier & la Cité des sciences et d'industrie, Paris (France), 183 p. (in French).

Gives an overview of the development of the scientific production and practices realized in the Arabic language from the 8th to the 16th century. The chapters deal with mathematics, astronomy, geography, medicine, chemistry, mechanics and the appropriation of the Arab sciences in Europe.

DJE-05c

2005 Djebbar, Ahmed (Ed.): *Le catalogue de l'exposition "L'âge d'or des sciences arabes"* [Catalogue of the exhibition "The golden age of the Arabic sciences"], Institut du Monde Arabe - Actes Sud, Paris (France) (in French).

DJE-05d

2005 Djebbar, Ahmed: Savoirs mathématiques et pratiques métrologiques arabes [Mathematical knowledge and Arab metrological practices], in: L. Moulinier, L. Sullimann, C. Verna & N. Weill-Parot (Eds.), *La juste mesure, quantifier, évaluer, mesurer, entre Orient et Occident (VIIIe-XVIIIe siècle)*, Presses Universitaires de Vincenne, Paris (France), 59-78 (in French).

DJE-05e
2005 Djebbar, Ahmed: Les sciences arabes et leur circulation autour de la Méditerranée [Arab sciences and their circulation around the Mediterranean], in: *Figures de la science*, Editions Parenthèses, Marseille (France), 174-186 (in French).

DJE-05f
2005 Djebbar, Ahmed**:** Universalité et localité dans les pratiques scientifiques des pays d'Islam [Universality and locality in the scientific practices of the countries of the Islam], *Alliage*, Nice (France), No. 55-56, 35-42.

DJE-05g
2005 Djebbar, Ahmed: Les poèmes mathématiques arabes [The Arab mathematical poems], *Pour la science*, Paris (France), Dossier No. 47, 42-43 (in French).

DOU-84
1984 Doumbia, Salimata (Ed.): *Mathématiques dans l'environnement socio-culturel Africain* [Mathematics in the African social-cultural environment], Vol. 1: *Jeux* [Games], Institut de Recherches Mathématiques, Abidjan (Côte d'Ivoire), 240 p. (in French).

Studies mathematical aspects of traditional games of Ivory Coast / Côte d'Ivoire:
1. Verbal games: memory and counting games (S. Doumbia, J. Garin & T. Nguyen);
2. Simple calculation games: Lokoto and Abikou (T. Nguyen);
3. Board games: Awalé (S.Doumbia), Tiouk-Tiouk (F. Carpentier & T. Nguyen), Dili (T. Nguyen), Kpanê and Kro Konono Kpanê (S. Doumbia & T.Nguyen);
4. Gambling games: Kélio (F. Carpentier & S. Doumbia);
5. Games of chance: weight games (T. Nguyen), Nigbé, a cowry game (S. Doumbia).

DOU-89a
1989a Doumbia, Salimata: Review of Beart's *Jeux et jouets de l'ouest africain* (BEA-55), *AMUCHMA Newsletter*, Maputo (Mozambique), No. 3, 6-8.

DOU-89b

1989b Doumbia, Salimata: Mathematics in traditional African games, in: C. Keitel, P. Damerow, A. Bishop, P. Gerdes (Eds.), *Mathematics, Education, and Society*, UNESCO, Paris (France), 174-175.

The Mathematical Research Institute of Abidjan (IRMA, Ivory Coast) classified the traditional games of the country into five categories: verbal games, games of memory, calculating games, games on a checkerboard and games of chance. IRMA studies the mathematics involved in these games and looks for ways to integrate this mathematics into the curriculum. As an illustration the knowledge of probabilities in the *Nigbe Alladian* game is described.

DOU-91

1991 Doumbia, Salimata: *Jeux africains et mathématiques* [African games and mathematics], IRMA, Abidjan (Côte d'Ivoire), 10 p. (mimeo) (in French).

Paper presented at the 3rd Pan African Congress of Mathematicians, (Nairobi, Kenya, 1991), explaining the research program on African games and mathematics at IRMA (Mathematical Research Institute) of Abidjan (Côte d'Ivoire).

DOU-92

1992 Doumbia, Salimata & Pil, J.C.: *Les jeux de cauris* [Cowrie shells games], CEDA & Institut de Recherches Mathématiques d'Abidjan, Abidjan (Côte d'Ivoire), 74 p. (in French).

Describes games with cowrie shells: *nigbé* (as played by the Alladian, Côte d'Ivoire), *nigbé* (as played by the Godié, Côte d'Ivoire), *ediprè* (Ebrié, Côte d'Ivoire), *tiatia* (Bambara, Mali), *koue* (Gourounsis, Burkina Faso), *kar* (Dogon, Mali), *tcha-tcha* djirokémé (Benin), *kô* (Wès, Côte d'Ivoire), and equivalent games of chance like *nama* and *piaf* (Mali), *horbido* (Lébous, Senegal), *sonrai* and *bozo* (Mali), *paradis* (Mali), *abbia* (Gabon, Cameroon). Also analyses the mathematical aspects of these games and shows how the rules of some of the games like *nigbé* (Alladian) give all participants equal opportunity (chance) to win, i.e. they reflect an empirical knowledge of the involved probabilities. The book presents computer simulations of some games and argues for the uses of these games in mathematics education.

DOU-94a
1994 Doumbia, Salimata: Dossier jeux, mathématiques et sociétés [File on games, mathematics and societies], *Plot*, Orléans (France), Vol. 69, 1-31 (in French).

Contains an introduction on 'Mathematics in the African socio-cultural environment' with information about and examples from the traveling exhibition 'Games, Mathematics and Societies.'

DOU-94b
1994b Doumbia, Salimata & N'guessan, D.: Les jeux de cauris [Cowrie shells games], in TOU-94 (in French).

DOU-95
1995 Doumbia, Salimata: L'experience en Côte d'Ivoire de l'étude de jeux traditionnels africains et de leur mathématisation [The experience of Côte d'Ivoire in the study of traditional African games and their mathematization], in: IREM-95, 549-555 (in French).

DOU-97
1997 Doumbia, Salimata: Maths et Cultures: Pythagore en Afrique [Mathematics and cultures: Pythagoras in Africa], *Bulletin sur l'Harmonisation des Programmes de mathématiques des pays francophones d'Afrique et de l'Océan Indien*, Abidjan (Côte d'Ivoire), Vol. 3, 6-11 (in French).

Gives examples of Pythagoric figurate numbers in West Africa and presents some ideas of Gerdes' book *African Pythagoras* on African crafts and the Pythagorean theorem.

DRAC-50
1950 Drachmann, A. G.: Heron and Ptolemaios, *Centaurus*, Copenhagen (Denmark), Vol. 1, 117-131 (in German).

DRA-86
1986 Draisma, Jan; Tembe, Albasine; Kuijper; Jelske & Neeleman, Wim: Mathematics Education in Mozambique, in: *Proceedings of the 4th Symposium of the Southern Africa Mathematical Sciences Association (SAMSA)*, University of Swaziland, Kwaluseni (Swaziland), 56-96.

Presents an overview of the development of mathematics education in Mozambique.

DRA-93
1993 Draisma, Jan: How to handle, in (teacher) education, the theorem 8 + 5 = 13?, in: Julie, Cyril; Angelis, Desi & Davis, Zain (Eds.), *Political Dimensions of Mathematics Education 2. Curriculum reconstruction for society in transition*, Maskew Miller Longman, Cape Town (South Africa), 196 – 207

DRA-96
1996 Draisma, Jan: *Written subtraction in Mozambican schools*, in: Puig, Luis & Gutiérrez, Angel: *Proceedings of the 20th Conference of the International Group for the Psychology of Mathematics Education*, University of Valencia (Spain), Vol. 2, 321-328.

Paper gives information on written subtraction algorithms used in Mozambican schools and on how primary teachers do subtraction and interpret the procedures.

DRA-99
1999 Draisma, Jan: Numeração falada e gestual como recursos na aprendizagem inicial da matemática [Spoken and gesture numeration as resources for the early learning of mathematics], in: *Actas do ProfMat 99*, Associação de Professores de Matemática, Lisbon (Portugal), 253-269 (in Portuguese).

DRA-00
2000 Draisma, Jan: Gesture and oral computation as resources in the early learning of mathematics, in: Nakahara, Tadao & Koyama, Masataka: *Proceedings of the 24th Conference of the International Group for the Psychology of Mathematics Education*, Hiroshima University, Higashi-Hiroshima (Japan), Vol. 2, 257–264.

Report of an experimental program of gesture and oral computation realized in a semi-rural primary school in the centre of Mozambique. Portuguese, Ndau, Sena and Chuwabo are the four languages spoken by the pupils. Modified number words based on ten and five were introduced in Portuguese, Ndau and Sena, in order to give them the

same regular structure as they have in Chuwabo and have these correspond directly to gesture computation.

DRA-06a
2006 Draisma, Jan: *Teaching gesture and oral computation in Mozambique: four case studies*, doctoral thesis, Monash University, Clayton, Victoria (Australia).

DRA-06b
2006 Draisma, Jan: Ensinar cálculo gestual e oral em Moçambique: o Programa Experimental de Matemática (Beira, 1999-2003) [Teaching gesture and oral computation in Mozambique], *Matemática & Educação*, Beira (Mozambique), No. 3, 50-60.

DUR-94
1994 Duranti, Gian Carlo: *Codici del Pentateuco e matematica egizio-platonica* [Codex of the Pentateuch and Platonic-Egyptian mathematics], L'Arcipelago, Genova (Italy), 68 p. (in Italian).

DUV-99
1999 Duvillié, Bernard: *Sur les traces de l'Homo mathematicus: les mathématiques avant Euclide: Mésopotamie - Egypte – Grèce* [On the trace of the Homo Mathematicus: mathematics before Euclid: Mesopotamia – Egypt – Greece], Ellipses, Paris (France), 461 p. (in French).

DZI-95
1995 Dzielska, Maria: *Hypatia of Alexandria*, Harvard University Press [*Revealing Antiquity*, Vol. 8], Boston MA (USA), 157 p.

Translation by F. Lyra of an unpublished manuscript in Polish *Hypatia z Aleksandrii*. Contents the following chapters: The literary legend of Hypatia; Hypatia and her circle; The life and death of Hypatia.

Review: DEA-96.

E

EBE-92
1992 Ebeid, William: *Research in Mathematics Education in Egypt*, Aïn Shams University, Cairo (Egypt), 8 p. (mimeo)

Paper presented at the First AMU Symposium on Mathematics Education in Africa for the 21st Century (5-10 September 1992, Cairo, Egypt), giving an overview on the 240 theses (171 M.Ed. and 69 Ph.D.) in Mathematics Education defended at Egyptian universities in the period 1954-1990.

EGL-89
1989 Eglash, Ron & Broadwell, P.: Fractal Geometry in Traditional African Architecture, *The Dynamics Newsletter*, Santa Cruz CA (USA), July issue, 3-9.

A 2-dimensional Fourier transform is used to show fractal structure in an aerial photo of a Songay village in Mali.

EGL-94
1994 Eglash, Ron; Christian Sina Diatta & Nfally Badiane: Fractal structure in Jola material culture, *Ekistics*, Athens (Greece), Vol. 61, No. 368, 367-371.

Discusses self-similarity in altar, house, and village structures among the Jola in the Lower Casamance region in southern Senegal.

EGL-95a
1995a Eglash, Ron: Scaling hexagons in a Bassari initiation mask, *Mathematics Teacher*, Reston VA (USA), Vol. 88, No. 7, 618-620.

Short note that analyses the presence of a scaling series of hexagons in a mask from the Bassari (eastern Senegal) and compares it with the use of the number six in other contexts (time reckoning, string tallies, divination).

EGL-95b
1995b Eglash, Ron: Fractal geometry in African material culture, *Symmetry: Culture and Science*, Budapest (Hungary), Vol. 6, No. 1, 174-177.

The author reviews his findings of fractals in African material culture, and notes that we should resist making assumptions about the social dynamics associated with these structures, since they vary widely.

EGL-95c

1995c Eglash, Ron: African influences in cybernetics, in: Gray, Chris (Ed.), *The Cyborg Handbook*, Routledge, New York (USA), 17-28.

Many of the fundamental concepts of cybernetics (self-organization, the binary code) have connections with the history of the black diaspora.

EGL-97a

1997a Eglash, Ron: Bamana Sand Divination – Recursion in Ethnomathematics, *American Anthropologist*, Arlington VA (USA), Vol. 99, No. 1, 112-122.

Reflecting on his fieldwork realized among Bamana (or Bambara) diviners (Mali), the author compares their use of recursion, where the iterative function is addition modulo 2, with Cantor's recursion (cantor set), and hypotheses that an African concept of self-generated fecundity is the shared origin of both the Bamana divination and transfinite set theory.

EGL-97b

1997b Eglash, Ron: The African heritage of Benjamin Banneker, *Social Studies of Science*, London (UK), Vol. 27, 307-315.

"Benjamin Banneker (1731-1806) is well known for his accomplishments in early American applied science, as well as for his seminal role in African-American science history. Historical and linguistic evidence suggests that his grandfather was of Wolof origin, and that his father was from the area between what is now Ghana and Nigeria. This cultural heritage may have emerged in some of his mathematical thinking" (p.307).

EGL-98a

1998a Eglash, Ron: Geometric algorithms in Mangbetu design, *Mathematics Teacher*, Reston VA (USA), Vol. 91, No. 5, 376-381.

Analyses an ivory hat pin from the Mangbetu (northeastern Congo / Zaire) and the geometric algorithm involved in its production. The top of the pin is composed of four scaled, similar heads (forming isosceles right triangles in photographic projection).

EGL-98b
1998b Eglash, Ron: Fractals in African settlement architecture, *Complexity*, New York (USA), Vol. 4, No. 2, 21-29.

A comparison of fractals in African material culture and fractals in complexity theory.

EGL-98c
1998c Eglash, Ron & Gloria Gilmer: *Ethnomathematics and African hairstyle designs*, paper presented at 76th Annual Meeting of the National Council of Teachers of Mathematics (2-4 April 1998, Washington DC, USA).

EGL-99
1999 Eglash, Ron: *African Fractals: Modern Computing and Indigenous Design*, Rutgers University Press, Piscataway (USA), 258 p.

This beautifully illustrated book "introduces readers to fractal geometry and explores the ways it is expressed in African cultures. Drawing on interviews with African designers, artists, and scientists, Eglash investigates fractals in African architecture, traditional hairstyling, textiles, sculpture, painting, carving, metalwork, religion, games, practical craft, quantitative techniques, and symbolic systems. He also examines the political and social implications of the existence of African fractal geometry."

Review: PETE-99.

EIS-77
1877 Eisenlohr, August: *Ein mathematisches Handbuch der alten Ägypter "Papyrus Rhind"* [Rhind papyrus: A mathematical handbook of the ancient Egyptians], reprint: Martin Sändig, Walluf b. Wiesbaden (Germany), 1972 (in German).

Introduction to and translation of the Ahmose papyrus.

EKU-75
1975 Ekundayo, S. A.: Vigesimal number derivational morphology: Yoruba grammatical competence epitomized, Linguistics Department, University of Ife, Ife (Nigeria).

ELA-90
1990 El-Abbadi, Mostafa: The life and fate of the ancient Library of Alexandria, UNESCO, Paris (France), 250 p. (also published in French and Arabic).

Describes the background and the history of the Library of Alexandria: from its creation in the early third century BC to the destruction of the Royal Library in 48 BC and of the Daughter Library in 391. Particular attention is given to the type of scholarship cultivated at Alexandria. Eratosthenes of Cyrene, author of 'On the Measurement of the Earth', was the chief librarian from 245 to 204/1 BC. Other mathematicians that are referred to, are Euclid (86), Heron (90), Claudius Ptolemy (141), Theon and Hypathia (159).

ELS-78
1978 El Sawi, M.: Change in Mathematics Education since the late 1950's - ideas and realisation: Sudan, *Educational Studies in Mathematics*, Dordrecht (Netherlands), Vol. 9, No. 3, 317-330.

ELT-79a
1979a El Tom, Mohamed: The proliferation and popularization of mathematical results: the needs of the underdeveloped countries, in: Booss, Bernhelm & Niss, Mogens (Eds.), *Mathematics and the real world*, Birkhauser, Basel (Switserland), 54-57.

ELT-79b
1979b El Tom, Mohamed: On Future Mathematics in Underdeveloped Countries, in: Booss, Bernhelm & Niss, Mogens (Eds.), *Mathematics and the real world*, Birkhauser, Basel (Switserland), 112-115.

ELT-79c
1979c El Tom, Mohamed (Ed.): Developing Mathematics in Third World Countries. Proceedings of the international conference

held in Khartoum, March 6-9, 1978, North-Holland Publishing Company, Amsterdam (Netherlands), 207 p.

Includes three papers by African mathematicians:
* Mohamed El Tom (Sudan): The conference: Its background and work (3-22)
* A. A. Ashour (Egypt): Strategies and priorities in mathematical education and research in developing countries (25-31)
* Henri Hogbe-Nlend (Cameroon): La situation actuelle et les potentialités mathématiques de l'Afrique [Today's situation and the mathematical potential of Africa] (157-164).

ELT-83
1983 El Tom, Mohamed: Problems of curriculum development in Sudan, *Proceedings of the Fourth International Congress on Mathematical Education*, Boston MA (USA), 366-368.

ENG-85
1985 Engels, Hermann: Quadrature of the circle in Ancient Egypt, *Historia Mathematica*, New York (USA), Vol. 3, 137-140 [reproduced in: Berggren, Lennart; Borwein, Jonathan & Borwein, Peter (Eds.), *Pi: A Source Book*, Springer, New York, 2004 (3rd edition), 3-6]

Presents an hypothesis on how the Ancient Egyptian formula for the determination of the area of a circle could have been obtained.

ENG-00
2000 Engels, Hermann: Über Kreisquadraturen der Antike [On Quadratures of the Circle in Antiquity], *Mitteillungen aus dem mathematischen Seminar Giessen*, Giessen (Germany), Vol. 243, 51-77 (in German).

Notes a connection between an Egyptian and an Indian approximation of π and contains an analysis of the first Archimedean bounds for π and a reconstruction of the second Archimedean bounds mentioned by Heron of Alexandria.

ENU-79
1979 Enukoha, I. O.: *The mathematical heritage of the Igbos*, M.Ed. project, Ahmadu Bello University, Zaria (Nigeria).

ENU-86

1986 Enukoha, I. O.: *Counting and geometry in traditional Ibibio and Efik societies* (paper presented at the 2nd Pan-African Congress of Mathematicians, March 1986, Jos, Nigeria).

Describes the Efik-Ibibio counting words system, which is a mixture of base five and base ten, and the local concepts of lines and shapes.

ENU-92

1992 Enukoha, Obinna & Nwaiwu, Sunny I.: *Teaching mathematics in the elementary schools*, Institute of Education, University of Calabar (Nigeria), 132 p.

ERN-80

1980 Ernest, Paul: On the Adequacy of the Egyptian Representation of Fractions, *Bulletin of the Institute of Mathematics and its Applications*, Southend-on-Sea (UK), Vol. 16, No. 10, 219-221.

"This discusses different algorithms for representing rational numbers as sums of unit fractions (1/n), referring to and inspired by the Ancient Egyptian representation of fractions as such sums."

ERN-81

1981 Ernest, Paul: Egyptian Fractions in the Classroom, *Mathematics in School*, Leicester (UK), No. 66, 19-20. (Reprinted in Cornelius, M. (Ed.), *The Best of Mathematics in School*, Longman, London, 1989, 73-74.)

"This describes a school based project for getting pupils to better appreciate mathematics through its history, and focuses on the Ancient Egyptian representation of fractions."

ESH-74

1974 Eshiwani, George: Mathematics and science education in Kenya: issues and problems, *Kenya Educational Review*, University of Nairobi, Nairobi (Kenya), June.

The author asserts that neither the aims nor the practice of mathematics and science education in Kenya is attuned to the needs of society or of the individual learner. The traditional worldview differs radically from that of the western world. At primary level, traditional concepts of reality and causality are ignored and the problems of

linguistic transfer not appreciated. Practical reforms are suggested, with a particular emphasis on teaching mathematics and science in conjunction with traditional technologies.

ESH-75

1975 Eshiwani, George: Sex differences in the learning of mathematics among Kenyan high school students, *Kenya Educational Review*, University of Nairobi, Nairobi (Kenya), December, 111-119.

Study aimed to find out whether there is a significant difference in achievement and retention in mathematics between boys and girls in Kenyan secondary schools, and to identify factors which are significant predictors of achievement. It was found that teaching methods were a significant differential predictor between the sexes, but that attitudes towards mathematics, and expectations of their sex roles, were not.

ESH-79

1979 Eshiwani, George: The goals of mathematics teaching in Africa: a need for re-examination, *Prospects*, UNESCO, Paris (France), Vol. IX, No. 3, 346-352.

ESH-80

1980 Eshiwani, George S.: *The death of new mathematics in Kenya*, Bureau of Educational Research, Kenyatta University College, Occasional paper No. 3042, Nairobi (Kenya), 8 p.

ESH-83a

1983a Eshiwani, George: *A study of the goals of mathematics education in Africa*, Bureau of Educational Research, Kenyatta University College, Nairobi (Kenya), 37 p.

ESH-83b

1983b Eshiwani, George: *A study of women's access to higher education in Kenya, with a special reference to mathematics and science education*, Bureau of Educational Research, Kenyatta University College, Nairobi (Kenya), 75 p.

ESH-93
1993 Eshiwani, George S.: *School, mathematics and work: a study of external efficiency of primary schools in Kenya*, Bureau of Educational Research, Kenyatta University, Nairobi (Kenya), 50 p.

ETI-86
1986 Étienne, E. & Roels, J.: Deux aspects particuliers du problème des moyennes dans Pappus d'Alexandrie [Two particular aspects of averages in Pappus of Alexandria], *Revue des Questions Scientifiques*, Paris (France), Vol. 157, No. 2, 179-198.

ETU-67
1967 Etuk, Elisabeth: *The development of number concepts, an examination of Piaget's theory with Yoruba Nigerian children*, doctoral thesis, Columbia Teachers College, New York (USA).

EUC-26
1926 Euclid of Alexandria: *The Thirteen Books of Euclid's Elements (translated with introduction and commentary by Sir Thomas L. Heath*), Dover Publication, New York (USA), Vol. 1 (Books I and II), 432 p., Vol. 2 (Books III-IX), 436 p., Vol. 3 (Books X-XIII), 546 p. (New Edition, 1956).

EUC-69
1969 Euclid of Alexandria: *Die Elemente* [The Elements], Wissenschaftliche Buchges., Darmstadt (Germany), 479 p. (in German).

EUC-90
1990 Euclid of Alexandria: *Les Élements (traduits du texte de Heiberg)*[The Elements translated from the Heiberg text]: Vol. 1, Livres I-IV: *Géométrie plane* [Books I-IV: Plane geometry], Presses Universitaires de France, Paris (France), 531 p. (in French).

Contains a general introduction by Maurice Caveing (13-148) and a translation and commentaries by Bernard Vitrac on the first four books of Euclid's *Elements* on plane geometry based on the text by Heiberg (149-519).

EUC-93a
1993 Euclid of Alexandria: *Les oeuvres d'Euclide* [The works of Euclid], Blanchard, Paris (France), 627 p. (in French).

Reprint of the translation of Euclid's works by F. Peyard with a new introduction by Jean Itard.

EUC-93b
1993 Euclid of Alexandria: *The Data of Euclid*, Union Square Press, Baltimore (USA), 207 p.

Translation from the text of H. Menge (1896) by George L. Mc Dowell & Merle A. Sokolik.

EUC-94
1994 Euclid of Alexandria: *Les Eléments* [The Elements], Volume 2, *Livres V-IX* [Books V-IX], Presses Universitaires de France, Paris (France), 572 p. (in French).

This is the French translation, by Bernard Vitrac, of Books V to IX of Euclid's *Elements* based on Heiberg's edition. The translation is preceded by an introduction and is accompanied by a number of commentaries.

EUC-98
1998 Euclid of Alexandria: *Les Eléments* [The Elements], Vol. 3: *Livre X: Grandeurs Commensurables et Incommensurables. Classification des Lignes Irrationelles* [Book X: Commensurable and incommensurable magnitudes. Classification of irrational lines], Presses Universitaires de France, Paris (France), 433 p. (in French).

An annotated translation by Bernard Vitrac of book X of Euclid's *Elements*.

Review: GUG-99.

EUC-01a
2001a Euclid of Alexandria: *Euclid's Elements of Plane Geometry* (with appendix and supplements by William Desborough Cooley), Elibron, Boston MA (USA), 189 p. (paperback and electronic versions).

Reprint of the 1840 edition of Cooley's edition of the *Elements*, which was intended primarily for educational purposes.

EUC-01b

2001b Euclid of Alexandria: *The Elements of Euclid for the Use of Schools and Colleges* (with notes, appendix, and exercises by Isaac Todhunter), Elibron, Boston MA (USA), 421 p. (paperback and electronic versions)

Reprint of the 1864 edition of Todhunter's edition of the *Elements*; contains the first six books and portions of books XI and XII.

EUC-01c

2001c Euclid of Alexandria: *Les Eléments* [The Elements], Volume 4, *Livres XI-XIII* [Books XI-XIII], Presses Universitaires de France, Paris (France), 482 p. (in French).

It is the last volume (Books XI-XIII) of the project of new French translation by Bernard Vitrac of Euclid's *Elements*, based on the Heiberg edition.

F

FAG-90
1990 Fagborun, J. Gbenga: Yoruba counting verses: a linguistic approach to oral tradition, *African Languages and Cultures*, London (UK), Vol. 3, No. 2, 167-180.

Analyses the way in which wordplay is used as a device to aid the memorization of counting mnemonics in Yoruba (Nigeria).

FAI-85
1985 Fainzang, Sylvie: Les sexes et leur nombres - Sens et fonction du 3 et du 4 dans une societé burkinabé [The sexes and their numbers. The meaning and function of 3 and 4 in a Burkinabe society], *L'Homme, revue française d'anthropologie*, Paris (France), Vol. XXV, No. 96, 97-109 (in French).

"The author analyzes in sociological terms the widespread West-African tendency to associate the numbers 3 and 4 with man and woman respectively, practice usually attributed to certain aspects of male and female anatomy. An analysis of Bisa society (Burkina Faso) shows how the meaning and function of this symbolism are directly related to representations of the person on the one hand, and to social space as defined by residence rules on the other. The author suggests that the discourse implied by this symbolism serves to found social relations between the sexes and to legitimate male domination" (109).

FAK-80
1980 Fakuade, R. A.: The controversy about mathematical education in Nigeria, *West African Journal of Education*, Ibadan (Nigeria), Vol. 21, No. 2, p. 29-41.

FAT-91
1991 Fataki, Kawalie Massane: Mathematics in the daily lives of Afrikans, personal recollections, *Research Notes on Africa*, Institute for Independent Education, Washington DC (USA), Vol. 3, 28-33.

Presents examples from measurement, games and riddles from Uganda and Congo / Zaire.

FAV-91

1991 Favilli, Franco & Villani, Vinicio: *Disegno e definizione del cubo: un'esperienza didattica in Somalia* [Drawing and definition of a cube: a didactical experience in Somalia], Universitá di Pisa, Pisa (Italy), 18 p. (mimeo) (in Italian)

Analyses the comparative findings of a test on defining and visualizing a cube realized among Somalian and Italian students.

FED-91

1991 Federspiel, Michel: Sur la définition euclidienne de la droite [On the Euclidean definition of the straight line], in RAS-91a, 115-130.

FEDE-90

1990 Federici Vescovini, G.: La fortune de l'optique d'ibn al-Haitham : le livre 'De aspectibus (Kitab al-Manazir)' dans le moyen-âge latin [The fortune of the optics of Ibn al-Haytham: the book 'De aspectibus (Kitab al-Manazir)' in the Latin Middle Ages], *Archives Internationales d'Histoire des Sciences*, Rome (Italy), Vol. 40, No. 125, 220-238 (in French).

FEM-97a

1997a FEMSA (Ed.), *Extracurricular and out of school factors affecting girls' participation in science, mathematics and technology subje*cts, Forum for African Women Educationalists (FAWE), Nairobi (Kenya), Report No. 5, 17 p.

FEM-97b

1997b FEMSA (Ed.), *Teacher Training, Qualification and Working Conditions*, Forum for African Women Educationalists (FAWE), Nairobi (Kenya), Report No. 8, 13 p.

FEM-97a and FEM-97b are examples of a series of dissemination reports of the Female Education in Mathematics and Science in Africa (FEMSA) project, based on country profiles compiled by Rose Eboutou Mfou (Cameroon), Georgina Quaisie (Ghana), Verdiana Masanja (Tanzania), and Jane Mulemwa (Uganda).

FIB-03

2003 *Leonardo Fibonacci: Matematica e società nel Mediterraneo nel secolo XIII* [Leonardo Fibonacci: Mathematics and society in the Mediterranean in the 13th century; Special double issue of the Italian journal *Bolletino di Storia delle scienze matematiche*, Bologna (Italy): Vol. 2, 2003; Vol. 1, 2004], 272 p.

Directly related to North Africa are the chapters by Djamil Aïssani and Dominique Valerian "Mathematics, commerce and society in Béjaïa (Bugia) at the time of the stay of Leonardo Fibonacci (12th – 13th century)" (in French), by Roshid Rashed "Fibonacci and the Latin continuation of Arabic mathematics" (in French), and by Ivo Schneider "The solutions of the two main problems concerning games of chance in the late European Middle Ages and the possibility of Islamic sources."

FIN-93

1993 Finch, Charles S.: *Africa and the Birth of Science and Technology*, Khenti, Decatur GA (USA).

FIN-98

1998 Finch, Charles S.: *The Star of Deep Beginnings, The Genesis of African Science and Technology*, Khenti, Decatur GA (USA), 284 p.

"The proto-technology of the modern world is traceable to iron ore mining 43,00 years ago in southern Africa and to the emergence of proto-mathematics from Africa's Great Lakes region over 25,000 years ago. From these Paleolithic beginnings, science and technology underwent a steady development in Africa, and the remotest origins of formal mathematics, astronomy, engineering, architecture, navigation, and map-making can be found there."

FINK-80

1980 Fink, D. R.: The Bono concept of measure: an essential factor in formal and nonformal educational programs, in: D. Brokensha, D.; Warren, D.M. & Werner, O. (Eds.), *Indigenous knowledge systems and development*, University Press of America, Lanham (USA), 243-267.

"The author has found among the Bono of Ghana terms which function as nouns and verbs and which reveal indigenous categories of linear distance, volume, weight, and time measures. The areas of applicability for mathematics and science teaching in local schools are pointed out."

FIS-79
1979 Fischler, R.: A remark on Euclid II, 11, *Historia Mathematica*, New York (USA), Vol. 6, No. 4, 418-422.

FLE-04
2004 Fleming, Steven: Review of Gerdes' *Awakening of Geometrical Thought in Early Culture* (GER-03a), *Nexus Network Journal,* Florence (Italy), Vol. 6, No. 1 (online available at: http://www.nexusjournal.com/reviews_v6n1-Fleming.html).

FLET-97
1997 Fletcher, Jonathan Arko: *A study of the appraisal of mathematics teachers in Ghana*, doctoral thesis, University of London (UK).

FOL-87
1987 Folkerts, M.: Adelard's versions of Euclid's 'Elements', in: Burnett, Charles (Ed.), *Adelard of Bath, an English scientist and Arabist of the early 12th century*, Warburg Institute, University of London, London (UK), 55-68.

FOL-93
1993 Folkerts, Menso & Hogendijk, Jan (Eds.), *Vestigia Mathematica, Studies in medieval and early modern mathematics in honour of H. L. L. Busard*, Rodopi B.V., Amsterdam (Netherlands), 473 p.

The following chapters, written by African historians and/or related to the history of Mathematics in Africa, are included:

* A. Djebbar: Deux mathématiciens peu connus de l'Espagne du XIe siècle: al-Mu'taman et Ibn Sayyid [Two little known mathematicians from Spain in the 11th century: al-Mu'taman and Ibn Sayyid] (79-91)
* J. Hogendijk: The Arabic version of Euclid's *On Division*

* R. Lorch: Abû Kâmil on the pentagon and decagon
* B. Rosenfeld: "Geometric trigonometry" in treatises of al-Khwârizmî, al-Mâhânî and Ibn al-Haytham
* J. Sesiano: The medieval Latin version of the *Algebra* of Abû Kâmil
* P. Kunitzsch, "The peacock's tail": on the names of some theorems of Euclid's 'Elements' (205-214).

FOW-80
1980 Fowler, D. H.: Book II of Euclid's 'Elements' and a pre-Eudoxan theory of ratio, *Archive for History of Exact Sciences*, Berlin (Germany), Vol. 22, Nos. 1-2, 5-36.

FOW-82
1982 Fowler, D. H.: Book II of Euclid's 'Elements' and a pre-Eudoxan theory of ratio. II, Sides and diameters, *Archive for History of Exact Sciences*, Berlin (Germany), Vol. 26, No. 3, 193-209.

FOW-83
1983 Fowler, D. H.: Investigating Euclid's Elements, *British Journal for the Philosophy of Science*, Oxford (UK), Vol. 34, 57-70.

FOW-92
1992 Fowler, David H.: An invitation to read Book X of Euclid's 'Elements', *Historia Mathematica*, New York (USA), Vol. 19, No. 3, 233-264.

FOW-99
1999 Fowler, David H. & Taisbak, Chr. Marinus: Did Euclid's Circles Have Two Kinds of Radius ?, *Historia Mathematica*, New York (USA), Vol. 26, 361-364.

"It is often asserted that Euclid had no single word for "radius," but rather used the description "the line drawn from the center." We examine the linguistic practice of Euclid, Archimedes, and Appolonius and find that it is more subtle than that."

FRA-72
1972 Fraser, Peter M.: *Ptolemaic Alexandria*, Claredon Press, Oxford (UK), 3 vol., 812 p.

FRE-92
1992 Freitas, Lima de: Notes on some pentagonal "mysteries" in Egyptian and Christian iconography, in: Hargittai, Istvan (Ed.), *Fivefold Symmetry*, World Scientific, Singapore, 307-332.

FRI-05
2005 Friberg, Jören: *Unexpected links between Egyptian and Babylonian mathematics*, World Scientific, Hackensack NJ (USA), 308 p.

"Mesopotamian mathematics is known from a great number of cuneiform texts, most of them Old Babylonian, some Late Babylonian or pre-Old-Babylonian, and has been intensively studied during the last couple of decades. In contrast to this Egyptian mathematics is known from only a small number of papyrus texts, and the few books and papers that have been written about Egyptian mathematical papyri have mostly reiterated the same old presentations and interpretations of the texts. In this book, it is shown that the methods developed by the author for the close study of mathematical cuneiform texts can also be successfully applied to all kinds of Egyptian mathematical texts, hieratic, demotic, or Greek-Egyptian. At the same time, comparisons of a large number of individual Egyptian mathematical exercises with Babylonian parallels yield many new insights into the nature of Egyptian mathematics and show that Egyptian and Babylonian mathematics display greater similarities than expected."

FUR-03
2003 Furlong, David: *Sekeds and the Geometry of the Egyptian Pyramids* (online available at: www.kch42.dial.pipex.com/articles_frame_earth_sekeds.htm)

G

GAF-87
1987 Gafai, M. M.: *Basic mathematical knowledge of unschooled adults of Katsina State*, M.Ed. project, Ahmadu Bello University, Zaria (Nigeria).

GAI-01
2001 Gairín Sallán, José María: Una interpretácion de las fracciones egipcias desde el Recto del Papiro Rhind [An interpretation of Egyptian fractions based on the Recto of Rhind's Papyrus], *LLULL, Revista de la Sociedad Española de Historia de las Ciencias y de las Técnicas*, Zaragoza (Spain), Vol. 24, 649-684 (in Spanish).

"Accepting as a premise that numerical entities must be associated to the social reality in which they appear, this article exposes that ancient Egyptian fractions are considered to be expressions of the magnitude quantities which have been obtained after being equally shared-out. Taking into account this view, an exhausting analysis of the different cases collected in the table, which appears in the Recto of the Rhind's papyrus has allowed as the reconstruction of the shared-out processes used by the scribe Ahmes. Such a process has been undoubtedly complex, due to the fact that, for each one of the situations collected in this table, the scribe must make those decisions, which will help the realization of a real share-out under the most suitable conditions. This reconstruction has enabled us to interpret Egyptian fractions as the addition of the partial results obtained when the share-out must be carried out following consecutive stages, as well as to devise two possible alternatives about the way in which the scribe would execute the numerical calculations associated to the share-out process."

GAM-80
1980 Gama Amaral, Manuel: A contagem entre os Wayao [Counting among the Yao], in: M. Gama Amaral, *O povo Yao, Subsídios para o estudo de um povo do noroeste de Moçambique* [The Yao people, a contribution to the study of a people from the Northwest of Mozambique], Instituto de Investigação Científica Tropical, Lisbon (Portugal), 437-441 (in Portuguese).

Describes finger counting and the spoken numeration system of the Yao.

GAN-50

1950 Ganay, Solange de: Graphies bambara des nombres [Bambara graphical representation of numbers], *Journal de la Société des Africanistes*, Paris (France), Vol. 20, No. 2, 295-305.

Presents an overview of various graphic signs developed among the Bambara (Mali) to represent numbers.

GANN-64

1964 Gannoun, Abdallah: The "Graft of the spirits for the utilization of the dust ciphers" of Ibn al-Yâsamîn, *Majallat al-bahth al-ᶜilmî*, Rabat (Morocco), No. 1, 181-190 (in Arabic).

GANN-65

1965 Gannoun, Abdallah: Ibn al-Bannâ the number theorist, *Majallat al-bahth al-ᶜilmî*, Rabat (Morocco), No. 11-12, 89-105 (in Arabic).

GARD-91

1991 Gardies, J.-L.: La proposition 14 du livre V dans l'économie des 'éléments' d'Euclide [Proposition 14 of book V in the economy of Euclid's Elements], *Revue d'Histoire des Sciences*, Evry (France), Vol. 44, Nos. 3-4, 457-467 (in French).

GARD-94

1994 Gardies, J.-L.: L'organisation du livre XII des 'éléments' d'Euclide et ses anomalies [The organization of book XII of Euclid's Elements and its anomalies], *Revue d'Histoire des Sciences*, Evry (France), Vol. 47, No. 2, 189-208 (in French).

GARE-94

1994 Garegae-Garekwe, Kgomotso: *Cultural games and mathematics teaching in Botswana*, M. Ed thesis, University of Alberta, Edmonton, Alberta (Canada), 130 p.

"The purpose of this study was to find out the extent to which teachers of lower primary (Grades 2 & 3) used cultural games in teaching mathematics, and how they integrated such games in their instructional

practices. The study showed that teachers have little experience in using cultural games in mathematics teaching. However, it was shown that teachers use cultural games in teaching Setswana Language and Social Studies. The limitation of usage in mathematics lessons was due to lack of guidance on how to use them. Geometry was one of the topics in which cultural games such as 'mhele' and 'morabaraba' were used. The study is based on the survey of 145 teachers, ten (10) of which were interviewed."

GARE-96

1996 Garegae-Garekwe, Kgomotso: Multiplication and division of numbers using cultural games: The case of 'Diketo', *Mathematical Association of Botswana Newsletter*, Gaborone (Botswana), No. 81, 10 -12.

"The article presents a lesson plan on how a teacher could teach addition and multiplication of whole numbers using one of the cultural games called 'diketo'. This game is played by girls and boys (mostly played by girls) from age 5."

GARE-02

2002 Garegae-Garekwe, Kgomotso: *Teachers' Beliefs about mathematics, its teaching and learning and the communication of these beliefs to students: A case study in Botswana*, doctoral thesis, University of Manitoba, Winnipeg, Manitoba (Canada), 302 p.

"The study focused on teachers' beliefs about mathematics, its teaching and learning and the communication of these beliefs to students. It is a qualitative case study of three Junior Secondary School mathematics teachers. Data collection techniques included classroom observations, interviews, concept maps, personal essays, and perusal of official documents. In addition to responding to open-ended questionnaires, students constructed concept maps about their teachers' views about the teaching and learning of mathematics."

GAR-54

1954 Garnier, P.: Les noms de nombre en bambara [The number words in Bambara], *Notes africaines*, Paris (France), Vol. 62, p. 50 (in French).

Short comment on the words in Bambara (Mali) for 7, 9 (related to the duration of a pregnancy), 20 (related to the word for human being),

and 40 (related to the word for mat). As 7 is a secret number, the author does not know an expression for it other than the indirect 'wuoron-fla', that is, the 'second six'.

GARR-81
1981 Garrouste-Berte, Anne-Marie: *Observation dans les classes sur le développement de l'activité mathématique chez les élèves (1er cycle de l'enseignement secondaire au Niger)* [Observation in classes concerning the development of mathematical activity among the pupils (first cycle of the secondary school in Niger], doctoral thesis, Université de Paris VII (France) (in French).

GAY-67
1967 Gay, John & Cole, Michael: *The new mathematics and an old culture, a study of learning among the Kpelle of Liberia*, Holt, Rinehart & Winston, New York (USA), 100 p.

This classical study of the Kpelle people in central Liberia reveals the extent to which mathematical ideas and techniques are built into their culture and language: where daily life demands it, a mathematical skill becomes highly developed. In the 'western-style' school these skills are mostly ignored. The final chapter of the book presents recommendations. The basic recommendation for the teacher is "to reverse the present pattern of education. Instead of using the traditional Kpelle authoritarian method of rote memory and imitation as means of introducing the Western content, the teacher should use the Western, scientific method for comprehending, clarifying and organizing content drawn *directly* from the child's familiar, daily experiences" (p. 93).

GAY-71
1971 Gay, John & Welmers, William: *Mathematics and logic in the Kpelle language*, Institute of African Studies, University of Ibadan (Nigeria), Occasional Publication No. 21, 184 p.

Presents an analysis of mathematical terms in the Kpelle language of Liberia and indicates the range of concepts and skills with which Kpelle children arrive at primary school.

GEE-44

1944 Geevers, Theodor Friedrich: *The syllabus of Transvaal secondary school mathematics: an historical and critical study*, doctoral thesis, University of Pretoria (South Africa).

GER-80a

1980a Gerdes, Paulus: Mathematics Education in the People's Republic of Mozambique, *Materialien zur Analyse der Berufspraxis des Mathematikers*, Bielefeld (Germany), Vol. 25, 127-142.

Describes the development of mathematics education in Mozambique in the first years after the independence of the country (1975-1980).

GER-80b

1980b Gerdes, Paulus: Mathematik in Mozambique: Bildung und Mathematik-unterricht [Mathematics in Mozambique: Education and mathematics teaching], *Materialien zur Analyse der Berufspraxis des Mathematikers*, Bielefeld (Germany), Vol. 25, 143-275 (in German).

Introduction to the history of mathematics education in primary and secondary schools and in higher education in Mozambique during the colonial time and the first years after the independence of the country in 1975.

GER-81

1981 Gerdes, Paulus: Changing mathematics education in Mozambique, *Educational Studies in Mathematics*, Dordrecht (Netherlands), Vol. 12, 455-477.

Revised version of GER-80a.

GER-84

1984 Gerdes, Paulus: The first Mathematics Olympiads in Mozambique, *Educational Studies in Mathematics*, Dordrecht (Netherlands), Vol. 15, 149-172.

Describes the first Mathematics Olympiads in Mozambique and includes biographies of the winners.

GER-85

1985 Gerdes, Paulus: Three alternate methods of obtaining the ancient Egyptian formula for the area of a circle, *Historia Mathematica*, New York (USA), Vol. 12, 261-268.

New conjectures on the origin of the ancient Egyptian formula for the area of a circle are formulated on the basis of an examination of old African craft techniques, e.g. the transformation of an elongated rectangle in the form of a coiled rope into a circle.

GER-86a

1986 Gerdes, Paulus: How to recognize hidden geometrical thinking: a contribution to the development of anthropological mathematics, *For the Learning of Mathematics*, Montreal (Canada), Vol. 6, No. 2, 10-12, 17.

Deals with a method for recognizing geometrical thinking 'hidden' in the forms of traditional (African) objects, like baskets, pots, fish traps, houses.

GER-86b

1986 Gerdes, Paulus: On culture, mathematics and curriculum development in Mozambique, in: Stieg Mellin-Olsen & M. J. Hoines (Eds.), *Mathematics and Culture, a seminar report*, Caspar Forlag, Radel (Norway), 15-42.

GER-88a

1988 Gerdes, Paulus: On possible uses of traditional Angolan sand drawings in the mathematics classroom, *Educational Studies in Mathematics*, Dordrecht (Netherlands), Vol. 19, No. 1, 3-22

"Following a brief description of the drawing tradition of the Cokwe people (Angola), some possible uses of their pictograms in the mathematics classroom are suggested. The examples given in this paper range from the study of arithmetical relationships, progressions, symmetry, similarity, and Euler graphs to the determination of the greatest common divisor of two natural numbers."

Translation: GER-89a.

GER-88b

1988 Gerdes, Paulus: On some possible uses of traditional Angolan sand drawings in the mathematics classroom, *Abacus, the*

Journal of the Mathematical Association of Nigeria, Ilorin (Nigeria), Vol. 18, No. 1, 107-125.

Reproduction of GER-88a.

GER-88c
1988 Gerdes, Paulus: On culture, geometrical thinking and mathematics education, *Educational Studies in Mathematics*, Dordrecht (Netherlands), Vol. 19, 137-162.

"This article confronts a widespread prejudice about mathematical knowledge, that mathematics is 'culture-free', by demonstrating alternative constructions of Euclidean geometrical ideas developed from the traditional culture of Mozambique."

Reproduced in: POW-97.

GER-89
1989 Gerdes, Paulus: Desenhos tradicionais na areia em Angola e seus possíveis usos na aula de matemática, *BOLEMA*, Rio Claro (Brazil), Special No.1, 51-77 (in Portuguese).

Translation of GER-88a.

GER-90a
1990 Gerdes, Paulus: *Lusona: Recreações geométricas de África* [*Lusona*: Geometrical recreations of Africa], Instituto Superior Pedagógico, Maputo (Mozambique), 110 p. (in Portuguese).

Presents a brief introduction to the Cokwe sand drawings from Angola and presents geometrical recreations of the "Find the missing figures" type, inspired by the variation of dimensions while maintaining the geometrical algorithm as practiced in the Cokwe tradition.

Translations: GER-91c, GER-97b.
New edition: GER-02a.

GER-90b
1990b Gerdes, Paulus: *Vivendo a matemática: desenhos da África* [Living mathematics: Drawings of Africa], Editora Scipione, São Paulo (Brazil), 64 p. (in Portuguese).

Booklet for children age 8-14 on the mathematics of the Cokwe *sona* drawings from Angola.

GER-90c, GER-91a, GER-91b

1990c Gerdes, Paulus: On Mathematical Elements in the Tchokwe "Sona" Tradition, *For the Learning of Mathematics*, Montreal (Canada), Vol. 10, No. 1, 31-34.

1991a Gerdes, Paulus: On mathematical elements in the Tchokwe drawing tradition, *Discovery and Innovation, Journal of the African Academy of Sciences*, Nairobi (Kenya), Vol. 3, No. 1, 29-36.

1991b Gerdes, Paulus: On Mathematical Elements in the Tchokwe 'Sona' tradition, *Afrika Mathematika, Journal of the African Mathematical Union*, Ibadan (Nigeria), Series 2, Vol.3, 119-130.

The related papers GER-90c, GER-91a, and GER-91b present an introduction of the author's research findings on mathematical ideas in the sand drawing (*sona*) tradition of the Cokwe people (Angola): symmetries and monolinearity, classes and geometrical algorithms, rules for the construction of monolinear *sona*; and discuss the educational and mathematical potential of this tradition. The examples given in the papers vary.

GER-91c

1991 Gerdes, Paulus: *Ethnogeometrie. Kulturanthropologische Beiträge zur Genese und Didaktik der Geometrie* (Ethnogeometry. Cultural-anthropological contributions on the genesis and the didactics of geometry), Franzbecker Verlag, Bad Salzdethfurth (Germany), 360 p. (in German).

Studies the historical relationship between (the development of) geometrical knowledge and socially important activities (in Africa), such as mat and basket weaving, pot making and house building. In the second part of the book hypotheses on the early development of geometrical thinking are formulated, e.g. on the discovery of the 'Pythagoras' Theorem' and of the ancient Egyptian formula for the volume of a truncated pyramid. The last part presents examples of didactical experimentation with the aforementioned incorporation. Peter Damerow wrote the preface, entitled 'Ethnomathematics and Curriculumexport.'

Reprint: 2003.
Translations (partial): GER-91d, GER-92a, GER-03a.

GER-91d

1991 Gerdes, Paulus: *Cultura e o despertar do pensamento geométrico* [Culture and the Awakening of Geometrical Thinking], Instituto Superior Pedagógico, Maputo (Mozambique), 146 p. (in Portuguese).

Translation into Portuguese of GER-91c, excluding the didactical experimentation.

GER-91e

1991 Gerdes, Paulus: *Récréations géométriques d'Afrique - Lusona - Geometrical recreations of Africa*, Instituto Superior Pedagógico, Maputo (Mozambique), 110 p. (bilingual edition in French and English).

Translation into English and French of GER-90a. Presents examples of traditional sand drawings, called *(lu)sona*, from north-eastern Angola and geometrical recreations inspired by them. In the "Find the missing figures" activities the reader is given certain figures in the style of the 'sona' and invited to draw / create the missing figure(s) in the sequence.

New edition: GER-97b.

GER-91f

1991 Gerdes, Paulus: Fivefold Symmetry and (basket) weaving in various cultures, in: Istvan Hargittai (Ed.), *Fivefold symmetry in a cultural context*, World Scientific Publishing, Singapore, 243-259.

Discusses the appearance of fivefold symmetry in traditional craft work, especially from Mozambique.

GER-92a

1992 Gerdes, Paulus: *Sobre o despertar do pensamento geométrico* [On the Awakening of Geometrical Thinking], Universidade Federal de Paraná, Curitiba (Brazil), 105 p. (in Portuguese).

Brazilian reproduction of GER-91d with a preface by Ubiratan D'Ambrosio.

GER-92b

1992 Gerdes, Paulus: On the history of mathematics in Africa south of the Sahara, *AMUCHMA Newsletter*, Maputo (Mozambique), No. 9, 3-32.

Overview presented at the 3rd Pan-African Congress of Mathematicians (Nairobi, 1991) of research findings and of sources on or related to mathematics in the history of Africa south of the Sahara. Topics such as counting and numeration systems, mathematical games and puzzles, geometry, graphs, and continental and international connections are included.

Translation into Portuguese: GER-92d.
Updated version: GER-94f.

GER-92c

1992 Gerdes, Paulus: *Pitágoras Africano: Um estudo em cultura e educação matemática* [African Pythagoras: A study in Culture and Mathematics Education], Instituto Superior Pedagógico, Maputo (Mozambique), 102 p. (in Portuguese).

Includes two chapters related to the history of mathematics in Africa: 'Did Egyptian artisans know how to construct a square equal in area to the sum of the areas of two given squares?' (6-14) and 'A new proof related to an Ancient Egyptian decoration technique' (97-99). The other chapters show how diverse African designs may be used to discover and find proofs for the theorem of Pythagoras.

Translation: GER-94j.

GER-92d

1992 Gerdes, Paulus: Sobre a História da Matemática na África ao Sul da Sahara, *AMUCHMA, Revista sobre a História da Matemática em África*, Maputo (Mozambique), No. 1, 5-36 (in Portuguese).

Translation of GER-92b.

GER-93a

1993 Gerdes, Paulus (Ed.): *A Numeração em Moçambique* [Numeration in Mozambique], Instituto Superior Pedagógico, Maputo (Mozambique), 159 p. (in Portuguese).

Analyses the development of numeration systems in Mozambique and includes the following chapters:

* Paulus Gerdes & Marcos Cherinda: African systems of numeration (8-28);
* Paulus Gerdes: On the history of verbal numeration (29-34);
* Written sources on numeration and counting in Mozambique [languages: Makonde, Yao, Nyanja, Nyungwe, Makhuwa, Sena, Shona, Tshwa, Chope, Changana, Ronga, Swazi, Zulu] (35-106);
* Oral sources on numeration and counting in Mozambique (107-120), including: Abdulcarimo Ismael & Daniel Soares: Popular counting methods in Mozambique (114-120);
* Abílio Mapapá & Evaristo Uaila: Comparative tables and maps about spoken numeration in Mozambique (121-132);
* Jan Draisma: Spoken numeration as a resource in the learning of arithmetic (134-150);
* Some reflections to stimulate debate and research (151-159).

GER-93b

1993 Gerdes, Paulus & Marcos Cherinda: Words, gestures and symbols, *The UNESCO Courier*, Paris (France), November issue, 37-39 (also published in Arabic, French, Spanish, etc.).

Abridged version of a paper on numeration systems in Africa.

GER-93c

1993 Gerdes, Paulus: *L'ethnomathématique comme nouveau domaine de recherche en Afrique: quelques réflexions et experiences du Mozambique*, Instituto Superior Pedagógico, Maputo (Mozambique), 84 p. (in French).

Analyses ethnomathematics as a new a new research field in Africa and presents some reflections based on experiences in Mozambique.

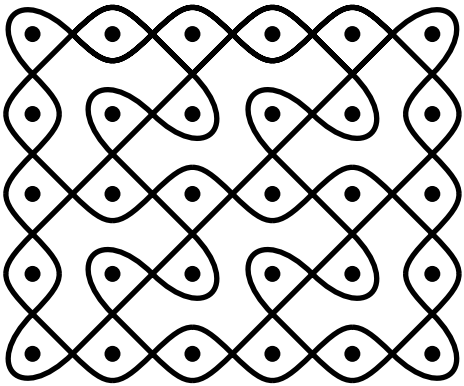
GER-93d

1993 Gerdes, Paulus: *Geometria Sona: Reflexões sobre uma tradição de desenho em povos da África ao Sul do Equador* [Sona Geometry: Reflections on a Drawing Tradition among Peoples of Africa South of the Equator], Instituto Superior Pedagógico, Maputo (Mozambique), Vol. 1, 200 p. (in Portuguese).

Volume 1 is dedicated to the analysis and reconstruction of mathematical elements in the sand drawing tradition of the Cokwe and

neighboring peoples in Angola, Congo / Zaire, and Zambia. Symmetries, classes and algorithms for the execution of the drawings (called 'sona'), and rules for the systematic construction of monolinear 'sona' are among the themes analyzed.

Translations: GER-94i, GER-95a, GER-97a, GER-06.



Example of a monolinear *(lu)sona*
(cf. GER-06, p. 71)

GER-93e
1993 Gerdes, Paulus: *Geometria Sona: Reflexões sobre uma tradição de desenho em povos da África ao Sul do Equador*, Instituto Superior Pedagógico, Maputo (Mozambique), Vol. 2, 169 p. (in Portuguese).

The second volume examines the educational and mathematical potential of the reconstructed 'sona' tradition.

Translations: GER-95a, GER-97a.

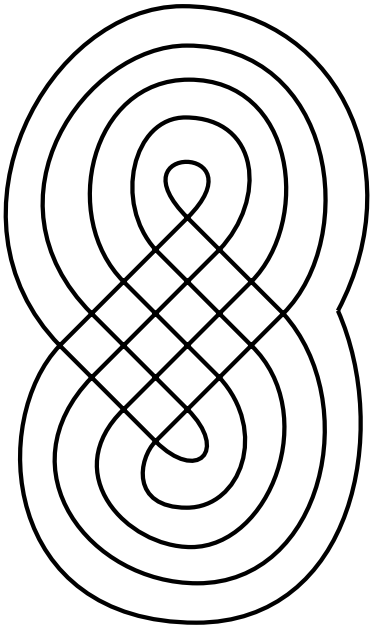
GER-93f
1993 Gerdes, Paulus: Exploring Angolan sand drawings (*sona*): stimulating cultural awareness in mathematics teachers, *Radical Teacher*, Cambridge MA (USA), Vol. 43, 18-24.

GER-94a
1994 Gerdes, Paulus: *Geometria Sona: Reflexões sobre uma tradição de desenho em povos ao Sul do Equador* [Sona Geometry: Reflections on a drawing tradition of peoples in Africa South of

the Equator], Instituto Superior Pedagógico, Maputo (Mozambique), Vol. 3. 123 p. (in Portuguese).

The third volume presents a comparative analysis, studying traditions from other parts of Africa and the world and/or of other periods that are technically similar to the 'sona' tradition. It contains the following chapters: 9. On geometrical algorithms in Ancient Egypt, 10. On monolinear motifs in Ancient Mesopotamia, 11. On some geometrical algorithms in India, 12. Short excursion to other continents, and 13. Back to Africa.

Translations: GER-95a, GER-97a.



Example of a monolinear engraving from Ancient Egypt
(cf. GER-95a, p. 479)

GER-94b

1994 Gerdes, Paulus & Gildo Bulafo: *Sipatsi: Tecnologia, Arte e Geometria em Inhambane*, Instituto Superior Pedagógico, Maputo (Mozambique), 102 p. (in Portuguese).

Analyses the technological and geometrical knowledge of basket weavers in Mozambique's Inhambane province. Presents a catalogue of decorative strip patterns on woven handbags (*sipatsi*) and some

suggestions are made for an educational and mathematical exploration of 'sipatsi'.

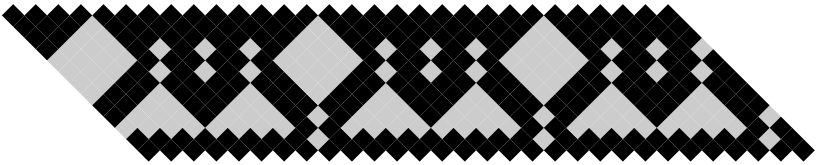
Translations: GER-94c, GER-94d.
Expanded edition: GER-03d.

GER-94c
1994 Gerdes, Paulus & Gildo Bulafo: *Sipatsi: Technology, Art and Geometry in Inhambane*, Instituto Superior Pedagógico, Maputo (Mozambique), 102 p.

Translation of GER-94b by Arthur B. Powell.
Review: ARO-95, VAQ-99.

GER-94d
1994 Gerdes, Paulus & Gildo Bulafo: *Sipatsi: Technologie, Art et Géométrie à Inhambane*, Instituto Superior Pedagógico, Maputo (Mozambique), 102 p. (in French).

Translation of GER-94b into French.



Example of a twill woven band on a *gipatsi*
(cf. GER-94)

GER-94e
1994 Gerdes, Paulus (Ed.): *Explorations in Ethnomathematics and Ethnoscience in Mozambique*, Instituto Superior Pedagógico, Maputo (Mozambique), 76 p.

The following chapters deal with mathematics and culture:
* Abdulcarimo Ismael: On the Origin of the Concepts of "Even" and "Odd" in Makhuwa culture (9-15);
* Marcos Cherinda: Mathematical-educational exploration of traditional basket weaving techniques in a children's "Circle of Interest" (16-23);
* Daniel Soares & Abdulcarimo Ismael: Popular counting methods in Mozambique (24-29);

* Jan Draisma: How to handle the theorem 8+5=13 in (teacher) education (30-48);
* Abílio Mapapá: Symmetries and metal grates in Maputo (49-55);
* Daniel Soares: Symmetric ornamentation on wooden spoons from Sofala Province (56-58);
* Marcos Cherinda: Strip patterns on wooden spoons from Inhambane Province (59-61).

GER-94f

1994 Gerdes, Paulus: On Mathematics in the History of Sub-Saharan Africa, *Historia Mathematica*, New York (USA), Vol. 21, 345-376.

Updated version of GER-92b.

GER-94g

1994 Gerdes, Paulus: Afrikanische Geometrien in Mathematik-unterricht [African Geometries in Mathematics Education], in: Schönbeck, J. et al. (Eds.), *Der Wandel im Lehren und Lernen von Mathematik und Naturwissenschaften*, Deutscher Studien Verlag, Weinheim (Germany), 192-202 (in German).

Two examples of southern African geometries are briefly presented: the originally female geometry of the ornamentation of 'sipatsi' handbags in Mozambique's Inhambane Province, and the male geometry of 'sona' sand drawings mostly of Eastern Angola and North-Western Zambia. The potential of these geometries for mathematics education is described.

GER-94h

1994 Gerdes, Paulus: Recherche ethnomathématique: une réponse à l'un des plus grands défis lancés à l'enseignement des Mathématiques en Afrique, *Édition Francophone de l'ISGEm Newsletter*, Dijon (France), No. 5, 9-12 (in French).

Short paper on ethnomathematical research as an answer to one of the most important challenges to mathematics education in Africa.

GER-94i

1994 Gerdes, Paulus: *Sona Geometry: Reflections on the tradition of sand drawings in Africa South of the Equator*, Instituto Superior Pedagógico, Maputo (Mozambique), Vol. 1. 200 p.

Translation of GER-93d into English by Arthur B. Powell. New edition: GER-06.

GER-94j

1994 Gerdes, Paulus: *African Pythagoras: A study in Culture and Mathematics Education*, Instituto Superior Pedagógico, Maputo (Mozambique), 102 p.

Contains the following chapters: 1. Did ancient Egyptian artisans know how to find a square equal in area to two given squares?, 2. From woven buttons to the Theorem of Pythagoras, 3. From fourfold symmetry to 'Pythagoras', 4. 'Pythagoras', similar triangles and the elephants-defense pattern of the BaKuba (Congo / Zaire), 5. A widespread decorative motif and the Theorem of Pythagoras, 6. From mat weaving patterns to 'Pythagoras' and magic squares, 7. A new proof by means of limits, 8. A new proof related to an ancient Egyptian decoration technique.

Translation of GER-92c.

GER-95a

1995 Gerdes, Paulus: *Une tradition géométrique en Afrique — Les dessins sur le sable* [A geometrical tradition in Africa — The sand drawings], L'Harmattan, Paris (France), 3 volumes, 594 p.

French language edition of GER-93d, GER-93e, and GER-94a.

GER-95b

1995 Gerdes, Paulus: *Women and Geometry in Southern Africa: Some Suggestions for Further Research*, Universidade Pedagógica, Maputo (Mozambique), 201 p.

The main objective of the book is to call attention to some mathematical aspects and ideas incorporated in the patterns invented by women in Southern Africa. It is meant as a contribution to the valuing, revival and development of traditions that may otherwise vanish. The themes treated in the book are: decorated handbags (Mozambique), coiled baskets (Swaziland), mat weaving, string figures, decorated pottery, grass brooms (Lesotho), tattooing and body painting, bead ornaments (Angola, Mozambique, South Africa), and mural decoration (Lesotho, South Africa).

Translation: GER-96b.
New Edition: GER-98d.

GER-95c
1995 Gerdes, Paulus: *Ethnomathematics and Education in Africa*, Institute of International Education, University of Stockholm, Stockholm (Sweden), 184 p.

Collection of papers published earlier in journals: 1. Introduction (1-4), 2. Ethnomathematical research (5-11), 3. On the concept of ethnomathematics (12-20), 4. How to recognize hidden geometrical thinking (21-29), 5. On culture, geometrical thinking and mathematics education (30-52), 6. A widespread decorative motif and the Pythagorean theorem (53-62), 7. 'Pythagoras' and patterns from the Bakuba (Congo / Zaire) (63-76), 8. On possible uses of traditional Angolan sand drawings in the mathematics classroom (77-102), 9. Exploration of the mathematical potential of 'sona' sand drawings (103-128), 10. Technology, art, games and mathematics education (129-134), 11. On the history of mathematics in Africa south of the Sahara (135-156), References (157-184).

GER-95d
1995 Gerdes, Paulus: L'ethnomathématique en Afrique [Ethnomathematics in Africa], *Plot*, Orléans (France), No. 70, 21-25 (in French).

Reproduction of the introduction to GER-93c.

GER-96a
1996 Gerdes, Paulus: On Ethnomathematics and the Transmission of Mathematical knowledge in and outside schools in Africa South of the Sahara, in: R. Waast (Ed.), *Les Sciences hors d'Occident au XXème Siècle* [Science outside the West in the 20th century], Vol.5: (Ed. M. Barrere): *Sciences et développement*, ORSTOM / UNESCO, Paris (France), 229-246.

Reflects on ethnomathematics and the teaching and learning of mathematics.

GER-96b
1996 Gerdes, Paulus: *Femmes et Géométrie en Afrique Australe*, L'Harmattan, Paris (France), 219 p. (in French).

French language edition of GER-95b.
Review: DAMB-98.

GER-96c

1996 Gerdes, Paulus: Sobre Matemática na História da África Sub-Sahariana, *Proceedings - Actes - Actas "História e Educação Matemática", ICME-8 satellite meeting of the International Study Group on the Relations between History and Pedagogy of Mathematics (HPM), Deuxième Université d'Été Européenne sur l'Histoire et l'Épistémologie dans l'Éducation Mathématique*, Associação de Professores de Matemática, Braga (Portugal), Vol. 1, 23-34 (in Portuguese).

Presents an introductory overview of mathematics in the history of Africa South of the Sahara.

GER-96d

1996 Gerdes, Paulus: On Women and Geometry (education) in Southern Africa, in: T. Kjaergard, A. Kvamme, N. Linden (Eds.), *Numeracy, Race, Gender, and Class — Proceedings of the Third International Conference on the Political Dimensions of Mathematics Education*, Gaspar Forlag, Landas (Norway), 207-217.

Suggests the incorporation of aspects of traditional female activities in geometry teaching.

GER-96e

1996 Gerdes, Paulus: *Lunda Geometry — Designs, Polyominoes, Patterns, Symmetries*, Universidade Pedagógica, Maputo (Mozamique), 149 p.

Develops the geometry of Lunda-designs, invented in the context of analyzing mathematically a class of sand drawings from northeastern Angola, a region called Lunda.

GER-97a

1997 Gerdes, Paulus: *Ethnomathematik dargestellt am Beispiel der Sona Geometrie* (Ethnomathematics through the Example of the Sona Geometry), Spektrum Verlag, Heidelberg (Germany), 433 p.

German language edition of the three volumes (GER-93d, GER-93e, and GER-94a) on the geometry of the 'sona' tradition in southern-central Africa. Preface by Harald Scheid and Erhard Scholz.

Reviews: HOY-98, KRAU-98, SCHM-98.

GER-97b
1997 Gerdes, Paulus: *Récréations géométriques d'Afrique - Lusona - Geometrical recreations of Africa*, L'Harmattan, Paris (France), 127 p. (Bilingual edition in French and English).

New edition of GER-91.

GER-98a
1998 Gerdes, Paulus: On culture and mathematics teacher education, *Journal of Mathematics Teacher Education*, Dordrecht (Netherlands), Vol. 1, No. 1, 33-53.

Presents a short history of mathematics teacher education in Mozambique since independence in 1975, highlighting the multicultural context and the role of the history of mathematics and of ethnomathematics in teacher education.

GER-98b
1998 Gerdes, Paulus: On some Geometrical and Architectural Ideas from African Art and Craft, in: Kim Williams (Ed.), *Nexus II: Architecture and Mathematics*, Edizioni dell'Erba, Florence (Italy), 75-86.

Presents some examples of geometrical ideas in traditional African building, as well as some further suggestions for architectural shapes inspired by African art and craft.

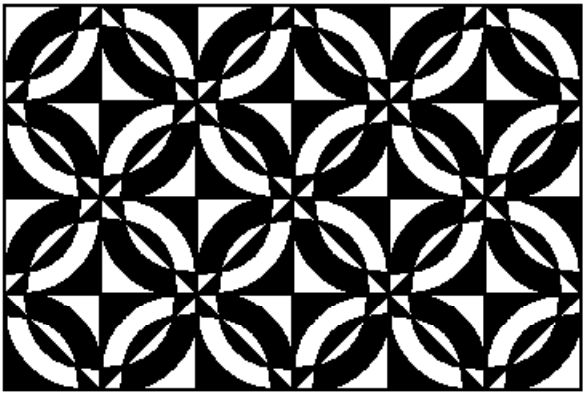
GER-98c
1998 Gerdes, Paulus: The Study of African Sona Geometry as an Example of Ethnomathematical Research, *Ethnologie Heute*, Münster (Germany), Vol. 2
[available online at: www.uni-muenster.de/EthnologieHeute]

Presents an introduction to studies on 'sona' geometry (Southern Central Africa).

GER-98d
1998 Gerdes, Paulus: *Women, Art and Geometry in Southern Africa*, Africa World Press, Lawrenceville NJ (USA) / Asmara (Eritrea), 244 p.

New edition of GER-95b, with an appendix by Salimo Saide on "The Geometry of Pottery Decoration by Yao Women (Nyassa Province)" (203-230).

Review: DAMB-98, SIZ-99.



Example of a *litema* wall decoration (Lesotho)
(cf. GER-98d, p. 156; GER-99a, p. 92)

GER-98e

1998 Gerdes, Paulus: On culture and mathematics education in (southern) Africa, in: Bernard Hodgson et al. (Eds.), *8th International Congress on Mathematical Education. Selected Lectures*, S.A.E.M. Thales, Sevilla (Spain), 221-231.

Presents a short overview of research on culture, mathematics, and mathematics education in Africa south of the Sahara, concentrating on southern Africa.

GER-99a

1999 Gerdes, Paulus: *Geometry from Africa: Mathematical and Educational Explorations*, The Mathematical Association of America, Washington DC (USA), 210 p.

Presents geometrical ideas from Africa south of the Sahara, with suggestions how they can be explored both mathematically and in mathematics education (secondary school, teacher education, university). The book is organized in the following parts:
Preface (Geometrical and educational explorations inspired by African cultural activities); Part 1: On geometrical ideas in Africa south of the Sahara [overview, p.2-53]; Part 2: From African designs to

discovering the Pythagorean Theorem [p.54-87]; Part 3: Geometrical ideas in crafts and possibilities for their educational exploration [Explores ideas from house building, wall decoration, mat and basket weaving, p.88-155]; Part 4: The 'sona' sand drawing tradition and possibilities for its educational use [p.156-204]. Contains a foreword by Arthur B. Powell.

Reviews: MIC-99, PETER-99, ASC-00, ASH-00, INO-00, JOH-00, ZAS-00c, BARRO-01,

GER-99b

1999 Gerdes, Paulus: On the production of mathematical knowledge in central and southern Africa, *Communications of the Centre for Advanced Studies of African Society* (CASAS), Cape Town (South Africa), Occasional paper, No. 7, 18 p.

Text of a paper presented at the Fourth World Archaeological Congress, January 10-14, 1999, University of Cape Town, South Africa.

GER-00a

2000 Gerdes, Paulus: Gerade und Ungerade – Zu einigen mathematischen Aspekten der mattenflechterei der Yombe-Frauen am unteren Kongo [Even and odd – On some mathematical aspects of the plaiting of mats by Yombe women in the Lower Congo], in: J. Blankenagel & W. Spiegel (eds.), *Mathematikdidaktik aus Begeisterung fuer die Mathematik. Festschrift fuer Harald Scheid*, Ernst Klett Verlag, Stuttgart (Germany), 83-93 (in German).

Analysis of mathematical aspects of the mats plaited by women of the Yombe people in the Lower Congo area at the end of the 19th century and the beginning of the 20th century.

GER-00b

2000 Gerdes, Paulus: *Le cercle et le carré: Créativité géométrique, artistique et symbolique de vannières et vanniers d'Afrique, d'Amérique, d'Asie et d'Océanie* [The circle and the square: Geometric, artistic and symbolic creativity of female and male basket weavers from Africa, America, Asia, and Oceania], L'Harmattan, Paris (France), 301 p. (in French).

Presents, on the one hand, a comparative and structural analysis of a type of plaited circular tray or basket cover, produced in several regions of Africa, America, Asia and Oceania, and, on the other hand, some elements of a catalogue, complemented by comments on the cultural context, the techniques and some implied geometrical ideas. Chapters 2 to 5 deal with Africa: The Bedik in Senegal (Chapter 2, 23-76); The Twsa, the Tonga and the Chope in south-east Mozambique (Chapter 3, 77-100); The Makonde and Makhuwa in north-east Mozambique (Chapter 4, 101-130), Varia Africana (Chapter 5, 131-148). Preface by Maurice Bazin.

Makhuwa circular tray with woven multiple spiral structure
(cf. GER-00b, p. 122)

GER-00c

2000 Gerdes, Paulus: Africa: South of the Sahara, in: Arne Hessenbruch (Ed.), *Reader's Guide to the History of Science*, Fitzroy Dearborn Publications, London (UK), 13-14.

Brief presentation of books on the history of science in Sub-Saharan Africa (Paper written in 1996).

GER-00d

2000 Gerdes, Paulus: Ethnomathematics, in: Arne Hessenbruch (Ed.), *Reader's Guide to the History of Science*, Fitzroy Dearborn Publications, London (UK), 227-229.

Brief presentation of books on ethnomathematics, in particular related to Sub-Saharan Africa (Paper written in 1996).

GER-00e

2000 Gerdes, Paulus: On mathematical ideas in cultural traditions of Central and Southern Africa, in: Helaine Selin (Ed.), *Mathematics across Cultures: A History of Non-Western Mathematics*, Kluwer, Dordrecht (Netherlands), 313-343.

An introduction to mathematical ideas in cultural traditions of Central and Southern Africa.

GER-01a

2001 Gerdes, Paulus: Ethnomathematics as a new research field, illustrated by studies of mathematical ideas in African history, in: Juan José Saldaña (Ed.), *Science and Cultural Diversity: Filling a Gap in the History of Science*, Cuadernos de Quipu, No. 5, Mexico City (Mexico), 11-36.

Paper presented at the conference 'New Trends in the History and Philosophy of Mathematics' (Roskilde, Denmark, 1998). Reproduced in GER-04a.

GER-01b

2001 Gerdes, Paulus: On the 'African Renaissance' and Ethnomathematical Research, in: Inocente Mutimucuio (Ed.), *Proceedings of the 9th Conference of the Southern African Association for Research in Mathematics, Science and Technology Education*, SAARMSTE, Maputo (Mozambique), Vol. 1, 1-14

Opening address at the 2001 annual conference of the Southern African Association for Research in Mathematics, Science and Technology Education (SAARMSTE).

GER-01c

2001 Gerdes, Paulus: Intrecci culturali [Cultural interweavings], in: P. Bellingeri, M. Dedò, S. di Sieno, C. Turrini (Eds.), *Il ritmo*

delle forme, Itenerario matematico (e non) nel mondo della simmetria, Mimesis, Milano (Italy), 121-124 (in Italian).

Describes some geometrical aspects of basket weaving in Mozambique.

GER-01d

2001 Gerdes, Paulus: Fantasie geometrico-simmetriche nell'artigianato africano [Geometrical-symmetrical imagination in African craft], in: Michele Emmer (Ed.), *Matemática e Cultura 2001*, Springer, Milano (Italy), 3-10 (in Italian).

Illustrates some geometrical-symmetrical aspects of African craft.

GER-01e

2001 Gerdes, Paulus: Exploring the Game of Julirde, *Teaching Children Mathematics*, NCTM, Reston VA (USA), Vol. 7, No. 6 (Focus issue: Mathematics and Culture), 321-327.

Illustrates how the 'julirde' game from the Fulbe in Cameroon may be explored in the teaching of geometry.

GER-02a

2002 Gerdes, Paulus: *Lusona: Recreações geométricas de África* [Lusona: Geometrical Recreations from Africa], Moçambique Editora, Maputo (Mozambique) / Texto Editora, Lisbon (Portugal), 128 p. (in Portuguese).

New edition of GER-90a. Preface by Jaime de Carvalho e Silva.

GER-02b

2002 Gerdes, Paulus: Mathematics in Mozambique, *The Mathematical Intelligencer*, New York (USA), Vol. 24, No. 2, 26-29.

Short overview of the development of mathematical activity in Mozambique since the Independence of the country in 1975.

GER-02c

2002 Gerdes, Paulus: Sobre a Produção de Conhecimentos Matemáticos em Países da África Central e Austral [On the production of mathematical knowledge in Central and Southern Africa], in: Mariana Leal Ferreira (Ed.), *Ideias Matemáticas de*

Povos Culturalmente Distintos, Global Editora, São Paulo (Brazil), 206-220 (in Portuguese).

Translation by Mariana Leal Ferreira of GER-99b.

GER-03a
2003 Gerdes, Paulus: *Awakening of Geometrical Thought in Early Culture*, MEP-Publications, Minneapolis (USA), 200 p.

Partial translation of GER-91c with a preface by Dirk J. Struik. Includes the following chapters: (1) Mathematicians on the origin of elementary geometrical concepts, (2) How did people learn to geometrize?, (3) Early geometrical concepts and relationships in societal activities, (4) Social activity and the formation of ancient geometry, (5) Conclusion.

Reviews: DAR-03, FLE-04, LUM-03a, ZAS-03b.

GER-03b
2003 Gerdes, Paulus: Origins of Geometrical Thought in Human Labor, *Nature, Society, and Thought*, Minneapolis (USA), Vol. 14, No. 4, 391-418.
[available online at: http://umn.edu/home/marqu002 by going to the NST link].

Slightly modified excerpt constructed from the first, second, and third chapters of GER-03a.

GER-03c
2003 Gerdes, Paulus: Plaited strip patterns on Tonga handbags in Inhambane (Mozambique) – An update, *Visual Mathematics*, Belgrade (Serbia), March 2003, Vol. 5, No. 1 [available online at: members.tripod.com/vismath/pap.htm].

The paper presents an update on strip patterns found on twill-plaited handbags and baskets made by Tonga artisans, mostly women. It includes a catalogue of 58 new strip patterns. All seven symmetry classes are represented. Attention is drawn to two particular types of strip patterns characterized by special plaiting structures.

GER-03d
2003 Gerdes, Paulus: *Sipatsi: Cestaria e Geometria na Cultura Tonga de Inhambane* [Sipatsi: Basketry and Geometry in the

Tonga Culture of Inhambane], Moçambique Editora, Maputo (Mozambique), 176 p. (in Portuguese).

Expanded edition of GER-94b. The book explains how artisans produce beautiful hand bags, called 'sipatsi' in Gitonga, a language spoken in the Mozambican province of Inhambane. The activity of weaving 'sipatsi' is originally a female activity. The book presents a catalogue of 362 decorative strip patterns plaited into the 'sipatsi', resulting from collecting 'sipatsi' for more than twenty-five years. It also includes suggestions for the mathematical-educational use of 'sipatsi', varying from the study of composition and symmetries to the study of progressions and pentagons. The book concludes with the presentation of some new phenomena in the production of 'sipatsi', underlining the geometric-artistic creativity of the basket weavers, and a comparison of 'sipatsi'-patterns with some woven strip patterns from other cultures (Northeast of Mozambique, Mexico and Brazil). Alcido Nguenha, the Minister of Education of Mozambique, wrote the preface.

GER-03e

2003 Gerdes, Paulus: Pensée mathématique et exploration géométrique en Afrique et ailleurs [Mathematical thinking and geometrical exploration in Africa and elsewhere], *Revue Diogène*, UNESCO & Presses Universitaires de France, Paris (France), No. 202, 126-144 (in French; also published in the Arabic, Chinese, English and Spanish versions of the journal).

English language version: GER-04c.

GER-03f

2003 Gerdes, Paulus: Symmetry-Geometry Aspects of *Mavuku* Baskets among the Makhuwa (Mozambique), *Symmetry: Culture and Science*, Vol. 12, No. 1-2, Budapest (Hungary), 87-114.

Presents an analysis of *mavuku* containers produced by Makhuwa basket weavers in the Northeast of Mozambique. The containers consist of two twill woven circular trays. The paper analyses the symmetries and the geometric structure of the weaving designs. The know-how of the old master-weaver Mulaliha from Rapale receives particular attention.

GER-03g

2003 Gerdes, Paulus: Exploring Plaited Plane Patterns among the Tonga in Inhambane (Mozambique), *Symmetry: Culture and Science*, Vol. 12, No. 1-2, Budapest (Hungary), 115-126.

Discusses a class of plane patterns encountered on twill-plaited baskets recently made by Tonga artisans, mostly women, where colored strands alternate with groups of natural-colored strands. Within the conditions considered by the basket weavers and taking into account the symmetries, they discovered all possible solutions.

GER-03h

2003 Gerdes, Paulus: From African 'sona' drawings to the discovery of new symmetries and matrices, in: Nouzha El Yacoubi *et al*. (Eds.), *Proceedings of the 13th Pan African Mathematics Olympiad*, Ministério da Educação, Maputo (Mozambique), 51-64.

GER-04a

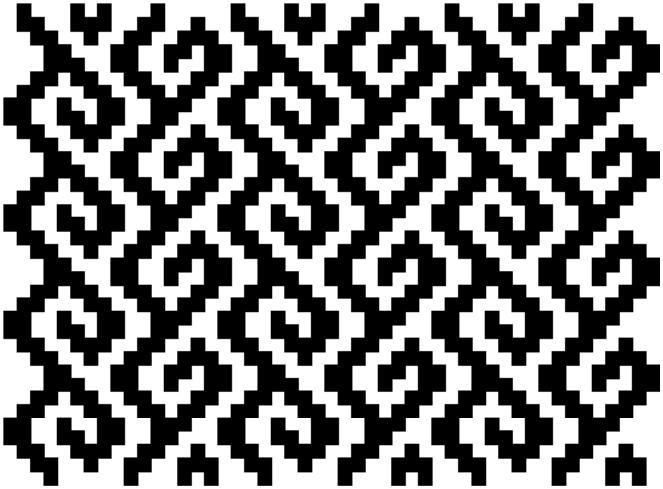
2004 Gerdes, Paulus: Ethnomathematics as a new research field, illustrated by studies of mathematical ideas in African history, in: Tinne Hoff Kjeldsen, Stig Andur Pedersen & Lise Mariane Sonne-Hansen, *New Trends in the History and Philosophy of Mathematics,* University Press of Southern Denmark, Odense (Denmark), 135-161.

Reproduction of GER-01a.

GER-04b

2004 Gerdes, Paulus: Symmetries on mats woven by Yombe women from the Lower Congo area: On the interplay between cultural values and mathematical-technical possibilities, in: Dorothy K. Washburn & Donald W. Crowe (Eds.), *Symmetry Comes of Age, The Role of Pattern in Culture*, University of Washington Press, Seattle (USA), 81-99.

Analyses mathematical ideas involved in the designing and production of mats by Yombe women from the Lower Congo area at the end of the 19th century and the beginning of the 20th century.



Example of a Yombe woven plane pattern
(cf. GER-04b, p. 92)

GER-04c
2004 Gerdes, Paulus: Mathematical thinking and geometrical exploration in Africa and elsewhere, *Diogenes*, UNESCO & Sage Press, London (UK), No. 202, 107-122.

English language version of GER-03e. Paper originally written for presentation at the international conference "Towards an encounter of rationalities" (Porto Novo, Benin, 2002).

GER-04d
2004 Gerdes, Paulus: Interweaving Art and Mathematics in African Design, *International Review of African American Artists* (USA), Vol. 19, No. 3, 45-47.

GER-04e
2004 Gerdes, Paulus: Vinte cinco Anos de Estudos Histórico-Etnomatemáticos em África ao Sul da Sahara [Twentyfive Years of Historical-Ethnomathematical Studies in Africa South of the Sahara], *LLULL, Revista Española de História de las Ciencias y de las Técnicas*, Zaragoza (Spain), Vol. 26, No. 56, 491-520 (in Portuguese).

An overview of the bibliography organized by region and country.

GER-04f
2004 Gerdes, Paulus: *Basketry, Geometry, and Symmetry in Africa and the Americas*, E-book, Visual Mathematics, Belgrade (Serbia) [on-line available at: www.mi.sanu.ac.yu/vismath/].

GER-04g
2004 Gerdes, Paulus: Weaving Polyhedra in African Cultures, *Symmetry: Culture and Science*, Budapest (Hungary), Vol. 13, No. 3-4, 339-355.

GER-05a
2005 Gerdes, Paulus: About Culture and Geometrical Thought, in: Giandomenico Sica (Ed.), *What is Geometry?*, Polimetrica, Milano (Italy), 53-64.

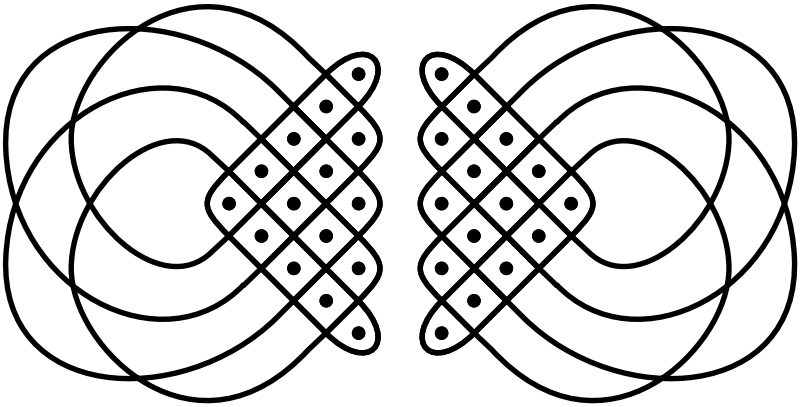
GER-05b
2005 Gerdes, Paulus: Ethnomathematics, geometry and educational experiences in Africa, in: Theophilus Okere, Chukwudi Njoku & René Devisch (Eds.), *All knowledge is first of all local knowledge*, Special issue of the *Africa Development Journal*, CODESRIA, Dakar (Senegal), Vol. XXX, No. 3, 48-65.

GER-05c
2005 Gerdes, Paulus: *Nirrosula*, an African musical instrument as a source of inspiration for mathematical exploration, in: Rosemond, Frances A. & Copes, Larry (Eds.), *Educational Transformations: Changing our lives through mathematics; A tribute to Stephen Ira Brown*, AuthorHouse, Bloomington Indiana (USA), 367-378.

A *nirrosula* is composed of a series of plaited nonahedra. Each nonahedron is made with only one plant strip.

GER-06
2006 Gerdes, Paulus: *Sona Geometry from Angola. Mathematics of an African Tradition*, Polimetrica International Science Publishers, Monza (Italy), 232 p. [Preface by Arthur B. Powell]

New edition of GER-94i. Includes an appendix on "Mathematical research inspired by the *sona* tradition: the example of mirror curves, Lunda-designs and cycle matrices" (217-232).



Example of a symmetric *(lu)sona* composed of two monolinear halves (cf. GER-06, p. 122)

GER-07

2007 Gerdes, Paulus: *African Doctorates in Mathematics: A Catalogue*, Lulu, New York (USA), 383 p. [Preface by Mohamed Hassan].

Presents a catalogue of over 2000 doctoral theses by Africans in all fields of mathematics, including applied mathematics, mathematics education and history of mathematics. The catalogue is organized by African country of birth and/or citizenship. The introduction explains the purpose, criteria for inclusion, data collection, and scope of the catalogue. Equally information is given about distribution by country of the doctorate holders, localization of doctorate awarding institutions, distribution of doctorates by period and by gender, mathematical density of African countries, mobility of African mathematicians inside and outside the continent, mathematical families and the first male and female African mathematicians who earned a doctorate. The appendices contain lists of female doctorate holders, of holders of doctorates in mathematics education, and of doctorates awarded by African universities to non-Africans and of doctoral theses by non-Africans about mathematics in Africa. The last appendix gives an overview of activities of African mathematicians at the service of their wider communities, inclusive lists of African mathematicians serving as presidents of their universities and as ministers in national governments.

GERH-85

1985 Gerhardt, Ludwig: *Zahlensysteme in der nigrianischen Plateausprachen – Import und Export in Naira und Schilling* [Numeral systems in Nigerian Plateau languages – Import and export in Naira and Shilling], manuscript, Würzburg (Germany), 1985 (in German).

GERH-87

1987 Gerhardt, Ludwig: Some remarks on the numerical systems of Plateau languages, *Afrika und Übersee*, Berlin (Germany), Vol. 70, 19-29.

Discusses the transition from duodecimal to decimal numeration in some languages belonging to the eastern Kainju group (like Eggon) and some Western and Central Plateau languages (Nigeria).

GERI-84

1984 Gericke, Helmuth: *Mathematik in Antike und Orient* [Mathematics in Antiquity and the East], Springer, Berlin (Germany), 292 p. (in German).

Includes sections on mathematics in ancient Egypt (47-65) and in the Islamic countries (196-214).

GET-99

1999 Getz, Chonat: Computer generation of geometric designs woven into the *izimbenge* using algorithmic processes developed in the field of fractal geometry, *South African Journal of Science*, Johannesburg (South Africa), Vol. 95, 434-439.

"Geometric designs woven into copper wire baskets (*izimbenge*) by the Zulu people of South Africa have been analyzed and regenerated on a computer using algorithmic processes developed mainly in the field of fractal geometry. The mathematical concept of self-similarity is used to facilitate the comprehension of several aspects of fractal geometry. The algorithmic processes used are the deterministic algorithm, the random iteration algorithm and the escape time algorithm."

GIA-76a

1976 Giacardi, Livia & Tullio Viola: Il calcolo del volume del tronco di piramide nella matematica egizia (Discussione sulle ipotesi piu importanti gia proposte), *Atti della Accademia delle Scienze di Torino*, Torino (Italy), Vol. 111, 1976-1977, 441-453 (in Italian).

Contains a brief analysis of the hypotheses of Gunn and Peet, Vogel, Neugebauer, Van der Waerden and Gillings on the origin of the ancient Egyptian formula for the volume of a truncated pyramid.

GIA-76b

1976 Giacardi, Livia & Tullio Viola: Saggio su un possibile calcolo dei volumi di alcuni poliedri nella matematica egizia, *Atti della Accademia delle Scienze di Torino*, Torino (Italy), Vol. 111, 1976-1977, 523-537 (in Italian).

Proposes a new deduction of the ancient Egyptian formula for the volume of a truncated pyramid, based on the successive determination of the volumes of particular pyramids and prisms.

GIA-78

1978 Giacardi, Livia & Silvia C. Roero: *La matematica delle civiltà arcaiche. Egitto, Mesopotamia, Grecia* [Mathematics in ancient civilisations Egypt, Mesopotamia, Greece], Stampatori, Torino (Italy), 321 p. (in Italian).

GIB-96

1996 Gibbs, William & Sihlabela, Mprophet: String figures, *Mathematics in School*, Leicester (UK), Vol. 25, No. 3, 24-27.

Examples of string figures from Bhutan (Asia) and Kenya, Zambia and Swaziland are presented, and suggestions for their exploration in mathematics education are presented.

GIE-50

1950 Giese, W.: Review of "J. Delgado - Sistema de numeración norteafricano (Madrid, 1949)" [Review of "J. Delgado – North-African system of numeration"], *Revista de Historia*, La Laguna (Canary Islands, Spain), Vol. 89, 89-94.

GILL-27
1927 Gillain, O.: *La science égyptienne; l'arithmétique au moyen empire* [Egyptian science: Arithmetic during the Middle Kingdom], Fondation Egyptologique Reine Elisabeth, Brussels (Belgium), 326 p.

GIL-59
1959 Gillings, Richard J.: The Egyptian 2/3 table for fractions, *The Australian Journal of Science*, Sydney (Australia), Vol. 22, No. 6, 242-250.

GIL-61
1961 Gillings, Richard J.: 'Think of a number' problems 28, 29 of the Rhind Mathematical Papyrus (B.M. 10057-8), *The Mathematics Teacher*, Washington DC (USA), Vol. 54, No. 2, 97-100.

GIL-62a
1962 Gillings, Richard J.: Problems 1-6 of the Rhind Mathematical Papyrus, *The Mathematics Teacher*, Washington DC (USA), Vol. 55, No. 1, 61-69.

GIL-62b
1962 Gillings, Richard J.: The Egyptian mathematical leather roll (B.M. 10250), *The Australian Journal of Science*, Sydney (Australia), Vol. 24, No. 8, 339-344.

GIL-64
1964 Gillings, Richard J.: The volume of a truncated pyramid in Ancient Egyptian papyri, *The Mathematics Teacher*, Washington DC (USA), Vol. 57, No. 8, 552-555.

GIL-65
1965 Gillings, Richard J.: The addition of Egyptian unit fractions, *The Journal of Egyptian Archeaology*, London (UK), Vol. 51, 95-106.

GIL-66a
1966 Gillings, Richard J.: Mathematical fragment from the Kahun Papyrus, *The Australian Journal of Science*, Sydney (Australia), Vol. 29, No. 5, 126-130.

GIL-66b
1966 Gillings, Richard J.: The remarkable mental arithmetic of the Egyptian scribes, *The Mathematics Teacher*, Washington DC (USA), Vol. 59, No. 4, 372-381 (Part 1); Vol. 59, No. 5, 476-484 (Part 2).

GIL-66c
1966 Gillings, Richard J.: The volume of a cylindrical granary in Ancient Egypt, *The Australian Mathematics Teacher*, Sydney (Australia), Vol. 22, No. 1, 1-4.

GIL-67a
1967 Gillings, Richard J.: Mathematical fragment from the Kahun Papyrus IV, 3, *The Australian Mathematics Teacher*, Sydney (Australia), Vol. 23, No. 3, 126-130.

GIL-67b
1967 Gillings, Richard J.: The area of the curved surface of hemisphere in Ancient Egypt, *The Australian Journal of Science*, Sydney (Australia), Vol. 30, No. 4, 113-116.

GIL-68
1968 Gillings, Richard J.: Sum of *n* terms of an arithmetical progression in Ancient Egypt, *The Australian Journal of Science*, Sydney (Australia), Vol. 31, No. 1, 47-50.

GIL-69
1969 Gillings, Richard J. & Rigg, W. J. A.: The area of the circle in Ancient Egypt, *The Australian Journal of Science*, Sydney (Australia), Vol. 32, No. 5, 197-200.

GIL-72
1972 Gillings, Richard J.: *Mathematics in the time of the Pharaohs*, MIT, Cambridge MA (USA), 288 p. [Dover reprint, New York (USA), 1982].

GIL-74
1974 Gillings, Richard J.: The recto of the Rhind Mathematical Papyrus : How did the ancient Egyptian scribe prepare it?, *Archive for History of Exact Sciences*, Berlin (Germany), Vol. 12, 291-298.

GIL-79
1979 Gillings, Richard J.: The Recto of the Rhind Mathematical Papyrus and the Egyptian mathematical leather roll, *Historia Mathematica*, New York (USA), Vol. 6, No. 4, 442-447.

GIL-81
1981 Gillings, Richard J.: The Egyptian Mathematical Leather Role - line 8 : How did the scribe do it?, *Historia Mathematica*, New York (USA), Vol. 8, No. 4, 456-457.

GIN-78
1978 Ginsburg, Herbert: Poor children, African mathematics, and the problem of schooling, *Educational Research Quarterly*, Los Angeles (USA), Vol. 2, No. 4, 26-44.

Examines the development of mathematical thinking in two ethnic groups in Ivory Coast, Baoulé and Dioula, and "analyses the role of culture in the development of thinking. Findings reveal that practical arithmetic procedures like addition and subtraction can develop without formal schooling and are heavily influenced by cultural conditions" (p. 26).

GIR-96
1996 Girndt, Uwe: Einige Untersuchungen zur altägyptischen Grundeinheit der Längen-messung [Some research on the ancient Egyptian basic unit for length measurement], *Göttinger Miszellen*, Göttingen (Germany), Vol. 151, 53-66.

GIV-70
1970 Givón, Talmy: The magical number two, Bantu pronouns and the theory of pronominalization, *Studies in African Linguistics*, University of California, Los Angeles CA (USA), Vol. 1, No. 3, 279-300.

GLA-27
1927 Glanville, Stephen R. K.: The mathematical leather roll in the British Museum, *The Journal of Egyptian Archaeology,* London (UK), No. 13, 232-239.

GLAV-94
1994 Glavas, Christos B.: *The place of Euclid in ancient and modern mathematics*, Korfi, Athens, (Greece), 167 p.

GLU-44
1944 Gluckman, Max: The logic of African science and witchcraft, *Rhodes-Livingstone Journal,* Lusaka (Zambia), No. 1, 61-71.

GNAE-98
1998 Gnaedinger, Franz: *Im Haus der Seschat, Vol. 1, Geometrie und Mathematik in alten Aegypten. Von den Errungenschaften der vordynastischen Aera über die Pyramiden zum Papyrus Rhind* (Geometry and Mathematics in ancient Egypt. From the results of the predynastic period via the pyramids to the Rhind papyrus), Private edition, Zürich (Switzerland), 134 p. (in German).

Includes a discussion of the use of Pythagorean triples in Egyptian architecture, and hypotheses on the calculation with unit fractions and the approximation of the area of a circle.

GNA-81
1981 Gnanvo, Cyprien: Plaidoyer pour la decimalisation [Argument for the decimalization], *Langues africaines et échange des connaissances*, UNESCO / Conseil Interafricain de Philosophie, Cotonou (Benin) (in French).

GNA-85
1985 Gnanvo, Cyprien: L'enseignement des mathématiques dans les langues africaines: cours de géométrie en Fon [Mathematics teaching in African languages: a geometry course in Fon], in: *Colloque sur Langues Africaines et Philosophie*, Cotonou (Benin) (in French).

GNA-86
1986 Gnanvo, Cyprien: L'enseignement des mathématiques dans les langues africaines (problémes théoriques et linguistiques): le cas du Bénin [], *Actes du Colloque International sur Langues Africaines et Philosophie*, Cotonou (Benin), 230-233 (in French).

GON-50
1950 González Echegaray, Carlos: Los sistemas de numeración y los numerales en los pueblos de la Guinea Española [The numeration systems and numerals among the peoples of

Spanish Guinea (Equatorial Guinea)], *Archivos del Instituto de Estudios Africanos*, Vol. IV, No. 12, 19-29 (in Spanish).

Describes counting methods using fingers, knots, pebbles, etc., and number words (mostly decimal, some with auxiliary base five) among the peoples of Equatorial Guinea.

GRAN-73

1973 Grandet, Eliane: La numération cardiale dans quelques langues de Côte d'Ivoire, *Annales de l'Université d'Abidjan*, Abidjan (Côte d'Ivoire), Série H, Linguistique, Vol. 6, No. 1, 47-102 (in French).

Analyses cardinal numeration in several language groups from Côte d'Ivoire (Mandé, Voltaïque, Krou, and Kwa, including Akan).

GRA-94

1994 Grattan-Guinness, Ivor (Ed.): *Companion Encyclopedia of the History and Philosophy of the Mathematical Sciences*, Routledge, London (UK), 2 volumes, 1721 p.

Two chapters deal directly with mathematics in Africa:

* C. S. Roero: Egyptian mathematics (30-45);
* C. Zaslavsky: Mathematics in Africa: explicit and implicit (85-92).

GRA-96

1996 Grattan-Guinness, Ivor: Numbers, magnitudes, ratios, and proportions in Euclid's 'Elements': how did he handle them?, *Historia Mathematica*, New York (USA), Vol. 23, No. 4, 355-375.

GRA-03

2003 Grattan-Guinness, Ivor & B. S. Yadav (Eds.): *History of the Mathematical Sciences*, Hindustan Book, New Delhi (India), 244 p.

Proceedings of the International Conference on the History of the Mathematical Sciences held at New Delhi from December 21-23, 2001. Contains two chapters dealing with mathematics in Africa:

* Milo Gardner: The Egyptian mathematical leather roll (119-134);
* Gregg De Young: A new source of evidence for the lost Arabic translations of Euclid's *Elements* (149-162).

GRIA-38
1938 Griaule, Marcel: Numération secrète [Secret numeration], in: M. Griaule, *Jeux Dogons*, Institut d'Ethnologie, Paris (France), p. 222. (in French).

In his book on children's games of the Dogon in Mali, Griaule presents two examples of a secret numeration (one to ten) used (and maybe invented) by the children of the Pamyon and Guinna neighbourhoods and often not understood by children from other neighbourhoods.

GRIA-51
1951 Griaule, Marcel: Systèmes graphiques des Dogons [Graphical systems of the Dogon], in: Griaule, M. & Dieterlen, G. (Eds.), *Signes graphiques soudanais*, Hermann, Paris (France), 7-30 (in French).

GRI-26
1926 Grimme, Hubert: Nachtrag zu A. Klingenhebens Studie über die Berberischen Zählenmethoden [Comments on A. Klingenheben's study of the Berber methods of counting], *Zeitschrift für Eingeborenensprachen*, Hamburg (Germany), Vol. 17 (1926/27), 230-234 (in German).

Complements Klingenheben's paper (KLI-26) with further information on vigesimal numeration.

GUEG-83
1983 Guégan, Dominique: *Enseignement et mathématiques en langues africaines: expériences connues et problématique de l'enseignement du calcul* [Teaching and mathematics in African languages: known experiences and problematics of the teaching of arithmetic], Agence de coopération culturelle et technique, Paris (France), 181 p.

GUE-90
1990 Guergour, Youcef: *The life and work of the Maghrebian mathematician Ibn Qunfudh*, "Magistère" thesis, École Normale Supérieure, Algiers (Algeria), 760 p. (in Arabic).

GUE-91

1991 Guergour, Youcef: The mathematical writings of Ibn Qunfudh al-Qasantînî (1339-1407), *Cahier du séminaire Ibn al-Haytham*, E.N.S. de Kouba, Algiers (Algeria), No. 1, 15-21 (in Arabic).

GUE-96

1996 Guergour, Youcef: Les travaux mathématiques d'Ibn Qunfudh al-Qasantînî (1339-1407) et leurs relations avec les écrits de son temps [The mathematical works of Ibn Qunfudh al-Qasantînî (1339-1407) and their relationship with the writing of his time], in: *Actes du premier colloque national sur l'histoire des mathématiques arabes (Ghardaïa, Algérie, 5-7 avril 1993)*, Association Algérienne d'Histoire des Mathématiques, Algiers (Algeria), 39-70 (in French).

GUE-98

1998 Guergour, Youcef: The phenomenon of mathematical commentaries in the Islamic West in the 14th century and its consequences, in: *Proceedings of the 6th International Colloquium on the History of Arab Science (Ras al-Khayma, United Arab Emirats, 16-20 December 1996)*, Institute for the History of Arab Sciences, Alep (Syria) (in Arabic).

GUE-99

1999 Guergour, Youcef: Les différents systèmes de numération au Maghreb, à l'époque ottomane: l'exemple des chiffres rûmî [The different systems of numeration in the Maghreb during the Ottoman epoch: Example of the rumi ciphers], *Cahier du Séminaire Ibn al-Haytham*, Alger (Algeria), No. 9, April, 11-22 (in French).

Presents a system of numeration, called "rumi ciphers" or "ciphers of Fez" or "register ciphers", composed of 27 distinct symbols. This system was used in the Extreme Maghreb (today's Morocco) in administration and in accounting.

GUE-00

2000 Guergour, Youcef: Les différents systèmes de numération au Maghreb: l'exemple des chiffres rumi [The different numeration systems in the Maghreb: The example of the rumi

ciphers], in: E. Ihsanoglu, A. Djebbar & F. Günergun (Eds.), *Science, Technology and Industry in the Ottoman World. Proceedings of the XXth International Congress of History of Science (Liège, 20-26 July 1997)*, Brepols Publisher, Turnhout (Belgium), Vol. VI, 67-74 (in French).

The author presents and analyses a type of a non-positional numeration system with 27 symbols used by the administration in the Western Maghreb.

GUE-06

2006 Guergour, Youcef: *La géométrie euclidienne chez al-Mu'taman Ibn Hûd (m. 478/1085): Contributions à l'étude de la tradition géométrique arabe en Andalus et au Maghreb* [Euclidian geometry by al-Mu'taman Ibn Hûd (d. 478/1085): Contributions to the study of the Arab geometrical tradition in Andalusia and in the Maghreb], doctoral thesis, Université de Annaba (Algeria) (in French).

GUG-77

1977 Guggenheimer, H.: The axioms of betweenness in Euclid, *Dialectica, International Journal of Philosophy*, Bern (Switzerland), Vol. 31, Nos. 1-2, 187-192.

GUG-99

1999 Guggenheimer, H.: Review of Vitrac's translation of Book X of Euclid's *Elements* (EUC-98), *Mathematical Reviews*, Lancaster PA (USA), 99m:01004.

GUI-92

1992 Guillemot, M.: Les notations et les pratiques opératoires permettent-elles de parler de "fractions égyptiennes"? [Do notational and operational practices allow us to speak of "Egyptian fractions"?], in: BEN-92, 53-70 (in French).

Having examined different domains where the ancient Egyptians might have been able to express a concept of "general fraction", the author asserts that this concept was unknown to them. He shows also that the only domain where they operated with them was that of the "Egyptian numbers", that is, essentially, the sums of whole numbers and days.

GUL-58
1958 Gulliver, P. H.: Counting with the fingers by two East African tribes, *Tanganyika Notes and Records*, Dar es Salaam (Tanzania), No. 51, 259-262.

"When they are counting or mentioning numbers in their conversation some African peoples have certain conventional ways of indicating and emphasizing the numbers referred to by the manipulation of the fingers. In this note the conventional finger actions are described for two Nilo-Hamitic tribes – the Arusha of northern Tanganyika and the Turkana of north-western Kenya."

GUT-96
1996 Gutenberg, J. & Imhausen, Annette: Das Zahlensystem der Ägypter -k-ein dezimalsystem? [The numeration system of the Egyptians: (not) a decimal system], *Discussions in Egyptology*, Oxford (UK), No. 36, 49-51 (in German).

GWA-67
1967 Gwarzo, Hassan Ibrahim: The theory of chronograms as expounded by the 18th century Katsina astronomer-mathematician Muhammad B. Muhammad, *Research Bulletin of the Center of Arabic Documentation*, University of Ibadan, Ibadan (Nigeria), Vol. 3, No. 2, 116-123.

H

HAA-67
1967 Haag, V. H.: An African mathematics programme, *Ghana Teachers' Journal*, Accra (Ghana), Vol. 53, 55-60.

HAD-89
1989 Hadfi, Hmida: *Mathematics in Ifraiqya during the Middle Ages: Jerba*, Diplôme d'Etudes Approfondies, Faculté des Lettres et des Sciences Humaines, Université de Tunis 1, Tunis (Tunisia), 258 p. (in Arabic).

This Masters thesis deals with mathematical activities in the eastern Maghreb and in divided into six parts: 1. Introduction (5-18), 2. Mathematics in Ifriqya (20-51), 3. Mathematics at Jerba (53-62), 4. catalogue of mathematicians from Ifriqya (63-200), 5. Mathematical and historical analysis of the mathematical contents of the *Muqaddima* of Ibn Khaldun (202-232), 6. General bibliography 234-253).

HADI-06
2006 Hadibi, Mohamed: *Le Groupe d'Études sur l'Histoire des Mathématiques à Béjaïa. Une association indépendante à la recherche du patrimoine d'une ville et de sa province dans l'Algérie d'aujourd'hui* [The Beja Study group on the History of Mathematics, an independent association for research of the heritage of a city and its province in today's Algeria], doctoral thesis, Ecole des Hautes Etudes en Sciences Sociales, Paris (France) (in French).

HAG-64
1964 Haggerty, John: *Kalah*: An ancient game of mathematical skill, *The Arithmetic Teacher*, Reston VA (USA), Vol. 11, 326-330.

Describes the *kalah* (mancala) game as useful for arithmetic teaching.

HAN-99
1999 Hansen, Keven: Teaching mathematics (and history) with Egyptian fractions, in: D. Curtin, D. Otero & J. Wine (Eds.), *Combined Proceedings of the Sixth and Seventh Midwest*

History of Mathematics Conferences, University of Wisconsin, La Crosse (USA), 218-230.

Presents a survey of computational and representational methods for Egyptian fractions up to the present. The author provides examples to include in a discrete mathematics course.

HARA-00

2000 Harakat, Ibrahim: *Madkhal ilâ târîkh al-^{c}ulûm bi al-Maghrib al-muslim hattâ al-qarn 9/15* [Introduction to the history of science in the Islamic Maghreb until the 15th century], Dâr al-Rashâd al-hadîtha, Casablanca (Morocco), 3 Vol., 1147 p. (in Arabic).

Volume 1 contains a chapter about the history of mathematics and astronomy in the Maghreb until the 15th century (429-443).

HAR-97

1997 Harbili, Anissa: *Mathematics education in Tlemcen in the 14th century through the Commentary of al-^{c}Uqbani (d. 811 / 1408)*, Magister thesis, École Normale Supérieure, Algiers (Algeria), 407 p.

The first part of the thesis presents the life and work of the Maghrebian mathematician al-^{c}Uqbani and the mathematical activities in Tlemcen (Algeria) at his time. The second part is dedicated to a mathematical analysis of his work, that is a commentary of the famous *Manual of arithmetic operations* of Ibn al-Bannâ (d. 1321). The third and last part is a critical edition of the only surviving copy of the work of al-^{c}Uqbani.

HART-97

1997 Hartshorne, Robin: *Companion to Euclid: a course of geometry based on Euclid's Elements and its modern descendants*, AMS, Providence RI (USA), 362 p.

HART-00

2000 Hartshorne, Robin: *Geometry: Euclid and Beyond*, Springer, New York (USA), 526 p.

HAZ-83
1983 Hazoume, Marc-Laurent: La numération en Gun, Gen et en Bariba [Numeration in Gun, Gen and Bariba], *Langues Africaines et échange des connaissances*, Conseil Interafricain de Philosophie, Cotonou (Benin) (in French).

HEA-64
1964 Heath, Thomas: *Diophantes of Alexandria; a study in the history of Greek algebra*, Dover, New York (USA), 387 p.

HEBE-89
1989 Hébert, Elisabeth (Ed.): *Decouvrir les mathématiques arabes* [Discover Arab mathematics], IREM de Rouen, Rouen (France), 149 p. (in French).

A document elaborated by a group of Moroccan students at the Institute for Research in Mathematics Education (IREM) in Rouen under the responsibility of E. Hébert. Describes the development of numeration, algebra, number theory, combinatorics, trigonometry, geometry, numerical analysis and calculus in the Islamic countries and its diffusion to Europe.

HEBE-95
1995 Hébert, Elisabeth; Aïssani, Djamil; Boufrioua, Abdelaziz; Bensmina, Youssef; Boréani, Jacqueline; Nordon, Nicole & Trotoux, Didier: *Quelques aspects des mathématiques d'Ibn al-Bannâ de Marrakech (1256 – 1321)* [Some aspects of the mathematics of Ibn al-Bannâ of Marrakesh (1256-1321), IREM de Rouen, Rouen (France), 130 p. (in French).

Based on the recent studies of Ahmed Djebbar and Mohamed Aballagh, the book presents to non-specialists some chapters of work of Ibn al-Bannâ. Presenting in parallel complete texts (given in their French translations) and multiple mathematical and cultural comments, it allows the reader to become acquainted with some aspects of mathematical knowledge of the 13th and 14th century.

HEB-58
1958 Hebga, Meinrad P.: Plaidoyer pour les logiques d'Afrique noire [Arguments for the logics of black Africa], in: *Aspects de la culture noire*, Présence Africaine, Paris (France), 104-116 (in French).

HEND-75
1975 Hendy, M. D.: Euclid and the fundamental theorem of arithmetic, *Historia Mathematica*, New York (USA), Vol. 2, 189-191.

HEN-86
1986 Henry-Carmichael, Alberta: *The development of mathematical concepts and skills among unschooled Nupe children*, doctoral thesis, Ahmadu Bello University, Zaria (Nigeria).

A Piagetian-type study of children of the Nupe group of Central Nigeria. Interviewing 336 children, the author found skills developing better than concepts, and some differences by sex and urban-rural distinctions.

HER-39
1939 Herskovitch, Melville Jean: The numerical system of the Kru, *Man*, London (UK), Vol. 39, 154-155.

HERT-84
1984 Hertz-Fischler, Roger: What are propositions 84 and 85 of Euclid's Data all about ?, *Historia Mathematica*, New York (USA), Vol. 11, 86-91.

HIT-92
1992 Hitchcock, Gavin: The "Grand Entertainment": Dramatising the birth and development of mathematical concepts, *For the Learning of Mathematics*, Montreal (Canada), Vol. 12, No. 1, 21-27.

Mathematician from Zimbabwe proposes dramatic replays of the mathematical journeys of the past as a tool and an art form worth exploring in mathematics education.

HOF-52
1952 Hoffmann, Carl: Zur Verbreitung der Zahlwortstämme in Bantu-sprachen [On the distribution of number word roots in Bantu languages], *Afrika und Übersee*, Berlin (Germany), Vol. 37, No. 2 (1952/3), 65-80 (in German).

Discusses the distribution of the roots for the number words 1 to 10. For the number words 2 to 5 and 10 the uniformity is greater than for the number words for 1, and 6 to 9 (p. 78).

HOG-85
1985 Hogbe-Nlend, Henri: *L'Afrique, berceau de la mathématique mondiale?* [Africa, the cradle of world mathematics?], United Nations University, Nairobi (Kenya), 11 p. (mimeo) (in French).

This paper by the first president of the African Mathematical Union (1976-1986) is intended as an introduction to the contribution of Ancient Africa to world mathematics. After underlying the Black-African character of Pharaonic Egypt and analyzing the dialectics of intuitive and deductive reasoning, it is stated that mathematics in Pharaonic Africa was intuitive, demonstrative and rational; Africa is the mother of Geometry.

HOGE-85
1985 Hogendijk, Jan P.: *Ibn al-Haytham's Completion of the Conics. (Introduction, Critical edition, translation and analysis)*, Springer, New York (USA), 417 p.

The book is an enriched and revised version of a Doctoral thesis defended at the University of Utrecht (Netherlands) in 1983. It contains a history of conics since the works of Apollonius, a biography of Ibn al-Haytham, a critical edition with translation and analysis of an important mathematical text: the tentative reconstitution, by Ibn al-Haytham, of the contents of Book VIII of the *Conics* of Apollonius, that the Arab mathematicians of the Middle Ages had not been able to find, and which is still today considered lost.

HOGE-87a
1987 Hogendijk, Jan P.: On Euclid's Lost Porisms and Its Arabic Traces, *Bolletino di Storia delle Scienze Matematiche*, Bologna (Italy), Vol. VII, Fasc. 1, 93-115.

HOGE-87b
1987 Hogendijk, Jan P.: Observations on the icosahedron in Euclid's 'Elements', *Historia Mathematica*, New York (USA), Vol. 14, No. 2, 175-177.

HOGE-01
2001 Hogendijk, Jan: Review of Mansfeld's *Pappus* (MANS-98), *Mathematical Reviews*, Lancaster PA (USA), 2001d:01003.

HOL-88
1988 Holgate, Philip: Summation of factorial series by the Egyptians, *The Mathematical Gazette*, London (UK), Vol. 72, 41-43.

Suggests an explanation of how the summation method in 'Demotic mathematical papyrus PMD 10520 (British Museum)' could have been obtained.

HOUN-94
1994 Houndonougbo, Victor: Processus stochastique du Fâ: une approche mathématique de la géomancie des côtes du Bénin, in: HOU-94, 139-157 (in French).

Analyses Fâ divination practices in the coastal zones of Benin from a mathematical point of view (theory of probability).

HOU-87
1987 Houtondji, Paulin J.: *Bilan de la Recherche Philosophique Africaine. Repertoire Alphabetique* [Philosophical Research in Africa. A bibliographic survey], Conseil Interafricain de Philosophie, Cotonou (Benin), Part 1: 1900-1985, Vol. 1: A-M, 339 p.

HOU-94
1994 Hountondji, Paulin (Ed.): *Les savoirs endogènes: pistes pour une recherche* [Endogenous knowledge: Research trails], CODESRIA, Dakar (Senegal), 345 p. (in French).

Includes HOUN-94 and TCH-94.
Translation: HOU-97.

HOU-97
1997 Hountondji, Paulin (Ed.), *Endogenous knowledge: Research trails*, CODESRIA, Dakar (Senegal), 376 p.

Translation of HOU-94.

HOY-89
1989 Hoyrup, Jens: Egyptian mathematics, in: J. Hoyrup: *Mathematics, Algebra and Geometry: the Mathematical Context of the Bible*, Roskilde University Centre, Roskilde (Denmark), 20-30.

Gives an overview of ancient Egyptian mathematics and discusses its diffusion: "The full range of Egyptian mathematics was probably never diffused to the Palestinian area. From the time when the Israelite Kingdoms began approaching a redistributive economy, however, and when the royal scribes came in need of computational tools, epigraphic evidence shows that they took over the Egyptian hieratic numbers. ...They must have been imported together with at least part of that wider mathematical culture which they served. In all probability, the administration in the Divided Kingdom will thus have been effected by means of Egyptian routines and techniques."

HOY-97

1997 Høyrup, Jens: Hero, Ps.-Hero, and Near Eastern practical geometry. An investigation of Metrica, Geometrica, and other treatises, *Antike Naturwissenschaft und ihre Rezeption*, Trier (Germany), Vol. 7, 67-93.

The author intends to "firstly, that Hero's geometry depends to a greater extent than usually assumed on Near Eastern practical geometry or its descendant traditions in the classical world, and that the conventional image [of Hero] as the transformer of theoretical into applied mathematics is only a half-truth; secondly, that much of what is shared by Hero's *Metrica* and the pseudo-Heronian collections assembled by Heiberg as *Geometrica* are shared borrowings from the same tradition..." (p. 67).

HOY-98

1998 Hoyrup, Jens: Review of Gerdes' *Ethnomathematik dargestellt am Beispiel der Sona Geometrie* (GER-97a), *Zentralblatt Mathematik*, Berlin (Germany), No. 908.01001.

HUY-95

1995 Huylebrouck, Dirk: Traditional scientific knowledge in Burundi and Rwanda, *Mathematics in School*, Leicester (UK), Vol. 24, No. 5, 28-31.

"Presents a first try-out about ethno-mathematics in Rwanda and Burundi, regions about which no such studies were done before, contrary to for instance West or Southern Africa or the Arab countries. Includes collaborations with Pierre Nzohabonayo (Univ. Burundi) and Désiré Karangwa (KIST, Rwanda)."

HUY-96a

1996a Huylebrouck, Dirk: Puzzles, Patterns, Drums: the Dawn of Mathematics in Rwanda and Burundi, *Humanistic Mathematics Network Journal*, Claremont (USA), Vol. 14, 9-22.

Presents mathematical ideas involved in the 'igisoro' board game (four-row mancala type game), displays decorative patterns from basketry, and analyses mathematical structures in drum music.

HUY-96b

1996b Huylebrouck, Dirk: The bone that began the space odyssey, *The Mathematical Intelligencer*, New York (USA), Vol. 18, No. 4, 56-60.

Describes the Ishango bone (Congo / Zaire) as a Mesolithic mathematical artifact, some interpretations of the notches, and uses. Shallit remarks in a letter to the editor (Vol. 19, No. 3, p. 7) that papers by A. S. Brooks present a date of 20,000 years ago (not 11,000 years ago as stated by Huylebrouck) for the bone. Yet, no polemics was engaged to set its true age at 22,000 years ago (20,000 B.C.) in an additional letter to the editor.

HUY-97

1997 Huylebrouck, Dirk: Counting on hands in Africa and the origin of the duodecimal system, *Wiskunde en Onderwijs*, Antwerp (Belgium), No. 89, 53-57.

"Summarizes the popular 'astronomical' and 'arithmetical' explanations about the origin of the base 12 and the related use of the number 60, to reject them in favor of the ethnomathematical counting hypothesis the phalanxes of one hand with the thumb."

HUY-98

1998 Huylebrouck, Dirk: The Ishango bone: from Africa to space, *EOS-magazine*, Ghent (Belgium), No. 7/8, 46-50.

"A vulgarization about the Ishango rod, the oldest mathematical object, in a glossy science magazine for a wide audience. Includes large color pictures."

HUY-00a

2000a Huylebrouck, Dirk: The parabolic flight of the Ishango artifact: the oldest mathematical find, *Wiskunde en Onderwijs*, Antwerp (Belgium), No. 101, 4-16.

"In the 1968 movie *2001, A Space Odyssey* the opening scene shows a human ancestor throwing his first discovery, the use of a bone as a tool, into space. As to realize Kubrick's metaphor, the Ishango rod was brought in zero gravity during a parabolic flight of the European Space Agency. By carrying this oldest mathematical object with him, moviemaker Georges Kamanayo (Rwanda) became the first weightless 'African-European'."

HUY-00b

2000b Huylebrouck, Dirk: The oldest mathematical artifact, but not in Flanders, *Umubano, Journal of the Flanders-Rwanda Association*, Ninove (Belgium), autumn, 4-7.

"A rather assertive paper in which the author expresses his disappointment for the lack of interest in Flanders (Belgium) for African mathematics."

HUY-01

2001 Huylebrouck, Dirk: The Ishango bone, the oldest mathematical object, *Kadath, Chroniques des Civilisations Disparues*, Brussels (Belgium), No. 98, 25-32.

"Presents details on the Ishango rod, original pictures of the excavation site, a paper model of the object, explanations about the difficulties with the dating of the object, and, in particular, the new interpretation of the notches by V. Pletser."

HUY-03

2003 Huylebrouck, Dirk: *Afrika en wiskunde. Etnowiskunde in zwart Afrika, vanaf de koloniale tijd terug naar de oudste wiskundige vondst van de mensheid: het Ishangobeen* [Africa and mathematics. Ethnomathematics in black Africa from the colonial times backwards to the oldest mathematical find of humanity: the Ishango bone], Author's edition, Schaarbeek (Belgium), 246 p. (Preface by Vladimir Pletser) (in Flemish).

This book is the by-product of more than hundred lectures given around Belgium for pupils of high schools and candidates for working

in developing countries. It contains the following parts and chapters: Chap. 1 "Ethno-mathematics: why?" (p. 9-17); Chap. 2 "Sources for African ethno-mathematics" (19-26); Part 1 "Introductory mathematical voyage," Chap. 3 "Narrative and musical introduction" (29-44); Chap. 4 "Creative counting in Africa" (45-62); Chap. 5 "Drawing" (63-83); Chap. 6 "Reasoning without writing" (85-111); Chap. 7 "Multiplication following the Yoruba and Ethiopian way" (113-128); Part 2 "The Ishango bone", Chap. 8 "The Ishango site" (131-142); Chap. 9 "Mathematical notches" (143-153); Chap. 10 "Missing link" (155-169); Chap. 11 "Not out of Africa" (171-180); Part 3 "Multicultural mathematics, from Africa to space," Chap. 12 "Black mathematics" (183-212); Chap. 13 "An imaginative idea" (213-226); References (227-236).

HUY-05

2005 Huylebrouck, Dirk: *Afrika + Wiskunde* [Africa + Mathematics], VUB Press, Brussels (Belgium), 304 p. (in Flemish).

New edition of HUY-03.

HUY-06

2006 Huylebrouck, Dirk: Mathematics in (central) Africa before colonization, *Anthropologica et Præhistorica*, Brussels (Belgium), Vol. 117, 135 - 162.

"The paper provides a summary of (black) African ethnomathematics, with a special focus on results of possible interest to eventual mathematical properties of the Ishango rod(s). The African diversity in number names, gestures and systems (including base 2 of the Bushmen, probably related to the early Ishango people) shows frequent decompositions of numbers in small groups (similar to the carvings on the rod), while the existence of words for large numbers illustrates counting was not merely done for practical reasons. A particular case is the base 12, with it straightforward finger counting method on the hands, and used in Nigeria, Egypt and the Ishango region. Geometric representations are found in traditional sand drawings or decorations, where lines and figures obey abstract rules. Number lines drawn in the sand (using small and long lines as on the rod) make anyone immediately remind the Ishango carvings. Knotted strings and carved counting sticks (even looking like exact wooden copies of the Ishango rod) illustrate an African counting practice, as confirmed in written sources of, for instance, a gifted American slave.

Finally, mancala mind games, Yoruba and Egyptian multiplication (using doublings as on the Ishango rod) or kinship studies mathematical language, ever since."

HUY-07a
2007 Huylebrouck, Dirk: Wat had de vroege mens op zijn kerfstok? [What was the early human counting on a rod?], *EOS Magazine*, Antwerp (Belgium), No. 3, 36-41 (in Dutch).

HUY-07b
2007 Huylebrouck, Dirk: L'Afrique est le berceau des mathématiques [Africa is the cradle of mathematics], *EOS, le magazine des sciences*, Antwerp (Belgium), No. 3, 24-29 (in French).

Translation of HUY-07a. Analyses the various interpretations of the first Ishango rod and presents a first analysis of a second counting rod found at Ishango in 1959 by Marcel Spinglaer, a collaborator of Jean De Heinzelin who had found and studied the first rod.

I

IBN-83

1983 Ibn al-Haytham: *Kitâb al-Manâzir, al-maqâlât 1-2-3, al-Ibsâr ^{c}alâ al-istiqâma* [The Work on Optics, Books 1-2-3, on direct vision] (Critical edition by A. I. Sabra), Koweit, 779 p. (in Arabic).

Contains the first three books of the famous work of Ibn al-Haytham (d. 1039) on geometrical Optics: "The manner vision is realized in general" (Book I), "Census of elements that vision observes, their causes and the way to perceive them" (Book II), "The errors of direct vision and their causes" (Book III). This edition is preceded by an Introduction that presents the life of Ibn al-Haytham, his different contributions to Optics and the influence of his work on later studies in Optics in the Arabic tradition, and in Europe.

IBN-89

1989 Ibn al-Haytham: *The Optics, Books I-III, On Direct Vision* (Translation by A. I. Sabra), The Warburg Institute-University of London, London (UK), 2 volumes, 613 p.

Comprises the English translation of the first three books of *The Optics* of Ibn al-Haytham. This translation is complemented by an introduction with commentaries, and an Arabic-Latin glossary.

IBN-90

1990 Ibn al-Haytham: *On the Configuration of the World* (Translated with critical commentary by Y. Tzvi Langermann), Garland, New York (USA), 392 p.

IBN-02

2002 Ibn al-Haytham: *Kitâb al-Manâzir, al-maqâlatân 4-5, Fî in^{c}ikâs al-adwâ' wa mawâdi^{c} al-khayâlât al-mubsara bi l-in^{c}ikâs* [The Work on Optics, Books 4-5, On Reflection and Image seen by Reflection] (Translation by A. I. Sabra), Koweit, 2 volumes, 723 p. (in Arabic).

IGB-67

1967 Igboko, P. M.: Improving school mathematics, *West African Journal of Education*, Vol. 11, No. 2, 85-88.

IHS-00
2000 Ihsanoglu, Ekmeleddin; Djebbar, Ahmed & Günergun, Feza (Eds.), *Science, Technology and Industry in the Ottoman World. Proceedings of the XXth International Congress of History of Science (Liège, 20-26 July 1997),* Brepols, Turnhout (Belgium), Vol. VI, 152 p.

Includes the papers ABA-00, DJE-00b and GUE-00.

IMH-96
1996 Imhausen, Annette: *Probleme ägyptischer Mathematik am Beispiel des mathematischen Papyrus Moskau* [Problems of Egyptian mathematics through the example of the mathematical Moscow Papyrus], Masters thesis (Staatsexamensarbeit), Mainz University, Mainz (Germany) (in German).

IMH-99a
1999 Imhausen, Annette: Die Mathematisierung von Brot und Bier [The mathematical handling of bread and beer], in: Danny Beckers, Katja Peters, Carsen Vollmers (Eds.), *9. Novembertagung zur Geschichte der Mathematik, Nijmegen, 29.X–1.XI 1998*, KUN, Nijmegen (Netherlands), 12–21 (in German).

IMH-99b
1999 Imhausen, Annette: Aufgabe 16 des mathematischen Papyrus Moskau – Rechenfehler oder Ligatur? [Moscow mathematical papyrus, problem 16 – miscalculation or ligature?], in: *Göttinger Miszellen*, Göttingen (Germany), Vol. 168, 45–48 (in German).

IMH-01
2001 Imhausen, Annette: Die aHa-Aufgaben der ägyptischen mathematischen Texte und ihre Lösungen" [The aHa-problems in Egyptian mathematical texts and their solutions], in: C.-B. Arnst *et al*. (Eds.), *Begegnungen. Antike Kulturen im Niltal*, Verlag Helmar Wodtke und Katharina Stegbauer, Leipzig (Germany), 213-220 (in German).

IMH-02
2002 Imhausen, Annette: The Algorithmic Structure of the Egyptian Mathematical Problem Texts", in: John Steele and Annette

Imhausen (Eds.), *Under One Sky: Astronomy and Mathematics in the Ancient Near East, Proceedings of the Conference held in the British Museum, London, June 25–27, 2001*, Alter Orient und Altes Testament Vol. 297, Ugarit Verlag, Münster (Germany), 147-166.

IMH-03a
2003 Imhausen, Annette: Ägyptische Algorithmen. Eine Untersuchung zu den mittelägyptischen mathematischen Aufgabentexten [Egyptian Algorithms. A Study of Middle Egyptian Mathematical Problem Texts], Verlag J.B. Metzler, Wiesbaden (Germany), 388 p. (in German).

"This technical analysis of ancient Egyptian mathematical algorithms is based on a catalogue of Middle Egyptian mathematical texts. The study considers the presentation of Egyptian mathematics as collections of algorithms, and the application of mathematical texts in everyday life and business. The catalogue presents the texts in hieroglyphs with a transcription and commentary."

IMH-03b
2003 Imhausen, Annette: Egyptian Mathematical Texts and Their Contexts, *Science in Context*, Cambridge (UK), Vol. 16, 367-389.

"The extant sources for ancient Egyptian mathematics are extremely limited. It is therefore necessary to read the few sources carefully and use additional information from further Egyptian sources in order to achieve the most detailed picture possible. Traditional approaches to Egyptian mathematics have provided only a superficial account of mathematical practices and almost no information about the role of mathematics within Egyptian culture. To enlarge our knowledge it is crucial to use a different methodological approach in the analysis of ancient mathematical techniques. In addition, it is indispensable to contextualize the mathematical problems with sources that are not specifically mathematical per se. In this article I discuss several possibilities for these additional sources, such as administrative texts, reliefs found in tombs, and other archaeological evidence. I exemplify the use of these sources with two problems from the Moscow mathematical papyrus."

IMH-03c

2003 Imhausen, Annette: Calculating the Daily Bread: Rations in Theory and Practice, *Historia Mathematica*, New York (USA), Vol. 30, 3-16.

"This article discusses the handling of rations in Middle Kingdom Egypt (2119-1794/93 BC) as it is displayed in three types of texts: mathematical problem texts, administrative ration texts ("real" ration texts), and literary texts. The example of handling rations is used to examine the relation between mathematical problem texts-which served according to the 'opinio communis' to educate scribes- and administrative texts, the actual documents from the professional life of scribes. Using one specific example, the use of a mathematical technique from the problem texts within a ration text is demonstrated. The presentation is complemented by passages from literary texts referring to rations."

IMH-03d

2003 Imhausen, Annette: Zahl, II. Ägypten [Number, II, Egypt], in: Hubertus Cancik & Helmuth Schneider (Eds.), *Der Neue Pauly. Enzyklopädie der Antike*, Vol. 12/2 Ven-Z, Stuttgart (Germany), 668-669 (in German).

IMH-04a

2004 Imhausen, Annette & Ritter, James: Mathematical fragments: UC 32114, UC 32118, UC 32134, UC 32159-UC32162, in: Mark Collier & Stephen Quirke (Eds.), *The UCL Lahun Papyri*, Archaeopress, Oxford (UK), Vol. 2, 71-96.

IMH-04b

2004 Imhausen, Annette: *Mathematical Fragments from Lahun*, University College of London, London (UK) (online available at: www.petrie.ucl.ac.uk/digital_egypt/lahun/mathintro.html).

"Among the Lahun papyri a small number of fragments can be identified as mathematical texts, i.e. texts that have been written to record a mathematical procedure or used to carry out a mathematical procedure. Very few sources of ancient Egyptian mathematical texts are still extant. Of these, the mathematical fragments of the Lahun papyri hold a significant place. They contain both table texts and problem texts. While they are in many respects like the two major sources, the Rhind (mathematical) papyrus and the Moscow

(mathematical) papyrus, they also show a number of significant details that are not seen in any other text."

INO-00
2000 Inoue, Noriyuki: Review of P. Gerdes' *Geometry from Africa* (GER-99a), *Newsletter of the International Study Group on Ethnomathematics*, New York (USA), Vol. 15, No. 1, 9-10.

IRE-77
1977 IREM de Niamey (Ed.): *Mathématique, langues africaines et français* [Mathematics, African languages and French], Institut de Recherche sur l'Enseigment des Mathématiques, Université de Niamey, Niamey (Niger), 154 p. (in French).

IRE-95
1995 IREM de Montpellier (Ed.): *Proceedings of the First European Summer University "History and Epistemology in Mathematics Education" / Actes de la Première Université d'Été Européenne "Histoire et Épistémologie dans l'Éducation Mathématique"*, Université de Montpellier II, Montpellier (France), 598 p.

The following are contributions by Africans and / or deal with mathematics in the history of Africa:

* Bebbouchi, Rachid: À propos de la continuité [About continuity] (85-89);
* Assem, Ali: Relations entre l'enseignement et les facteurs culturels — Qu'en est-il des mathématiques élémentaires en Algérie? [The relationship between education and culture — what is the case of elementary mathematics education in Algeria?] (305-307);
* Aissani, Djamil: Bougie médiévale — centre de transmission méditerranéen [Medieval Béjaïa — centre of Mediterranean transmission] (499-506);
* Doumbia, Salimata: L'experience en Côte d'Ivoire de l'étude de jeux traditionnels africains et de leur mathématisation [The experience of Côte d'Ivoire in the study of traditional African games and their mathematization] (549-555).

IRU-84

1984 Irumu, Agozia-Kario: Le système numéral 'logo' face au système numéral 'bangala': un cas d'emprunt linguistique [The Logo numeral system compared with the Bangala numeral system: a case of linguistic borrowing], *Bulletin de l'AELIA (Association d'études linguistiques interculturelles africaines)*, Bureau européen de l'AUPELF, Paris (France), No. 7 (in French).

ISM-06

2006 Ismael, Abdulcarimo: Ideias probabilísticas em jogos: considerações didácticas [Probabilistic ideas in games: Didactical considerations], *Matemática & Educação*, Beira (Mozambique), No. 2, 24-31.

Presents examples of probabilistic ideas in games played in Mozambique and suggest ways to use them in teaching mathematics.

ISO-92

1992 Isoun, T.: Mathematics and Africa, *Discovery and Innovation, Journal of the African Academy of Sciences*, Nairobi (Kenya), Vol. 4, No. 1, 4-6.

Editorial on the place of mathematics in the history of Africa and in contemporary Africa which expresses the "need for mathematicians in Africa to write textbooks to reflect our cultural background, and ensure that mathematics is firmly grounded within our environment" (p. 6).

ITA-62

1962 Itard, Jean: *Les livres arithmétique d'Euclide* [Euclid's arithmetical books], Hermann, Paris (France), 230 p.

ITO-80

1980 Ito, Shuntaro: *The Medieval Latin Translation of the 'Data' of Euclid*, Tokyo University Press, Tokyo (Japan) & Birkhauser, Boston (USA), 256 p.

J

JAC-69
1969 Jacobsen, Edward Carl: *Recommendations for the implementation of a modern mathematics program in Botswana*, doctoral thesis, University of Kansas (USA).

JAC-84
1984 Jacobsen, Edward: What goals for mathematics teaching in African schools?, *Educafrica*, Dakar (Senegal), Vol. 10, 118-134.

JAM-99
1999 Jama, Jama Musse: The role of ethnomathematics in mathematics education: Cases from the Horn of Africa, *ZDM, International Reviews on Mathematical Education*, Karlsruhe (Germany), Vol. 99, No. 3, 92-95.

Presents examples of cultural elements from Somalia that may be explored in mathematics education.

JAN-05
2005 Jansen, Jan: *De lessen van Namagan Kanté: Maninka zanddivinatie (Mali-Guinée)* [The lessons of Namagan Kanté: Maninka sand divination (Mali – Guinea)], Universiteit Leiden (Netherlands), 92 p. (in Dutch).

Analyses aspects of Maninka sand divination (geomancy), in particular, the arithmetic and logic of several formal operations.

JAO-86
1986 Jaouiche, Khalil: *La théorie des parallèles en pays d'Islam* [The theory of parallels in the Islamic countries], Vrin, Paris (France), 266 p. (in French).

This is the French part of a publication in two volumes that includes the analysis, critical edition and French translation of the principal investigations of Arabic mathematicians on the 5th Postulate in Book I of Euclid's *Elements*, i.e. on the Parallel Postulate. In the first part the author analyses and translates 12 texts, in particular those of an-Nayrîzî (10th C.), al- Jawharî (10th C.), Thâbit Ibn Qurra (d.901), Ibn

al- Haytham (d.1040), al-Khayyâm (d. 1131) and Nasîr ad-Dîn at-Tûsî (d.1274).

JAO-88
1988 Jaouiche, Khalil: *Nazariyyat al-mutawâziyyât fî l-handasa al-islâmiyya* [The theory of parallels in Islamic geometry], Bayt al-Hikma, Carthage (Tunisia), 256 p. (in Arabic).

Critical edition of texts published in 1986 in a French translation (JAO-86), preceded by an introduction and a presentation of the used manuscripts.

JOH-65
1965 Johnson, Gabriel Kuavi: *Numérotation en langue Gen or Gengbe-mina-popo du bas Togo, essai d'un nouveau mode de comptage* [Numeration in the Gen or Gengbe-mina-popo language of the lower Togo, essay on a new counting method], Institut Togolais de Sciences Humaines, Lomé (Togo), 16 p. (in French).

JOHN-00
2000 Johnson, Julia: Review of Gerdes' *Geometry from Africa* (GER-99a), *Crux Mathematicorum*, Ottawa (Canada), September, 278-279.

JOS-91
1991 Joseph, George Gheverghese: *The Crest of the Peacock: non-European Roots of Mathematics*, Tauris Publishers, London (UK) & New York (USA), 368 p.

The author states in chapter 1 that the "standard treatment of the history of non-European mathematics exhibits a deep-rooted historiographical bias in the selection and interpretation of facts, and that mathematical activity outside Europe has as a consequence been ignored, devalued or distorted" (p.3). In the subsequent chapters he contributes to an alternative perspective. With respect to Africa, it is noted that "Much research needs to be done..." (p.22). Information is given on the Ishango bone (23-27), on Egyptian mathematics (57-90, 125-129), on the Zulu counting system (43-44) and on Yoruba arithmetic (44-46).

JUL-89
1989 Julie, Cyril (Ed.): *Proceedings of a Conference on the Politics of Mathematics Education*, NECC Mathematics Commission, University of Western Cape, Cape Town (South Africa), 38 p.

Contains the keynote address by Paulus Gerdes and discussion contributions from Cyril Julie, Yousuf Gabru, Daya Reddy, Brent Walters and Jan Persens.

JUL-91a
1991 Julie, Cyril (Ed): *People's Mathematics: Early Ideas and Debates*, University of the Western Cape, Cape Town (South Africa).

JUL-91b
1991 Julie, Cyril: Equation of Inequality: Challenging the School Mathematics Curriculum, *Perspectives in Education*, Johannesburg (South Africa), 1991/1992, Vol. 13, No. 1, 53-60.

JUL-96
1996 Julie, Cyril: Mathematics Curricula for Social Justice: Quo Vadis?' Tore, K. *et al.* (Eds.), *Proceedings of the Third International Conference on Political Dimensions of Mathematics Education*, Caspar Forlag, Landas (Norway).

JUL-98
1998 Julie, Cyril: Ideal and Reality: Cross-curriculum work in school mathematics in South Africa, *ZDM, International Reviews on Mathematical Education*, Karlsruhe (Germany), Vol. 98/4, 110-115.

"Within various school mathematics dispensations in South Africa the intention for cross-curriculum work is expressed in the official documents describing the intended school mathematics curriculum. The paper traces this expressed intention from 1962 to 1998. The view is adopted that textbook authors are the major interpreters of the intended curriculum and therefore the manifestations of the cross-curricular ideal in school textbooks for the various periods are described and commented on."

K

KAN-82
1982 Kane, Elimane Abdoulaye: Topologie archaique [Archaic topology], *Revue Senegalaise de Philosophie*, Dakar (Senegal), Vol. 1, 75-90 (in French).

KAN-87
1987 Kane, Elimane Abdoulaye: *Les systèmes de numération parlée des groupes ouest-atlantiques et Mandé. Contribution à la recherche sur les fondements et l'histoire de la pensée logique et mathématique en Afrique de l'Ouest*, doctoral dissertation ('Thèse d'Etat'), Université de Lille III (France), 2 volumes (in French).

Studies spoken numeration systems in about twenty languages in Senegal, of which some are spoken only by some tens of people (like the Bapé, Bassari, Bédik and Koânagi languages). Analyses the understanding of the reforms that took place in these numeration systems, in particular of the spectacular evolution of some of them, like those of the Mandé group. It shows that the spoken numeration systems are susceptible to reform and evolution. Volume 1 deals essentially with cardinal numeration. Volume 2 is dedicated to the symbolic numeration systems.

Review: DJE-89a.

KAN-91
1991 Kane, Abdoulaye Elimane: Systèmes de comptage africains et préarithmétique: de l'opération à la categorization [African counting systems and pre-arithmetic: from operation to categorization], *Épistème, revue sénégalaise d'histoire, sociologie, philosophie des sciences et techniques*, Dakar (Senegal), No. 2, 83-91 (in French).

KANG-05
2005 Kang, Henry: *Stakeholders' receptiveness to an ethnomathematics curriculum foundation: The case of Cameroon*, doctoral thesis, University of British Columbia (Canada).

KANI-86

1986 Kani, Ahmad Mohammad: *The history of 'Ilm al-Hisab' (Arithmetic) in Nigeria with emphasis on Kanem-Borno and Hausaland to 1860* (paper presented at the 2nd Pan-African Congress of Mathematicians, Jos, Nigeria, mimeo)

Presents aspects of mathematics in Islam, especially as studied by the Islamic scholars of pre-colonial northern Nigeria, and notably by Muhammed ibn Muhammed al Katsinawi (c.1740) who worked on 'magic squares' and numerological patterns.

KANI-92a

1992a Kani, Ahmad Mohammad: Mathematics in the Central Biläd Al-Sudän, in: THOM-92a, 17-36.

KANI-92b

1992b Kani, Ahmad Mohammad: Arithmetic in the Pre-Colonial Central Sudan, in: THOM-92b, 33-39.

Considers *ᶜIlm al-Hisab* (arithmetic) as part of the Islamic sciences introduced some time after the 11th century in Nigeria, first in Kanem-Borno and later, probably 15th century in Hausaland. Arithmetic being taught in both 'secular' and *Islâmiyya* schools, was used in the courts (calculation of inheritance), collecting and distributing *zakât* (poordues), business and land surveying. Scholars of Hausaland and Borno consulted Coptic Solar Calendars in determining their economic activities, especially agricultural ones. The author concludes his paper with the following remarks: "Despite the availability of a great deal of literature on medicine, astrology, arithmetic and other related sciences, written in Arabic, Fulfulde, Hausa and other languages, little effort has been made to systematically study these sciences within the historical perspective. The intellectual output of the *ᶜUlamâ* (scholars) in this area has been wrongly classified by our contemporary historians and social scientists under the rubric of 'mysticism'. A serious investigation into the literary output of the scholars of the Western and Central Sûdân, however, may reveal the fact that these scholars had explored agricultural, medicinal, astronomical and mathematical sciences long before the advent of colonial rule" (p.38).

KANO-00
2000 Kanouté, Mamadou Lamine: **Mathématiques et langue nationale en milieu scolaire bambara [Mathematics and national language in the bambara school environment],** *Nordic Journal of African Studies, Uppsala (Sweden), Vol. 9, No. 3, 80-97*

"This article studies the teaching of mathematics in bilingual education in the Bambara-speaking town of Ségou. The bilingual principles of convergent pedagogy are examined in the light of teaching material, teacher training and classroom practice. It shows that neither the textbooks nor the training enable the teachers to follow the pedagogical principles that have been laid down, and that the transition from Bambara to French, which takes place in the 4th grade as far as mathematics is concerned, still represents a great problem in the 5th grade. However, the fieldwork, which took place in 1997, revealed an interesting method that seemed to be a local invention. Building on the Bambara play of riddles, different groups challenge each other both in creating and solving mathematical problems, and the children participate eagerly in this game. This type of teaching fits in well with active pedagogy and could be introduced into convergent pedagogy at a general level."

KAP-01
2001 Kaphesi, Elias: *The use of language in mathematics teaching in primary schools in Malawi: bringing language to the surface as an explicit feature in the teaching of mathematics*, doctoral thesis, University of Nottingham (UK).

KAR-99
1999 Karuhije, Eric: *Mathematics education in Uganda in historical perspective*, paper presented at the Columbia Workshop on Mathematics and Mathematics Education in Africa, Columbia University, New York (USA), November 13.

KASA-92
1992 Kasanda, Choshi D.: The Zambia mathematics pre-service programme: its ability to impart teaching strategies and classroom management skills as perceived by its graduates, *Zimbabwe Journal of Educational Research*, Harare (Zimbabwe), Vol. 4, No. 3, 285-293.

KAS-77
1977 Kaseka, Madiambu: *L'histoire de la formation mathématique des enseignants du secondaire au Zaïre* [The history of the mathematical education of secondary school teachers in Zaire (DR Congo)], masters thesis, Université de Montréal (Canada) (in French).

KAT-96
1996 Katz, Victor: Egyptian Mathematics, *Proceedings - Actes - Actas "História e Educação Matemática", ICME-8 satellite meeting of the International Study Group on the Relations between History and Pedagogy of Mathematics (HPM)*, Associação de Professores de Matemática, Braga (Portugal), Vol.1, 45-53.

Presents an introductory overview of mathematics in Ancient Egypt.

KAT-07
2007 Katz, Victor (Ed.): *The mathematics of Egypt, Mesopotamia, China, India, and Islam: a sourcebook*, Princeton University Press, Princeton (USA).

KAZ-83
1983 Kazadi, Corneille wa Mashinda: *Sur quelques difficultés dans l'enseignement des entiers négatifs aux élèves du 1er cycle de l'enseignement secondaire au Zaïre* [On some difficulties of the teaching of negative numbers to pupils of the first level of the secondary school in Zaire (DR Congo)], doctoral thesis, Université de Paris 7 (France) (in French).

KAZ-88
1988 Kazadi, Corneille wa Mashinda: *Some logical and linguistic problems met by African pupils (in Zaire)* (paper presented at the 6th International Congress on Mathematics Education, Budapest, Hungary, mimeo).

KAZI-02
2002 Kazima, Mercy: *Malawian students understanding of probability*, doctoral thesis, University of Leeds (UK).

KHA-86a
1986a Al-Khattabi, Mohamed Larbi.: The epistle of Ibn al-Bannâ on the universal plane astrolabe of az-Zarqâlî, *Revue Da^{c}wat al-haqq,* Rabat (Morocco), No. 241, 20-25; No. 242, 19-24 (in Arabic).

KHA-86b
1986b Al-Khattabi, Mohamed Larbi: Two epistles of Ibn ar-Raqqâm and Ibn al-Bannâ on the science of measurement, *Revue Da^{c}wat al-haqq*, Rabat (Morocco), No. 256, 39-47 (in Arabic).

KHA-87
1987 Al-Khattabi, Mohamed Larbi: Commentary on the elixir of the science of measurement of Abû ^{c}Abdallah Ibn al-Qâdî, *Revue Da^{c}wat al-haqq*, Rabat (Morocco), No. 258, 77-87 (in Arabic).

KHU-97
1997 Khuzwayo, Herbert: *Mathematics Education in South Africa: A Historical Perspective from 1948-1994*, Research Report No.7, Department of Mathematics, Physics, Chemistry and Informatics, Royal Danish School of Educational studies, Copenhagen (Denmark).

KHU-98
1998 Khuzwayo, Herbert: "Occupation of our minds": A dominant feature in mathematics education in South Africa (on-line available at: www.nottingham.ac.uk/csme/meas/papers/khuzwayo.html).

"As South Africa moves forward with new curricular initiatives which are aimed at the elimination of many disparities, questions about what needs to be done in order to address the inequities to mathematics arising from the education system under the apartheid regime are also being asked. Such disparities certainly extend to mathematics. I believe questions about what is / has been taught in various subjects should be an important consideration. Reconstruction of the educational system must be in accordance with national policy that education should be non-sexist, non-racist and committed to equal access."

KHU-00
2000 Khuzwayo, Herbert: *Mathematics education in South Africa: A historical perspective from 1948 to 1994*, doctoral thesis, University of Aalborg (Denmark).

KHU-04
2004 Khuzwayo, Herbert: A history of mathematics education research in South Africa: The apartheid years, in: 5-VITH-04.

KIB-80
1980 Kibasomba, Man Byemba: *Sur la logique africaine: procès de formalisation* [On African logic: formalization process], University of Lubumbashi, Lubumbashi (Congo / Zaire) (in French).

KIE-55
1955 Kielland, Else Christie: *Geometry in Egyptian Art*, Alec Tiranti, London (UK), 214 p.

Presents a brief survey of Egyptian geometry based on the papyri that have been found, followed by the interpretations which scholars placed on the geometric marks found on the Egyptian works of art. Finally, Lange's law of frontality is discussed, with its revision by Schäfer.

KIES-87
1987 Kiese, M'boka: Un commentaire sur les fondements des mathématiques d'après l'introduction de Cheikh Anta Diop [A commentory on the foundations of mathematics after the introduction of Cheikh Anta Diop], *Revue Ethiopiques*, Dakar (Senegal), No. 1-2, 43-55 (in French).

KIES-90
1990 Kiese, M'boka: Mathématiques et Langue kikongo: le raisonnement parallèle [Mathematics and the (ki)kongo language: parallel reasoning], *Revue Paari*, Paris (France), No. 2, 16-18 (in French).

KIES-91
1991 Kiese, M'boka: Mathématiques et Langue kikongo (Suite) [Mathematics and the (ki)kongo language (continuation)], *Revue Paari*, Paris (France), No. 4, 89-91 (in French).

KIES-01
2001 Kiese, M'boka: Phénoménologie de l'inauguralité: L'épistémologie de Cheikh Anta Diop et les mathématiques [Phenomenology of the inaugurality: The epistemology of Cheikh Anta Diop and mathematics], in: Hommage à Cheikh Anta Diop, Editions Paari, Paris (France), 107-144 (in French).

KIL-05
2005 Kilani, Imed Ben: *Les effets didactiques des différences de fonctionnement de la négation dans la langue arabe, la langue française et le langage mathématique*, doctoral thesis, ISEFC, Le Bardo (Tunisia) (in French).

KIN-97
1997 King, Vanessa: The impact of Dogon religious beliefs on their concept of numbers, *Pythagoras*, Cape Town (South Africa), No. 44, 24-26.

Article based on information contained in M. Griaule's *Conversations with Ogotemmeli* (Oxford University Press, 1965).

KLE-88
1988 Klein, Herbert Arthur: *The science of measurement, a historical survey*, Dover, New York (USA), 736 p.

Contains little information on measurement in Africa: Egyptian length measures ('cubit' and 'foot', 59-61); Egyptian weigth 'ratl' (86); 'Cape foot' from South Africa (63).

KLEP-72
1972 Klepzig, Fritz: *Kinderspiele der Bantu* [Games of Bantu children], Verlag Anton Hain, Meisenheim am Glan (Germany), 563 p. (in German).

Includes games of chance, string figures, board games and riddles.

KLI-26
1926 Klingenheben, August: Zu den Zählenmethoden in den Berbersprachen [On the counting methods in the Berber methods], *Zeitschrift für Eingeborenen-sprachen*, Hamburg (Germany), Vol. 17 (1926/27), 40-51 (in German).

Analyses different numeration systems in Berber languages in northwest Africa: mostly decimal, sometimes quinar-trigesimal

(Nefusa language), sometimes vigesimal (Sus region [Morocco]) and the interaction with Arabic.

KLU-37

1937 Kluge, Theodor: *Die Zahlbegriffe der Sudansprachen, ein Beitrag zur Geistesgeschichte der Menschen* [The number concepts in the Sudanese languages, a contribution to the spiritual history of man], edition of the author, Berlin-Steglitz (Germany), 260 p., 17 maps (in German).

Presents the number words in 976 Sudanese languages and dialects, organized in 16 regional groups from the Senegal-Guinea to the Nile-Chad. A comparative analysis of the languages in each group is included. The sources (mostly grammars and dictionaries) used by the author are indicated.

KLU-38

1938 Kluge, Theodor: *Die Zahlbegriffe der Australier, Papua und Bantuneger nebst einer Einleitung über die Zahl; ein Beitrag zur Geistesgeschichte des Menschen* [The number concepts of the Australian, the Papua and the Bantu Negroes together with an introduction to number; a contribution to the spiritual history of man], edition of the author, Berlin-Steglitz (Germany), 304 p. (in German)

The section on Bantu languages presents first the number words in 274 Bantu languages (and dialects), organized by geographical region (197-276), followed by a comparative analysis of the number word root and structure (277-300). The sources used by the author are not indicated.

KNO-76

1976 Knorr, Wilbur: Problems in the interpretation of Greek number theory: Euclid and the "fundamental theorem of arithmetic", *Studies in History and Philosophy of Science*, Exeter (UK), Vol. 7, No. 4, 353-368.

KNO-85

1985 Knorr, Wilbur R.: Euclid's tenth book: an analytic survey, *Historia Scientiarum*, Tokyo (Japan), Vol. 29, 17-35.

KNO-91a
1991 Knorr, Wilbur R.: On the principle of linear perspective in Euclid's 'Optics', *Centaurus*, Copenhagen (Denmark), Vol. 34, No. 3, 193-210.

KNO-91b
1991 Knorr, Wilbur R.: What Euclid meant: on the use of evidence in studying ancient mathematics, in: Alan C. Bowen (Ed.), *Science and philosophy in classical Greece*, Garland, New York (USA), 119-163.

KNO-92
1992 Knorr, Wilbur R.: When circles don't look like circles: an optical theorem in Euclid and Pappus, *Archive of the History of Exact Sciences*, Berlin (Germany), Vol. 44, No. 4, 287-329.

KNO-93
1993 Knorr, Wilbur: *Arithmêtike stoicheiôsis*: On Diophantus and Hero of Alexandria, *Historia Matematica*, New York (USA), Vol. 20, No.2, 180-192.

"Two ancient works, cited in ancient sources as the "Preliminaries to the *Arithmetic Elements*" and the "Preliminaries to the *Geometric Elements*" - of which the former is no longer extant, while the latter is an alternative designation of the *Definitions*, now commonly attributed to Hero of Alexandria - are here argued to be companion works by the same author, namely Diophantus of Alexandria. This attribution has implications for the dating of Diophantus."

KON-91
1991 Kondangba, Yembeline: Structure des numéraux en bantu (lingcmbè) et en non-bantu (ngbaka minagende, ngbandi, ngbundu, mcnc, mbanza) [The structure of numerals in Bantu (lingcmbè) and non-Bantu languages (ngbaka minagende, ngbandi, ngbundu, mcnc, mbanza)], *Annales aequatoria*, Mbandaka (DR Congo), Vol. 12, 307-319 (in French).

Describes and compares the numeration systems of five languages spoken in the Equator administrative region of the DR Congo.

KOU-99

1999 Kouidri, Khadidja: *The method of false position in the Arab mathematical tradition*, magister thesis, École Normale Supérieure, Algiers (Algeria), 88 p. (in Arabic).

Contains an analysis of a certain number of Arab texts produced between the 10th and the 14th century, which deal with the solution of linear equations and of systems of linear equations by means of the methods of false position.

KRA-83

1983 Krause, Marina: *Multicultural Mathematics Materials*, National Council of Teachers of Mathematics, Reston VA (USA) (6th printing 1993), 76 p.

Chapter 1 deals with Africa (1-7): Egyptian match, Egyptian numeration system, Senet (Egypt) and Wari (West Africa) games.

KRAU-98

1998 Krause, Henning: Review of P. Gerdes' *Ethnomathematik dargestellt am Beispiel der Sona Geometrie* (GER-97a), *Spektrum der Wissenschaft*, Berlin (Germany), September, 118-120 (in German).

KRE-89

1989 Kreith, K.: Euclid turns to probability, *International Journal of Mathematics Education in Science and Technology*, London (UK), Vol. 20, No. 3, 345-351.

KUB-86

1986 Kubik, Gerhard: African graphic systems, *Muntu, revue scientifique et culturelle du Centre International des Civilisations Bantu* (CICIBA), Libreville (Gabon), Vol.4-5, 71-135.

"In pre-colonial times, a varied range of graphic systems existed in Sub-Saharan Africa. The author presents the results of his own investigations made in Tanzania, Malawi, Gabon, Cameroon, Angola and Zambia between 1962 and 1984." The author analyses also tusona-luchazi-ideographs. "The forefathers of the Eastern Angolan peoples discovered higher mathematics and a non-Euclidian geometry on an empirical basis applying their insights to the invention of these [tusona] unique configurations" (p.108).

KUB-87a

1987a Kubik, Gerhard: African space/time concepts and the *tusona* ideographs in Luchazi culture with a discussion of possible cross-parallels in music, *African Music*, Grahamstown (South Africa), Vol. 6, 53-89.

In this paper the author deals with those pictographs of Eastern Angolan culture that are characterized by a highly geometrical construction and examines their space/time relationships. He shows that these drawings "flourish upon abstract principles of a mathematical nature similar to those in some older traditions of African music."

KUB-87b

1987b Kubik, Gerhard: Tusona/Sona - an ideographic script found among the Luchazi and Cokwe of eastern Angola and adjacent areas, in: T. Obenga (Ed.), *Les peuples Bantu: migrations, expansion et identité culturelle*, L'Harmattan, Paris (France), 443-483.

KUB-87c

1987c Kubik, Gerhard: *Tusona-Luchazi ideographs, a graphic tradition as practised by a people of West-Central Africa*, Verlag Stiglmayr, Fohrenau (Austria), 311 p.

Describes and analyses the *tusona* tradition as practiced among the (Va)Luchazi in northwestern Zambia. In the introduction to chapter 14, entitled "The mathematics of the 'tusona' tradition" (195-227), the author states that "The majority of 'tusona' is based on the combination and the geometrical relationships of regular numerical components. Some of these relationships are so highly complex that we can presuppose the existence of empirical mathematical knowledge in the history of the eastern Angolan / northwestern Zambian culture area." He discusses the "rules of behaviour" for the drawing of a particular class of 'tusona'.

KUB-90

1990 Kubik, Gerhard: *Visimu vya mukatikati* – dilemma tales and 'arithmetical puzzles' collected among the Valuchazi, *South African Journal of African Languages*, Pretoria (South Africa), Vol. 10, No. 2, 59-68.

Dilemma tales are discussed on the basis of recordings and cinematographic documentation of narrative performances from eastern Angola and northwestern Zambia. In the oral literature of the Valuchazi, use is sometimes made of explanatory visual symbols, or ideographs, drawn in the sand. This is exemplified by Chindamba Ngunga's dilemma tale, transcribed and analyzed in this paper: "This particular dilemma tale is about three women and three men who want to cross a river in order to attend a dance on the other side. With the river between them there is a boat with the capacity for taking only two people at one time. However, each of the men wishes to marry all the three women himself alone. Regarding the crossing, they would like to cross in pairs, each man with his female partner, but failing that any of the other men could claim all the women for him self. How are they crossing?" (p. 62).

KUK-93

1993 Kuku, Aderemi O.: *Mathematical research and education in Africa: problems and prospects* (Invited address at the joint AMS, CMS, MAA meeting at the University of British Columbia [Vancouver, Canada] on 16.08.1993), AMU.

Includes a short history of the African Mathematical Union.

L

LAA-90

1990 Laabid, Ezzaim: *Arithmétique et Algèbre d'héritage selon l'Islam, deux exemples: Traité d'al-Hubûbî (X^e-XIe s.) et pratique actuelle au Maroc* [Arithmetic and algebra of inheritance according to the Islam, two examples: a treatise of al-Hubûbî (10th-11th century) and today's practice in Morocco], masters thesis, University of Quebec, Montreal (Canada), 234 p. (in French).

LAA-06

2006 Laabid, Ezzaim: *Les techniques mathématiques dans la résolution des problèmes de partages successoraux dans le Maghreb médiéval: l'exemple du 'Mukhtasar' d'al-Hûfî (m. 588/1192)* [Mathematical techniques in the solution of inheritance problems in the medieval Maghreb: the example of the *Mukhtasar* of al-Hûfî (d. 588 / 1192)], doctoral thesis, Université Mohammed V – Agdal (Rabat, Morocco) (in French).

LAB-81

1981 Labatut, Roger: Réflexions sur la numération peule [Reflections on the Fulbe numeration], *Itinérances en pays peul et ailleurs; mélanges réunis à la mémoire de P.F. Lacroix*, Mémoires de la Société des africanistes, Paris (France), Vol. 1, p. 91-102.

LAG-68

1968 Lagercrantz, Sture: African tally-strings, *Anthropos*, Sankt Augustin (Germany), Vol. 63, 115-128.

Gives an overview of the ethnographic literature on mnemonic aids in counting in Sub-Saharan Africa. Map on p.126 displays the distribution of tally-strings over the continent. The most important tallies of higher age are the "memorial cairns (i.e. the custom that every one passing a place where someone for instance has suffered a violent death throws down a stone or a stick)."

LAG-73
1973 Lagercrantz, S.: Counting by means of tally sticks or cuts on the body in Africa, *Anthropos*, Sankt Augustin (Germany), Vol. 68.

LAM-68
1968 Laman, Karl: Arithmetic, in: K. Laman, *The Kongo*, Studia Ethnographica Upsaliensia, Upsala (Sweden), Vol. IV, 8-9.

Describes briefly counting and measuring among the Sundi. Accounts are kept by means of stones, palm nuts, knots, tally sticks, etc. In games the score may be kept by putting aside certain objects, by tying knots in a string, or by chanting a jingle (examples are given).

LAMB-81
1981 Lamrabet, Driss: *La mathématique maghrébine au moyen-âge* [Maghrebian mathematics during the Middle Ages], thesis (Mémoire de Post-Graduation), Free University of Brussels (Belgium), 160 p.

LAMB-94
1994 Lamrabet, Driss: *Introduction à l'Histoire des Mathématiques maghrébines* [Introduction to the history of Maghrebian mathematics], Imprimerie El-maârif al-Jadida, Rabat (Morocco), 302 p. (in French).

This book is in three parts. In the first part the author presents a short introduction to mathematical activity in Egypt, Babylonia, India and ancient Greece (p. 2-9); followed by a chapter on "The birth of Arab mathematics: the Islamic East" (p.10-19) and by a third chapter on "The mathematics of the Islamic West: Andalusia" (p.20-41). The second part contains bio-bibliographical files of Maghrebian mathematicians and presents the contents of some mathematical works produced in the Maghreb. The third part contains "extracts of Maghrebian mathematicians" relative to Arithmetics, Algebra and Geometry.

LAMB-03
2003 Lambaret, Driss: Some mathematicians in ancient North Africa until the beginning of the 14th century, in: Kinani, A. El (Ed.), *Non-normed topological algebras. Procedings of the international conference on topological algebras and*

applications, École Normale Supérieure, rabat (Morocco), 154-168.

Presents examples of mathematicians in North Africa or the Maghreb until the beginning of the 14th century. Firstly, examples are given from the pre-Islamic period like Theodorus (465-398 B.C.), Eratosthenes (276-194 B.C.) and Nicotelese (c. 250 B.C.) of Cyrene, Theodoses of Tripoli (2nd century B.C.) and Apuleius of Madaura (124-170 A.D.). Secondly, examples are presented from the Islamic period. Particular attention is given to al-Hassar (c. 1150 A.D.), Ibn Mun^{c}im (d. 1228) and his work in combinatorics, and Ibn al-Banna (1256-1321). The paper concludes with examples from mathematical notation and algorithms developed in the Maghreb.

LAN-89
1989 Langdon, Nigel: Cultural starting points for mathematics: a view from Ghana, *Science Education Newsletter*, British Council, London (UK), Vol. 87, 1-3.

LANG-95
1995 Lange, Robert; Maurice Bazin and Modesto Tamez: Playing games: Madagascar solitaire, in: Bazin, Maurice & Modesto Tamez (Eds.): *Math across cultures*, Exploratorium Teacher Activity Series, San Francisco CA (USA), 15-22 (reproduced in BAZ-02).

Suggestions for teachers on how to use a solitaire board game from Madagascar in the mathematics classroom.

LAR-04
2004 Laridon, Paul, Mogari, David and Mosimege, Mogege David: (2002). Ethnomathematics Research in South Africa, in: C. Keitel, J. Adler & R. Vithal (Eds.), *Mathematics Education Research in South Africa: Perspectives, Practices and Possibilities*, Human Sciences Research Council, Pretoria (South Africa).

LAS-75
1975 Lassa, Peter Ntasiri: *A study of the mathematics programs for elementary school teachers in Nigeria*, doctoral thesis, University of Wisconsin, Madison (USA).

LAS-80
1980 Lassa, Peter: *The problems of teaching and of learning mathematics in a second language (African experiment)* (paper presented at ICME IV, Berkeley CA, USA), 18 p. (mimeo).

LAS-84
1984 Lassa, Peter N.: *The sorry state of mathematics education in Nigeria* [Inaugural address delivered at the University of Jos on 20th January, 1984], University of Jos (Nigeria), 21 p.

LAS-86a
1986a Lassa, Peter: *Problems and prospects of mathematics teaching and learning in African schools* (paper presented at the 2nd Pan-African Congress of Mathematicians, University of Jos, Nigeria), 15 p. (mimeo).

LAS-86b
1986b Lassa, Peter: *Les problémes et l'avenir de l'enseignement et l'apprentissage de mathématiques aux écoles africaines* (paper presented at the 2nd Pan-African Congress of Mathematicians, University of Jos, Nigeria), 18 p. (mimeo).

Translation of LAS-86a into French.

LEA-87a
1987a Lea, Hilda: Botswana baskets, *Mathematics Teaching*, London (UK), Vol. 118, 56-57.

LEA-87b
1987b Lea, Hilda: Traditional mathematics in Botswana, *Mathematics Teaching*, London (UK), Vol. 119.

Reports on an investigation into traditional mathematics in Botswana carried out by University students. Old people were interviewed to ascertain how mathematical activities were carried out in the past, and how some older people do mathematics today. Contains information on counting, arithmetical operations, geometrical forms, and measurement of length, volume and time.

LEA-89a
1989a Lea, Hilda: Informal mathematics in Botswana, *Proceedings of the 41st CIEAEM Meeting of the International Commission for*

the Study and Improvement of Mathematics Teaching, Brussels (Belgium), 43-53.

LEA-89b
1989b Lea, Hilda: Traditional mathematics in Botswana, *Botswana Notes and Records*, Gaborone (Botswana), Vol. 20, 143-146.

LEA-90a
1990a Lea, Hilda: *Informal Mathematics in Botswana: Mathematics in the Central Kalahari*, Faculty of Education, University of Botswana, Gaborone (Botswana), 9 p.

"A good example of what mathematical ideas were used before recorded history, can be seen today in the daily activities of Bushman society. They carry out mathematical activities suitable for their traditional way of life, and their highly developed spatial abilities are very necessary for survival in their harsh environment" [p.1]. The paper describes counting (one, two, two-one, two-two, two-two-one etc.), measurement, time reckoning, classification, tracking and mathematical ideas in technology and craft. "Bushmen have the oldest pattern of life found in the world today... A hunting and gathering community does not have need of counting precise measurement though requires basic skills for survival, and very special skills to interpret the environment. They need very good visual discrimination and visual memory" [p.7].

LEA-90b
1990b Lea, Hilda: *Informal Mathematics in Botswana: Spatial concepts in the Kalahari*, Faculty of Education, University of Botswana, Gaborone (Botswana), 9 p.

"Hunters and herdsmen in the Kalahari, who have never been to school and who have lived in very remote areas all their lives, were interviewed on two occasions to ascertain how far their spatial concepts have developed. When asked how they recognized animal footprints, and how they found their way in the desert, they were seen to have a very good visual memory, and to be aware of the minutest detail in recognizing shapes. When given a visual thinking test, they performed with a high degree of skill on items related to their environment."

LEA-90c
1990c Lea, Hilda: Spatial concepts in the Kalahari, *Proceedings of the 14th International Conference on Psychology of Mathematics Education*, Oaxtepec (Mexico), Vol. 2, 259-266.

LEGEN-58
1958 Legendre, Marcel: *Survivance des mesures traditionnelles en Tunisie* [Survival of traditional measures in Tunisia], Presses universitaires de France, Paris (France), 90 p. (in French).

LEG-89
1989 Legon, John A.R.: The Geometry of the Great Pyramid, *Göttinger Miszellen*, Göttingen (Germany), No. 108, 57-64

LEG-90
1990 Legon, John A.R.: The Geometry of the Bent Pyramid, *Göttinger Miszellen*, Göttingen (Germany), No. 116, 65-72.

LEG-92
1992 Legon, John A.R.: A Kahun Mathematical Fragment, *Discussions in Egyptology* Oxford (UK), Vol. 24, 21-24.

LEG-94a
1994 Legon, John A.R.: *Nbj*-Rod Measures in the Tomb of Senenmut, *Göttinger Miszellen*, Göttingen (Germany), Vol. 143, 97-104.

LEG-94b
1994 Legon, John A. R.: Measurement in Ancient Egypt, *Discussions in Egyptology*, Oxford (UK), No. 30, 87-100.

LEG-94c
1994 Review of ROI-93, *Discussions in Egyptology*, Oxford (UK), No. 30, 87-100.

LEG-96
1996 Legon, John A.R., The Quest for the true *nbj* measure, *Discussions in Egyptology*, Oxford (UK), No. 36, 69-78.

"In the ongoing discussion of the *nbj* measure the author deals with his interpretation of the *nbj* in the Senenmut ostraca as a measurement of volume, a cubic measure, whereas Roik defends a linear measure.

With respect to the canon of proportions he also replies to the points raised by Roik with regard to the claimed use of the units of the rod-65."

LEVE-66

1966 Levey, Martin: *The Algebra of Abû Kâmil*, University of Wisconsin Press, Madison (USA), 226 p.

LEV-29

1929 Lévy-Bruhl, Lucien: La numération chez les Bergdama [Numeration among the Bergdama], *Africa, Journal of the International Institute of African Languages and Cultures*, London (UK), Vol. II, No. 2, 162-173 (in French).

Compares aspects of (finger) counting of the Bergdama (Berg Damara) of South Africa and Namibia with the (verbal) counting of their neighbors, the Nama.

LIE-90

1990 Liebenberg, Louis: *The Art of Tracking: The origin of Science*, David Philip Publ., Claremont (South Africa), 176 p.

Studies first the evolution of hunter-gatherer subsistence in general, and thereafter the hunter-gatherers of the Kalahari in southern Africa in particular. Principles of tracking, classification of signs, and spoor interpretation are analyzed. The author asserts "it is possible that the development of tracking played a significant role in the evolution of the scientific faculty" (p.48). "The critical attitude of contemporary Kalahari Desert trackers, and the role of critical discussion in tracking suggest ... that the rationalist tradition of science may well have been practiced by hunter-gatherers long before the Greek philosophic schools were founded" (p.45).

LIN-08

1908 Lindblom, Gerhard: The magic significance of numbers, in: G. Lindblom, *The Akamba in British East Africa, an ethnological monograph* (Reprint: Negro University Press, New York (USA), 1969, 607 p.), 306-310.

"... Odd numbers are generally considered disastrous or at least unlucky..." A contrary state of affairs is encountered "at a medicine man's divination, as the pebbles that fall out of his calabash are a good omen if they are odd and vice versa..." (p. 306).

LOB-03
2003 Lobry, Claude: *La recherche mathématique en Afrique: une nécessité pour le développement* [Mathematical research in Africa: a necessity for development], L'Harmattan, Paris (France), 156 p. (in French) (preface: Jean-Pierre Kahane).

LOO-90
1990 Loomis, D. E.: Euclid: rhetoric in mathematics, *Philosophia Mathematica*, Toronto (Canada), Vol. 5, Nos.1-2, 56-72.

LOR-87
1987 Lorch, R.: Some remarks on the Arabic-Latin Euclid, in: Burnett, Charles (Ed.), *Adelard of Bath, an English scientist and Arabist of the early 12th century*, Warburg Institute, University of London, London (UK), 45-54.

LOR-95
1995 Lorch, R. P.: Ptolemy and Maslama on the transformation of circles into circles in stereographic projection, *Archive for History of Exact Sciences*, Berlin (Germany), Vol. 49, No. 3, 271-284.

LOU-82
1982 Loucou, Jean-Noël: La mathématique chez les Baoulé de Côte d'Ivoire [Mathematics among the Baoulé of Côte d'Ivoire], *Annales de l'Université d'Abidjan*, Abidjan (Côte d'Ivoire), Série 1, Histoire, No. 10, 59-86 (in French).

Analyses mathematical knowledge embedded in oral tradition and on games and gold weights.

LUB-00
2000 Lubisi, R.: *An investigation into mathematics teachers' perceptions and practices of classroom assessment in South African lower secondary schools*, doctoral thesis, University of Nottingham (UK).

LUM-79
1979 Lumpkin, Beatrice: *A young genius in old Egypt*, Dusable Museum Press, Chicago IL (USA), 24 p.

Booklet for children with information on ancient Egyptian number symbols and arithmetical procedures (New edition: LUM-92a).

LUM-80a
1980 Lumpkin, Beatrice: The Pyramids - Ancient Showcase of African Technology, *Journal of African Civilizations,* New York (USA), 1980, Vol. 2, Nos. 1 & 2, 10-26.

LUM-80b
1980 Lumpkin, Beatrice: The Egyptians and Pythagorean triples, *Historia Mathematica*, New York (USA), Vol. 7, No. 2, 186-187.

LUM-81
1981 Lumpkin, Beatrice & Zitler, Siham: Mathematicians of the Cairo Science Academy in the Middle Ages, *Journal of African Civilizations*, New York (USA), Vol. 3, No. 2, 25-38.

Shows that Egypt and North Africa continued during the Middle Ages "their tradition of leadership in science and mathematics, a tradition then already 4,000 years old". Criticizes "most European historians (and North Americans)" who "have denied that Muslim scholars created anything new, merely crediting them with preserving Greek (European) learning during the Middle Ages" (p.1).

LUM-83a
1983 Lumpkin, Beatrice: Africa in the mainstream of mathematics history, in SER-83, 100-109.

"For thousands of years, Africa was in the mainstream of mathematics history. This history began with the first written numerals of ancient Egypt, a culture whose African origin has been reaffirmed by the most recent discoveries of archaeology. With a longer period of scientific work than any other area of the world, progress in mathematics continued on the African continent through three great periods, ancient Egyptian, Hellenistic and Muslim." "Although all peoples and continents have played a role in the history of mathematics, the contributions of Africa are still unacknowledged by western historians."

Reproduced in POW-97 with postscript (101-117).

LUM-83b
1983 Lumpkin, Beatrice: *Senefer and Hatshepsut*, Dusable Museum Press, Chicago IL (USA), 130 p.

Novel about Egypt in the time of Hatshepsut (1500 BC) with information on ancient Egyptian mathematics (numerals, arithmetic, measurement, progressions).

LUM-83c
1983 Lumpkin, Beatrice: The Pyramids: ancient showcase of African science and technology, in SER-83, 67-83.

"The pyramids and other great monuments of Egypt and the Sudan are the product of a long development of African science and technology. Their development is traced from the mud brick beginning to the great pyramids and temples. Planning of the monuments is described; examples are given of written plans, and the level of mathematics and technology required for pyramid building are discussed. Possible methods of construction of the pyramids are considered."

LUM-87
1987 Lumpkin, Beatrice: *African and African American Contributions to Mathematics*, Baseline Essays, Multinomah School District, Portland Oregon (USA).

LUM-88
1988 Lumpkin, Beatrice: Hypatia and Women's Rights in Ancient Egypt, in: I. Van Sertima (Ed.), *Black Women in Antiquity,* Transaction Books, New Brunswick, N.J. (USA).

LUM-92a
1992 Lumpkin, Beatrice: *Senefer: A young genius in Old Egypt*, Africa World Press, Trenton NJ (USA), 32 p.

New edition of LUM-79.

LUM-92b
1992b Lumpkin, Beatrice: *Multiculturalism in Mathematics, Science and Technology. Ten articles*, Addison-Wesley, Menlo Park (USA), 208 p.

Contains the several papers about African Americans in science, including 'Benjamin Banneker' (25-30) and about mathematics in Africa: 'The Ancient Egyptians I' (57-60), 'The Ancient Egyptians II'

(61-64), 'The Ancient Egyptians III' (65-68). 'Eratosthenes' (69-72), and 'Hypatia' (77-79).

LUM-95a

1995a Lumpkin, Beatrice: *African cultural materials for elementary mathematics*, Educational Equity Services, Illinois State Board of Education, Chicago IL (USA), 122 p. (field test copy).

Contains the following chapters: 1. Beginnings of mathematics, 2. Meaning of addition and subtraction, 3. Adding whole numbers, 4. Subtracting whole numbers, 5. Measuring time, capacity, and mass, 6. Multiplication properties, 7. The meaning of division, 8. Fractions, 9. Mixed numbers and decimals, 10. Measure of capacity and mass, 11. Geometry, 12. Multiplication of whole numbers, 13. Dividing whole numbers, 14. Learn with African games. The chapters give examples of how mathematical ideas and practices from ancient Egypt and from other African regions may be used in elementary mathematics education.

LUM-95b

1995b Lumpkin, Beatrice & Arthur B. Powell: *Math: a rich heritage*, Globe Fearon Educational Publisher, Upper Saddle River NJ (USA), 48 p.

Booklet intended to motivate African-Americans to study mathematics. It explores "the African roots of modern mathematics" and explains "how math influenced the contributions and achievements of several African American in math-related careers" (p. 5).

LUM-95c

1995c Lumpkin, Beatrice & Dorothy Strong: *Multicultural Science and Math Connections — Middle School Projects and Activities*, Weston Walch Publisher, Portland, Maine (USA), 193 p.

Part 1 (From Africa to the Arctic) includes the following chapters related to Africa: 1. Nubia (3-16); 2. Egypt (17-37); 4. Mozambique (44-52); 5. Kenya (53-61). Part 2 (Lives in Science and Math) includes the following chapters related to mathematics in Africa: Thomas Fuller (140-143) and Hypathia of Alexandria (144-149).

LUM-96

1996 Lumpkin, Beatrice: From Egypt to Benjamin Banneker: African origins of false position solutions, in: Ronald Calinger (Ed.), *Vita Mathematica, Historical Research and Integration with Teaching*, Mathematical Association of America Notes, Washington DC (USA), Vol. 40, 279-289.

Describes the use of the rule of false positions in ancient Egypt, in the work of later Alexandrian mathematicians, like Diophantus (c. 250), and of Abu Kamil (born 850), the influence on mathematicians in Europe and later on Benjamin Banneker (1731-1806), one of the first African American who dedicated himself to mathematics.

LUM-02

2002 Lumpkin, Beatrice: Mathematics Used in Egyptian Construction and Bookkeeping, *The Mathematical Intelligencer*, New York (USA), Vol. 24, No. 2, 20-25.

Presents "examples from ancient construction and bookkeeping practices [which] indicate that the development of relatively modern concepts, such as recognition of zero as a quantity and the metricizing of space, has a long history, going back at least 4,700 years in ancient Egyptian mathematics. The examples include a bookkeeping balance sheet with many columns containing zero remainders and numbered construction lines at pyramids and mastabas. The same symbol, 'nfr', was used for the zero remainders and the zero reference point on the construction guidelines. A third example was a very interesting architect's diagram that gave vertical coordinates for points located on a curve. The horizontal spacing of the points appears to be one cubit apart."

LUM-03a

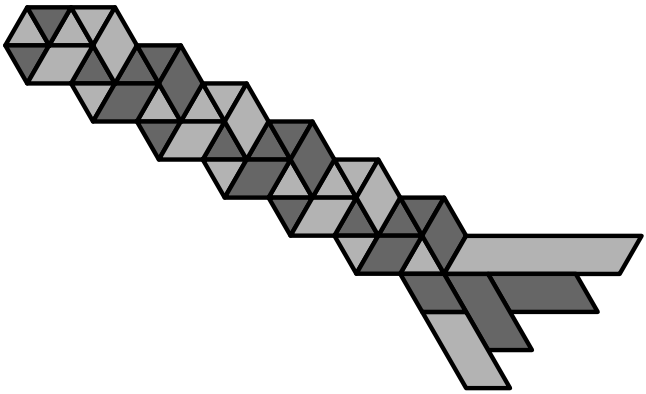
2003a Lumpkin, Beatrice: Review of Gerdes' Awakening of Geometrical Thought in Early Culture (GER-03a), Political Affairs: Ideology, Politics and Culture, New York (USA), Vol. 82, No. 10, 40-41.

LUM-03b

2003b Lumpkin, Beatrice: Ancient Egyptian Mathematics and Forerunners: Some Hints from Work Sites, in: A. K. Eyma & C. Bennett (Eds.), *A Delta-man in Yebu*, Occasional Volume of

the Egyptologists' Electronic Forum, Universal Publishers / uPublish.com, No.1, 210-214.

LUN-45
1945 Lundsgaard, Erik: Aegyptisk matematik [Egyptian mathematics], J. H. Schultz, Copenhagen (Denmark) (in Danish).



Hexagonal woven strip from Benin, Kenya, Mozambique, Nigeria
(cf. GER-99a, p. 111)

M

MAD-86
1986 Mada, Nalimbi: *Étude de la numération dans certaines langues centrafricaines et applications dans l'enseignement élémentaire* [Study of the numeration in certain Central African languages and applications in primary schools], doctoral thesis, Université Paris VII (France) (in French).

MAEO-82
1982 Mae Ohuche, Nancy: *The development of the horizontal-vertical co-ordinate reference system by some Nigerian (Igbo) children and young adults*, doctoral thesis, University of Nigeria, Nsukka (Nigeria).

MAE-10
1910 Maes, M. J.: La numération chez les peuplades du Lac Léopold II [Numeration among the peoples of Lake Leopold II], *La Revue Congolaise*, Brussels (Belgium), Vol. III, 275 ff. (in French).

MAG-78
1978 Magdalena, Henri: *L'expression du nombre en Empire Centrafricain* [The expression of number in the Central-African Empire], IREM de Bangui, Bangui (Central-African Republic), 39 p.

MAGI-02
2002 Magide Fagilde, Sarifa Abdul: *Towards a characterisation of communication and gender patterns in secondary mathematics classrooms in Mozambique*, doctoral thesis, University of the Western Cape, Bellville (South Africa).

MAH-98
1998 Mahlomaholo, Geoffrey: *Signification of African cultural identity, individual African identity and performance in mathematics among some standard nine African pupils in Mangaung high schools*, doctoral thesis, University of the Western Cape, Bellville (South Africa).

MANC-90
1990 Mancha, J. L.: Ibn al-Haytham's homocentric epicycles in Latin astronomical texts of the XIVth and XVth centuries, *Centaurus*, Copenhagen (Denmark), Vol. 33, No.1, 70-89.

MAN-86
1886 Mann, Adolphus: Notes on the Numeral System of the Yoruba Nation, *Journal of the Anthropological Institute of Great Britain and Ireland*, London (UK), Vol. 16, 59-64.

Explains how the addition, multiplication and subtraction principles are used to form the Yoruba (Nigeria) numerals: 15 = 5 less 20, 40 = 20 x 2, 170 = (20 x 9) - 10, 185 = (200 - 10) - 5, 5000 = 200 x 25, etc. The author suggests that the origin of this system may be found in "the way in which large sums of money (cowries) are counted."

MANO-65
1965 Al-Manouni, Mohamed: The professors and the authors of Geometry in the Sa^{c}adian Maghreb, *Da^{c}wat al-haqq*, Rabat (Morocco), No. 3, 101-104 (in Arabic).

MANO-77
1977 Al-Manouni, Mohamed: *Sciences, literature and arts in the epoch of the Almohads*, Dar al-Maghrib, Rabat (Morocco), 325 p. (in Arabic).

MANO-79
1979 Al-Manouni, Mohamed: *Leaflets on the Moroccan civilization in the epoch of the Merinids*, Imprimatlas, Rabat (Morocco), 376 p. (in Arabic).

MANO-84
1984 Al-Manouni, Mohamed: Note on the activities related to the study of mathematics and astronomy in Meknes, *Al-Manâhil*, Rabat (Morocco), No. 30, 32-87 (in Arabic).

MANO-85
1985 Al-Manouni, Mohamed: Mathematical study activities in Morocco of the fourth period of the Middle Ages (Merinid's period), *Al-Manâhil*, Rabat (Morocco), No. 33, 77-115 (in Arabic).

MANO-89
1989 Al-Manouni, Mohamed: *The civilization of the Almohads*, Tubqal, Casablanca (Morocco), 218 p. (in Arabic).

MANS-98
1998 Mansfeld, J.: *Pappus, Mathematicus en een beetje Filosoof* [Pappus, Mathematician and a bit of a Philosopher], Koninklijke Nederlandse Akademie van Wetenschappen, Amsterdam (Netherlands), 20 p. (in Dutch).

A brief analysis of some philosophical passages in Books III and V of the *Mathematical collection* of Pappus of Alexandria and in Pappus' commentary on Book X of Euclid's *Elements*.

Review: HOGE-01.

MAP-96
1996 Mapapá, Abílio: Children's games and toys in mathematics education in Mozambique, in: T. Kjaergard et al. (Eds.), *Numeracy, Race, Gender, and Class — Proceedings of the Third International Conference on the Political Dimensions of Mathematics Education*, Gaspar Forlag, Landas (Norway), 221-228.

Explores possibilities of using traditional Mozambican games and toys in mathematics education.

MAP-97
1997 Mapapá, Abílio: Barns leker og spill i matematikk undervisningen, in: *Tangenten — Tidsskrift for Matematikk-undervisning*, Landas (Norway), No. 4, 5-11.

Translation into Norwegian of MAP-96.

MARC-88
1988 Marcos, Berthe Elisabeth: *Pédagogie de l'initiation aux mathématiques* [Pedagogy for initiation in mathematics], CEDA, Abidjan (Côte d'Ivoire), 72 p. (in French).

Didactical suggestions for preschool and early school mathematics teaching, using, in particular, games from the cultural environment of children in Ivory Coast.

MAR-64
1864 Marre, Aristide: Le Talkhys d'Ibn al-Bannâ [The *Talkhys* of Ibn al-Bannâ], *Atti dell'Accademia Pontificia de Nuovi Lincei*, Rome (Italy), Vol. 17, 289-319 (in French).

MART-65
1965 Martin, William Ted (Ed.): *A report of an African Education Program*, Educational Services, Watertown MA (USA), 51 p.

Report of an educational program in ten African countries (Ethiopia, Ghana, Kenya, Liberia, Malawi, Nigeria, Sierra Leone, Tanzania, Uganda, Zambia), being the African Mathematics Program (AMP) its first part. Contains the following contributions related to AMP:

* Onyerisara Ukeje: The Entebbe Mathematics Workshop, Summer 1962 (17-19),
* John Oyelese: The Entebbe Mathematics Workshop, Summer 1963 (20-24),
* Cyril Okosi: The Entebbe Mathematics Workshop, Summer 1964 (25-29),
* Stanley Weinstein: AMP, Tutor and Teacher Training Institutes (36-46).

MARTI-92
1992 Martinson, Annemarie: *The role of rock art in mathematics education*, University of Witwatersrand, Johannesburg (South Africa), 6 p. (mimeo).

Suggests that South African rock art may be explored in the mathematics classroom.

MAS-87
1987 Masinga, L. C.: The development of mathematics education in Swaziland in the past two decades, *Proceedings of the 6th Symposium of the Southern Africa Mathematical Sciences Association*, University of Dar es Salaam (Tanzania), 215-228.

MAT-17
1917 Mathews, H.F.: Notes on the Nungu tribe, Nassawara Province, Northern Nigeria, and the neighboring tribes which use the duodecimal system of numeration, *Harvard African Studies*, Cambridge MA (USA), Vol. 1, 83-93 (Pages 92-93 are

reproduced in *AMUCHMA Newsletter*, Maputo (Mozambique), No. 27, 11-13.

Describes (pages 92-93) the numeration systems used by the Nungu and by neighboring peoples like the Ninzam on the north, the four clans known as the Artum, Barrku, Burrza, and Upye on the east, and the people known collectively as the Mada on the south.

MAT-64
1964 Mathews, H.F.: Duodecimal numeration in Northern Nigeria, *The Nigerian Field*, Vol. 34, No. 4, 181-191 (original paper from 1916).

MED-71
1971 Medvedev, F. A.: Les quadratures et les cubatures chez Pappus d'Alexandrie [Quadrature and cubature by Pappus of Alexandria], in: *Actes XIIe Congrès International d'Histoire des Sciences Histoire des Mathématiques et de la Mécanique*, Paris (France), Vol. IV, 107-110.

MEH-75
1975 Mehész, Kornél Zoltán: *Secretos de la Matematica Egípcia, Griega y Hindu* [Secrets from Egyptian, Greek and Hindu mathematics], Editorial Diogenes, Corrientes (Argentina) (in Spanish).

Deals mostly with mathematics from the Hellenistic period and links with other cultures, particularly with regard to cube roots, unsolved geometrical problems, and the regular pentagon.

MEI-15
1915 Meinhof, Carl: Rezension von M. Schmidl 'Zahl und Zählen in Afrika', *Zeitschrift für Kolonialsprachen*, Berlin (Germany), Vol. 6 (1915-1916), 251-252 (in German).

Review of SCH-15.

MEI-17
1917 Meinhof, Carl: Rezension von K. Sethe 'Von Zahlen und Zahlworten bei den alten Ägyptern und was für andere Völker und Sprachen daraus zu lernen ist', *Zeitschrift für Kolonialsprachen*, Berlin (Germany), Vol. 8 (1917-1918), 268-270 (in German).

Review of SET-16. Sethe is criticized for the fact that he advances with a comparison with Semitic languages, but forgets to study the relationship with African languages.

MEI-23

1923 Meinhof, Carl: Ein magisches Quadrat auf einem Hausa-Amulett [A magic square on a Hausa amulet], *Zeitschrift für Eingeborenensprachen*, Hamburg (Germany), Vol. 14, 224-226.

Reconstructs and analyses a 7x7 magic square on a Hausa amulet (Nigeria), reproduced in C. Robinson's '*Specimens of Hausa-Literature*' (Cambridge, 1896). We are dealing with a bordered or concentric magic square: taking away the successive borders, the smaller squares remain magic. Meinhof calls it a Stifelius' square after Michael Stifel, who discussed this type of magic square in his *Arithmetica integra* (1544).

10	45	44	7	11	12	46
9	19	34	17	20	35	41
8	18	24	23	28	32	42
49	37	29	25	21	13	1
48	36	22	27	26	14	2
47	15	16	33	30	31	3
4	5	6	43	39	33	40

MEM-92

1992 *Memorial issue for Professor Adegoke Olubummo*, *Journal of the Nigerian Mathematical Society*, University of Ibadan, Ibadan (Nigeria), Vol. 11, No. 2., 131 p.

MERE-95

1995 Mereku, Kofi Damian: *A comparison of the official primary mathematics curriculum in Ghana with the way in which it is implemented by teachers*, doctoral thesis, University of Leeds (UK).

MEU-79

1979 Meunier, Dominique: Note sur la survivance des poids anciens à Tombouctou [Note on the survival of the old weights in Timbuktu], *Revue d'histoire maghrébine*, Zaghouan (Tunisia), Vol. 6, No. 15/16, 93-105 (in French).

MIC-96
1996 Michalowicz, Karen: Fractions of Ancient Egypt in the contemporary classroom, *Mathematics Teaching in the Middle School*, NCTM, Reston VA (USA), Vol. 1, No. 10, 786-789.

Presents suggestions of how using ancient Egyptian fractions in the mathematics classroom.

MIC-99
1999 Michalowicz, Karen Dee: Review of Gerdes' *Geometry from Africa* (GER-99a) (online available at: www.maa.org/ reviews/gerdes.html).

MICH-74
1974 Michau, J. M. Z.: Problem areas in the acquisition of mathematical concepts by black children in South Africa, *Journal of Education of the University of Natal*, Vol. 10, 21-29.

Suggests two possible causes for poor achievement in mathematics in 'black' high schools in South Africa: the effects of differing cultural backgrounds, and the effect of the change of language medium from a mother tongue to English.

MID-97
1997 Middleton, John (Ed.): *Encyclopedia of Africa South of the Sahara*, Charles Scribner's Sons, New York (USA), 4 volumes.

The encyclopedia contains two short articles by Paulus Gerdes: 'Geometries' (Vol. 2, 224-227) and 'Number systems' (Vol. 3, 346-348).

MIL-92
1992 Millroy, Wendy: *An ethnographic study of the mathematical ideas of a group of carpenters*, NCTM, Reston VA (USA), 210 p.

The author conducted an ethnographic study as an apprentice carpenter in Cape Town, South Africa, to document the mathematical ideas that are embedded in the everyday woodworking activities of a group of carpenters.

MIZ-71
1971 Mizony, Michel: Les jeux stratégiques camerounais et leurs aspects mathématiques [Cameroonian strategic games and their

mathematical aspects], *Annales de la Faculté des Sciences du Cameroun*, Yaoundé (Cameroon), Vol. 6, 19-38 (in French).

Presents a classification and regional distribution of strategic games from Cameroon and suggest that they be used "to understand many mathematical notions."

MMA-65

1965 Mmari, Geoffrey: *An analytical study of foreign textbooks of mathematics used in Tanganyika secondary schools*, M.A. thesis, University of Northern Iowa (USA).

Analysis of three foreign textbooks (algebra, arithmetic and geometry), in widespread use in Tanganyika (today Tanzania) schools in the early 1960's. It shows how learning mathematics was made more difficult for African children by the cultural gulf between themselves and the authors of the books.

MMA-74

1974 Mmari, Geoffrey: *Tanzania's experience in, and efforts to resolve, the problem of teaching mathematics through a foreign language*, UNESCO (ED-74/CONF.808/12), Paris (France).

Paper presented at the symposium 'Interactions between Linguistics and Mathematical Education' held in Nairobi (Kenya, 1-11 September 1974). Having decided to adopt (Ki)Swahili as the medium of instruction, Tanzania has been faced with the problem of enriching the language in order to be used in school mathematics education. The ways in which this is being tackled are described.

Reproduction in CASM-75, 32-43.

MMA-78

1978 Mmari, Geoffrey: The United Republic of Tanzania: mathematics for social transformation, in: Frank Swetz (Ed.), *Socialist Mathematics Education*, Burgundy Press, Southampton PA (USA), 301-350.

Analyses the history of mathematics education in Tanzania before and after Independence.

MMA-80
1980 Mmari, Geoffrey: Secondary Mathematics in the United Republic of Tanzania, *Studies in Mathematics Education,* UNESCO, Paris (France), Vol. 1, 106-126.

MMA-91
1991 Mmari, Geoffrey: *On the history of the Mathematical Association of Tanzania*, Mathematics Association of Tanzania (MAT), Dar es Salaam (Tanzania) (mimeo).

MOI-85
1985 Moiso, Bokula & Ngandi Litanga: Numération cardinale dans les langues Bantu du Haut-Zaire [Cardinal numeration in the Bantu languages of Upper-Zaire], *Annales Aequatoria*, Mbandaka (Congo / Zaire), Vol. 6, 189-196 (in French).

MOI-91
1991 Moiso, Bokula: Etude comparée du système de numérotation de 1 à 10 dans quelques langues non-Bantu du Haute-Zaire [Comparative study of the system of numeration from 1 to 10 in some non-Bantu languages from Upper-Zaire], *Annales Aequatoria*, Mbandaka (Congo / Zaire), Vol. 12, 475-479 (in French).

MOR-70
1970 Morrow, Glenn Raymond. (Ed.), *A commentary on the first book of Euclid's 'Elements'*, Princeton University Press, Princeton NJ (USA), 355 p.

MOS-96
1996 Mosimege, Mogege David: Ethnomathematical activities in South Africa: some developments, reflections and possibilities, in: T. Kjaergard et al. (Eds.), *Numeracy, Race, Gender, and Class — Proceedings of the Third International Conference on the Political Dimensions of Mathematics Education*, Gaspar Forlag, Landas (Norway), 229-241.

Explores possibilities of using South African string figure patterns, games, architecture, flag, and counting methods in the classroom.

MOS-97

1997 Mosimege, Mogege David: The Use of Games in Mathematics Classrooms, in: M. Sanders (Ed.), *Proceedings of the 5th Conference of the Southern African Association for Research in Mathematics and Science Education*, University of the Witwatersrand, Johannesburg (South Africa).

MOS-98a

1998a Mosimege, Mogege David: Culture, games and mathematics education: An exploration based on string figures, in: OLI-98, Vol. 3, 279-286.

MOS-98b

1998b Mosimege, Mogege David: Culturally Specific Games in the Mathematics Classrooms: The Impact of their Use in the Learning of Mathematics, *Journal of the Southern African Association for Research in Mathematics and Science Education*, Cape Town (South Africa), Vol. 2, No. 1, 52 - 60.

MOS-00a

2000a Mosimege, Mogege David: *The potential of the use of culturally specific games in school mathematics*, doctoral thesis, University of the Western Cape (South Africa), 337 p.

Studies the potential of the use of culturally specific games, in particular, string figures, in secondary school mathematics classrooms in the North, West and the Northern Provinces of South Africa.

MOS-00b

2000b Mosimege, Mogege David & Lebeta, V.: An Ethnographic Study of Mathematical Activities at the Basotho Cultural Village, in: S. Mahlomaholo (Ed.) *Proceedings of the 8th Conference of the Southern Association for Research in Mathematics and Science Education*, University of Port Elizabeth, Port Elizabeth (South Africa).

MOS-02

2002 Mosimege, Mogege David: History and Cultural Specificity of Ethnomathematical Activities in the Mathematics Classrooms, in: C. Malcolm and C. Lubisi (Eds.), *Proceedings of 10th Annual Conference of the Southern African Association for*

Research in Mathematics, Science and Technology Education, University of Natal.

MOS-03
2003 Mosimege, Mogege David: Research Methods in Indigenous Mathematical Knowledge: An Example of a Research Model Based on Indigenous Games. *Indilinga: African Journal of Indigenous Knowledge Systems*, Pietermaritzburg (South Africa), Vol. 2, No. 1, 11-24.

MPE-99
1999 Mpey-Nka, Richard Ngub'usim: La symbolique et la mystique du nombre '9' chez le peuple Yansi traditionel [The symbolics and mystics of the number '9' among the traditional Yansi people], *Congo-Afrique*, Kinshasa (DR Congo), No. 337, 417-434.

The number '9' plays important ritual role in specific therapies and evokes fecundity. In order to analyze and understand the Yansi (DR Congo) symbolic system, the author starts with comparing it with numeric symbolisms from other cultures.

MPO-93
1993 Mpofana, Wilberforce Siyabonga: *Aspects of pre-service and in-service training of mathematics teachers in KwaZulu: a didactical survey*, doctoral thesis, University of the Free State, Bloemfontein (South Africa).

MTE-91
1991 Mtetwa, David Kufakwami Jani: *An investigation of Zimbabwean secondary school students' mathematical beliefs and classroom contexts*, doctoral thesis, University of Virginia, Charlottesville (USA).

MTE-92a
1992 Mtetwa, David: Mathematics and ethnomathematics – Zimbabwean students' view, *International Study Group on Ethnomathematics Newsletter*, Albuquerque NM (USA), Vol. 7, No. 1, 1-3.

MTE-92b

1992 Mtetwa, David: Females can just as good in math: Zimbabwean school girls proclaim, *Women & Mathematics Education*, Vol. 14, No. 2, 2.

MTE-95

1995 Mtetwa, David K. J., & Jaji, G.: School mathematics, out-of-school mathematics, and Zimbabwean youngsters, *Journal of Qualitative Studies in Education*, London (UK), Vol. 8, No. 4, 387-391.

Findings from an "exploratory study investigating beliefs about mathematics held by Zimbabwean secondary school students indicate that the students believe 'traditional' ethnomathematics exists; is legitimate mathematics; is the foundation upon which school mathematics expanded; but is too elementary, basic, and routine to be regarded as serious mathematics. Such beliefs, of course, need to be interpreted within the context of the student's own epistemic worldview of mathematics as a form of knowledge."

MTE-99

1999 Mtetwa, David K. J.: *Interactive teaching and learning of mathematics*, Zimbabwe Open University, Harare (Zimbabwe), 62 p.

Text book designed and suitable for use as reference material for a teaching certification course for mathematics teachers.

MTE-00a

2000 Mtetwa, David K. J.: On the nature of mathematical knowledge: Teaching students both mathematics and about mathematics, *Southern Africa Journal of Mathematics and Science Education*, Gaborone (Botswana), Vol. 3, No. 1&2, 53-62.

"This paper draws from the views expressed by some Zimbabwean secondary school and in-service practicing teachers on a topic discussing some implications of students' beliefs for learning and instruction. An important caveat that emerges from the discussion is that teacher preparation for school mathematics teaching should begin with an extensive consideration of the socio-epistemological aspects of mathematics as a cultural activity, rather than dwell exclusively on "methods of teaching" particular topics (the techniques) as is often the

practice, - if an overall goal is to empower learners through mathematics education. Students need to learn both mathematics and about mathematics."

MTE-00b
2000 Mtetwa, David K. J. & Thompson, J. J.: The dilemma of mentoring in mathematics teaching: Implications for teacher preparation in Zimbabwe, *Journal of In-service Education*, Oxford (UK), Vol. 26, No. 1, 139-162.

MUB-88
1988 Mubumbila, Mfika: *Sur le sentier mystérieux des nombres noirs* [On the mysterious path of black numbers], L'Harmattan, Paris (France), 187 p. (in French).

The first part analyses oral and possible graphic numeration systems from Congo / Zaire. The second part deals with the symbolic expression of numbers in Luba cosmogeny (Congo / Zaire), e.g. the significance of even and odd, the use of 'numbers of peace': 4 and 12,24,48, 96... The author stresses that "the explanation of the origin of life by numbers [is] practically equal to that of Pythagoras" (p.153).

MUB-92a
1992a Mubumbila, Mfika: *Sciences et traditions africaines: les messages du Grand Zimbabwe* [Sciences and African traditions: the messages from Great Zimbabwe], L'Harmattan, Paris (France), 108 p. (in French).

The author intends to reveal "some scientific knowledge of the pre-colonial Bantu world" (p.91), in particular of numeration and geometric figures in the Great Zimbabwe civilization.

MUB-92b
1992b Mubumbila, Mfika & Bum, Silas: *De la pyramide à la case, le secret du bâtiment africain*, Association Culturelle et Philosophique Bantu, Strasbourg (France), 34 p. (in French).

Booklet on African architecture giving particular attention to shape and geometric form.

MUE-69
1969 Mueller, Ian: Euclid's 'Elements' and the axiomatic method, *British Journal for Philosophy of Science*, Oxford (UK), Vol. 20, 289-309.

MUE-81
1981 Mueller, Ian: *Philosophy of Mathematics and Deductive Structure in Euclid's Elements*, MIT Press, Cambridge (USA), 378 p.

MUE-91a
1991 Mueller, Ian: Sur les principes des mathématiques chez Aristote et Euclide [On the principles of mathematics in Aristotle and Euclid], in RAS-91a,101-113.

MUE-91b
1991 Mueller, Ian: On the notion of a mathematical starting point in Plato, Aristotle, and Euclid, in: Alan C. Bowen (Ed.), *Science and philosophy in classical Greece*, Garland, New York (USA), 59-97.

MUG-1991
1991 Mugambi, Paul: *On mathematics in Uganda, personal reminiscences* (paper presented at the 3rd Pan-African Congress of Mathematicians, Nairobi, Kenya, mimeo).

MUKA-71
1971 Mukarovsky, Hans G.: Die Zahlwörter "eins" bis "zehn" in den Mandesprachen [The numerals "one" to "ten" in the Mande languages], in: Six, V.; Cyffer, Norbert & Wolff, E. (Eds.), *Afrikanische Sprachen und Kulturen; ein Querschnitt*, Deutsches Institut für Afrika-Forschung, Hamburg (Germany), 142-153 (in German).

Compares the numeration in the Mande languages (West Africa) with that of the Cushitic languages.

MUK-02
2002 Mukono, Tendai: Review of Gerdes' *Geometry from Africa* (GER-99a), *Indigenous Knowledge World Wide Newsletter*, The Hague (Netherlands), March, 7 (available on the web: www.nuffic.nl/ik-pages/ikww/index.html).

MUL-53
1953 Müller, W.: Das isoperimetrische Problem im Altertum mit einer Übersetzung der Abhandlung des Zenodoros nach Theon von Alexandrien [The isoperimetric problem in Antiquity with a translation of the treatise of Zenodoros according to Theon of Alexandria], *Sudhoffs Arch*iv, *Zeitschrift für Wissenschafts-geschichte*, Leipzig (Germany), Vol. 37, 39-71 (in German).

MURA-89
1989 Murata, T.: A tentative reconstruction of the formation process of Book XIII of Euclid's 'Elements', *Commentarii Mathematici Universitatis Sancti Pauli*, Tokyo (Japan), Vol. 38, No. 1, 101-127.

MURA-92
1992 Murata, T.: Quelques remarques sur le Livre X des 'Eléments' d'Euclide [Some remarks on book X of Euclid's Elements], *Historia Scientiarum*, Tokyo (Japan), Series 2, Vol. 2, No. 1, 51-60 (in French).

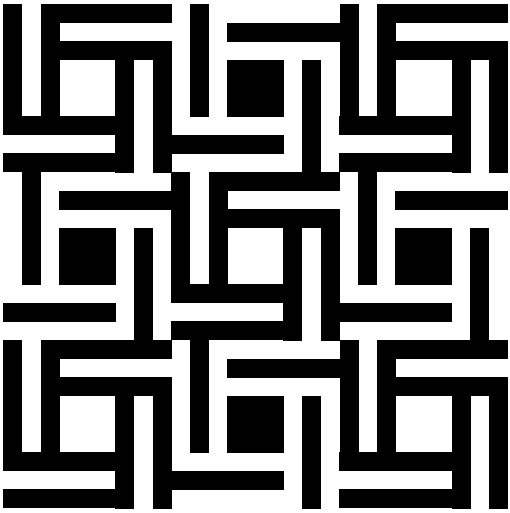
MURR-84
1984 Al-Murrâkushi, al-Hasan: *Comprehensive Collection of Principles and Objectives in the Science of Timekeeping*, Reprint F. Sezgin (Ed.), Institut für Geschichte der Arabisch-Islamischen Wissenschaften, Frankfurt (Germany), Vol. I-II, 755 p. (in Arabic).

MUS-87
1987 Musa, Mamman: *The mathematical heritage of the Hausa people: a resource guide for mathematics teaching*, masters thesis, Ahmadu Bello University, Zaria (Nigeria).

Summarizes mathematics in daily life, measures, art, religion, etc. for the Hausa culture of northern Nigeria.

MWA-00
2000 Mwakapenda, Willy: *On using everyday experiences in teaching secondary mathematics in Malawi: Possibilities and constraints for change*, doctoral thesis, Deakin University, Melbourne (Australia).

MWI-85
1985 Mwika, Kayembe: *Comportements d'élèves zaïrois face à des expressions fractionnaires (dans le 1er cycle de l'enseignement secondaire)*, doctoral thesis, Université Paris 7 (France) (in French).



Example of a Kuba two-colour design (Congo)
(cf. GER-99a, p. 14)

N

NDI-95

1995 Ndigi, Oum: L'expression des cardinaux et des ordinaux en égyptien et en basaa [The expression of cardinal and ordinal numbers in Egyptian and in Basaa], *Discussions in Egyptology*, Oxford (UK), No. 33, 57-72 (in French).

Comparative study of numerals in Ancient Egypt and in the Basa language of Cameroon, also discussed in the author's doctoral thesis *Les Basa du Cameroun et l'antiquité pharaonique égypto-nubienne: Recherche historique et linguistique comparative sur leurs rapports culturels à la lumière de l'égyptologie* [The Basa of Cameroon and Egyptian-Nubian Pharaonic Antiquity: Comparative historical and linguistic research on their cultural links in the light of Egyptology], Lyon, 1997.

NDI-03

2003 Ndigi, Oum: Notes sur la grammaticalisation du cardinal "un", w^c, en égyptien ancien [Notes on the grammaticalization of the cardinal number "one", w^c, in ancient Egyptian], *Cahiers Caribéens d'Egyptologie*, Martinique (France), No. 5, 179-185 (in French).

NEB-95

1995 Nebout Arkhurst, Patricia: *La signification contextuelle dans les processus de transposition didactique: l'exemple de l'enseignement de la géométrie au niveau du collège en Côte d'Ivoire* [The contextual signification in the process of didactic transposition: the example of the teaching of geometry at the high school level in Côte d'Ivoire], doctoral thesis, Université de Paris 5 (France) (in French).

NES-98

1998 Ness, Daniel: *Ethnomathematics and Asante kete drumming*, paper presented at the 76th Annual Meeting of the National Council of Teachers of Mathematics (2-4 April 1998, Washington DC, USA).

NEU-31
1931 Neugebauer, Otto: Die Geometrie der ägyptischen mathematischen Texte [The geometry of the Egyptian mathematical texts], *Quellen und Studien zur Geschichte der Mathematik, Astronomie und Physik*, Springer, Berlin (Germany), Vol. 1, 413-451 (in German).

NEU-34
1934 Neugebauer, Otto: Ägyptische Geometrie [Egyptian geometry], in: Neugebauer, Otto, *Vorlesungen über Geschichte der Antiken Mathematischen Wissenschaften*, Vol. 1: *Vorgriechische Mathematik* [Pre-Greek mathematics], Springer, Berlin (Germany), Vol. 1, 122-137 (in German).

NEU-57
1957 Neugebauer, Otto: *The exact sciences in Antiquity*, Munksgaard, Copenhagen (Denmark) (reprinted by: Brown University Press, Providence RI (USA)), 240 p.

NEV-72
1972 Neville, Mary C.: *Study of the major events that influenced the introduction of modern mathematics curricula in selected primary schools in Zambia*, doctoral thesis, The American University, Washington DC (USA).

NEWC-81
1981 Newcomb, V. N.: *Practical calculations for business studies: problems and applications for students in Africa*, Wiley, Chichester N.Y. (USA), 156 p.

NGC-91
1991 Ngcobo, Minenhle: *Historical perspectives in the teaching of maths teachers in Swaziland*, M.Ed. thesis, University of Leeds, Leeds (UK), 67 p.

NGUE-02
2002 N'Guessan-Depry, A.: Mathématiques et environnement socioculturel africain (MESCA): problématique de la vocation mathématique [Mathematics and the African socio-cultural environment], *Éthiopiques*, Dakar (Senegal), No. 68, 141-159 (in French).

Describes the rationale and experience of the 'Mathematics in the socio-cultural context' (MESCA) workshop of the Mathematical Research Institute of Abidjan (IRMA, Côte d'Ivoire) with stimulating interest in mathematics by using various African verbal games as a preparation for mathematics education.

NHL-93
1993 Nhlengetfwa (Lafakudze), Thuli: *The impact of elementary schools Maths / Science Inservice Teacher Education on the Manzini region (Swaziland) schools*, doctoral thesis, Ohio University (USA).

NIANE-03
2003 Niane, Mary Teuw: *La numération dans les langues nationales au Sénégal* [Numeration in the national languages of Senegal], Université Gaston Berger, Saint-Louis (Senegal), 17 p. (in French).

Proposes a representation compatible with the way to express numbers in four national languages of Senegal (Joola, Pulaar, Sereer, and Wolof), spoken by more than 90% of the population.

NIA-71
1971 Niang, Souleymane: Négritude et Mathématique [Negritude and mathematics], *Présence Africaine*, Paris (France), No. 78, 27-47 (in French).

NIAN-84
1984 Niangoran-Bouah, Georges: *L'univers Akan des poids a peser l'or* [The Akan universe of gold weights], Vol.1: *Les poids non figuratifs* / *The Akan world of gold weights*, Vol. 1: *Abstract design weights* (bilingual edition), Les Nouvelles Editions Africaines, Abidjan (Côte d'Ivoire), 311 p.

Two chapters of the first volume of this beautifully edited trilogy deal explicitly with mathematics of the Akan (Ghana, Côte d'Ivoire):

* Akan mathematical writing (250-269): The author explains how numbers, addition, multiplication and division are symbolically represented on the weights. He also analyses two series of monetary values. The first is decimal; the second has a binary structure (7 units, from 12 *ba*, 24 *ba*, .. to 384 *ba*).
* Weights and the practical applications of geometry (270-277): The Akans constructed certain figurines in such a way that they

represent signs, symbols and ideograms whether seen from in front or in profile.

NICH-77
1977 Nicholson, John & Seddon, G. M.: The understanding of pictorial spatial relationships by Nigerian secondary school students, *Journal of Cross-Cultural Psychology*, Beverly Hills CA (USA), Vol. 8, No. 4, 381-399.

NIC-68
1968 Nicolas, Guy: Un système numérique symbolique: le quatre, le trois et le sept dans la cosmologie d'une société hausa (vallée de Maradi) [A symbolic numerical system: four, three and seven in the cosmology of a Hausa society (Maradi valley)], *Cahiers d'études africaines*, Paris (France), Vol. VIII, No. 3, 566-616 (in French).

The numbers four (*hudu*), three (*uku*) and seven (*bakwai*) play an important role in ritual, economic and social life among the Hausa in the Maradi valley (Niger). This role is described, analyzed and discussed.

NJO-76
1976 Njock, Georges Edward: Document on co-operation presented at the First Pan-African Congress of Mathematicians held in Rabat, Morocco in 1976, *Association of African Universities Bulletin*, Accra (Ghana), Vol. 3, No. 1, 135-147.

NJO-79
1979 Njock, Georges Edward: *Langues africaines et non africaines dans l'enseignement des mathématiques en Afrique* [African and non-African languages in mathematics education in Africa] (paper presented at the Interafrican Seminar on the Teaching of mathematics in Primary and secondary Schools in Africa, April 1979, Accra, Ghana).

NJO-85
1985 Njock, Georges Edward: Mathématiques et environnement socio-culturel en Afrique Noire [Mathematics and the social-cultural environment in Black Africa], *Presence Africaine*, Paris (France), New Bilingual Series No. 135, 3rd Quarterly, 3-21 (in French).

Stresses that it is very urgent to study the history of mathematics in Africa, as colonialism and neo-colonialism neglected the existence of mathematics in Black Africa. "Pure mathematics is the art of creating and imagining. In this sense black art is mathematics." The author gives a summary of the development of numeration systems, arithmetic and mathematical games in Africa.

NKH-05

2005 Nkhwalume, Alakanani Alex: *A study of the motivational orientations of six girls towards mathematics as directed by their social context: a sociological and critical dimension of gender differentials in mathematics education in Botswana*, doctoral thesis, University of Nottingham (UK).

NTA-97

1997 Ntambue Tshimbulu, Raphael: *La logique formelle en Afrique noire: problématique, enseignement et essais* [Formal logic in black Africa], Academia Bruylant, Louvain-la-Neuve (Belgium) (in French).

NTE-04

2004 Ntenza, S. Philemon: Teachers' perceptions of the benefits of children writing in mathematics classrooms, *For the Learning of Mathematics*, Kingston (Canada), Vol. 24, No. 1, 13-19.

Article arises from a study that investigates the benefits and forms of mathematical writing and written text produced by pupils in junior high schools in KwaZulu-Natal (South Africa).

NUL-80

1980 NUL (Ed.): *Language in the mathematics and science lesson*, National University of Lesotho (NUL), Roma (Lesotho), 25 p.

Report of a workshop for science and mathematics teachers in Lesotho, convened to analyze problems faced in school by pupils who are learning in English, but whose mother tongue is Sesotho.

NYI-94

1994 Nyikahadzoyi, Maroni Runesu: *Comparison of attitudes of third year and first year student teachers toward mathematics and the teaching of mathematics [in Zimbabwe]*, masters thesis, University of Zimbabwe, Harare, 1994, 112 p.

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OAV-36

1936 O., A. V.: The Hima method of counting, *Uganda Journal*, Kampala (Uganda), Vol. 4, No. 1, 91.

OBE-73

1973 Obenga, Théophile: Système opératoire négro-africain, in: *T. Obenga, L'Afrique dans l'Antiquité. Egypte pharaonique - Afrique noire*, Présence Africaine, Paris (France), 333-353 (in French).

The author analyses the numeration system and arithmetics (including the use of fractions) and 'cosmical numbers' of the Mbosi (Congo) and makes a comparison with the mathematics of ancient Egypt.

OBE-74

1974 Obenga, Théophile: Science et langage en Afrique [Science and language in Africa], *Présence Africaine*, Paris (France), No. 92, 149-160.

OBE-90

1990 Obenga, Théophile: *La Philosophie africaine de la période pharaonique, 2780-330 avant notre ère* [The African philosophy of the Pharaonic period], L'Harmattan, Paris (France), 567 p.

Chapter 11 deals with mathematics (p. 355-427). It includes the following sections: Egyptian conception of mathematics; Knowledge of the technique of algebraic reckoning; The notion of Pharaonic mathematical logic; Metrology; Calculation of the area of a triangle; Calculation of the area of a circle; Surface of a semi-sphere; Calculation of the volume of the cylinder; Volume of a truncated pyramid; Calculation of the angle of inclination of a pyramid; Proof of the calculation of the angle of inclination of a pyramid.
Each section contains the reproduction of an Egyptian text, Obenga's translation and his commentaries. The section on metrology includes a comparison with the measures and numeration used by the Duala (Cameroon), Fang (Cameroon, Equitorial Guinee, Gabon), Yoruba (Nigeria), Ganda, BaNgongo (Congo) among others. Also of interest

to the history of Mathematics is the chapter on Astronomy (265-301), with the following sections: Astronomical and geometrical orientation of buildings; Egyptian calendars. The last section includes a comparative description of Ancient Egyptian, Fang, Mbochi (Congo), Borana (Ethiopia) and Dogon (Mali) Astronomy.

OBE-95

1995 Obenga, Théophile: *La Géométrie Égyptienne — Contribution de l'Afrique antique à la Mathématique mondiale* [Egyptian geometry — Contribution of ancient Africa to world mathematics], L'Harmattan, Paris & Khepera, Gif-sur-Yvette (France), 335 p.

This book by the Congolese linguist and Egyptologist Obenga, presents an overview of geometrical knowledge of ancient Egypt, stressing the relationship of this knowledge with know-how developed in other parts of Africa. He also underlines the influence of Egyptian geometry on the development of mathematics in ancient Greece, criticizing eurocentric views on the history of mathematics.

OCO-04

2004 O'Connor, John & Robertson, Edmund: *Ancient Egyptian mathematics* (online available at: www-history.mcs.st-andrews.ac.uk/Indexes/Egyptians.html).

An introduction and bibliography.

OFI-97

1997 Ofiaja, Nicholas: *The Existence of Mathematics and Science in Aspects of African History*, paper presented at the conference "The History of Mathematics and Science and Its Uses in Teaching: A Multicultural Approach", City University of New York (USA), March 14.

OHU-73

1973 Ohuche, R. Ogbonna: Geometry, estimation and measurement in traditional Sierra Leone, in: *Report of Research in Commonwealth Countries*, Education Division, The Commonwealth Secretariat, London (UK).

OHU-75
1975 Ohuche, R. Ogbonna: The uses of real numbers in traditional Sierra Leone, *West African Journal of Education*, Vol. 19, 329-338

Indicates the concepts of number with which children are expected to develop in the every-day activities of society, independently of formal schooling.

OHU-78
1978 Ohuche, R. Ogbonna: Change in Mathematics Education since the late 1950's - ideas and realisation: Nigeria, *Educational Studies in Mathematics*, Dordrecht (Netherlands), Vol. 9, 271-282.

OIS-91
1991 Oiso, B.: Étude comparée du système de numération de 1 à 10 dans quelques Langues non-Bantu du Haut-Zaïre [Comparative study of the system of numeration from 1 to 10 in several non-Bantu languages from Upper-Zaire], *Annales Aequatoria*, Mbandaka (DR Congo), Vol. 12, 475-479.

OJO-88
1988 Ojoade, J. Olowa: The number 3 in African Lore, *Abacus, the Journal of the Mathematical Association of Nigeria*, Ilorin (Nigeria), Vol. 18, No. 1, 21-43.

Describes the "frequent occurrence of the number 3 in African lore, making comparisons, where necessary with other world lore. Additionally the paper highlights the sacredness, mysticism, taboos, and superstitions attached to the number."

OKO-70
1970 Okonji, Michael O.: Culture and children's understanding of geometry, *Provisional council for the social sciences in East Africa*, Dar es Salaam (Tanzania), Vol. 4, 13-29.

OKO-71
1971 Okonji, Michael O.: Culture and children's understanding of geometry, *International Journal of Psychology*, Paris (France), Vol. 6, No. 2, 121-128.

"Attempt to replicate Piaget's investigation of the development of geometric concepts among 358 children in Ankole district of Uganda where there are no traditional precision measuring instruments either geometric or others with a view to throwing some light on the extent to which schooling experience affected this development. The children's understanding of three geometric concepts was investigated, the conservation of length, angular measurement and coordinate systems as indicated by the ability to locate a point in a rectangular sheet of paper." "The findings do suggest that certain concepts of geometry may depend almost entirely on skills acquired through formal education and not on biologically based maturing logical structures of the child."

OLA-77
1977 Oladimeji, F.: *A brief study of Yoruba traditional mathematics*, B.Ed. project, Ahmadu Bello University, Zaria (Nigeria).

OLIV-03
2003 Oliver, Jack: Fractions in Ancient Egyptian Times, *Mathematics in School*, Leicester (UK), 32(1), 14-16

Introduction to fractions in Ancient Egypt in a 'History Special' of *Mathematics in School*, edited by John Earle of the British Society for the History of Mathematics.

OLI-98
1998 Olivier, Alwyn & Newstead, Karen (Eds.): *Psychology of Mathematics Education: Proceedings of the 22nd Conference of the International Group for the Psychology of Mathematics Education*, University of Stellenbosch (South Africa), 4 volumes.

The proceedings contain the following contributions and abstracts, related mathematics and culture in Africa:

* Draisma, Jan: On verbal addition and subtraction in Mozambican Bantu languages, Vol. 2, 272-279;
* Mosimege, Mogege David: Culture, games and mathematics education: An exploration based on string figures (South Africa), Vol. 3, 279-286;

* Mogari, David: Some geometrical constructs and pupil's construction of miniature wire toy cars (South Africa), Vol. 4, 284;
* Soares, Daniel: On the geometry involved in the building of traditional houses with rectangular base in Mozambique, Vol. 4, 307;
* Mucavele, João: The *mathakuzana* game as a didactical resource for the development of number sense and oral arithmetic (Mozambique), Vol. 4, 345.

OMO-00
2000 Omotunde, Jean-Philippe: *L'Origine négro-africaine du savoir grec* [The Black-African origin of Greek knowledge], Editions Menaibuc (France) (in French).

OMO-03a
2003 Omotunde, Jean-Philippe: *Les nègres : inventeurs du zéro en mathématique* [The Blacks: inventors of zero in mathematics] (online available at: www.africamaat.com/article.php3?id_article=105)

OMO-03b
2003 Omotunde, Jean-Philippe: *L'Afrique reste le berceau des sciences mathématiques* [Africa remains the cradle of the mathematical sciences] (online available at: www.africamaat.com/article.php3?id_article=117).

Discusses the Blombos stone (80 000 BC, South Africa), and the Lebombo (35 000 BC, Swaziland) and Ishango (20 000 BC, Congo) bones, and the ancient Nile civilizations Nubia and Egypt.

ONYU-96
1996 Onyumbe, Tshonga & Kabasele, Malumba: Mesures et poids aux marchés de Mbandaka [Measures and weights at the markets of Mbandaka], *Annales aequatoria*, Mbandaka (DR Congo), Vol. 17, 417-422.

Analyses the emergence of new measurement unities, smaller than the traditional ones, as a consequence of diminishing buying-power.

OPO-04
2004 Opolot-Okurut, Charles: *Attitudes towards mathematics, achievement in mathematics aptitude problems and concomitant teacher practices in Ugandan secondary schools*, doctoral thesis, University of the Western Cape, Bellville (South Africa).

OSH-95
1995 Oshin, Babatunde Adetokunbo: *Brief History of Mathematics*, TWD Publications, Ijebu-Igbo (Nigeria), 88 p.

Short history of mathematics for teachers. Its references to Africa are to Ancient Egypt.

OTA-71
1971 Otaala, Barnabas: *The development of operational thinking in primary school children: an examination of some aspects of Piaget's theory among the Iteso children in Uganda*, doctoral thesis, Columbia University, New York (USA).

OYED-96
1996 Oyedeji, O. A.: Assessing gender factor in some secondary mathematics textbooks in Nigeria, *Zimbabwe Journal of Educational Research*, Harare (Zimbabwe), Vol. 8, No. 1, 45-54

Examines the gender factor in seven common mathematics textbooks used in Nigerian secondary schools. Significant differences were found on the number of items and illustrations that were male or female related, to the detriment of females.

OYE-99
1999 Oyeneyin, A. M.; Salau, M. O. & Ayodele, E. A.: *Accessibility of women to science, technology, and mathematics (STM) education in Nigeria*, Development Policy Centre, Ibadan (Nigeria), 109 p.

P

PAG-87
1987 Page, Donna: Two, three, four: multiples in African art, Kahan Gallery, New York (USA), 36 p.

"Forty objects of African art, mostly from the Yoruba (Nigeria) are analyzed in function of the involved repetitions. The twofold objects evoke the most usual dichotomies: good/bad, life/death; the threefold objects evoke sometimes a hierarchy; the fourfold objects may be associated with the directions in space" [summary reproduced from: *Afrique Contemporaine*, Paris (France), 1989, No. 149, p. 94].

PAL-90
1990 Palmquist, S. R.: Kant on Euclid: geometry in perspective, *Philosophia Mathematica*, Toronto (Canada), Series 2, Vol. 5, Nos. 1-2, 88-113.

PAN-69
1969 Pankhurst, Richard: A preliminary history of Ethiopian measures, weights and values, *Journal of Ethiopian Studies*, Institute of Ethiopian Studies, Addis Ababa (Ethiopia), Vol. 7, No. 1 and 2.

PAP-83
1983 Pappademos, John: An outline of Africa's role in the history of physics, in: SER-83, 177-196.

Revises conventional assumptions about the role of Africans in the history of physics by outlining some of their contributions to measurement, mechanics, optics, astronomy, and metallurgy.

PAPP-82
1982 Pappus of Alexandria: La collection mathématique (traduite par Paul Ver Eecke) [The mathematical collection (translated by Paul Ver Eecke)], Blanchard, Paris (France), 2 Vol., 883 p. (in French).

PAPP-86
1986 Pappus of Alexandria: *Book 7 of the Collection* (Translation and Commentary by Alexander Jones), Springer, New York (USA), 748 p.

PAR-72
1972 Parker, Richard A.: *Demotic Mathematical Papyri*, Brown University Press, Providence RI (USA), 86 p.

Publication and analysis of five mathematical papyri from Hellenistic Egypt.

Review: WAE-74.

PAS-94
1994 Passalacqua, L.: The Collections of Pappus: editorial polemics and circulation of manuscripts in the correspondence of Francesco Barozzi with the Duke of Urbino, *Bollettino di Storia delle Scienze Matematiche*, Bologna (Italy), Vol. 14, No. 1, 91-156 (in Italian).

PAT-90
1990 Pater, C. de: Was Augustine Mathematics-Hostile?, *Nieuw Archief voor Wiskunde*, Amsterdam (Netherlands), Vol. 8, No. 1, 43-45.

Criticizes an article by H. Beckers [1988] in the same journal, in which it is asserted that Augustine (354-430), bishop of Hippo (North Africa), warned that "good Christians should beware of mathematicians, because the danger exists that they have made a pact with the devil." On the contrary says the author Augustine warned against astrologers: the Latin 'mathematicus' also means astrologer. Augustine considered geometry and arithmetic as useful disciplines.

PATE-03
2003 Patel, Ramila: Symmetry in Patterns on Swazi grass mats, *Symmetry: Culture and Science*, Vol. 12, No. 1-2, Budapest (Hungary), 127-157.

Explores "the presence of symmetry in patterns on Swazi grass mats made by women in Swaziland. The fundamental aim is to elucidate and present basic explanations of the presence of symmetry in the patterns on the Swazi grass mats. … Symmetry in patterns on other

parallel Swazi material culture that admits patterning in clay beer pots, beaded necklaces, grinding mats and more recently baskets…" are illustrated.

PAU-71
1971 Paul, Sigrid: Afrikanische Konzentrationsspiele [African games of concentration], in: *Afrikanische Sprachen und Kulturen*, Deutsches Institut für Afrika-Forschung, Hamburg (Germany), 358-367 (in German).

Presents a classification of games that presuppose a high level of mental concentration.

PEE-23
1923 Peet, Thomas Eric: *The Rhind Mathematical Papyrus: British Museum 10057 and 10058*, The University Press of Liverpool, Liverpool / Hodder & Stoughton, London (UK), 135 p. [reprinted: Kraus reprint, Nendeln (Liechenstein)]

PEE-31
1931 Peet, Thomas Eric: A problem in Egyptian geometry, *Journal of Egyptian Archaeology*, London (UK), Vol. 17, 100-106.

PER-84
1984 Pereira da Silva, C.: Diophantus de Alexandria, *Boletim da Sociedade Paranaense de Matemática*, Curitiba (Brazil), Vol. 5, No. 1, 1-10 (in Portuguese).

PETER-99
1999 Peterson, Ivars: Review of P. Gerdes' *Geometry from Africa* (GER-99a) and R. Eglash's *African Fractals* (EGL-99), *Science News* Washington DC (USA) (available online at: www.sciencenews.org/sn_arc99/11_27_99/mathland.htm)

PETE-84
1984 Peterson, Wayne (Ed.): *Multicultural Mathematics Posters and Activities*, National Council of Teachers of Mathematics, Reston VA (USA), 52 p.

The following topics deal with Africa: Egyptian numerals, Oware game, Egyptian multiplication, “Nine Men’s Morris”, African string puzzle, Egyptian rope stretchers, Shongo networks.

PET-78
1978 Petitto, Andrea: *Knowledge of arithmetic among schooled and unschooled tailors and cloth merchants in Ivory Coast*, doctoral thesis, Cornell University, Ithaca NY (USA).

PET-82a
1982a Petitto, Andrea L.: Practical arithmetic and transfer, a study among West African tribesmen, *Journal of Cross-Cultural Psychology*, Beverly Hills CA (USA), Vol. 13, No. 1, 15-28.

Study of the transfer of mathematical problem-solving ability among adult unschooled Dioula tailors and cloth merchants in Ivory Coast.

PET-82b
1982 Petitto, Andrea & Ginsburg, Herbert: Mental arithmetic in Africa and America: strategies, principles, and explanations, *International Journal of Psychology*, Paris (France), Vol. 17, 81-102.

Comparative study of the mental arithmetic abilities among unschooled Dioula adults (Ivory Coast) and USA college students. “Both groups showed accurate mental arithmetic strategies related to the base ten structure of their native counting systems.”

PETR-71
1971 Petracek, Karel: Die Zahlwörtersysteme der zentral-saharanischen Sprachen [The numeral systems in the central-Saharan languages], in: Six, V.; Cyffer, Norbert & Wolff, E. (Eds.), *Afrikanische Sprachen und Kulturen; ein Querschnitt*, Deutsches Institut für Afrika-Forschung, Hamburg (Germany), 246-252 (in German).

PHY-71
1971 Phythian, J. E.: Mathematical *kujitegemea* in Tanzania, *Educational Studies in Mathematics*, Dordrecht (Netherlands), Vol. 4, No. 2, 187-200.

Examines the way in which the principle of *kujitegemea* (Swahili expression for 'self-reliance') is being applied in the development of school mathematics.

PIE-79
1879 Pietschmann, R.: Über die Kanarischen Zahlworte [On the Canarian number words], *Zeitschrift für Ethnologie*, Berlin (Germany), Vol. XI, 377-391 (in German).

PLE-99
1999 Pletser, Vladimir & Dirk Huylebrouck: *Does the Ishango bone indicate knowledge of the base 12? An interpretation of a prehistoric discovery, the first mathematical tool of humankind* (online available at: http://etopia.sintlucas.be/3.14/).

PLO-50
1950 Plooij, Edward Bernard: *Euclid's conception of ratio and his definition of proportional magnitudes as criticized by Arabian commentators*, doctoral thesis, Rijksuniversiteit Leiden (Netherlands).

POS-79
1979 Posner, Jill & Baroody, A.: Number conservation in two West African societies, *Journal of Cross-Cultural Psychology*, Beverly Hills CA (USA), Vol. 10, No. 4, 479-496.

POS-78
1978 Posner, Jill K.: *The development of mathematical knowledge among Baoulé and Dioula children in Ivory Coast*, doctoral thesis, Cornell University, Ithaca NY (USA).

POS-82
1982 Posner, Jill: The development of mathematical knowledge in two West African societies, *Child Development*, Chicago (USA), Vol. 53, 200-208.

Investigates the development of mathematical concepts among children from two groups in central Ivory Coast, an agricultural population (Baoule) and a merchant society (Dioula). "The advancement of quantitative understanding appears to be dependent on

certain kinds of experiences which both schooling and a merchant culture afford."

POW-97a

1997a Powell, Arthur B. & Frankenstein, Marilyn (Eds.): *Ethnomathematics: Challenging Eurocentrism in Mathematics Education*, State University of New York Press, Albany (USA), 440 p.

The following chapters or parts of them relate to mathematics in Africa:

* Martin Bernal: Animadversions on the origins of western science (83-99) [originally 1992]:
  Presents "arguments for the existence of rich mathematical – particularly geometrical – and astronomical traditions in Egypt by the time Greek scholars came in contact with Egyptian learned priests" (p. 95);
* Reproduction of LUM-83 with postscript (101-117);
* Herbert Ginsburg: The myth of the deprived child (129-154) [originally 1986 with postscript]:
  Includes references to the author's research on the development of mathematical thinking among the Dioula and Baoulé (Côte d'Ivoire);
* Reproduction of GER-88c (223-247);
* Claudia Zaslavsky: World cultures in the mathematics class (307-320) [originally 1991];
* Paulus Gerdes: Survey of current work on ethnomathematics (331-371) [originally 1993].

POW-97b

1997b Powell, Arthur B.: *A biographical sketch of Caleb Gattegno, an African Mathematician and Educator*, Rutgers University, Newark NJ (USA), preprint, 8 p.

Presents a biographical sketch of the Egyptian born Caleb Gattegno (1911-1988), who moved in 1945 to Europe and later to the USA.

POW-07

2007 Powell, Arthur B.: Caleb Gattegno: A famous mathematics educator from Africa, *Revista Brasileira de História da Matemática*, Rio Claro (Brazil) (in press).

PRE-93

1993 Presmeg, Norma C.: *Mathematics in Multicultural Classrooms: Uthongathi Students' Voices* (on-line available at: www.coe.uga.edu/quig/proceedings/Quig93_Proceedings/presmeg.93.html)

Describes "the attitudes of students in a multiracial school in South Africa about the interplay between their different cultures and mathematics. The school is deliberately multicultural. Students still experience language problems, but seem to enjoy the mathematics more when it is related to their everyday experiences."

PRU-86

1986 Prussin, Labelle: *Hatumere, Islamic design in West Africa*, University of California Press, Berkeley CA (USA), 306 p.

Includes examples of the use of magic squares.

PTO-88

1988 Ptolemy of Alexandria: *Composition mathématique* [Mathematical composition], Blanchard, Paris (France), 2 volumes, 1090 p. (in French).

Reprint of the translation of Ptolemy of Alexandria's *Mathematical Composition* by Abby Halma published in 1813 and 1816 with notes by Delambre (Ptolemy = Claudius Ptolemeus, c. 85 – c. 165).

PYE-93

1993 Pyenson, Lewis: *Civilizing Mission: Exact Sciences and French Overseas Expansion, 1830-1940*, The John Hopkins University Press, Baltimore (USA), 377 p.

Section 2.4 "Algeria: The Overseas projection of Metropolain Terrain" (87-127), section 2.5 "Tunisia and Morocco: The Antebellum Satrapies (128-154), and section 3.7 "Lebanon and Madagascar: Peripheral Territories" (207-240) deal with French colonial policy towards mathematics and the natural sciences in Africa.

R

RAM-89

1989 Rambaran, Anirud: *The relationship between environmental factors and performance in mathematics of Indian pupils in the junior secondary phase*, doctoral thesis, University of South Africa, Pretoria (South Africa).

RAS-68

1968 Rashed, Roshdi: Le "Discours de la lumière" d'Ibn al-Haytham (Alhazen) : Traduction francaise critique [The 'Discourse on light' of Ibn al-Haytham (Alhazen): Critical French translation], *Revue d'Histoire des Sciences Appliquées*, Evry (France), Vol. 21, 197-224.

RAS-74

1974 Rashed, Roshdi: Les travaux perdus de Diophante I [The lost works of Diophantus I], *Revue d'Histoire des Sciences*, Evry (France), Vol. 27, 97-122.

RAS-75

1975 Rashed, Roshdi: Les travaux perdus de Diophante. II [The lost works of Diophantus II], *Revue d'Histoire des Sciences*, Evry (France), Vol. 28, 3-30.

RAS-78

1978 Rashed, Roshdi: L'extraction de la racine n^{ième} et l'invention des fractions décimales (XIe-XIIe siècles) [The extraction of the nth root and the invention of decimal fractions], *Archive for History of Exact Sciences*, Berlin (Germany), Vol. 18, No. 3, 191-243 (in French).

RAS-79

1979 Rashed, Roshdi: Ibn al-Haytham's construction of the regular heptagon, *Journal for the History of Arabic Science*, Aleppo (Syria), Vol. 3, No. 2, 387-309 (in Arabic).

RAS-80
1980 Rashed, Roshdi: Ibn al-Haytham et le Théorème de Wilson [Ibn al-Haytham and the Theorem of Wilson], *Archive for History of Exact Science*, Berlin (Germany), Vol. 22, 305 (in French).

RAS-81
1981 Rashed, Roshdi: Ibn al-Haytham and the measurement of the paraboloid, *Journal for the History of Arabic Science*, Aleppo (Syria), Vol. 5, Nos. 1-2, 262-191 (in Arabic).

RAS-84
1984 Rashed, Roshdi: *Entre Arithmétique et Algèbre. Recherches sur l'Histoire des Mathématiques Arabes* [Between Arithmetic and Algebra. Studies on the history of Arabic mathematics], Les Belles Lettres, Paris (France), 321 p. (in French).

Collection of papers published between 1973 and 1980. They deal with certain aspects of Algebra, of Numerical Analysis, of Combinatorics, and of Number Theory in the medieval Arabic mathematical tradition. On pp. 259-299 the author discusses the contribution of the Maghrebian mathematician Ibn al-Banna (1256-1321) to Combinatorics and Number Theory.

Translation: RAS-94.

RAS-89
1989 Rashed, Roshdi: Ibn al-Haytham et les nombres parfaits [Ibn al-Haytham and perfect numbers], *Historia Mathematica*, New York (USA), Vol. 16, No. 4, 343-352 (in French).

RAS-91a
1991 Rashed, Roshdi (Ed.), *Mathématiques et Philosophie de l'Antiquité à l'Âge classique*, Éditions du CNRS, Paris (France), 315 p. (in French).

Contains the papers FED-91, MUE-91, and RAS-91b.

RAS-91b
1991 Rashed, Roshdi: L'analyse et la synthèse selon Ibn al-Haytham [Analysis and synthesis according to Ibn al-Haytham], in: RAS-91a, 131-162 (in French).

The author analyses certain aspects of the contents of two works of the 11th century mathematician Ibn al-Haytham, entitled *Maqâla fî t-tahlîl wa tarkîb* (Book on Analysis and Synthesis) and *Kitâb al-maᶜlûmât* (Book of the Known). The paper is concluded by an appendix that contains the French translation of the introduction by Ibn al-Haytham to *his Book on Analysis and Synthesis* (150-162).

RAS-92

1992 Rashed, Roshdi (Ed.): *Optique et mathématique: Recherches sur l'histoire de la pensée scientifique en arabe* [Optics and Mathematics: Research on the History of scientific thinking in Arabic], Variorum, Ashgate (UK), 340 p. (in French).

RAS-93

1993 Rashed, Roshdi: *Géométrie et dioptrique au Xe siècle: Ibn Sahl, Al- Qûhî et Ibn Al-Haytham* [Geometry and dioptric in the 10th century], Les Belles Lettres, Paris (France), 315 p. (in French).

RAS-94a

1994 Rashed, Roshdi: *The Development of Arabic Mathematics. Between Arithmetic and Algebra*, Kluwer, Dordrecht (Netherlands), 372 p.

Updated translation of the RAS-84.

RAS-94b

1994 Rashed, Roshdi: Notes sur la version arabe des trois premiers livres des 'Arithmétiques' de Diophante, et sur le problème 1.39 [Notes on the Arabic version of the first three book of the 'Arithmetica' of Diophantus and on problem 1.39], *Historia Scientiarum*, Tokyo (Japan), Series 2, Vol. 4, No. 1, 39-46.

RAS-96

1996 Rashed, Roshdi: *Encyclopedia of the History of Arab Science*, Routledge, London (UK), 3 volumes, 1128 p. [Vol.1: Astronomy; Vol.2: Mathematics and the Physical Sciences; Vol.3: Technology, Alchemy and Life Sciences].

Volume 2 includes the following chapters includes the following chapters: Numeration and Arithmetic (A. Saidan); Algebra (R.

Rashed); Combinatory analysis, Numerical analysis, Diophantine analysis and number theory (R. Rashed); Geometry (B. Rosenfeld, A. Youschkevitch); Trigonometry (M. Debarnot); The influence of Arab mathematics in the medieval West (A. Allard); Music science (J. Chabrier); Statics (M. Rozhanskaya); Geometrical optics (R. Rashed); The emergence of physiological optics (G. Russell); The Western reception of Arab optics (D. Lindberg).

RAT-91

1991 Ratteray, Joan Davis: African-based themes for mathematics classrooms, *Research Notes on Africa*, Institute for Independent Education, Washington DC (USA), Vol. 3, 16-27.

Presents suggestions for the use of African games and drawings and some ideas from Ancient Egypt in mathematics classrooms.

RAU-38

1938 Raum, Otto: *Arithmetic in Africa*, Evans Brothers, London (UK), 94 p.

The book is addressed to those responsible for teaching arithmetic to speakers of Bantu languages. The author suggests, "for teaching the African child to handle the system of numbers and to carry out operations in it, tribal activities, both adult and juvenile, with numerical bearing, are the most suitable media". Several examples of such activities, including games, are given. Furthermore, he suggests, "if generalizations and abstractions are to be acquired by the pupils as lasting instruments of thought, advanced arithmetical processes must be developed from the numerical problems of their own cultural background." Presents examples mostly from South Africa and Tanzania.

REB-88

1988 Rebstock, Ulrich; Rainer Osswald & Abdalqadir Wuld: *Katalog der arabischen Handschriften in Mauretanien* [Catalogue of Arabic manuscripts in Mauritania], F. Steiner Verlag, Wiesbaden (Germany), 160 p. (in German and Arabic).

Contains a catalogue of Arabic manuscripts in Mauritania, including some repertories on mathematics.

REB-89

1989 Rebstock, Ulrich: *Sammlung arabischer Handschriften aus Mauretanien: Kurzbeschreibungen von 2239 Handschriften-einheiten mit Indices* [Collection of Arabic manuscripts from Mauritania], O. Harrassowitz, Wiesbaden, 278 p. (in Arabic; prefatory material and notes in German).

The catalogue includes mathematical manuscripts.

REB-92

1992 Rebstock, Ulrich: *Rechnen im islamischen Orient* [Calculation in the Islamic East], Wissenschaftliche Buchgesellschaft, Darmstadt (Germany), 328 p. (in German).

Discusses the different methods of calculation used in the Arab mathematical tradition from the East (arithmetic, algebra, heritage, land measuring, metrology, etc.), with many references to the Arab mathematical tradition of the Maghreb.

REB-95

1995 Rebstock, Ulrich: Der Muamalat-Traktat des ibn al-Haitam [The Muamalat-tractate of Ibn al-Haytham], *Zeitschrift für Geschichte der Arabisch-Islamischen Wissenschaften*, Frankfurt (Germany), Vol. 10 (1995/96), 61-121 (in German).

RED-06

2006 Reddy, Vijay (Ed.): *Mathematics and science achievement at South African schools in TIMSS 2003*, HSRC Press, Cape Town (South Africa), 129 p.

REDJ-77

1977 Redjeb, Souad: *Le developpement des structures logico-mathématiques élémentaires chez des enfants tunisiens de milieux sociaux différents* [The development of logical-mathematical structures among Tunisian children from different social background], doctoral thesis, Université de Bordeaux 2 (France).

REH-82
1982 Rehder, W.: Die Analysis und Synthesis bei Pappus [Analysis and synthesis by Pappus], *Philosophia Naturalis*, Frankfurt am Main (Germany), Vol. 19, Nos. 3-4, 350-370 (in German).

REI-82
1982 Reineke, Walter-Friedrich: Die mathematischen Kenntnisse der ägyptischen Verwaltungsbeamten [The mathematical knowledge of the Egyptian managers], in: *L'égyptologie en 1979. Axes prioritaires de recherches*, Éditions du Centre national de la recherche scientifique, Paris (France), Vol. 2, 159-165 (in German).

REI-87
1987 Reineke, Walter-Friedrich: *Gedanken und Materialien zur Frühgeschichte der Mathematik in Ägypten* [Thoughts and materials on the early history of mathematics in Egypt], doctoral dissertation, Humboldt University, Berlin (Germany) (in German).

REN-32
1932 Renaud, Henri-Paul-Joseph: Additions et Corrections à Suter "Die Mathematiker und Astronomen der Araber" [Additions and corrections to Suter's 'The mathematicians and astronomers of the Arabs'], *Isis*, Madison WI (USA), Vol. 18, 166-83 (in French).

REN-33
1933 Renaud, Henri-Paul-Joseph: L'enseignement des sciences exactes et l'édition d'ouvrages scientifiques au Maroc avant l'occupation européenne [The teaching of the exact sciences and the publication of scientif works in Morocco before the European occupation], *Hespéris*, Paris (France), Vol. XVI, 78-89 (in French).

REN-37
1937 Renaud, Henri-Paul-Joseph: Sur les dates de la vie du mathématicien arabe marocain Ibn al-Bannâ (XIIIe-XIVe s. J.C.) [On the life dates of the Moroccan Arab mathematician

Ibn al-Bannâ (13th – 14th century)], *Isis*, Vol. XXVII, No. 2, 216-218 (in French).

REN-38a
1938a Renaud, Henri-Paul-Joseph: Ibn al-Bannâ de Marrakech, sufi et mathématicien (XIIIe-XIVe s. J.C.) [Ibn al-Bannâ of Marrakech, sufi and mathematician (13th – 14th century)], *Hespéris*, Paris (France), Vol. XXV, 13-42 (in French).

REN-38b
1938b Renaud, Henri-Paul-Joseph & Colin, J. S.: Note sur le muwaqqit marocain Abû Muqric-ou mieux Abû Miqrac al-Bat;.t;.īwī, *Hespéris*, Paris (France), Vol. XXV, 94-96 (in French).

REN-41
1941 Renaud, Henri-Paul-Joseph: Déterminations marocaines de l'obliquité de l'écliptique [Moroccan determinations of the obliquity of the eclips], *Bulletin de l'enseignement public*, October-December, No. 170, 321-336 (in French).

REN-42
1942 Renaud, Henri-Paul-Joseph: Astronomie et Astrologie marocaine [Moroccan astronomy and astrology], *Hesperis*, Paris (France), Vol. XXIX, 41-63 (in French).

REN-44
1944 Renaud, Henri-Paul-Joseph: Sur un passage d'Ibn Khaldûn relatif à l'histoire des mathématiques [On a passage of Ibn Khaldûn concerning the history of mathematics], *Hespéris*, Paris (France), Vol. XXXI, 35-47 (in French).

REN-45
1945 Renaud, Henri-Paul-Joseph: Sur les lunes du Ramadan [On the moons of the Ramadan], *Hespéris*, Paris (France), Vol. XXXII, 51-68 (in French).

REN-48
1948 Renaud, Henri-Paul-Joseph: *Le calendrier d'Ibn al-Bannâ de Marrakech* [The calendar of Ibn al-Bannâ of Marrakech], Larose Paris (France), 94 p. (in Arabic and French). Reproduced in SEZ-98a, 207-300.

REY-98
1998 Reyes García, Ignacio: *Estudio Etnolingüístico de los antiguos numerales canarios* [Ethnolinguistic study of the ancient Canarian numerals], Baile del Sol, Tenerife (Canary Islands, Spain), 120 p. (in Spanish).

This study is a philological analysis of the transmitted names of some cardinal numbers of the old Canarian numeration system. It combines a linguistic, ethnological and historic focus.

RIN-03
2003 Rincon, Paul: Greeks borrowed Egyptian Numbers, *BBC Science*, London (UK), September 2003 (online available at: http://news.bbc.co.uk/1/hi/sci/tech/3109806.stm).

RIS-74
1974 Rising, G. R.: The Egyptian use of unit fractions for equitable distribution, *Historia Mathematica*, New York (USA), Vol. 1, No. 1, 93-94.

RIT-89
1989 Ritter, James: Chacun sa verité: les Mathématiques en Egypte et en Mésopotamie [To each his truth: mathematics in Egypt and Mesopotamia], in: Michel Serres (Ed.), *Eléments d'histoire des sciences*, Bordas, Paris (France), 39-62 (in French).

The paper contains a comparative study of the Babylonian and Egyptian computing techniques as they appear in the documents that survived.

Translation: RIT-95

RIT-93
1993 Ritter, James: *Pratique de la raison en Mésopotamie et en Egypte au IIIe et IIe millénaires* [Praxis of reasoning in Mesopotamia and Egypt during the 3rd and 2nd millennia],

doctoral thesis, Université de Paris Nord, Paris (France), 446 p. (in French).

The thesis contains five parts: 1. Introduction (5-42), 2. Rational practices (43-95), 3. The delimitation of a rational field: the case of medicine (96-111), 4. The evolution of a rational field: the case of mathematics (112-201). 5. References, tables, general bibliography, index (202-446).

RIT-95
1995 Ritter, James: Measure for measure: mathematics in Egypt and Mesopotamia, in: Michel Serres (Ed.), *A History of Scientific Thought*, Blackwell, Oxford (UK), 44-72.

Translation of RIT-89.

RIT-00
2000 Ritter, James: Egyptian Mathematics, in SEL-00, 115-136.

The paper is structured in the following sections: Sources, Writing and Metrology, the Mathematical Texts, Fractions and Tables, Notes, Bibliography.

RIT-03
2003 Ritter, James: Closing the Eye of Horus: The Rise and Fall of Horus-Eye Fractions, in: Steele, John & Imhausen, Annette (Eds.), *Under One Sky*, Ugarit-Verlag, Münster, 298-323.

ROB-85
1985 Robins, Gay & Charles C. Shute: Mathematical bases of ancient Egyptian architecture and graphic art, *Historia Mathematica*, New York (USA), Vol. 12, 107-122.

"Deals with the trigonometric basis of pyramid architecture and disposes of the erroneous notion that pyramidal dimensions intentionally incorporate irrational numbers."

ROB-87
1987 Robins, Gay & Charles Shute, *The Rhind Mathematical Papyrus: An Ancient Egyptian text*, British Museum Publications, London (UK), 60 p., 23 drawings, 24 pl. [Reprint: Dover, New York (USA), 1990].

Review: BRU-90b

ROB-94
1994 Robins, Gay: *Proposition and Style in Ancient Egyptian Art*, University of Texas Press, Austin (USA), 279 p.

"It has long been known that much Egyptian art executed in two dimensions as painting or relief was conceived and carried out on a squared grid, which helped to determine the proportions of the human figure. Although there have been several previous studies of the Egyptian grid, these have been almost entirely limited to single standing or seated male figures... In this book I have attempted to base my own ideas ... primarily on observations carried out on the actual monuments. I have considered female figures as well as male, other postures besides standing and sitting... I show that the squared grid had an important influence on the composition of scenes as a whole and in helping to determine the characteristic style of a particular period. I consider the effects of the major change in the grid that occurred in the twenty-fifth dynasty and persisted thereafter, and elaborate my discovery of the grid system adopted during the Amarna period" (Preface, p. vii).

ROE-94
1994 Roero, C. Silvia: Egyptian Mathematics, in: GRA-94, 30-45.

ROI-93
1993 Roik, Elke: *Das Längenmaßsystem im Alten Ägypten* [The system of length measurement in ancient Egypt], Christian-Rosenkreutz-Verlag, Hamburg (Germany), 404 p. (in German).

Review: LEG-94

ROSE-76
1976 Rosenfeld, B. A.: The list of physico-mathematical works of Ibn al-Haytham written by himself, *Historia Mathematica*, New York (USA), Vol. 3, 75-76.

ROS-01
2001 Rossi, Corinna: Dimensions and Slope in the Nineteenth and Twentieth Dynasty Royal Tombs, *Journal of Egyptian Archaeology*, London (UK), Vol. 87, 73-80.

ROS-02
2002 Rossi, Corinna & Tout, Christopher A.: Were the Fibonacci Series and the Golden Section Known in Ancient Egypt, *Historia Mathematica*, New York (USA), Vol. 29, 101-113.

"The Fibonacci series and the Golden Section have often been used to explain the proportions of ancient Egyptian art and architecture. All such theories, however, are based on our modern mathematical system. They have never been examined in the realm of ancient Egyptian mathematics, as we understand it from studying the surviving mathematical sources. This article analyses the compatibility of the Fibonacci series with ancient Egyptian mathematics and suggests how an ancient scribe could have handled it. The conclusion is that concepts such as φ and the convergence to φ have little in common with the surviving ancient Egyptian mathematical documents and that they are quite far from the ancient Egyptian mentality" (p. 101).

ROS-04
2004 Rossi, Corinna: *Architecture and mathematics in ancient Egypt*, Cambridge University Press, Cambridge (UK), 280 p.

ROU-97
1997 Rouxel, Bernard & Aïssani, Djamil: Le géomètre Albert Ribaucour (1845-1896) à Bougie [The Geometer Albert Ribaucour (1845-1896) in Béjaïa], in: *Proceedings of the International Conference "Béjaïa et sa région à travers les âges :Histoire, Société, Sciences, Culture"* [Béjaïa and its region through the ages: history, society, sciences, culture], Gehimab, Béjaïa (Algeria), 261-268 (in French).

The object of this paper is to determine the mathematical contribution of Ribaucour during his Algerian stay (in particular, his conflict with Gaston Darboux).

RYA-78
1978 Ryan, W. J.: Teaching measurement in an African village, *The Arithmetic Teacher*, Reston VA (USA), Vol. 26, No. 2, 18-19.

Describes how a teacher made measurement more meaningful to children in an African village school by using local examples.

S

SAB-97
1997 Sabra, Abdelhamid I.: One Ibn al-Haytham or two? An Exercise in Reading the Bio-Bibliographical Sources, *Zeitschrift für Geschichte der Arabisch-Islamischen Wissenschaften*, Frankfurt (Germany), Vol. 11, 1-50.

Presents a criticism of R. Rashed's hypothesis that works attributed to Ibn al-Haytham are actually the result of the confusion of two different historical characters, one a mathematician and the other a physician.

SAI-84
1984 Saidan, Ahmad S.: *History of Arithmetic among the Arabs. Part 3. Arithmetic in Andalusia and the Magheb,* Dâr al-Furqân, Amman (Jordan) (in Arabic, with English summary).

Contains the edition of the arithmetical book, entitled "The four epistles" of the Maghrebian mathematician Ibn al-Bannâ (d. 1321).

SAI-86
1986 Saidan, Ahmad S.: The science of Algebra in the Arab Maghreb, in: *History of the Science of Algebra in the Arab World*, National Council of Culture, Koweit, Vol. 2, 398-613 (in Arabic).

SAID-98
1998 Saide, Salimo: On the geometry of pottery decoration by Yao women (Nyassa Province), in: GER-98d, 203-230.

SAIT-85
1985 Saito, Ken: Book II of Euclid's 'Elements' in the light of the theory of conic sections, *Historia Scientiarum*, Tokyo (Japan), Vol. 28, 31-60.

SAIT-86
1986 Saito, Ken: Compounded Ratio in Euclid and Apollonius, *Historia Scientiarum*, Tokyo (Japan), No. 31,25-29.

SAIT-93
1993 Saito, Ken: Duplicate ratio in Book VI of Euclid's 'Elements', *Historia Scientiarum*, Tokyo (Japan), Series 2, Vol. 3, No. 2, 115-135.

SAIT-94
1994 Saito, Ken: Debate: Proposition 14 of Book V of the 'Elements' – a proposition that remained a local lemma. Comment on: "Proposition 14 of Book V in the organization of Euclid's 'Elements'", *Revue d'Histoire des Sciences*, Evry (France), Vol. 47, No. 2, 273-284.

SANC-43
1943 Sanchez Pérez, José Augusto: *La aritmética en Babilonia y Egipto* [Arithmetic in Babylonia and Egypt], Consejo Superior de Investigaciones Científicas, Madrid (Spain), 72 p. (in Spanish).

SAN-60
1960 Santos, Eduardo dos: Sobre a matemática dos Ouiocos de Angola, *Garcia da Orta*, Lisbon (Portugal), Vol. 3, No. 2, 257-271 (in Portuguese).

Paper on the numerals, arithmetical operations, measures, coins, time reckoning, and geometrical vocabulary of the Cokwe of North-East Angola.

SANZ-98
1998 Sanz, Nelson: *Problem solving and 'aha' calculation experiences with the Rhind mathematical papyrus*, paper presented at 76th Annual Meeting of the National Council of Teachers of Mathematics (2-4 April 1998, Washington DC, USA).

SAWY-70
1970 Sawyer, Harry & Todd, S. K.: The significance of the numbers 3 and 4 among the Mende of Sierra Leone, *Sierra Leone Studies: A Journal of the Arts and Sciences*, Freetown (Sierra Leone), Vol. 26, 29-36.

Discusses "the significance and incidence of the use of the figure three to symbolize female activity, and of the figure four to symbolize male participation among the Mende" (p. 30).

SCHE-98
1998 Scheerder, Jeroen & Renson, Roland: *Annotated Bibliography of Traditional Play and Games in Africa*, International Council of Sport Science and Physical Education (ICSSPE), Berlin (Germany).

SCHI-96
1996 Schillinger, Jolene Urquhart: *The ethnomathematics of the Senoufo women of Mali, West Africa*, doctoral thesis, The Union Institute (USA).

SCHM-98
1998 Schmeikal, Bernd: Review of P. Gerdes' *Ethnomathematik dargestellt am Beispiel der Sona Geometrie* (GER-97a), *Mathematical Reviews* Lancaster PA (USA), 6078-6080 [98j:01003].

SCH-15
1915 Schmidl, Marianne: Zahl und Zählen in Afrika [Number and numerals in Africa], *Mitteilungen der Anthropologischen Gesellschaft in Wien*, Vienna (Austria), Vol. 45, 165-209 (in German).

In the first part an overview and comparative analysis of counting systems in Sub-Saharan Africa is given. The second part deals with psychological and historical factors that influence the development of counting (systems).

Review: SET-15.

SCHW-79
1979 Schweigman, Caspar: *Doing Mathematics in a developing country: Linear programming with applications in Tanzania*, Tanzania Publications House, Dar es Salaam (Tanzania).

SCHW-85
1985 Schweigman, Caspar: *Operations research problems in agriculture in developing countries*, Tanzania Publications

House, Dar es Salaam (Tanzania) & Khartoum University Press, Khartoum (Sudan), 361 p.

The books SCHW-79 and SCHW-85 present applications of linear programming in developing countries, giving, in particular, examples from Tanzania and the Sahel.

SED-34

1834 Sédillot, Jean Jacques & Sédillot, Louis Amélie: *Traité des instruments astronomiques des arabes composé au treizième siècle par Aboul l-Hasan'Ali, de Maroc*, Imprimerie Royale, Paris (France), 630 p., New edition in facsimile by F. Sezgin, Institut für Geschichte der Arabisch-Islamischen Wissenschaften., Frankfurt (Germany), Series B 2, 1984 (in French).

This is the partial translation into French of the important astronomical treatise of the Maghrebian mathematician from the 13th century Abû l-Hasan al-Marrâkushî. The translation has been realized by Jean Jacques Sédillot and published by his son Louis Amélie. This treatise includes the description and the utilization of a whole series of astronomical instruments used in the countries of Islam between the 9th and the 13th century. The Arabic manuscript of al-Marrâkushî was published in facsimile [ed. F. Sezgin, I.G.A.I.W., Frankfurt (Germany), Series C 1, 1984, Vol. I, p., Vol. II, 376 p.].

SEG-01

2001 Segla, Dafon Aimé: *Appropriation des mathématiques dans une langue africaine: le yoruba* [The learning of mathematics in an African language: the case of Yoruba], doctoral thesis, Université de Paris 7 (France) (in French).

SEI-59

1959 Seidenberg, Abraham: On the eastern Bantu root for six, *African Studies*, Johannesburg (South Africa), Vol. 18, No. 1, 28-34.

"The almost universal stem for 3 in Bantu is -*tatu*, or a variant, in particular -*datu*. In the northeast the dominant form for 6 is -*tandatu*. It has been asserted that -*tandatu* is a duplication of -*datu*. This etymology is rejected. Instead the etymology -*tandatu* = 5+3 is suggested. Evidence is presented to show that -*tandatu* was originally in position 8 but then fell into position 6."

SEI-63

1963 Seidenberg, Abraham: On the eastern Bantu root for six: correction, *African Studies*, Johannesburg (South Africa), Vol. 22, No. 3, 116-117.

"In a previous article it was argued that the Eastern Bantu stem -*tandatu* for six originally meant 8, but later fell into position six. In the argument, the -*tan* of -*tandatu* was compared with the Bantu stem -*tano* for five. To this it has been (validly) objected that the t of -*tano* is of the palatal variety whereas the t of -*tanda* is not. The proposed comparison with -*tano* is abandoned, but the rest of the thesis maintained."

SEI-75

1975 Seidenberg, Abraham: Did Euclid's 'Elements, Book I,' develop geometry axiomatically?, *Archive for History of Exact Sciences*, Berlin (Germany), Vol. 14, No. 4, 263-295.

SEI-76

1976 Seidenberg, Abraham: *km*, a Widespread Root for Ten, *Archive for History of Exact Sciences*, Berlin (Germany), Vol. 16, No. 1, 1-16.

The word *kumi* (root *km*) is nearly universal as the word for 10 in the Bantu languages. In Africa, the equations km = 1, km =10 and km = 100 all occur. In Bantu, *kumi* = 10, *kama* = 100. *Keme* =1 occurs in Bagrim-ma. *Keme* = 100 occurs to the far west (Mande), e.g. *kome* = 1 occurs in Ga (Ghana). Tha author relates the *km* root to the ancient Indo-European sound *dekm* for ten and suggests a common origin. Also examples from other continents are given.

SEK-87

1987 Seka, Beniel R.: *History of mathematics in Tanzania*, Institute of Education, Dar es Salaam (Tanzania), 12 p. (mimeo).

Text of a paper presented at the Annual General Meeting of the Mathematics Association of Tanzania (May 1986). It describes the development of the mathematics curriculum in Tanzania since Independence. A distinction between three periods is made: "the traditional mathematics era, the modern era and the present era, which lends from both traditional and modern mathematics."

SEK-93a
1993 Seka, Beniel: *Jina Langu ni Sifuri* [My name is Zero], Diamond Publishers, Dar es Salaam (Tanzania), 17 p. (in Swahili).

SEK-93b
1993 Seka, Beniel: *Kipeo na Kipeuo Mahakamani* [Kipeo (root) and Kipeuo (square) together], Dar Es Salaam University Press, Dar es Salaam (Tanzania), 22 p. (in Swahili).

SEK-93a and SEK-93b are children's booklets that use the traditional story telling pedagogy to introduce and discuss mathematical ideas: the introduction of 0 in the first booklet, and of squares, square roots and the Pythagorean Proposition in the second.

SEL-97
1997 Selin, Helaine: *Encyclopedia of the History of Science, Technology, and Medicine in Non-Western Cultures*, Kluwer Academic Publishers, Dordrecht (Netherlands), 1117 p.

The following papers relate to the history of mathematics in Africa:

* Jacques Sesiano: Abu Kamil (4-5);
* Laurance Doyle & Edward Frank: Astronomy in Africa (96-100);
* Jehane Ragai & Gregg de Young: Calendars in Egypt (167-168);
* Ahmed Djebbar: Combinatorics in Islamic mathematics (230-232);
* Jan Hogendijk: Conics (235-236);
* Bala Achi: Construction techniques in Africa (236-240);
* Marcia Ascher: Ethnomathematics (326-330);
* Paulus Gerdes: Geometry in Africa: Sona Geometry (367-368);
* Emilia Calvo: Ibn al-Bannâ (404);
* Rosdi Rashed: Ibn al-Haytham (Alhazen) (405-408);
* Ahmed Djebbar: Ibn Al-Yâsamîn (414-415);
* Ahmed Djebbar: Ibn Mun^{c}im (427-428);
* Yousouf Guergour: Ibn Qunfudh (428-429);
* Jacques Sesiano: Magic squares in Islamic mathematics (536-538);
* Thomas Bassett: Maps and mapmaking in Africa (554-558);
* Paulus Gerdes: Mathematics in Africa: South of the Sahara (611-613);
* Ahmed Djebbar: Mathematics in Africa: The Maghreb (613-616);

* Salimata Doumbia: Mathematics in West Africa: Traditional mathematical games (616-619);
* James Ritter: Mathematics in Egypt (629-632);
* Jens Hoyrup: Practical and recreational mathematics (660-663);
* Lawrence Robbins: Namoratunga [archaeoastronomical site] (755);
* Paulus Gerdes: Numeration systems in Africa (781-784);
* Gregg de Young: Pyramids (828-829);
* Ahmed Djebbar: Al-Qalasâdî (830-832);
* Georges Niangoran-Bouah: Weights and measures in Africa: Akan gold weights (1005-1007);
* Ruth Willard: Weights and measures in Egypt (1012-1014).

SEL-00
2000 Selin, Helaine (Ed.), *Mathematics Across Cultures: The History of Non-Western Mathematics*, Kluwer Academic Publishers, Dordrecht (Netherlands), 479 p.

Concerning Africa, the book contains the papers GER-00e, RIT-00, SES-00 and VERR-00.

SELW-78
1978 Selwyn, J. B.: Why teach mathematics in Lesotho?, *Education in Lesotho*, Institute of Education, National University of Lesotho, Roma (Lesotho), Vol. 1, No. 1, 9-13.

SER-83
1983 Sertina, Ivan Van: *Blacks in science, ancient and modern*, Transaction Books, New Brunswick NJ (USA), 302 p.

Contains among other papers: LUM-83a, LUM-83c, PAP-83, 3-ADA-83a, 3-ADA-83b, 3-LYN-83.

SES-77
1977 Sesiano, Jacques: Les méthodes d'analyse indéterminée chez Abû Kâmil [The methods of indeterminate analysis of Abû Kâmil], *Centaurus*, Copenhagen (Denmark), Vol. 21, No. 2, 89-105 (in French).

SES-82
1982 Sesiano, Jacques: *Books IV to VII of Diophantus' "Arithmetica" in the Arabic Translation attributed to Qusta Ibn Luqa*, Springer, New York (USA), 502 p.

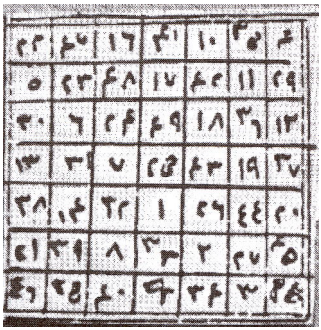
SES-89
1989 Sesiano, Jacques: Koptisches Zahlensystem und (griechisch-) koptische Multiplikationstafeln nach einem arabischen Bericht [The Coptic number system and Greek-Coptic multiplication tables as described in a short Arabic account], *Centaurus*, Copenhagen (Denmark), Vol. 31, 53-65 (in German).

Analyses a 15th century work devoted to presenting the old Coptic numeral system that used 27 Coptic letters to abbreviate calculations.

SES-94
1994 Sesiano, Jacques: Quelques méthodes arabes de construction des carrés magiques impairs [Some Arabic construction methods of odd magical squares], *Bulletin de la Société Vaudoise des Sciences Naturelles* (Switzerland), Vol. 83, No. 1, 51-76 (in French).

General construction methods of magic squares appeared in the countries of Islam in the 9th century, and the science of magic squares arrived there at its zenith in the 11th and 12th centuries. From the 13th century, magical and divinatory applications began to replace of mathematical study. Classical construction methods survived, however, in later treatises of a certain level, as in part of a work by Muhammad ibn Muhammad al-Fullani al-Kishnâwî (born in the north of Nigeria and died in Cairo in 1741), on the construction of magic squares of odd order. It is this chapter of the book of al-Kishnawi that is analyzed in the paper. In relationship to the contents of the chapter, the author of the paper states "We find here the explanation of different ways of disposing the numbers in the squares, and with diverse forms of magic. Although the majority of these constructions are already known from the classical period, they are often explained or applied in an easier way; time has, to a certain degree, served as a filter, and the reported methods are those whose use has been preserved by their simplicity or elegance. One finds also, at the end of the extract, the explanation of a topic that is new in relation to classical treatises (without doubt due to its magic use): that of magical squares of which one square is left unoccupied. All topics are presented by al-

Kishnâwî with great clarity. He certainly seems to be a person of worth: the biographical note dedicated to him by the historian al-Jabartî (1753-1825/6) in his *Chronicles* (Al-Jabartî 1888-89, II, 39-42) are full of praise for his capacities and merits. Al-Kishnâwî seems even to have been the authority in the new field of squares with holes, as he is mentioned elsewhere by the same al-Jabartî in relation to the properties of those squares of order 5."



A magical square in a manuscript of Al-Kishnâwî (vf. SES-94)

SES-96
1996 Sesiano, Jacques: Le *Kitab al-Misaha* d'Abû Kâmil [The *Kitab al-Misaha* of Abû Kâmil], *Centaurus*, Copenhagen (Denmark), Vol. 38, 1-21 (in French).

SES-00
2000 Sesiano, Jacques: Islamic mathematics, in SEL-00, 137-165.

The paper contains the following sections: Heritage (Mesopotamian, Indian, Greek), Arithmetic (reckoning, root extraction), Algebra (algebraic reckoning, geometrical illustration, other), Geometry (regular polygons [including Abû Kâmil], cubic equation, other), Number theory, and Magic Squares [including Ibn al-Haytham].

SETA-02
2002 Setati, Mamokgethi: *Language practices in intermediate multilingual mathematics classroom*, doctoral thesis, University of the Witwatersrand, Johannesburg (South Africa).

SET-16
1916 Sethe, Kurt: *Von Zahlen und Zahlworten bei den alten Ägyptern und was für andere Völker und Sprachen daraus zu*

lernen ist. Ein Beitrag zur Geschichte von Rechenkunst und Sprache [About numbers and number words among the Ancient Egyptians and what can be from them concerning other peoples. A contribution to the history of arithmetic and language], Trübner, Strassburg, 147 p. (in German).

SETI-65
1965 Setidisho, Noah: *An empirical study of mathematical ability in schoolchildren*, doctoral thesis, University of South Africa, Pretoria (South Africa).

SEZ-97a
1997 Sezgin, Fuat (Ed.): *Codex Leidensis 399,1. Euclidis Elementa ex interpretatione al-Hadschdschadschii cum commentariis al-Narizii. Arabice et latine editerunt notisque instruxerunt R. O. Besthorn et J. L. Heiber*, Institute for the History of Arabic-Islamic Science, Johann Wolfgang Goethe University, Frankfurt am Main (Germany), Collection "Islamic mathematics and science", Volume 14-15, 735 p. (in Arabic and Latin).

Reprint of the Edition Kopenhagen 1897-1905.

SEZ-97b
1997 Sezgin, Fuat (Ed.): *The Commentary of Pappus on Book X of Euclid's Elements. Arabic text and translation by William Thomson with remarks, notes and a glossary of technical terms by Gustav Junge and William Thomson*, Institute for the History of Arabic-Islamic Science, Johann Wolfgang Goethe University, Frankfurt am Main (Germany), Collection "Islamic mathematics and science", Volume 16, 298 p.

Reprint of the Edition Cambridge 1930.

SEZ-97c
1997 Sezgin, Fuat (Ed.), in collaboration with M. Amawi, C. Ehrig-Eggert, and E. Neubauer: *Euclid in the Arabic Tradition. Texts and Studies. Collected and reprinted*, *I*, Institute for the History of Arabic-Islamic Science, Johann Wolfgang Goethe University, Frankfurt am Main (Germany), Collection "Islamic mathematics and science", Volume 17, 340 p.

The first volume on Euclid contains papers by Franz Woepcke (1-31, in French); Ludwig Oftendinger (33-52, in German); Moritz Steinschneider (54-128, in German); Maximilian Curtze (129-134, in German); Hermann Weissenborn (135-160, in German); Antonio Favaro (161-186, in Italian); Johann Ludwig Heiberg (187-271, in German); Martin Klamroth (272-328, in German).

SEZ-97d

1997 Sezgin, Fuat (Ed.), in collaboration with M. Amawi, C. Ehrig-Eggert, and E. Neubauer: *Euclid in the Arabic Tradition. Texts and Studies. Collected and reprinted*, *II*, Institute for the History of Arabic-Islamic Science, Johann Wolfgang Goethe University, Frankfurt am Main (Germany), Collection "Islamic mathematics and science", Volume 18, 324 p.

The second volume on Euclid contains papers by Rasmus O. Besthorn (1-2, in German); Heinrich Suter (3-110, in German); Mansion (111-113, in French); Karl Lokotsch (115-141, in German); Raymond Archibald (143-236); Giuseppe Furlani (237-287, in German); Eilhard Wiedemann (288-296, in German); and Gotthelf Bergsträßer (297-324, in German).

SEZ-97e

1997 Sezgin, Fuat (Ed.), in collaboration with M. Amawi, C. Ehrig-Eggert, and E. Neubauer: *Euclid in the Arabic Tradition. Texts and Studies. Collected and reprinted*, *III*, Institute for the History of Arabic-Islamic Science, Johann Wolfgang Goethe University, Frankfurt am Main (Germany), Collection "Islamic mathematics and science", Volume 19, 310 p.

The third volume on Euclid contains papers by David Smith (1-6); Albert G. Kapp (8-121, in German); M.-A Kugener (122-124, in French); Claire Baudoux (125-129, in French); Gustav Junge (131-147, in German); Clemens Thaer (148-163, in German); A. S. Ünver (164-166); Edward B. Plooij (167-243); Marshall Clagett (244-270); and Abdalhamid Sabra (272-309, in Arabic).

SEZ-97f

1997 Sezgin, Fuat (Ed.), in collaboration with M. Amawi, C. Ehrig-Eggert, and E. Neubauer: *Abû Kâmil Shujâ' ibn Aslam (9th cent.). Texts and Studies. Collected and reprinted*, Institute for the History of Arabic-Islamic Science, Johann Wolfgang

Goethe University, Frankfurt am Main (Germany), Collection "Islamic mathematics and science", Volume 23, 262 p.

The volume on Abû Kâmil (Egypt) contains papers by Gustavo Sacerdote (Pentagon and decagon, 1-26, in Italian); Heinrich Suter (Pentagon and decagon; 27-54; Arithmetic, 56-76, in German); Louis Karpinski (Algebra, 78-106); and Josef Weinberg (Algebra, 107-251, in German).

SEZ-98a

1998 Sezgin, Fuat (Ed.), in collaboration with M. Amawi, C. Ehrig-Eggert, and E. Neubauer: *Ibn al-Bannâ al-Marrâkushî Abû l-'Abbâs Ahmad ibn Muhammad (d. 721/1321). Texts and Studies. Collected and reprinted*, Institute for the History of Arabic-Islamic Science, Johann Wolfgang Goethe University, Frankfurt am Main (Germany), Collection "Islamic mathematics and science", Volume 44, 312 p.

This volume on Ibn al-Bannâ (Morocco) contains papers by Aristide Marre (1-56, in French); Franz Woepcke (57-138, in French); Michel Chasles (139-147, in French); Moritz Steinschneider (149-150, in French); Giorgio Levi Della Vida (151-156, in Italian); Henri-Paul-Joseph Renaud (158-300, in French, reproduction of REN-37, REN-38a, REN-44, REN-48).

SEZ-98b

1998 Sezgin, Fuat (Ed.), in collaboration with M. Amawi, C. Ehrig-Eggert, and E. Neubauer: *Ibn al-Haytham al-Hasan ibn al-Hasan (d. 430 / 1039). Texts and Studies. Collected and reprinted. Vol. I*, Institute for the History of Arabic-Islamic Science, Johann Wolfgang Goethe University, Frankfurt am Main (Germany), Collection "Islamic mathematics and science", Volume 57, 363 p.

The first volume on Ibn al-Haytham (Egypt) contains papers by Louis-Amélie Sédillot (1-24, in French); Moritz Steinschneider (25-60, in Italian and French); Marcus Baker (61-65); Paul Bode (66-110, in German); Heinrich Suter (111-184, in German); Michael Jan de Goeje (168-188, in French); Eilhard Wiedemann (189-273 and 313-351, in German); and Johan Ludvig Heiberg & Eilhard Wiedemann (275-311, in German).

SEZ-98c
1998 Sezgin, Fuat (Ed.), in collaboration with M. Amawi, C. Ehrig-Eggert, and E. Neubauer: *Ibn al-Haytham al-Hasan ibn al-Hasan (d. 430 / 1039). Texts and Studies. Collected and reprinted. Vol. II*, Institute for the History of Arabic-Islamic Science, Johann Wolfgang Goethe University, Frankfurt am Main (Germany), Collection "Islamic mathematics and science", Volume 58, 331 p.

The second volume on Ibn al-Haytham (Egypt) contains papers by Eilhard Wiedemann (1-9, in German); Carl Schoy (11-93, in German); Karl Kohl (94-228, in German); Armand Abel (230-235, in French); Roberto Marcolongo (237-251, in Italian); José Maria Millás Vallicrosa (253-282, in Spanish); Henry J. Winter & W. Arafat (283-314); and Hâmid Dilgan (315-323, in French).

SHA-84
1984 Shawki, Galal: Formulation and development of Algebra by Muslim scholars, *Islamic Studies*, Islamabad (Pakistan), Vol. XXIII, No. 4, 337-352.

Highlights of some Muslim contributions to the development of algebra (8th –16th centuries) are pointed out: solution of quadratic, cubic and biquadratic equations, addition theorem of exponents, numerical approximation, introduction of algebraic symbolism, binomial theorem.

SHE-84
1984 Sheikh, Ahmed Shams El Din El: *School mathematics in Sudan*, doctoral thesis, University of Edinburgh (UK).

SHI-80
1980 Shirley, Lawrence: *Recent developments in mathematics education in Nigeria* (paper presented at the 4th International Congress on Mathematics Education, Berkeley CA, USA, 10 p., mimeo).

SHI-84
1984 Shirley, Lawrence: *Teacher Participation in Mathematics Curriculum Development and Implementation in Three Northern States of Nigeria*, doctoral thesis, Ahmadu Bello University, Zaria (Nigeria).

SHI-86a

1986 Shirley, Lawrence: History of mathematics in Nigerian mathematics classrooms: values and problems, *Abacus, the Journal of the Mathematical Association of Nigeria*, Ilorin (Nigeria), Vol. 12, 123-133.

Discusses the "problem of making the history of mathematics culturally relevant in the Nigerian setting when much of the recorded historical developments in mathematics have been Mediterranean, Arab and European."

SHI-86b

1986 Shirley, Lawrence: *Ethnomathematics and the history of African mathematics* (paper presented at the 2nd Pan-African Congress of Mathematicians, Jos, Nigeria, 8 p., mimeo).

"Although the value of studying and teaching the history of mathematics is clear, the European-centred content of standard history of mathematics may make it less relevant to African students." As a response, it is necessary to use 'a wider scope of mathematics; not simply the standard 'learned mathematics', but mathematics in daily life and culture, the so-called 'ethnomathematics'."

SHI-88a

1988 Shirley, Lawrence: *Historical and ethnomathematical algorithms for classroom use*, Ahmadu Bello University, Zaria (Nigeria), 12 p. (mimeo).

Paper presented at the 6th International Congress on Mathematics Education, Budapest. It gives an overview of studies on traditional Nigerian, arithmetical algorithms and suggested that such techniques could be used in classrooms as alternative algorithms: "... children might relate mathematics better to their home culture, by seeing techniques from their own traditional society being applied in the setting of their mathematics classroom."

SHI-88b

1988 Shirley, Lawrence: *Counting in Nigerian languages* (paper presented at the 6th International Congress on Mathematics Education, Budapest, mimeo).

SHI-95
1995 Shirley, Lawrence: Using Ethnomathematics to find multicultural mathematical connections, in: House, Peggy (Ed.), *Connecting Mathematics across the Curriculum*, National Council of Teachers of Mathematics, Reston VA (USA), chapter 4.

Includes suggestions from Africa (e.g. Mancala games, Adinkra textile patterns).

SHI-96
1996 Shirley, Lawrence: Activities from African Calendar and Time Customs, *Mathematics Teaching in the Middle School*, NCTM, Reston VA (USA), Vol. 1, No. 8, 616-620.

Presents suggestions of how using African "day-names" (examples are given from Ghana and Nigeria), and practices like the "sunrise clock" in the mathematics classroom.

SIC-05
2005 Sica, Giandomenico: *What mathematics from Africa?*, Polimetrica, Monza (Italy), 109 p.

Contains the following contributions:

* Aderemi Kuku: Mathematical sciences and the development of Africa, 9-16;
* Saliou Touré: La situation mathématique en Afrique [The mathematical situation in Africa], 17-24 (in French);
* Norbert Hounkounnou: Mathematics from Africa: Status, goals and responsibilities, 25-34;
* Nithaya Chetty & Ahmed Bawa: Developing computational mathematics in Africa, 35-52;
* Edward Lungu: Status of mathematics in Sub-Sahara Africa, 53-62;
* Kgomotso Garegae: Mathematics in different cultures and societies: the Botswana case, 63-82;
* Paulus Gerdes: Mathematical research inspired by African cultural practices: the example of mirror curves, Lunda-designs and related concepts, 83-100;
* Ron Eglash & Toluwalogo Odumosu: Fractals, complexity, and connectivity in Africa, 101-109.

SIMK-05
2005 Simkins, Charles & Paterson, Andrew: *Learner performance in South Africa: social and economic determinants of success in language and mathematics*, HSRC Press, Cape Town (South Africa), 76 p.

SIMO-92
1992 Simon, G.: L'*Optique* d'ibn al-Haytham et la tradition ptoléméenne, *Arabic Sciences and Philosophy*, New York (USA), Vol. 2, No. 2, 203-235.

SIMO-94
1994 Simon, G.: Aux origines de la théorie des miroirs: sur l'authenticité de la 'Catoptrique' d'Euclide [Towards the origin of the theory of mirrors: on the authencity of the Euclid's Catoptrics], *Revue d'Histoire des Sciences*, Evry (France), Vol. 47, No. 2, 259-272.

SIM-98
1998 Sims, John: *Designs from the Kuba (Congo) and the teaching of mathematics to arts students*, paper presented at 76th Annual Meeting of the National Council of Teachers of Mathematics (2-4 April 1998, Washington DC, USA).

SIZ-99
1999 Sizer, Walter: Review of Gerdes' *Women, Art and Geometry in Southern Africa* (GER-98a) (available online at: http://www.maa.org/reviews/wagsa.html).

SMI-82
1982 Smith, Arthur: Angles of elevation of the pyramids of Egypt, *Mathematics Teacher*, Reston VA (USA), Vol. 75, No. 2, 124-127.

Addresses the question 'Why did the Egyptians build pyramids using angles of elevation of approximately 43 1/2 or 52 degrees?'

SMIT-88
1988 Smith, A. Mark: The psychology of visual perception in Ptolemy's 'Optics', *Isis*, Madison WI (USA), Vol. 79, No. 297, 189-207.

SMIT-96
1996 Smith, A. Mark: *Ptolemy's theory of visual perception*, American Philosophical Society, Philadelphia PA (USA), 300 p.

SMIT-99
1999 Smith, A. Mark: *Ptolemy and the foundations of ancient mathematical optics: a source based guided study*, American Philosophical Society, Philadelphia PA (USA), 172 p.

SMITHJ-92
1992 Smith, J. D.: The remarkable Ibn al-Haytham, *The Mathematical Gazette*, London (UK), Vol. 76, 189-198.

SOA-91
1991 Soares, Daniel: *On popular counting practices in Mozambique* (paper presented at the 8th Symposium of the Southern Africa Mathematical Sciences Association, Maputo, mimeo).

SOA-96
1996 Soares, Daniel: The incorporation of the geometry of traditional house building in mathematics education in Mozambique, in: T. Kjaergard et al. (Eds.), *Numeracy, Race, Gender, and Class — Proceedings of the Third International Conference on the Political Dimensions of Mathematics Education*, Gaspar Forlag, Landas (Norway), 242-244.

Suggests the use of the geometry of house building techniques in mathematics education.

SOA-05
2005 Soares, Daniel: A construção de casas tradicionais e a resolução de problemas [The construction of traditional houses and problem solving], *Matemática & Educação*, Beira (Mozambique), No. 1, 32-35.

SOA-06
2006 Soares, Daniel: Métodos populares de construção do rectângulo [Popular methods of rectangle construction], *Matemática & Educação*, Beira (Mozambique), No. 2, 38-41.

Describes popular ways in the Sofala and Zambeze provinces of

Mozambique to construct the rectangular base of a traditional house.

SOA-07
2007 Soares, Daniel: *The incorporation of the geometry involved in traditional house building in Mathematics Education in Mozambique. The cases of the Zambezia and Sofala Provinces*, doctoral thesis, University of the Western Cape, Rondebosch (South Africa).

SOU-69
1969 Souissi, Mohamed: *Ibn al-Bannâ of Marrakech, The summary of operations of computation* (Edition, French translation and commentaries), Publications de l'Université de Tunis, Tunis (Tunisia), 197 p. (in Arabic and French).

SOU-72
1972 Souissi, Mohamed: An Andalusian-Tunisian scholar-mathematician, al-Qalasâdî, *Bulletin de l'Université de Tunis*, Tunis (Tunisia), No. 9, 33-49 (in Arabic).

SOU-73
1973 Souissi, Mohamed: Explanation of a page of the *Muqaddima* of Ibn Khaldûn on the arithmetical sciences, *Bulletin de l'Université de Tunis*, Tunis (Tunisia), No. 10, 87-93 (in Arabic).

SOU-75
1975 Souissi, Mohamed: *Un texte d'Ibn al Bannâ sur les nombres parfaits, abondants, deficients et aimables*, Hamdard National Foundation, Karachi (Pakistan), 14 p. (in French).

Paper presented at the International Congress of Mathematical Sciences, (Karachi, 14-20 July 1975), including a translation of a manuscript of Ibn al-Bannâ (1256-1321, Maghreb) on perfect, abundant, deficient and amicable numbers.

SOU-76
1976 Souissi, Mohamed: A text of Ibn al-Bannâ on perfect, abundant, deficient and amicable numbers, *Bulletin de l'Université de Tunis*, Tunis (Tunisia), No. 13, 193-209 (in Arabic).

Arabic version of SOU-75.

SOU-82a

1982a Souissi, Mohamed: Présentation et analyse du traité "Somme des principes et des conclusions" par le savant astronome Marocain al-Hasan al-Marrâkushî (était vivant en 1281), *Cahiers de Tunisie*, Tunis (Tunisia), Vol. XXX, 273-286 (in French).

Analyses the treatise "Summary of principles and conclusions" by the Moroccan astronomer al-Hasan al-Marrâkushî (13th century). This treatise may be considered the culmination of astronomic literature written in Arab. It gives a summary of the results obtained by al-Hasan's predecessors and adds his own observations and solutions.

SOU-82b

1982b Souissi, Mohamed: Analyses the treatise "Summary of principles and conclusions" by the Morrocan astronomer al-Hasan al-Marrâkushî (13th century), *Journal of the Institute of Arab Manuscripts*, Koweit, Vol. 1, No. 1, 63-71 (in Arabic).

Arabic version of SOU-82a.

SOU-83a

1983a Souissi, Mohamed: *Desire of the students on the commentary of the "Vow of calculators" of Ibn Ghâzi al-Miknâsi al-Fâsi*, Institute for the History of Arabic Science, Alep (Syria), 326 p. (in Arabic).

SOU-83b

1983b Souissi, Mohamed: The Maradinian light on the commentary of the poem of Ibn al-Yâsamin by al-Mâradini, National Council for Culture, Arts and Literature, Koweit, 77 p. (in Arabic).

SOU-84

1984 Souissi, Mohamed: The formulas of Ibn al-Bannâ for areas, *Journal of the Institute of Arab Manuscripts*, Koweit, Vol. 28, No. 2, 491-520 (in Arabic).

SOU-88a
1988a Souissi, Mohamed: *Al-Qalasâdî, Revelation of the secrets concerning the science of dust ciphers*, Arab Book, Tunis & Bayt al-Hikma, Carthage (Tunisia), 184 p. (in Arabic).

SOW-92
1992 Sowunmi, C. O. A.: Professor Adegoke Olubummo [1923-1992] - a multidimensional view, in MEM-92, iii-iv.

SSE-97
1997 Ssembatya, Vincent & Vince, Andrew: Mathematics in Uganda, *The Mathematical Intelligencer*, New York (USA), Vol. 19, No. 3, 27-32.

Overview of the development of mathematics at the Makerere University since its creation in 1922; brief information on the Uganda Mathematical Society established in 1972 under the leadership of Paul Mugambi – "the grandfather of mathematics in the country" (p.30).

STA-67
1967 Stappers, Leo: Het hoofdtelwoord in de Bantoe-talen [The cardinal number in the Bantu languages], *Africana Linguistica II*, *Annales du Musée Royal de l'Afrique Centrale*, Sciences Humaines, Tervuren (Belgium), Vol. 55, 175-198 (in Dutch).

Compares the prefixes used in the Bantu languages in connection with the cardinal numbers one to five. The paper analyses also 'abstract' counting (i.e. without reference to the objects), and 'distributive' ('two by two',...) and 'multiplicative' use of cardinals in the Bantu languages. Maps with information on the geographical distribution are included.

STEE-02
2002 Steele, John M. & Annette Imhausen (Eds.): *Under one sky: astronomy and mathematics in the ancient Near East*, Ugarit-Verlag, Münster (Germany), 496 p.

STE-77
1877 Steinschneider, Moritz: Rectification de quelques erreurs relatives au mathématicien arabe Ibn al-Bannâ [Rectification of some errors concerning the Arab mathematician Ibn al-Bannâ], *Bulletino di Bibliografia e di Storia Delle Scienze Matematiche*

e Fisiche (Boncompagni), Rome (Italy), Vol. 10, 313-314 (in French).

STEV-98
1998 Stevens, Anthony & Janet Sharp: *Learning about fractions and ratios by using African rhythms played on drums*, paper presented at 76th Annual Meeting of the National Council of Teachers of Mathematics (2-4 April 1998, Washington DC, USA).

STO-93
1993 Stott, L. & Lea, Hilda: *Common threads in Botswana*, British Council, Gaborone (Botswana), 82 p.

Presents suggestions about the use of baskets, hair braiding, and weaving designs in mathematics education.

STR-30
1930 Struve, V. V.: *Mathematische papyrus des Staatlichen Museums der Schönen Künste in Moskau* [Mathematical papyrus in the State Museum for Beautiful Art in Moscow], Springer Verlag, Berlin (Germany) (in German).

SUS-05
2005 Susuwele-Banda, William John: *Classroom assessment in Malawi: Teachers' perceptions and practices in mathematics*, doctoral thesis, Virginia Polytechnic Institute and State University (USA).

SUT-00
1900 Suter, Heinrich: *Die Mathematiker und Astronomen der Araber und ihre Werke* [The mathematicians and the astronomers of the Arabs and their works], Teubner, Leipzig (Germany), 277 p. (in German).

SUT-01
1901 Suter, Heinrich: *Das Rechenbuch des Abû Zakarîyâ* [The arithmetic book of Abû Zakariyâ], *Bibliotheca Mathematica*, Halle (Germany), Series 3, No. 2, 12-40 (in German).

SUT-10

1910 Suter, Heinrich: *Das Buch der Seltenheiten der Rechenkunst von Abû Kâmil el-Misrî* [The book of the particularities of the art of reckoning by Abû Kâmil el-Misr*î*], *Bibliotheca Mathematica*, Halle (Germany), Series 3, No. 11, 100-120 (in German).

SWI-56

1956 Swift, J. D.: Diophantus of Alexandria, *American Mathematical Monthly*, Washington DC (USA), Vol. 63, 163-170.

SZA-90

1990 Szabo, Arpad: Ein Satz über die mittlere Proportionale bei Euklid (Elem. III 36) [A theorem on the mean proportion in Euclid (Elem. III 36)], *Commentarii Mathematici Universitatis Sancti Pauli*, Tokyo (Japan), Vol. 39, No.1, 41-51 (in German).

T

TAF-87

1987 Tafla, Bairu: Some remarks on numerical idioms recurring in Ethiopian history, *Afrika und Übersee*, Berlin (Germany), Vol. 70, 73-98.

"… certain numbers [e.g. 2, 4, 40, 44, 80, 7] in the Semitic languages of Ethiopia form components of idiomatic expressions in which they lose their accurate mathematical significance and assume figurative meanings, or connotations which have no relation whatsoever to their original meaning. Some imply greatness, wholeness or totality; others indicate excessiveness of amount, or fantastical size" (p. 92).

TAH-95

1995 Tahir, H.: Pappus and mathematical induction, *Australian Mathematical Society Gazete*, Canberra (Australia), Vol. 22, No. 4, 166-167.

TAIS-82

1982 Taisbak, Christian Marinus: *Coloured Quadrangles, A Guide to the Tenth Book of Euclid's Elements*, Museum Tusculanum Press, Copenhagen (Denmark), 78 p.

TAIS-96

1996 Taisbak, Christian Marinus: Zeuthen and Euclid's *Data* 86. Algebra - or a Lemma about intersecting Hyperbolas?, *Centaurus*, Copenhagen (Denmark), Vol. 38, 122-139.

TAIS-03

2003 Taisbak, Christian Marinus: *Euclid's Data or the importance of being given*, Museum Tusculanum Press, University of Copenhagen, Copenhagen (Denmark), 271 p.

TAI-75

1975 Taiwo, C. O.: Teaching and learning mathematics in the Yoruba language, in CASM-75, 24-30.

The Yoruba Project extended the use of the Yoruba language as the medium of instruction in certain schools in Western Nigeria to the end

of primary school. Problems faced by the writers of the mathematics material are discussed, with examples of proposed solutions.

TAR-87
1987 Tarbo, B. T.: *A comparative study of mathematics concepts and skills possessed by Tiv and Idoma unschooled children in Bebue State, Nigeria*, Ahmadu Bello University, Zaria (Nigeria).

TCH-94
1994 Tchitchi, Toussaint Yaovi: Numérations traditionnelles et aritmétique moderne [Traditional numerations and modern arithmetic], in: Hountondji, Paulin (Ed.), *Les savoirs endogènes: pistes pour une recherche*, CODESRIA, Dakar (Senegal), 109-138.

Discusses traditional numeration in "àjá" (Benin) and possibilities of and experimentation with a decimalization.

TEM-38
1938 Tempels, Placidus: De tel-gebaren der Bashila [The number-gestures of the Bashila], *Congo-Overzee*, Antwerpen (Belgium), Vol. IV, No. 2, 49-53 (in Dutch).

Describes the number-gestures among the (Ba)Shila in Congo / Zaire. There are two series, one for counting from 1 to 10, and one for indicating individual numbers (cardinal numbers).

THA-33
1933 Thaer, Clemens: *Die Data von Euklid nach Heibergs Text aus dem Griechischen übersetzt* [The Data of Euclid according to Heiberg's Text translated from the Greek], Springer, Berlin (Germany), 78 p. (in German).

THA-62
1962 Thaer, Clemens: *Die Data von Euklid nach Menges Text aus dem Griechischen übersetzt* [The Data of Euclid according to Menges Text translated from the Greek], Springer, Berlin (Germany), 78 p. (in German).

THEI-78
1978 Theisen, W.: A note on John of Beaumont's version of Euclid's 'De visu', *British Journal for Philosophy of Science*, Oxford (UK), Vol. 11, No. 38, 151-155.

THEI-84
1984 Theisen, W.: Euclid, relativity, and sailing, *Historia Mathematica*, New York (USA), Vol. 11, No. 1, 81-85.

THE-90
1990 Theon of Alexandria: *Tables manuelles de Ptolémée, complétées par Théon d'Alexandrie* [Manual tables of Ptolemy, completed by Theon of Alexandria], reproduction of Volumes II and III of the 1822 and 1825 edition, 552 p.

THE-93
1993 Theon of Alexandria: *Commentaire sur les Livres I et II de la Syntaxe mathématique de Ptolémée* [Comments on the Syntax of Ptolemy (Greek text with French translation by Halma)], Blanchard, Paris (France), 461 p. (in French).

THO-20
1920 Thomas, N. W.: Duodecimal base of numeration, *Man*, London (UK), Vol. 20, 25-29.

About duodecimal systems of numeration in Nigeria.

THOM-87
1987 Thomas-Emeagwali, Gloria: Reflections on the development of science in the Islamic world and its diffusion into Nigeria before 1903, *Journal of the Pakistan Historical Society*, Karachi (Pakistan).

THOM-92a
1992a Thomas-Emeagwali, Gloria (Ed.): *The historical development of science and technology in Nigeria*, Edwin Mellen Press, Lewiston NY (USA), 192 p.

Analyses traditional methods of food processing, cassava-processing technology, textile technology, and pedagogy and science teaching in Nigeria. The text concentrates on the historical dimension but

approaches the subject in the context of multidisciplinary interpretation. The book includes KANI-92a.

THOM-92b
1992b Thomas-Emeagwali, Gloria (Ed.): *Science and technology in African history with case studies from Nigeria, Sierra Leone, Zimbabwe, and Zambia*, Edwin Mellen Press, Lewiston NY (USA), 204 p.

In science, the areas of focus include mathematics, medicine, and the sociology of medicine, as well as biologically-based warfare. In technology, iron, gold, diamond, and glass-making technologies dominate. Three of the cases of metallurgical development are centered on the pre-colonial periods. The book includes KANI-92b.

THOM-93
1993 Thomas-Emeagwali, Gloria (Ed.): *African systems of science, technology and art: the Nigerian experience*, Karnak House, London (UK), 143 p.

Includes chapters on methodological issues, textile technologies, traditional medicine, food processing, metal technology, mechanics and engineering.

TOB-90
1990 Tobin, R.: Ancient perspective and Euclid's 'Optics', *Journal of the Warburg and Courtauld Institutes*, London (UK), Vol. 53, 14-41.

TOO-90
1990 Toomer, G. J.: *Apollonius, Conics Books, V to VII, The Arabic Translation of the lost Greek Original in the version of the Banu Musa*, Springer-Verlag, New York (USA), 2 volumes, 888 p.

Contains the critical edition and the English translation of Books V, VI, and VII of the Conics of Apollonius, on the basis of the Arabic version translated from the Greek by Thâbit Ibn Qurra (d. 901) and corrected by the brothers Banû Mûsâ (9th century).

TOR-63
1963 Torrey, Volta: Old African number games?, *Science Digest*, Chicago IL (USA).

TOUH-79
1979 Touhoun, Benjamin: La numération décimale: le cas Aja [Decimal numeration: the case of Aja], *Actes du Séminaire National de Formation Linguistique*, CNL, Lokossa (Benin), 164-178.

TOU-94
1994 Touré, Saliou & Dona-Fologo, D. (Eds.): *Actes du Séminaire Interdisciplinaire Mathématique-Philosophie et Enseignement* [Proceedings of the Interdisciplinary Seminar Mathematics-Philosophy and Education], Ministère de l'Education Nationale, Abidjan (Côte d'Ivoire), 118 p.

Contains the proceedings of a seminar held at Yamoussoukro, January 25 to 29, 1993. The following sections deal with culture and mathematics:

* Tony Lévy: Euclid's *Elements*, text and history (10-13);
* Salimata Doumbia: Verbal games and traditional mathematics education in Africa (92-96);
* Salimata Doumbia: Cowrie games (97-101);
* Paulus Gerdes: Ethnomathematics as a new research area in Africa (101-106).

TOU-00
2000 Touré, Saliou: *Mathematical Life in Côte d'Ivoire from Independence until our days*, paper presented at the 2000 International Year of Mathematics ceremonies in Côte d'Ivoire, March 29 (in French).

TOU-01
2001 Touré, Saliou: *The evolution of mathematics since its origins until our days*, paper presented at the 2000 International Year of Mathematics ceremonies in Côte d'Ivoire, February 13, 2001 (in French).

TOU-02
2002 Touré, Saliou: L'enseignement des mathématiques dans les pays francophones d'Afrique et de l'Ocean Indien [Mathematics teaching in the French-speaking countries of Africa and the Indean Ocean], *Zentralblatt für Didaktik der Mathematik - International Reviews on Mathematical*

Education, Karlsruhe (Germany), Vol. 34, No. 4, 175-178 (in French).

"Mathematics teaching in the French-speaking countries of Africa and the Indian Ocean. We examine Mathematics teaching in the French-speaking countries of Africa and the Indian Ocean, starting from the consequences of the Colonial Period. At that time, education was mainly aimed at preparing the civil servants., and there was no organized structure for teaching. When they became independent, these countries started with the French system and methods, but they progressively realized that it was not totally adapted to the aims and specificities of such countries. So progressively new systems and curricula were designed. In this paper, we describe some examples, and give some trends in the development of Mathematics education in Africa and Indian Ocean, and perspectives for the future."

TOUS-93

1993 Toussaint, G.: A new look at Euclid's second proposition, *Mathematical Intelligencer*, New York (USA), Vol. 15, No. 3, 12-23.

TRA-06

2006 Traoré, Kalifa: *Étude des pratiques mathématiques développées en contexte par les Siamous au Burkina Faso* [Study of mathematical practices developed-in-context by the Siamous of Burkina Faso], doctoral thesis, Université de Quebec à Montréal (Canada) (in French).

TRE-50

1950 Treweek, A. P.: *A critical edition of the text of the Collection of Pappus of Alexandria*, doctoral thesis, University of London (UK).

TRO-80

1980 Tro, Gueyes: *Étude de quelques systèmes de numération en Côte d'Ivoire*, Author's edition, Abidjan (Côte d'Ivoire), 192 p.

Analyses the following numeration systems: Akan (Anyi, Baoule, Aboure, Attie, Ebrie, Aladian), Bete, Dida, Dan, Gouro, Kroumen, Koulango, Djan (Lobi), Malinke (Dioula), Senoufo, Tagwana, Wes. Discusses the characteristics of these numeration systems (base

twenty, base ten, mixed twenty-ten, base five) and proposes a numerical map of the country dividing it in four regions according to the characteristics of the systems.

TUC-95

1995 Tuchscherer, Konrad: *'Kikakui' tradition of writing among the Mende of Sierra Leone*, doctoral thesis, University of London, London (UK).

This is a phonographic script for writing the Mende language and a number writing system used to write Mende number words: "Like the dyllabic characters of the writing system, the numerals of the decimal based number writing system are written from right to left, from greater units to lesser units. Any number, other than zero, can be written in the system. Interestingly, while the numerals are decimal based, Mende number words are conceptualized largely on a vigesimal system of counting. The two systems overlap: numerals are written decimally and read aloud vigesimally" (*African Languages and Cultures*, Vol. 8, No. 2, 1995, p. 172).

TUC-99

1999 Tuchscherer, Konrad: The lost script of the Bagam, *African Affairs, The Journal of the Royal African Society*, London (UK), Vol. 98, No. 390, 55-77.

The paper presents new information on the Bagam script, an autochthonous writing system from Cameroon, which has now fallen into extinction. On page 73 are illustrated the numerals for one to ten. On page 77 the author notes the possible connection of the Bagan numerals to the Bamum numerals.

U

UAI-92
1992 Uaila, Evaristo: *Simetrias em ornamentos em cestos do tipo 'khuama'* [Symmetries in ornaments of baskets of the *khwama* type], 'licenciatura' thesis, Instituto Superior Pedagógico, Maputo (Mozambique).

Analyses symmetries of plaited ornaments in *khwama* baskets among the Changana in the South of Mozambique.

UKA-97
1997 Ukaegbu, Jon: *Proto-Mathematical Forms as Reflected in the Igbo Calendar*, paper presented at the conference "The History of Mathematics / Science and Its Uses in Teaching: A Multicultural Approach", City University of New York (USA), March 14.

UKP-84
1984 Ukpele, Ogbeche Pius: *An analysis of the mathematics curriculum and achievement in mathematics of the fifth form pupils in government secondary schools of Benue State, Nigeria*, doctoral thesis, Cardiff University (UK).

UNE-74
1974 UNESCO & UNICEF (Ed.): *Final report: Seminar on the Development of Science and Mathematics Concepts in Young Children in African Countries, Nairobi, September 17-27, 1974*, UNESCO, Nairobi (Kenya), 96 p.

Includes a survey of research in Africa involving conservation and classification together with a not annotated bibliography (84-92) on the development of science and mathematics concepts in African children.

UNE-75
1975 UNESCO (Ed.): *Interactions between Linguistics and Mathematical Education*, UNESCO, Paris (France).

Final report of the symposium 'Interactions between linguistics and mathematical education' held in Nairobi (Kenya, 1-11 September

1974). Analyses the relation between the learning of mathematics and the language through which it is learnt. The report analyses the situation in several countries of Anglophone Africa. Cf. CHI-74, COLL-74, MMA-74, YOH-74.

Includes the following papers:

* R. Morris: Linguistic problems encountered by contemporary curriculum development projects in mathematics (25-58)
* P. Strevens: Mutual concerns between teachers of mathematics and areas of linguistics (59-63)
* M. Halliday: Some aspects of sociolinguistics (64-73)
* R. Clark: Some aspects of psycholinguistics (74-81)
* J. Gay: Pedagogical implications (82-84).

V

VAH-94
1994 Vahabzadeh, B.: Two commentaries on Euclid's definition of proportional magnitudes, *Arabic Sciences and Philosophy*, New York (USA), Vol. 4, No. 1, 181-198.

VAQ-99
1999 Vaquero Martinez, José: Review of Gerdes' & Bulafo's *Sipatsi: Technology, Art and Geometry in Inhambane* (GER-94c), *LLULL, Revista de la Sociedad Española de Historia de las Ciencias y de las Técnicas*, Zaragoza (Spain), Vol. 22, No. 45, 943-944 (in Spanish).

VEL-82
1982 Vellard, Dominique: *Pratiques de calcul et opérations logiques en millieu traditionnel africain (exemples maliens et rwandais)* [Practices of computation and logical operations in a traditional African environment (Examples from Mali and Rwanda)], doctoral thesis, Université de Paris VII (France) (in French).

VEL-84
1984 Vellard, Dominique: *Counting Practices of Illiterate Country People in West-Africa*, paper presented at the 4th International Congress on Mathematical Education, Adelaide (Australia), 4 p.

Describes the use of the traditional 'bamane' counting system by illiterate Bambara (Mali). It is a mixed counting system with bases 10, 80 and 800.

VEL-88
1988 Vellard, Dominique: Anthropologie et sciences cognitives: une étude des procédures de calcul mental utilisées par une population analphabète, *Intellectica*, Orsay (France), Vol. 2, No. 6, 169-209 (in French).

Analyses the cognitive processes used by the Bambara population of Mali when solving problems of mental calculation.

VEL-93
1993 Vellard, Dominique: *Numeration systems and colonisation. A case study in Mali* (paper presented at the 19th International Congress of History of Science, Zaragoza, Spain) (in French).

VELP-04
2004 Velpry, Christiaan: *Euclide l'Africain ou la géométrie restituée – enquête mathématique et historique* [Euclid the African or geometry restored – a mathematical and historical enquiry], Éditions, Menaibuc, Paris (France), 113 p. (in French).

Contains a collection of reflections on the "themes of geometry and logic, Euclid's postulate, history of geometry and philosophy from Alexandria to our days."

VER-81
1981 Vergani, Teresa: *Analyse numérique des idéogrammes Tshokwe de l'Angola* [Numerical analysis of Cokwe ideograms of Angola], doctoral thesis, University of Geneva (Switzerland), 388 p.

VER-86
1986 Vergani, Teresa: Aplicação da análise factorial das correspondências aos desenhos iniciáticos do povo Cokwe de Angola [Application of factor analysis to initiatory drawings of the Cokwe people of Angola], *Revista Internacional de Estudos Africanos*, Lisbon (Portugal), Vol. 4, 281-301 (in Portuguese).

This paper gives an application of factor analysis to the study of the symbolical expression of numbers in the Cokwe drawing tradition (Angola).

VER-99
1999 Vergani, Teresa: Ethnomathematics and symbolic thought. The culture of the Dogon, *ZDM, International Reviews on Mathematical Education*, Karlsruhe (Germany), No. 2, 66-70.

The paper deals with "the following aspects of the culture of the Dogon (Mali): the specific mythological context and the related cognitive system; fundamental poles in the Dogon numerical symbology; the density of 5's significance; the spiral as a choreography of thought; ethnomathematics 'logosymbols' as 'event'

and social meaning; educational implications (transcultural expression of thought and feeling)."

VERH-92
1992 Verheyen, Hugo: The icosahedral design of the Great Pyramid, in: Hargittai, Istvan (Ed.), *Fivefold symmetry*, World Scientific, Singapore, 333-360.

VERN-51
1951 Vernet, Juan G.: *Contribucion al estudio de la labor astronomica de Ibn al-Bannâ* [Contribution to the study of the astronomic work of Ibn al-Bannâ], Editora Marroqui, Tetouan (Morocco), 230 p. (in Arabic and Spanish) (Fac simile by F. Sezgin, Islamic Mathematics and Astronomy, Frankfurt (Germany), Volume 43, 1998).

Partial critical edition in Arabic, Spanish translation and commentaries of "The student guide for the correction of the movements of the stars" of Ibn al-Bannâ.

VERN-58
1958 Vernet, Juan G.: Les manuscrits astronomiques d'Ibn al-Bannâ [The astronomical manuscripts of Ibn al-Bannâ], *Actes du VIIIe Congrès International d'Histoire des Sciences (Florence-Milan, 3-9 septembre 1956)*, Paris (France), 297-298.

VERR-00
2000 Verran, Helen: Accounting Mathematics in West Africa: Some Stories of Yoruba Number, in: SEL-00, 345-371.

VERR-01
2001 Verran, Helen: *Science and an African logic*, The University of Chicago Press, Chicago IL (USA), 277 p.

The author, who taught at Obafemi Awolowo University in Ile-Ife (Nigeria) between 1979 and 1986, reflects on how science, mathematics, and logic come to life in Yoruba primary schools. She describes how she "went from the radical conclusion that logic and math are culturally relative… to a new understanding of all generalizing logic."

Review: ASC-03.

VIS-85a
1985 Visser, Delene: *Vroue en wiskunde: fokus op geslagsverskille* [Women and mathematics: focus on gender differences], Raad vir Geesteswetenskaplike Navorsing, Pretoria (South Africa), 352 p. (in Afrikaans).

VIS-85b
1985 Visser, Judithe Delene: *Geslagsverskille in deelname aan wiskunde en wiskundeprestasie* [Gender differences in the participation in mathematics and mathematics achievement], doctoral thesis, University of South Africa, Pretoria (South Africa) (in Afrikaans).

VITH-93
1993 Vithal, Renuka: *The Construct of Ethnomathematics, an its implications for Curriculum Thinking in South Africa*, Masters thesis, University of Cambridge, Cambridge (UK).

VIT-93
1993 Vitrac, Bernard: *De quelques questions touchant au traitement de la proportionnalité dans les Eléments d'Euclide* [On some questions dealing with the treatment of proportionality in Euclid's Elements], doctoral thesis, Ecole des Hautes Etudes en Sciences Sociales, Paris (France), 1211 p. (in French).

The thesis is in seven parts: 1. Inventory of problems. Historiography (1-64). 2. Foundations of proportionality (65-224). 3. Manipulations and uses of proportions (225-572). 4. The history of the theory of proportions. Critical analysis (573-679). 5. Appendices, general bibliography and index (697-800). 6. Document 1: French translation of Books V to IX of Euclid's Elements (175 p.). 7. Document 2: Other translations (216 p.).

VIT-95a
1995 Vitrac, Bernard: Review of Aujac's "La Sphère" (AUJ-95), *Historia Mathematica*, New York (USA), Vol. 22, 196-202.

VIT-95b
1995 Vitrac, Bernard: Review of P. Tummers' "Anaritius' Commentary on Euclid. The Latin Translation, I-IV", *Historia Mathematica*, New York (USA), Vol. 22, 445-446 (in French).

VIT-95c

1995 Vitrac, Bernard: Euclide et Héron: Deux approches de l'enseignement des mathématiques dans l'Antiquité? [Euclid and Heron: Two approaches to mathematics education in Antiquity?], in: Gilbert Argoud (Ed.), *Science et vie intellectuelle à Alexandrie (I^e-IIIe siècle après J.C.)*, Centre Jean Palerne, Publications de l'Université de Saint-Etienne, Saint-Etienne (France), 121-145 (in French).

VIT-96

1996 Vitrac, Bernard: La Définition V. 8 des *Eléments* d'Euclide [Definition V.8 of Euclid's *Elements*], *Centaurus*, Copenhagen (Denmark), Vol. XXXVIII, No. 2-3, 97-121 (in French).

VIT-97

1997 Vitrac, Bernard: Théon d'Alexandrie et la Mesure du cercle d'Archimède [Theon of Alexandria and the circle measurement of Archimed], *Oriens-Occidens*, Paris (France), No. 1, 41-81 (in French).

VIT-99a

1999 Vitrac, Bernard: Les antécédents grecs du troisième chapitre du commentaire sur "Certaines prémisses problématiques du Livre d'Euclide" [The Greek antecedents of the third chapter of the commentary on "Certain problematic premises of Euclid's book"], *Farhang. Quarterly Journal of Humanities & Cultural Studies*, Teheran (Iran), Vol. 12, No. 29-32, 51-105 (in French).

VIT-99b

1999 Vitrac, Bernard: Review of Caveing's book "Essai sur le savoir mathématique dans la Mésopotamie et l'Egypte anciennes" (CAV-94), *Revue d'Histoire des Sciences*, Paris (France), Vol. 52, No. 2, 307-314 (in French).

VIT-00

2000 Vitrac, Bernard: Euclide, in: R. Goulet (Ed.): *Dictionnaire des philosophes antiques*, *III, d'Eccélos à Juvénal*, CNRS, Paris (France), 252-272 (in French).

VIT-02
2002 Vitrac, Bernard: Umar al-Khayyâm et l'anthyphérèse: Etude du deuxième Livre de son commentaire "Sur certaines prémisses problématiques du Livre d'Euclide" [Umar al-Khayyâm and the antithesis: Study of the second Book of his commentary "On certain problematic premises of Euclid's book"], *Farhang. Quarterly Journal of Humanities & Cultural Studies*, Teheran (Iran), Vol. 14, No. 39-40, 137-192 (in French).

VIT-04a
2004 Vitrac, Bernard: A propos des démonstrations alternatives et autres substitutions de preuves dans les Eléments d'Euclide [About alternative demonstrations and other substitutions of proofs in Euclid's *Elements*], *Archive for History of Exact Sciences*, Berlin (Germany) Vol. 59, No. 1, 1-44 (in French).

VIT-04b
2004 Vitrac, Bernard: Les géomètres de la Grèce antique [The geometers in Ancient Greece], in: *Les génies de la science*, Paris (France), No. 21, November 2004 - February 2005, 29-99.

Contains the following papers: Invention of geometry: an enigma (30-37); A first scandal in geometry? (38-45); The Alexandrian mathematical tradition (46-51); Euclid, the founder (52-59); Measure and prove (60-65); Construct and compare (66-71); Archimedes (72-81); The Roman conic and the contribution of Apollonius (82-89); The renewal of Alexandria (90-95); The end of the Alexandrian world (96-99).

VOG-30
1930 Vogel, Kurt: The truncated pyramid in Egyptian mathematics, *Journal of Egyptian Archaeology*, London (UK), Vol. 16.

VOG-59
1959 Vogel, Kurt: *Vorgriechische Mathematik* [Pre-Greek Mathematics], Vol. 1: *Vorgeschichte und Ägypten* [Prehistory and Egypt], H. Schroedel Verlag, Hannover (Germany) (in German).

VOG-70
1970 Vogel, Kurt: *Die Grundlagen der Ägyptischen Arithmetik, in ihrem Zusammenhang mit der 2/n tabelle des Papyrus Rhind*

[Foundations of Egyptian arithmetic in its relationship with the 2:n table of the Rhind Papyrus], M. Söndig, Wiesbaden (Germany), 211 p. (in German).

Originally a doctoral dissertation from 1929 (Beckstein, Munich, Germany).

VOGE-99
1999 Vogeli, Bruce: *US involvement in African Mathematics education development in historical perspective*, paper presented at the Columbia Workshop on Mathematics and Mathematics Education in Africa, Columbia University, New York (USA), November 13.

VOGEL-92
1992 Vogeli, Erich Daniel: *The ethnomathematics of southern Africa: Application in the middle school mathematics classroom*, doctoral thesis, Columbia University, New York (USA).

VOL-94
1994 Volmink, John: Mathematics by All, in: Lerman, S. (Ed.) *Cultural Perspective on the Mathematics Classroom*, Kluwer, Dordrecht (Netherlands), 51-68.

Analyses the South African mathematics education context.

VOR-83
1983 Vorbichler, Anton: Zahlensysteme des Balese-Obi und des Mamwu (Mangbetu-Efe-Gruppe der zentralsudanesischen Sprachen [Number systems of the Balese-Obi and Mamwu (Mangbetu-Efe-Group of the Central African languages)], *Afrika und Übersee*, Berlin (Germany), Vol. 66, 131-140 (in German).

Comparative study of numeration in the Balesi-Obi (decimal) and Mamwu languages spoken in Northeastern Congo (Zaire), based on data collected in the period 1954-1960. In the Mamwu language exist basic number words for 1, 2, 3, 4, 5, and 10, and by using also the terms *elí* (hand), *qarú* (foot) and *múdo* (human, 20), the cardinals are formed.

W

WAE-37
1937 Waerden, Bartel L. van der: Arithmetik und Rechentechnik der Ägypter [Arithmetic and technique of computation of the Egyptians], *Berichte der Sächischen Akademie*, Leipzig (Germany), Vol. 89, 171-172 (in German).

WAE-38
1938 Waerden, Bartel L. van der: Die Entstehungsgeschichte der ägyptischen Bruchrechnung [The genesis of the Egyptian arithmetic of fractions], *Quelen und Studien zur Geschichte der Mathematik*, Berlin (Germany), Vol. B4, 359-382 (in German).

WAE-54
1954 Waerden, Bartel L. van der: *Science Awakening*, Vol. 1: *Egyptian, Babylonian and Greek Mathematics*, Wolters, Groningen (Netherlands) [Reprint: Kluwer, Dordrecht (Netherlands), 1988], 306 p.

The following sections deal with directly with mathematics in Egypt: The Egyptians (15-36), The Alexandrian Era (330-200 BC) (201-263), The decay of Greek mathematics (264-291).

WAE-74
1974 Waerden, Bartel L. van der: Review of Parker's Demotic Mathematical Papyri (PAR-71), *Isis*, Vol. 65, No. 226, 110-111.

WAE-80
1980 Waerden, Bartel L. van der: The (2:n) Table in the Rhind Papyrus, *Centaurus*, Copenhagen (Denmark), Vol. 24, 259-274.

WAE-83
1983 Waerden, Bartel L. van der: *Geometry and Algebra in Ancient Civilizations*, Springer, Berlin (Germany), 223 p.

Several sections of the book deal with mathematics in Egypt: The Moscow papyrus (44), Diophantus and his predecessors (97-112), Egyptian problems (160-161), Mathematical papyri from Hellenistic

Egypt (164-170), An Ancient Egyptian rule for squaring the circle (170-172), Heron of Alexandria (181-188).

WAGN-83
1983 Wagner, R .J.: Euclid's intended interpretation of superposition, *Historia Mathematica*, New York (USA), Vol. 10, No. 1, 63-70.

WAL-65
1965 Wallman, Sandra: The communication of measurement in Basutoland, *Human Organization: Journal of the Society for Applied Anthropology*, Washington DC (USA), Vol. 24, No. 3, 11 p.

Analyses mal-communication of measurement (area, length) in Lesotho, involving Sesotho and English.

WAS-88
1988 Washburn, Dorothy & Crowe, Donald: *Symmetries of Culture, Theory and Practice of Plane Pattern Analysis*, University of Washington Press, Washington DC (USA), 312 p.

Shows how patterns from cultures from all over the world, can be classified according to the symmetries, which generate them. It examines a number of patterns from African contexts.

WAS-90
1990 Washburn, Dorothy K.: *Style, Classification and Ethnicity: Design Categories on Bakuba Raffia Cloth*, Transactions of the American Philosophical Society, Philadelphia (USA), Vol. 80, Part 3, 157 p.

"The study shows that while two kinds of features are used for category definition (object-specific features and basic perceptual properties) the style of a culture is primarily defined by the way the basic properties are specifically manipulated. This thesis is illustrated by a study of named pattern categories on Bakuba raffia cloth. One of the basic perceptual properties is symmetry. Chapter 5 details how a symmetry analysis of the raffia patterns can differentiate patterns produced by the different Bakuba groups."

WATE-93
1993 Waterhouse, William C.: Harmonic means and Diophantus I. 39, *Historia Mathematica*, New York (USA), Vol. 20, No. 1, 89-91.

WAT-86
1986 Watson, Helen: Applying numbers to nature: a comparative view in English and Yoruba, *The Journal of Culture and Ideas*, Vol. 2, No. 3, 1-26

WAT-87
1987 Watson, Helen: Learning to apply numbers to nature: a comparison of English speaking and Yoruba speaking children learning to quantify, *Educational Studies in Mathematics*, Dordrecht (Netherlands), Vol. 18, 339-357.

WAW-91
1991 Waweru, Gachuhi & Koske, J. K.: The understanding of the operations of addition, multiplication, division, and subtraction amongst high and low ability primary school students in Kenya, *Kenya Journal of Education*, Nairobi (Kenya), Vol. 5, No. 1, p. 75-83.

WEB-67
1967 Webb, N. G. G.: Some problems in the introduction of the School Mathematics Project of East Africa in Form I, *Tanzanian Mathematics Bulletin*, Dar es Salaam (Tanzania), Vol. 2, No. 1, 13-20.

Among the problems discussed are those of the pupil's background and the effect on pupils of the change of teaching method, both related to the culture and tradition of Tanzania.

WEI-78
1978 Weil, A.: Who betrayed Euclid? : Extract from a letter to the editor, *Archive for History of Exact Sciences*, Berlin (Germany), Vol. 19, No. 2, 91-93.

WEU-21
1921 Weule, Karl: *Die Anfänge der Naturbeherschung, Vol.1: Frühformen der Mechanik* [The beginnings of the control of

nature, Vol. 1: Early forms of mechanics], Kosmos, Stuttgart (Germany), 76 p. (in German).

Studies early forms of knowledge of mechanics as embodied for instance in the making traps. Includes, in particular, examples from East Africa.

WHI-88
1988 Whitcombe, Allan & Donaldson, Maureen: Shongo networks, a multicultural theme for the classroom, *Mathematics in School*, Leicester (UK), November, 34-38.

Suggests the use of graphs drawn traditionally in the sand by children of the Shongo – one of the [Ba]Kuba groups of Congo / Zaire – in the mathematics classroom.

WHIT-01
2001 White, Dorothy Y.: Kenta, Kilts, and Kimonos: Exploring Cultures and mathematics through Fabrics, *Teaching Children Mathematics*, NCTM, Reston VA (USA), Vol. 7, No. 6 (Focus issue: Mathematics and Culture), 354-361.

Shows, among other examples, how *kenta* cloth from West Africa may be explored in a geometry lesson.

WILD-75
1975 Wilder, Raymond: Review of C. Zaslavsky's *Africa Counts* (ZAS-73a), *Historia Mathematica*, New York (USA), Vol. 2, 207-210.

WIL-78
1978 Williams, Awadagin: Change in Mathematics Education since the late 1950's - ideas and realisation: Sierra Leone, *Educational Studies in Mathematics*, Dordrecht (Netherlands), Vol. 9, No. 3, 297-302.

WILA-71
1971 Williams, Grace Alele: The Entebbe Mathematics Project, *International Review of Education*, UNESCO, Hamburg (Germany), Vol. 17, No. 2, 210-214.

WILA-74
1974 Williams, Grace Alele: Dynamics of curriculum change in mathematics: Lagos State Mathematics Project, *West African Journal of Education*, Vol. 18, No. 2, 241-253.

WILA-76
1976 Williams, Grace Alele: The development of a modern mathematics curriculum in Africa, *The Arithmetic Teacher*, Reston VA (USA), No. 4, 254-261.

The papers WILA-71, WILA-74, and WILA-76 deal with the African Mathematics Program.

WILA-93
1993 Williams, Grace Alele: Mathematics and Administration: a curious mix for education leadership, in: *Science in Africa: Women Leading from Strength*, American Association for the Advancement of Science (AAAS), Washington DC (USA), 19-26.

WILL-43
1943 Williamson, John: Dabida numerals, *African Studies*, Johannesburg (South Africa), Vol. 2, 215-216.

"While searching for Dabida ways of using arithmetic, for the purpose of making the early studies of young children easier and more interesting, it was discovered that several sets of 'numerals' exist." These sets are described. "A counting system reputed to be much older than those in use today is still used by children in their early arithmetic work; they can sometimes be heard repeating these numbers with the aid of their fingers." The Dabida inhabit the Taita hills in Kenya.

WILLI-70
1970 Williamson, Kay & A. O. Timitimi: A note on Ijo number symbolism, *African Notes*, Institute of African Studies, Ibadan (Nigeria), Vol. 5, No. 93, 9-16.

"Among the Kolokuma Ijo of the Niger Delta odd numbers in general, and three in particular, are associated with men; while even numbers in general, and four in particular, are associated with women. The number seven is associated with the great divinities of the clan, such as Kolokuma Egbesu, and is therefore normally avoided." The paper gives examples.

WILS-80
1980 Wilson, Bryan: *A review of secondary school mathematics in English-speaking countries of Africa*, African Mathematical Union, Rabat (Morocco).

"This survey covers the period 1962-1978. It shows how the historical processes of decolonization and the emergence of a Commonwealth of sovereign nations has led to a remarkable uniformity of mathematics curricula in secondary schools in the Anglophone countries of east, West and Central Africa. Three curriculum projects of major influence – 'Entebbe', SMP and JSP – are considered in more detail. The emergence of the African Mathematical Union is welcomed as an attempt, on a professional plane, to bridge the political gulf between Anglophone and Francophone African countries."

WILS-81
1981 Wilson, Bryan: The African Education Program (Mathematics, Science) of the American Education Development Center, in: B. Wilson, *Cultural contexts of Science and Mathematics Education. A bibliographic guide*, Centre for Studies in Science Education, University of Leeds, Leeds (UK), 195-199.

Presents a short historical overview of the African Mathematics Program and related curriculum development programs and materials, like the Entebbe Mathematics Series (1963-1968), Kenya Primary Mathematics (1969-1976), Ghana Mathematics Series (1975-1979), and the New Mathematics Series (Sierra Leone, 1974-1979).

WILSO-94
1994 Wilson, Eva: The interlacing and geometrical art of the Kuba, in: Eva Wilson, *Ornament 8,000 years*, Harry N. Abrams, New York / British Museum Press, London (UK), 195-196.

Short note on (a)symmetries in Kuba art.

WOL-54
1954 Wölfel, D.: Les noms de nombre dans le parler Guanche des Isles Canaries [The number words of the Guanche of the Canary Islands], *Hespéris*, Paris (France), Vol. 41, 47-79 (in French).

Y

YAD-71
1971 Yadegari, Mohammad & Levey, Martin: *Abu Kamil's "On the pentagon and decagon*," History of Science Society of Japan, Tokyo (Japan), 53 p.

YAD-78
1978 Yadegari, Mohammad: The use of mathematical induction by Abû Kâmil Shuja ibn Aslam (850-930), *Isis*, Madison WI (USA), Vol. 69, No. 247, 259-262.

YAS-73
1973 al-Yasin M. H.: New proof of the Arabicity of the ciphers used in the Arab Maghreb, *Al-Lisân al-* ^{c}*arabî*, Rabat (Morocco), Vol. 10, Part 1, 231-233 (in Arabic).

YAS-80
1980 al-Yasin M. H.: The Arab ciphers in their state and in their circulation, *Al-Lisân al-* ^{c}*arabî*, Rabat (Morocco), No. 12, 42-49 (in Arabic).

YOH-74
1974 Yohannes, G. M.: *Linguistic problems in mathematics education in Ethiopia*, UNESCO (ED-74/CONF.808/18), Paris (France), 12 p.

Paper presented at the UNESCO Symposium on 'Interactions between Linguistics and Mathematical Education' (Nairobi, Kenya, 1-11 September 1974). Discusses the problems of learning mathematics in a system in which Amharic is the medium of primary education, and English that of secondary education.

YOU-76
1976 Youschkevitch, Adolf P.: *Les mathématiques arabes (VIII*^{e}*-XV*^{e} *siècles)* [Arab mathematics (8th – 15th century)], Vrin, Paris (France), 213 p.

YUS-95
1995 Yussupova, Gulnava: Zwei mittelalterliche arabische Ausgaben der *Sphaerica* des Menelaos von Alexandria, *Historia Mathematica*, New York (USA), Vol. 22, 64-66.

A description of two Arabic texts with commentaries on Menelaus' *Sphaerica*, one written by At-Tûsî (Persia) in the 13th century and the other by the 17th-century mathematician Al-Yazdî (Persia).

Z

ZAS-70a
1970a Zaslavsky, Claudia: Black African traditional mathematics, *The Mathematics Teacher*, Reston VA (USA), Vol. 63, No. 4, 345-356.

Overview of various number systems in Africa.

ZAS-70b
1970b Zaslavsky, Claudia: Mathematics of the Yoruba people and of their neighbors in Southern Nigeria, *Two-Year College Mathematics Journal*, Washington DC (USA), Vol. 1, 76-99.

ZAS-73a
1973a Zaslavsky, Claudia: *Africa Counts: Number and Pattern in African Culture*, Prindle, Weber & Schmidt Inc., Boston MA (USA), 328 p. (Paperback edition: Lawrence Hill, Westport, Connecticut (USA)).

Already classical introduction to the mathematical heritage of Africa south of the Sahara. Includes chapters on 'Numbers-words, gestures, significance', 'Numbers in daily life', 'Mathematical recreations', 'Pattern and shape', and two regional studies on southwest Nigeria and East Africa. Bibliography with 191 references.

Review: WILD-75.
Latest edition: ZAS-99a.
Translations: ZAS-84, ZAS-95.

ZAS-73b
1973b Zaslavsky, Claudia: Mathematics in the study of African culture, *The Arithmetic Teacher*, Reston VA (USA), Vol. 20, 532-535.

Presents some examples for classroom use from the (Ba)Kuba culture (Congo / Zaire) and from cowrie shells currency in West Africa.

ZAS-75
1975 Zaslavsky, Claudia: African network patterns, *Mathematics Teaching*, London (UK), Vol. 73, 12-13.

ZAS-76a
1976a Zaslavsky, Claudia: The Afro-American mathematical heritage, *Outlook*, Washington DC (USA), Vol. 20, 3-8.

ZAS-76b
1976b Zaslavsky, Claudia: African stones, *Teacher* (USA), Vol. 94, No. 2, 110-112.

ZAS-76c
1976c Zaslavsky, Claudia: African numbers, *Teacher* (USA), Vol. 94, No. 3, 91-96.

ZAS-79
1979 Zaslavsky, Claudia: Symmetry along with other mathematical concepts and applications in African life, in: *Applications in School Mathematics*, National Council of Teachers of Mathematics, Reston VA (USA), 82-97.

Examples of bilateral and rotational symmetries, repeated patterns on a strip, tessellations in the plane, occurring in African art, architecture and design (e.g. *adinkra* cloth of the Asante people, Ghana; *adire* cloth of the Yoruba people, Nigeria) are given and it is shown how these examples may be integrated in an interdisciplinary approach to the study of mathematics.

ZAS-80
1980 Zaslavsky, Claudia: *Count on your fingers African style,* Harper & Row, New York (USA), 33 p. (Latest edition: ZAS-99b).

The book "guides children (ages 6-9) through the animated activity of the marketplace, showing the traditional finger counting of various African peoples – the Maasai, the Kamba, and the Taita in Kenya; the Zulu of South Africa; and the Mende of Sierra Leone."

ZAS-81
1981 Zaslavsky, Claudia: Networks—New York subways, a piece of string, and African traditions, *The Arithmetic Teacher*, Reston VA (USA), Vol. 29 (October), 42-47.

Graph theoretical analysis for school children of the networks drawn by the Kuba of Congo.

ZAS-82
1982 Zaslavsky, Claudia: *Tic Tac Toe and other three-in-a-row games, from Ancient Egypt to the modern computer*, Harper & Row, New York (USA) & Fitzhenry & Whiteside, Toronto (Canada), 96 p.

"Games suitable for all ages, reading level ages 9-12." Includes several African versions: Achi (Ghana), Shisiba (Kenya), Murabaraba (Lesotho), Dara (Mali, Morocco, Niger, Nigeria), Akidada (Nigeria), Tsoro Yematatu (Zimbabwe).

ZAS-84
1984 Zaslavsky, Claudia: *Africa Szaniol*, Gondalet, Budapest (Hungary), 350 p.

Hungarian translation of ZAS-73a.

ZAS-89a
1989a Zaslavsky, Claudia: People who live in round houses, *The Arithmetic Teacher*, Reston VA (USA), September, 18-21.

Gives information on the tradition of round houses in Africa and other parts of the world with suggestions for incorporating this issue in the mathematics classroom.

ZAS-89b
1989b Zaslavsky, Claudia: Mathematical aspects of traditional African games, *AMUCHMA Newsletter*, Maputo (Mozambique), Vol. 3, 6.

ZAS-93
1993 Zaslavsky, Claudia: *Multicultural Mathematics: Inter-disciplinary Cooperative-Learning Activities*, J. Weston Walch, Portland ME (USA), 158 p.

Activities for middle grade students, involving ancient Egyptian numeration and computation, cowrie shell and other currency in West Africa, the African slave Thomas Fuller, Egyptian pyramids, probability with cowry shells and the Nigerian game of Igba Ita, and Chokwe and Kuba networks.

ZAS-94a
1994a Zaslavsky, Claudia: *Africa Counts* and Ethnomathematics, *For the Learning of Mathematics*, Montreal (Canada), Vol. 14, No. 2, 3-8.

A description of the motivation for and some of the research leading to the author's classic ZAS-73a.

ZAS-94b
1994 Zaslavsky, Claudia. Mathematics in Africa: Explicit and implicit, in GRA-94, Vol. 1, 85-92.

Mathematics in ancient Africa, African mathematics in the Arabic language, and mathematics "frozen" in the practices of many African societies.

ZAS-95
1995 Zaslavsky, Claudia: *L'Afrique compte! Nombres, formes et démarches dans la culture africaine*, Éditions du Choix, Argenteuil (France), 328 p.

French language edition of ZAS-73a.

ZAS-96
1996 Zaslavsky, Claudia: *The Multicultural Math Classroom: Bringing in the World*, Heinemann, Portsmouth (USA), 288 p.

Pleads for a multicultural mathematics curriculum and presents examples of mathematical activities for use in the classroom, including many examples from Africa.

ZAS-98
1998 Zaslavsky, Claudia: *Math Games and Activities from around the World*, Chicago Review Press, Chicago IL (USA), 146 p.

Book for children for ages 9 and up. Includes several examples of mathematical games or activities from Africa, like: [three-in-a-row games] *Shisima* from Kenya (4-5), *Tsoro yematatu* from Zimbabwe (8-9), *Dara* from Nigeria (18-19); [Mankala board games] Easy *oware* from Ghana (22-23), The real *oware* game from Ghana (24-25), *Giuthi* from Kenya (28-29); [More board games] *Yoté* from West Africa (42-43); [Games of chance] *Igba-ita* from Nigeria (52-53); [Puzzles with numbers] Magic squares from West Africa (64-65), Dividing the camels from North Africa (73-74), The Ishango bone from Congo

(75); [Puzzles without numbers] Crossing the river in Liberia (81), Crossing the river with jealous husbands from Kenya (82), The snake and the swallow's nest from Angola (84), The Chokwe story tellers from Angola (85-86), Decorations on the walls from Angola (87), How the world began from Angola (88-89), Children's networks from Congo (90-91); [Geometry all around us] Round houses in Kenya (100), Cone-cylinder houses in Kenya (101-102), The pyramids of ancient Egypt (105-106); [Repeating patterns] African patterns from Congo (127-129), *Adinkra* cloth from Ghana (133-134).

Translations: ZAS-00b, ZAS-02.

ZAS-99a
1999 Zaslavsky, Claudia, *Africa Counts: Number and Pattern in African Cultures*, Third edition, Lawrence Hill, Chicago IL (USA), 368 p.

Reprint of Claudia Zaslavsky's classical study ZAS-73a, updated with an additional chapter on ethnomathematics in Africa.

ZAS-99b
1999 Zaslavsky, Claudia: *Count on your fingers African style*, Black Butterfly Children's Books, New York (USA), 42 p. (illustrations by Wangechi Mutu).

New edition of ZAS-80.

ZAS-00a
2000 Zaslavsky, Claudia: African networks and African-American students, in: Marilyn Strutchens, Martin Johnson & William F. Tate (Eds.), *Changing the Faces of Mathematics: Perspectives on African Americans,* National Council of Teachers of Mathematics, Reston VA (USA), 157-166.

The appeal of such activities to African-American students at various grade levels, based on actual classroom experiences.

ZAS-00b
2000 Zaslavsky, Claudia: *Jogos e Atividades Matemáticas do Mundo Inteiro*, Editora Artes Médicas Sul, Porto Alegre (Brazil), 155 p.

Translation into Portuguese of ZAS-98 by Pedro Theobald.

ZAS-00c
2000 Zaslavsky, Claudia: Review of Gerdes' *Geometry from Africa* (GER-99a), *Humanistic Mathematics Network Journal*, Claremont CA (USA), Vol. 23, 55-57.

ZAS-02
2002 Zaslavsky, Claudia: *Math Games and Activities from around the World*, Yuan T. Lee Foundation, Taipei (Taiwan), 154 p. (in Chinese).

Chinese language edition of ZAS-98.

ZAS-03a
2003 Zaslavsky, Claudia: *More Math Games and Activities from Around the World*, Chicago Review Press, Chicago IL (USA), 160 p.

Sequel to ZAS-98. For children age nine and up. Includes the following games and activities from Africa: [Three-in-a-Row Games] *Achi* from Ghana (14-15), *Murabaraba* from South Africa and Lesotho (23-25); *Alquerque de Nueve* from Muslim Spain and North Africa and *Akidada* from Nigeria (18-20); [More Board Games: Mankala] Little Goat Game and Cow Game from Sudan (35-38), *Adi* from Ghana (39-41); [How People Use Numbers: Money] Beads, Shells and Gold from Africa (56-57); [Is There a Lucky Number?] Magic Squares from the Muslim World (70-71); [How People Measure] Standard Measures from Ancient Egypt (81); [Puzzles with Dots, String, and Paper Strips] *Julirde* from West Africa (91-93), Bead and String Puzzle from West Africa (94-95), Animal Picture and "Three Villages" Sand Drawings from Angola (98-103); [Symmetry and Similarity of Designs] *Akua Ba* Doll from Ghana (114-115); [Repeated Patterns] *Adire* Cloth from Nigeria (138-139).

ZAS-03b
2003 Zaslavsky, Claudia: Review of Gerdes' *Awakening of Geometrical Thought in Early Culture* (GER-03a), *AMUCHMA Newsletter*, Maputo (Mozambique), No. 27, 14-15; *History and Pedagogy of Mathematics Newsletter*, Romsey (UK), No. 53, 9-10.

ZAS-03c
2003 Zaslavsky, Claudia: The Influence of Ancient Egypt on Greek and Other Numeration Systems, *Mathematics Teaching in the Middle School,* NCTM, Reston VA (USA), Vol. 9, No. 3, 174-178.

"The article traces the development of the alphabetic numeration systems of the early Greeks, Hebrews, and Arabs to the concepts underlying ancient Egyptian hieratic numeration, and includes activities for students."

ZEL-00
2000 Zekele, Seleshi: Gender differences in mathematics achievement: a search for explanantions, *Zimbabwe Journal for Educational Research*, Harare (Zimbabwe), Vol. 12, No. 1, 100-118.

Based on the responses of secondary school students in a rural context - North Shoa, Ethiopia - the author investigates gender differences in mathematics achievement and attitudes. "…a significant gender difference was found in mathematics achievement, but not in attitude."

ZEL-01
2001 Zeleke, Seleshi: Gender differences in mathematics performance in the elementary grades: implications for women's participation in scientific and technical occupations, *Eastern Africa Social Science Research Review*, Addis Ababa (Ethiopia), Vol. 17, No. 2, 109-127.

Examines "gender differences in mathematics achievement among fifth and sixth grade students in Addis Ababa (Ethiopia) and identifies factors that account for variations in their performance."

ZEM-93
1993 Zemouli, Touhami: *Les écrits mathématiques d'Ibn al-Yâsamin (m. 1204* [The mathematical wrtings of Ibn al-Yâsamin (d. 1204)], "Magistère" thesis, École Normale Supérieure, Alger (Algeria), 349 p.

ZEP-82a
1982 Zepp, Raymond: Bilinguals' understanding of logical connectives in English and Sesotho, *Educational Studies in Mathematics*, Dordrecht (Netherlands), Vol. 13, 205-221.

ZEP-82b
1982 Zepp, Raymond: Correlation of English, mathematics, and science in a Lesotho high school, *Journal of Southern African Studies*, Roma (Lesotho), Vol. 1, 8-11.

ZEP-83a
1983 Zepp, Raymond: Inclusive disjunction in West African languages, *Perceptual and Motor Skills*, Missoula, Mont. (USA), Vol. 56, 322.

ZEP-83b
1983 Zepp, Raymond: A West African replication of the four card problem, *Journal of Cross-Cultural Psychology*, Beverly Hills CA (USA), Vol. 14, No. 3, 323-327.

ZEP-83c
1983 Zepp, Raymond: *L'apprentissage du calcul dans les langues de Côte d'Ivoire* [The learning of arithmetic in the languages of Ivory Coast], Institut de Linguistique Appliquée, Université d'Abidjan, Abidjan (Côte d'Ivoire), Vol. 99, 121 p.

ZHA-00
2000 Zhang, Xin Li: Ancient Egyptian Unit Fractions and their Calculation, *Journal of Liaoming Normal University. Natural Science*, Liaoming (China), Vol. 23, No. 3, 257-262 (in Chinese).

Presents an introduction to Egyptian unit fractions and their influence on other subjects.

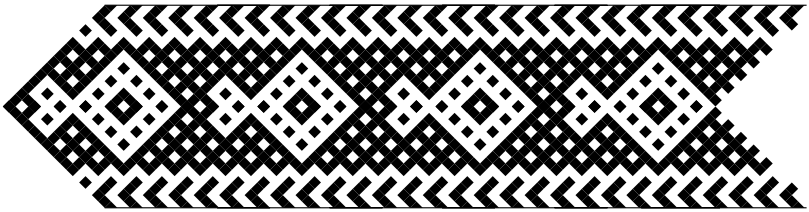
ZIT-97
1997 Zitarelli, David E.: *A collaborative approach to numbers*, Raymond-Reese Book Co., Wyncote PA (USA), 88 p.

Contains two modules on Africa: Number systems from Africa (1-8) and Unit fractions (37-44).

ZYL-42
1942 Zyl, Abraham Johannes van: *Mathematics at the cross-roads: a critical survey of the teaching of mathematics in the secondary schools of the Union of South Africa with suggestions for*

reorganization, doctoral thesis, Columbia University, New York (USA).

ZYL-43
1943 ZYL, Abraham Johannes Van (South Africa): *Mathematics at the cross-roads*, Maskew Miller, Cape Town (South Africa), 239 p.



Example of a woven strip design from Zanzibar (Tanzania)
(cf. GER-99a, p. 145)

APPENDICES

Appendix 1
On mathematicians of African descent / Diaspora

See also: EGL-95c, EGL-97b, LUM-87, LUM-92b, LUM-95b, LUM-95c, LUM-96.

1-AGW-03
2003 Agwu, Nkechi; Smith, Luell & Barry, Aissatou: Dr. David Blackwell, African American Pioneer, *Mathematics Magazine*, Washington DC (USA), Vol. 76, No.1, 3-14.

1-BAL-56
1956 Rouse Ball, W.: Calculating prodigies, in: J. Newman (Ed.), *The World of Mathematics*, Simon & Schuster, New York (USA), Vol. 1, 467-487.

Contains (p. 470) brief information on Thomas Fuller (1710-1790), born in Africa and brought as a slave to Virginia (USA) in 1724. Fuller was a prodigy in mental arithmetic. E.g. he could multiply nine-digit numbers.

1-BED-72
1972 Bedini, Silvio A.: *The Life of Benjamin Banneker*, Scribner, New York (USA), 434 p.

"Benjamin Banneker (1731-1806) was a famous member of the community of 'mathematical practitioners' in Colonial America. A landed freeman and tobacco planter, Banneker was introduced to astronomy and surveying during the 1780s, learning from the popular Newtonian texts of the period with the help of his neighbor George Ellicott. Banneker mastered methods for the calculation of ephemerides and incorporated his results in a series of almanacs published in Philadelphia, Baltimore, and other eastern cities between 1791 and 1796. During 1791 he served as astronomical assistant on the survey of the District of Columbia directed by Andrew Ellicott. Using all extant records concerning Banneker's life and a wide variety of other sources, Bedini has reconstructed the intellectual and social environment in which Banneker worked" [abstract published in:

American Studies, an annotated bibliography, Cambridge University Press, 1986].

1-CAMA-04

2004 Camara, Abdoulaye: Thomas Fuller (1710-1790) – Le grand calculateur [Thomas Fuller (1710-1790) – The great calculator] (online available at: www.africamaat.com article 92) (in French).

Short article based on 1-FAU-90a.

1-DEAN-98

1998 Dean, Nathaniel: *African Americans in Mathematics*, American Mathematical Society, Washington DC (USA), 205 p.

This book is the report of a Discrete Mathematics and Theoretical Computer Science (DIMACS) workshop (June 26-28, 1996). It includes the invited research talks by Jonathan D. Farley, Carolyn R. Mahoney, Curtis Clark, Walter M. Miller, Nathaniel Whitaker, Isom H. Hernon, Floyd L. Williams, and Scott W. Williams (cf. AMUCHMA 20:6.2), poster presentations, and the following historical articles:

* Lorch, Lee: Yesterday, today and tomorrow (157-168);
* Falconer, Etta: The challenge of diversity (169-182);
* Kenschaft, Patricia: What next? A meta-history of black mathematicians (183-186);
* Hill, Donald: A personal history of the origins of the National Association of Mathematicians' "Presentations by recipients of recent Ph.D.'s" (187-193);
* Agwu, Nkechi & Asamgah Nkwanta: Dr. J. Ernest Wilkins, Jr.: The man and his works (195-205).

1-DON-00

2000 Donaldson, James & Fleming, Richard: Elbert F. Cox: An Early Pioneer, *American Mathematical Monthly*, Washington DC (USA), Vol. 107, 105-128.

A biography of Elbert F. Cox, the first African-American to earn a Ph.D. in mathematics.

1-EGL-01

2001 Eglash, Ron & Bleecker, J.: The Race for Cyberspace: information technology in the black diaspora, *Science as*

Culture, Oxfordshire (UK), Vol. 10, No. 3 [available online at: www.rpi.edu/~eglash/eglash.dir/ethnic.dir/r4cyb.dir/r4cybh.htm]

"Focusing on the black Diaspora, this essay broadens the category of 'information technology' to show how traditions of coding and computation from indigenous African practices and black appropriations of Euro-American technologies have supported, resisted, and fused with the cybernetic histories of the west, and provide a strong source for changes in reconstructing identity, social position and access to power in communities of the black Diaspora."

1-FAU-90a

1990a Fauvel, John & Gerdes, Paulus: African Slave and Calculating Prodigy: Bicentenary of the Death of Thomas Fuller, *Historia Mathematica*, New York (USA), Vol. 17, 141-151.

Thomas Fuller (1710-1790) was an African, shipped to America as a slave in 1724. He had remarkable powers of calculation, and late in his life was discovered by antislavery campaigners who used him as a demonstration that blacks are not mentally inferior to whites. This paper describes what we know of Fuller, discusses the various uses made of his story since his death, and appeals for further study of the 18th-century African ethnomathematical context.

Translations: 1-FAU-90b, 1-FAU-92.

1-FAU-90b

1990b Fauvel, John & Gerdes, Paulus: Escravo africano e prodígio em cálculo: bicentenário da morte de Thomas Fuller, *Cadernos de História*, Maputo (Mozambique), No. 8, 103-116 (in Portuguese).

Translation of 1-FAU-90a.

1-FAU-92

1992 Fauvel, John & Gerdes, Paulus: Escravo africano e prodígio em cálculo: bicentenário da morte de Thomas Fuller, *AMUCHMA, Revista sobre a História da Matemática em África*, Maputo (Mozambique), No. 1, 1992, 37-48 (in Portuguese).

Translation of FAU-90a.

1-HAL-87
1987 Hall, E. R. & Post-Krammer, P.: Black mathematics and science majors: Why so few?, *The Career Development Quarterly* (USA), 207-219.

1-HAW-99
1999 Hawkins, William A.: *African and African-American Pioneers in Mathematics: Mathematics, the analysis of patterns and order, developed in Africa as did humankind*, SUMMA Program, The Mathematical Association of America, Washington DC (USA).

Poster with drawings of the first Ishango rod, photographs of the Ahmose' Papyrus ('Rhind' papyrus), and short biographies of Benjamin Banneker (1731-1806), Elbert Frank Cox (1895-1969), Evelyn Boyd Granville (b. 1924), Majorie Lee Browne (1914-1979), J. Ernest Wilkins, Jr. (b. 1923) and David H. Blackwell (b. 1919). The backside of the poster contains also a list of the earliest African-Americans with a doctorate in mathematics.

1-HER-29
1929 Herskovits, Melville: Adji-boto, an African game of the Bush-Negroes of Dutch Guiana, *Man*, London (UK), Vol. xxix, No. 90, 122-127 [reproduced in KOV-95].

1-HER-32
1932 Herskovits, Melville: Wari in the New World, *Journal of the Royal Anthropological Institute*, London (UK), Vol. 62, 23-37 [reproduced in KOV-95].

1-JOH-84
1984 Johnson, M. L.: Blacks in mathematics: A status report, *Journal for Research in Mathematics Education*, Reston VA (USA), Vol. 15, No. 2, 145-153.

1-KEN-81
1981 Kenschaft, Patricia: Black women in mathematics in the United States, *American Mathematical Monthly*, Washington DC (USA), Vol. 88, 592-602.

1-KEN-87

1987 Kenschaft, Patricia: Black men and women in mathematical research, *Journal of Black Studies*, Newburry Park CA (USA), Vol. 18, No. 2, 170-190.

The article is the result of numerous interviews with and letters from "leading black men and women in mathematics and their friends." It includes short biographies on the African-Americans who received a doctorate in mathematics.

1-KRAP-98

1998 Krapp, Kristine M. (Ed.): *Notable Black American Scientists*, Gale, Detroit (USA), 349 p.

1-NEW-80

1980 Newell, V., Gipson, J., Waldo Rick, L. & Stubblefield, B. (Eds.): *Black mathematicians and their works*, Dorrance & Company, Ardmore PA (USA), 327 p.

The first part of the book consists of scholarly articles published by North-American mathematicians of African descent. The second part is a biographical index of all mathematicians surveyed. Appendices are included, among them articles and letters concerning discrimination against blacks in the field of mathematics.

Review: 1-ZAS-83.

1-REDD-06

2006 Reddington, Luther V. (Ed.): *Trends in African Diaspora mathematics research*, Nova Science Publishers, Hauppauge, N.Y. (USA).

1-SPA-03

2003 Spangenburg, Ray *et al.*: *African Americans in Science, Math, and Invention*, Facts on File Inc., New York (USA), 254 p.

1-WILLIAM-99

1999 Williams, Scott W.: Black research mathematicians in the United States, *Contemporary Mathematics*, Vol. 252, Providence RI (USA), 165—168.

1-WILLIAM-03
2003 Williams, Scott: *Mathematicians of the African Diaspora* [available online at: www.math.buffalo.edu/mad/]

1-ZAS-83
1983 Zaslavsky, Claudia: Review of Newell's *Black Mathematicians and their works* (NEW-80), *Historia Mathematica*, New York (USA), Vol. 10, 105-115.

Appendix 2
Publications by African scholars on the History of Mathematics outside Africa (including reviews of these publications)

2-BOUD-98

1998 Boudine, Jean-Pierre & Djebbar, Ahmed: Omar Khayyam, le poète des maths [Omar Khayyam, the poet of mathematics], *Science et Vie Junior Special Math*, Paris (France), December 1998 - February 1999, 42-43 (in French).

2-DJE-93

1993 Djebbar, Ahmed: Deux mathématiciens peu connus de l'Espagne du XIe siècle: al-Mu'taman et Ibn Sayyid, in: Menso Folkerts & Jan Hogendijk (Eds.), *Vestigia Mathematica, Studies in medieval and early modern mathematics in honour of H. L. L. Busard*, Rodopi, Amsterdam (Netherlands), 79-91 (in French).

This paper contains the not previously published results of research conducted between 1982 and 1984, on the life and activities of two important mathematicians of Islamic Spain who were interested in Geometry and Number Theory.

2-DJE-98

1998 Djebbar, Ahmed: La folle histoire de l'algèbre [The extravagant history of algebra], *Science et Vie Junior Special Math*, Paris (France), December 1998 - February 1999, 34-47 (in French).

2-DJE-99a

1999a Djebbar, Ahmed: Les mathématiques dans l'Oeuvre d'Ibn Sina (Mathematics in the works of Avicenna), *Actes des Journées d'Etudes Avicenne (Marrakech (Maroc), 25-26 septembre 1998)*, Groupe d'Etude Ibn Sina (G.E.I.S.), Marrakech (Morocco), 51-70 (in French).

Presents the essential aspects of the contribution of the great philosopher and physician Avicenna (d. 1037) to the domains of mathematics and astronomy.

2-DJE-99b

1999b Djebbar, Ahmed: Les livres arithmétiques des *Éléments* d'Euclide dans le traite d'al-Mu'tanan du XIe siècle [The arithmetic books of Euclid's *Elements* in the study of al-Mu'tanan of the 11th century], *LLULL, Revista de la Sociedad Española de Historia de las Ciencias y de las Técnicas*, Zaragoza (Spain), Vol. 22, No. 45, 589-653 (in French).

"This paper studies the first chapter of *Kitab al-Istikmal*, a work of the 11th century by al-Mu'tanan Ibn Hud, a mathematician from al-Andalus who was the king of Zaragoza between 1081 and 1085. Different chapters of this remarkable work in the Arabic mathematical tradition have already been studied in the last decade, while others are still in progress."

2-DJE-00a

2000a Djebbar, Ahmed: Omar Khayyâm et les activités mathématiques en pays d'Islam aux XIe-XIIe siècles [Omar Khayyam and mathematical activities in the Islamic countries during the 11th –14th centuries], *Farhang*, Teheran (Iran), Vol. 12, No. 29-32, 1-31 (in French).

Paper dedicated to the life and work of Omar Khayam (d. 1131), in relationship with the scientific and cultural activities of his time.

2-DJE-00b

2000b Djebbar, Ahmed: Le nombre, la racine et le bien [Number, root and richness], *Les Cahiers de Science et Vie*, No. 56, 42-48 (in French).

The paper is addressed to high school and college students. It presents, in an anecdotal form, some information abut the birth of algebra and its development since the first Babylonian practices until the arrival of algebra in Europe from the 12th century onwards.

2-DJE-00c

2000c Djebbar, Ahmed: Un poète algébriste [An algebraist poet], *Les Cahiers de Science et Vie*, No. 56, 50-55 (in French).

The paper is addressed to young pupils and presents the life and works of the poet and mathematician Omar al-Khayyam (d. 1139).

2-DJE-02

2002 Djebbar, Ahmed: Le manuscrit (retrouvé) de Saragosse [The (rediscovered) manuscript of Saragossa], *Revue Alliage*, No. 47, 2002, 67-71 (in French).

Tells the story of the extraordinary fate of an important work by the mathematician and king of Zaragossa, al-Mu'taman (d. 1085), of its transmission from Europe to Asia passing through North Africa, and of its discovery — less than 20 years ago — by two researchers, Jan Hogendijk (Netherlands) and Ahmed Djebbar (Algeria).

2-DJE-05

2005 Djebbar, Ahmed: Kamâl Eddîn Fârsî, Physicien et mathématicien novateur [Kamâl Eddîn Fârsî, innovating physicist and mathematician], *Târikh-e'Elm*, Teheran (Iran), No. 3, 9-38 (in French).

2-GER-03

2003 Gerdes, Paulus: Níjtyubane — Sobre Alguns Aspectos Geométricos da Cestaria Bora na Amazónia Peruana [*Níjtyubane* — On some geometrical aspects of Bora basketry in the Peruvian Amazon], *Revista Brasileira de História da Matemática*, Rio Claro (Brazil), Vol. 3, No. 6, 3-22 (in Portuguese).

The paper discusses some geometrical aspects of Bora basketry in the Peruvian Amazon. In particular, twill-plaited, circular trays called 'níjtyubane' are analysed. Elements of their production and of the creation and transformation of geometric patterns are studied. An outline of their historical development is presented that stresses the similarity and the cultural diversity.

2-HIT-96

1996 Hitchcock, Gavin: A window on the history of mathematics, 1871: Reminiscences of De Morgan — A dramatic presentation, *Proceedings - Actes - Actas "História e Educação Matemática", ICME-8 satellite meeting of the International Study Group on the Relations between History and Pedagogy of Mathematics (HPM), Deuxième Université d'Été Européenne sur l'Histoire et Épistémologie dans l'Éducation Mathématique*, Associação de Professores de Matemática, Braga (Portugal), Vol. 2, 35-42.

Dramatic presentation of De Morgan's reminiscences at the end of his life, reflecting about the development of logic and algebra.

2-HIT-97

1997 Hitchcock, Gavin: Teaching the Negatives, 1870-1970: A Medley of Models, *For the Learning of Mathematics*, Vancouver (Canada), Vol. 17, No. 1, 17-25.

Six snapshots of important representative moments in the teaching of the negatives are represented in historical sequence as classroom scenes.

2-HOG-00

2000 Hogendijk, Jan: Review of R. Rashed & B. Vahabzadeh's *Al-Khayyam Mathématicien* (RAS-99), *Mathematical Reviews*, Lancaster PA (USA), 2000I:01013.

2-JAO-76

1976 Jaouiche, Khalil: *Le livre du Qarasan de Tabit Ibn Qurra: étude sur l'origine de la notion de travail et du calcul du moment statique d'une barre homogène*, doctoral thesis, Université de Paris 4 (France) (in French).

2-OGU-88

1988 Oguntebi, Z. K.: Some historical reflections on the function concept, *Abacus, the Journal of the Mathematical Association of Nigeria*, Ilorin (Nigeria), Vol. 18, No. 1, 74-79.

Examines a "few of the historical events and characters that contributed some works or discoveries in function-related concepts."

2-RAS-99

1999 Rashed, Rosdi & Vahabzadeh, B.: *Al-Khayyam Mathématicien*. Blanchard, Paris (France), 429 p.

Critical editions in French of al-Khayyam's works *The Algebra*, an untitled treatise written before the *Algebra*, and a commentary on the difficulties in the postulates of Euclid's *Elements*.

Review: 2-HOG-00.

Appendix 3
On Time-reckoning and Astronomy in African History and Cultures

See also BARR-93a, 94a, 96a, 97a, 97b, 99; BRU-65; BRUM-93a, 93b, 94; HARA-00; LOR-95; OBE-73, 90; SEL-97; UKA-97; VERN-52, 56.

3-ADA-83a
1983a Adams III, Hunter Havelin: African Observers of the Universe: the Sirius Question, in SER-83, 27-46.

3-ADA-83b
1983b Adams III, Hunter Havelin: New light on the Dogon and Sirius, in SER-83, 47-49 [Mali].

3-ANDE-87
1987 Andersen, K.: The central projection in one of Ptolemy's map constructions, *Centaurus*, Copenhagen (Denmark), Vol. 30, No. 2, 106-113.

3-BAS-88
1988 Bassi, M.: On the Borana calendrical system, a preliminary field report, *Current Anthropology*, Chicago IL (USA), Vol. 29, No. 4, 612-624 [Ethiopia].

3-BEI-63
1963 Beidelman, T. O.: Kaguru time reckoning: an aspect of the cosmology of an East African people, *Southwestern Journal of Anthropology*, University of New Mexico, Albuquerque N.M. (USA), Vol. 19, 9-20.

3-BERG-91
1991 Berggren, J. Lennart: Ptolemy's maps of earth and the heavens: a new interpretation, *Archive for History of Exact Sciences*, Berlin (Germany), Vol. 43, No. 2, 133-144.

3-BERG-92
1992 Berggren, J. Lennart: & Thomas, R. S. D.: Mathematical astronomy in the fourth century BC as found in Euclid's

'Phaenomena', *Physis Rivista Internazionale di Storia della Scienza*, Florence (Italy), Vol. 29, No. 1, 7-33.

3-BERG-96
1996 Berggren, J. Lennart: & Thomas, R. S. D.: *Euclid's 'Phaenomena': A translation and study of a Hellenistic treatise in spherical astronomy*, Garland, New York (USA), 132 p.

3-BRIT-69
1969 Britton, J. P.: Ptolemy's determination of the obliquity of the ecliptic, *Centaurus*, Copenhagen (Denmark), Vol. 14, 29-41.

3-BRIT-92
1992 Britton, J. P.: Models and precision: the quality of Ptolemy's observations and parameters, in: *Sources and Studies in the History and Philosophy of Classical Science*, New York (USA), Vol. 1.

3-BRUE-32
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The paper presents an overview of pre-colonial astronomical practices. The paper is structured in the following sections: sources of evidence, astronomical practices in the built environment, Khoisan sky lore, time reckoning in agricultural communities, cosmology and social cohesion, astronomical practice as an indicator of cultural exchange, colonialism and the decline of African astronomical practices.

3-SOP-82
1982 Soper, Robert: Archaeo-astronomical Cushites, *Azania*, Nairobi (Kenya), Vol. 17, 145-162

3-SWE-89
1989 Swerdlow, N. M.: Ptolemy's theory of the inferior planets, *Journal for the History of Astronomy*, Cambridge (UK), Vol. 20, No. 1, 29-60.

3-SWE-92
1992 Swerdlow, N. M.: The enigma of Ptolemy's catalogue of stars, *Journal for the History of Astronomy*, Cambridge (UK), Vol. 23, No. 3, 173-183.

3-TAB-94
1994 Tablino, Paul: The reckoning of time by the Borana Hayyantu, *Rassegna di Studi Etiopici*, Napoli / Roma (Italy), Vol. 38, 191-205 [Ethiopia].

3-TAIS-84
1984 Taisbak, Christian Marinus: Eleven eighty-thirds: Ptolemy's reference to Eratosthenes in 'Almagest' I.12, *Centaurus*, Copenhagen (Denmark), Vol. 27, No. 2, 165-167.

3-THOR-80
1980 Thornton, R.: *Space, time, and culture among the Iraqw of Tanzania*, Academic Press, New York (USA), 275 p.

3-TAB-88
1988 Tablino, Paolo: The calculation of time among the Gabra of Kenya, *Bulletin des études africaines de l'INALCO*, Paris (France), Vol. 8, No. 16, 97-107.

Analyses time reckoning among the Gabra and compares it with the calculation of time among the Borana.

3-TAB-94
1994 Tablino, Paolo: The reckoning of time by the Borana Hayyantu, *Rassegna di studi Etiopici*, Napoli / Roma (Italy), Vol. 38, 191-205.

Analyses time reckoning among the Borana (northern Kenya).

3-TIH-76
1976 Tihon, Anne: Notes sur l'astronomie grecque au Ve siècle de notre ère (Marinus de Naplouse - un commentaire au 'Petit commentaire' de Théon) [Notes on Greek astronomy in the 5th century BC], *Janus*, Amsterdam (Netherlands), Vol. 63, No. 1-3, 167-184 (in French).

3-TIH-85
1985 Tihon, Anne: Théon d'Alexandrie et les 'Tables faciles' de Ptolémée [Theon of Alexandria and the 'Handy Tables' of Ptolemy], *Archives Internationales d'Histoire des Sciences*, Rome (Italy), Vol. 35, 106-123 (in French).

3-TIH-87
1987 Tihon, Anne: Le livre V retrouvé du 'Commentaire à l'Almageste' de Théon d'Alexandrie [The rediscovered book V of the 'Commentary on the Almagest' of Theon of Alexandria], *L'Antiquité Classique*, Louvain (Belgium), Vol. 56, 201-218.

3-TOO-84
1984 Toomer, G. J.: *Ptolemy's Almagest*, Duckworth, London (UK), 679 p.

3-TOO-98
1998 Toomer, G. J.: *Ptolemy's Almagest*, Princeton University Press, Princeton NJ (USA), 693 p. (foreword by Owen Gingerich).

"Ptolemy's *Almagest* is one of the most influential scientific works in history. A masterpiece of technical exposition, it was the basic textbook of astronomy for more than a thousand years, and still is the main source for our knowledge of ancient astronomy. This translation, based on the standard Greek text of Heiberg, makes the work accessible to English readers in an intelligible and reliable form. It contains numerous corrections derived from medieval Arabic translations and extensive footnotes that take account of the great progress in understanding the work made in this century, due to the discovery of Babylonian records and other researches. It is designed to stand by itself as an interpretation of the original, but it will also be useful as an aid to reading the Greek text."

3-TUR-78
1978 Turton, D. & Ruggles, C.: Agreeing to disagree: the measurement of duration in a Southwestern Ethiopian Community, *Current Anthropology*, Vol. 19, 585-600.

3-VERG-37
1937 Vergiat, A. M.: Légendes sur les astres [Legends on the stars], in: *Moeurs et coutumes des Manjas*, Payot, Paris (France), 295-313 (in French).

3-VERN-98
1998 Vernet Ginés, Juan: *Contribución al estudio de la labor astronómica de Ibn al-Bannâ* [Contribution to the study of the astronomic work of Ibn al-Bannâ], Institute for the History of Arabic-Islamic Science, Johann Wolfgang Goethe University, Frankfurt am Main (Germany), Collection "Islamic mathematics and science", Volume 43, 220 p. (in Arabic and Spanish)

Reprint of the Edition Tetuan 1951-1952, edited by Fuat Sezgin.

3-WAE-57
1957 Waerden, Bartel L. van der: Tables for the Egyptian and Alexandrian Calendars, *ISIS*, Madison WI (USA). Vol. 47, 387-390.

3-WAE-58
1958 Waerden, Bartel L. van der: The astronomical Papyrus Ryland 27, *Centaurus*, Copenhagen (Denmark), Vol. 5, 177-191.

3-WAE-71
1971 Waerden, Bartel L. van der: Ägyptische Planetenrechnung [Egyptian planet computation], *Centaurus*, Copenhagen (Denmark), Vol. 16, 65-91.

3-WAR-96
1996 Warner, Brian: Traditional astronomical knowledge in Africa, in: Walker, Christopher (Ed.), *Astronomy before the telescope*, British Museum Press, London (UK), 304-317.

3-WILSON-84
1984 Wilson, C.: The sources of Ptolemy's parameters, *Journal for the History of Astronomy*, Cambridge (UK), Vol. 15, 37-47.

3-WLO-90
1990 Wlodarczyk, J.: Notes on the compilation of Ptolemy's catalogue of stars, *Journal for the History of Astronomy*, Cambridge (UK), Vol. 21, No. 3, 283-295.

3-ZAH-51
1951 Zahan, Dominique: Études sur la cosmologie des Dogon et des Bambara du Soudan français – I. La notion d'écliptique [Studies on the cosmology of the Dogon and the Bambara of the French Sudan – I. The notion of the eclipse], *Africa*, London (UK), Vol. XXI, No. 1, 18ff.

Analyses the notion of eclipse among the Dogon and the Bambara (Mali).

Appendix 4
String Figures in Africa

See also GER-95b, 96b, 98d; GIB-96; MOS-96, 97, 98a, 98b, 00a, 03.

4-CAN-93
1993 Cansdale, G. S.: Ghana string figures, *The Nigerian Field*, Vol. 58, 65-80.

4-CUN-06
1906 Cunnington, William: String figures and tricks from Central Africa, *Journal of the Royal Anthropological Institute*, London (UK), Vol. XXXVI, 121-131.

4-CUN-96
1996 Cunnington, William: The Moon Gone Dark, collected by W. Cunnington from the Marungu people of Congo / Zaire, *String Figure Magazine*, Pasadena CA (USA), Vol. 1, No. 4, 5-7.

4-CUN-99
1999 Cunnington, William: A bed, collected by William Cunnington at the south end of Lake Tanganyika, *String Figure Magazine*, Pasadena CA (USA), Vol. 4, No. 3, 16-18.

Partial reproduction of 4-CUN-06.

4-EAR-98
1998 Earthy, E. D.: Border between two countries, collected by E. D. Earthy from the Thonga people of Mozambique, Africa, *String Figure Magazine*, Passadena CA (USA), Vol. 3, No. 4 11-14.

Based on information contained in the book Earthy, E. D., *Valenga Women*, Oxford University Press, London (UK), 1953, 95-101.

4-EVA-55
1955 Evans-Pritchard, Edward: Zande string figures, *Folklore*, Autumn, 225-239 [Central Africa].

4-GRIA-38
1938 Griaule, Marcel: Jeux de ficelles [String figures], in: M. Griaule, *Jeux Dogon* [Games of the Dogon], Institut d'Ethnologie, Paris (France), 71-83.

Collection of string figures from the Dogon (Mali).

4-GRIA-97
1997 Griaule, Marcel: Nose Slip Trick, collected by M. Griaule from the Dogon people of Mali, *String Figure Magazine*, Passadena CA (USA), Vol. 2, No. 1, 5-6.

4-GRIF-25
1925 Griffith, C.: Gold Coast string games, *Journal of the Royal Anthropological Institute*, London (UK), Vol. LV, 271-302.

Collection of string figures from Ghana.

4-HAD-06
1906 Haddon, A.: String figures from South Africa. *Journal of the Royal Anthropological Institute*, London (UK), Vol. XXXVI, 142-149.

4-HADD-36
1936 Haddon, Kathleen & Treleaven, Hilda: Some Nigerian String Figures, *The Nigerian Field*, Vol. 5, No. 1, 31-38, and No. 2, 86-95.

4-HADD-50
1950 Haddon, Kathleen: Review of 4-LEA-29, *Man*, London (UK), Vol. 50, 93 [Angola].

4-HOR-28
1928 Hornell, James: The string games and tricks of Sierra-Leone, *Sierra Leone Studies*, Freetown (Sierra Leone), Vol. XIII, 3-9.

4-HOR-30
1930 Hornell, James: String figures from Sierra Leone, Liberia and Zanzibar, *Journal of the Royal Anthropological Institute*, London (UK), Vol. LX, 81-114.

4-HOR-40
1940 Hornell, James: String figures from the Anglo-Egyptian Sudan, *Sudan Notes and Records*, Khartoum (Sudan), Vol. 23, 99-122.

4-HOR-98
1998 Hornell, James: The Fishing Net, collected by James Hornell (1928) from the Mende of Sierra Leone, *String Figure Magazine*, Pasadena (USA), Vol. 3, No. 3, 12-15.

4- LAG-50
1950 Lagercrantz, Sture: *Contribution to the ethnography of Africa*, Studia Ethnographica Upsaliensia, Lund (Sweden) & Trubner, London (UK) [Reprint: Greenwood Press, Westport, Conn. (USA), 1979], 430 p.

Section on string figures (269-274) includes a map that illustrates the distribution of string figure making in Africa.

4-LEAK-49
1949 Leakey M. D. & Leakey, L. S. B.: Some string figures from North East Angola, *Subsídios para a História, Arqueologia e Etnografia dos Povos da Lunda*, Museu do Dondo, Lisbon (Portugal), 7-24.

Collection of 20 string figures collected among the Cokwe in January-February 1948. In a number of cases the Cokwe have "'serial' figures in which the successive stages seem to represent the illustrations of a story." As far as the authors know "such 'serial' figures are relatively scarce in Africa."

4-LIN-30
1930 Lindblom, Gerhard: *String figures in Africa*, Riksmuseets Etnografiska Avdelning, Smärre Meddelanden, Stockholm (Sweden), No. 9, 12 p.

4-PARK-06
1906 Parkinson, J.: Yoruba string figures, *Journal of the Royal Anthropological Institute*, London (UK), Vol. XXXVI, 132-141 [Nigeria].

4-REI-02
2002 Reichert, A.: Some string figures from modern Africa, *Bulletin of the International String Figure Association*, Pasadena CA (USA), Vol. 9, 241-248.

4-SMITH-99
1999 Smith, Carey C. K.: String Figures from the Congo, *Bulletin of the International String Figure Association*, Pasadena CA (USA), Vol. 4, 135-184.

"The article presents sixty-seven string figures gathered at Upoto in the former Belgian Congo by Mrs. Ethel M. Smith during the years 1910-1914. Among her informants were members of the Lingombe,

Lifoto, Ngombe, Ngwenzali, and Ngwengali tribes. Unlike F. Starr's Congo collection published in 1909, the Smith collection includes methods of construction for each figure. In an appendix to this article, the author presents methods for making thirty-nine of the sixty-two figures described by Starr."

4-SMI-00

2000 Smith, Carey C. K.: Some String Figures and Tricks from Sierra Leone and the Gold Coast, *Bulletin of the International String Figure Association*, Pasadena CA (USA), Vol. 7, 94-100 [Sierra Leone and Ghana].

4-SMITHE-98

1998 Smith, Ethel: 'Mangbongobo' or Flying Fox, collected by E. Smith from Congo / Zaire, *String Figure Magazine*, Pasadena CA (USA), Vol. 3, No. 1, 10-12.

4-STAR-09

1909 Starr, F.: Ethnographic notes from the Congo Free State, *Proceedings of Davenport Academy of Sciences*, Davenport, Iowa (USA), Vol. 12, 148-175.

4-STOR-03

2003 Storer, Tom: *String Figure Bibliography* (Available online at the website of the International String Figure Association (ISFA): www.isfa.org/biblio.htm)

General bibliography with a section on Africa that includes most references given in this appendix.

4-TES-12

1912 Tessmann, G.: Die Kinderspiele der Pangwe [Children's games of the Pangwe], *Bässler Archiv*, Basle (Switserland), Vol. 2, No. 5/6, 271-278 (in German).

4-TES-01

2001 Tessmann, G.; Reichert, A. & Sherman, Mark: Pangwe and Bubi String Figures, *Bulletin of the International String Figure Association*, Pasadena CA (USA), Vol. 8, 125-201.

Translation of 4-TES-12 with new illustrations and cultural notes.

4-TRA-36
1936 Tracey, Hugh: String Figures (*madandi*) found in Southern Rhodesia, *Southern Rhodesia Native Affairs Department Annual*, Salisbury (Harare, Zimbabwe), Vol. 14, 78-88.

4-TRA-99
1999 Tracey, Hugh: The eagle and its nest, collected by Hugh Tracey from the Mashona people of Southern Zimbabwe, *String Figure Magazine*, Pasadena CA (USA), Vol. 4, No. 1, 11-15

Partial reproduction of 4-TRA-36.

4-TRE-98
1998 Treleaven, Hilda: A Gun, collected by H. Treleaven from the people of Offa, Nigeria, *String Figure Magazine*, Pasadena CA (USA), Vol. 3, No. 2, 7-8.

4-WED-30
1930 Wedgwood, Camilla & I. Schapera: String figures from Bechuana Protectorate, *Bantu Studies*, Johannesburg (South Africa), Vol. IV, 215-268.

Collection of string figures from Botswana.

4-WED-99
1999 Wedgwood, Camilla: Oxen inspanned, *String Figure Magazine*, Pasadena CA (USA), Vol. 4, No. 4, 20-24.

Reproduction of the making of a string figure by the Kxatla people of Botswana, originally included in the paper WED-30.

4-WIR-00
2000 Wirt, W.: String figures from Southwestern Ethiopia, *Bulletin of the International String Figure Association*, Pasadena CA (USA), Vol. 7, 101-118.

The web-page of the International String Figure Association (ISFA) is:
www.isfa.org

Appendix 5
Examples of Books and Booklets Published by African Mathematicians

5-AHM-02

2002 Ahmad, Khalil [Morocco] (together with Pamy Manchanda & A.H. Siddiqi) (Eds.): *Current trends in industrial and applied mathematics*, Anamaya Publishers, New Delhi (India), 263 p.

5-ALVA-82

1982 Alvarinho, Ida [Mozambique](together with Serguei Vodopianov): *Geometria Euclidiana* [Euclidian Geometry], Textbook, Universidade Eduardo Mondlane, Maputo (Mozambique), 103 p.

5-ALV-00

2000 Alves, Manuel [Mozambique]: *Equações Diferenciais Funcionais Singulares de Segunda Ordem* [Second order singular functional differential equations], translation of doctoral thesis, Perm State University Press, Perm (Russia), 179 p. (in Portuguese).

5-ANI-00

2000 Animalu,, A. O. E.; Iyahen, Sunday O. & Tejumola, Haroon O. (Eds.) [Nigeria]: *Contributions to the development of mathematics in Nigeria*, National Mathematical Centre, Abuja (Nigeria), 302 p.

5-ASH-01

2001 Ashour, A. A. & Obada, A.-S. F. (Eds.) [Egypt]: *Mathematics and the 21st century: proceedings of the international conference, Cairo, Egypt, 15-20 January 2000*, World Scientific, Singapore & River Edge, NJ (USA), 398 p.

5-ASSA-03

2003 Assani, Idris [Benin] (together with Wiener Wintner): *Ergodic Theorems*, World Scientific, River Edge NJ (USA), 216 p.

5-BANY-97
1997 Banyaga, Augustin [Rwanda]: *The structure of classical diffeomorphism groups*, Kluwer, Boston (USA); Boston, 197 p.

5-BANY-99
1999 Banyaga, Augustin (Ed. together with H. Movahedi-Lankarani & R. Wells), *Topics in Low-Dimensional Topology*, World Scientific, Singapore.

5-BANY-02
2002 Banyaga, Augustin (Ed. together with J. Leslie, T. Robart), *Infinite Dimensional Lie groups in Geometry and Representation Theory*, World Scientific, Singapore.

5-BEI-82a
1982 Beirão, João Carlos [Mozambique] (together with Rodeon Alexandrov): *Problemas de análise matemática: Funções reais de várias variáveis* [Problems of mathematical analysis: Real functions of various variables], Textbook, Universidade Eduardo Mondlane, Maputo (Mozambique), 163 p. (in Portuguese).

5-BEI-82b
1982 Beirão, João Carlos (together with Rodeon Alexandrov): *Problemas de análise matemática: Séries* [Problems of mathematical analysis: Series], Textbook, Universidade Eduardo Mondlane, Maputo (Mozambique), 134 p. (in Portuguese).

5-BEI-83
1983 Beirão, João Carlos (together with Rodeon Alexandrov): *Problemas de análise matemática: Integrais múltiplas* [Problems of mathematical analysis: Multiple integrals], Textbook, Universidade Eduardo Mondlane, Maputo (Mozambique), 263 p. (in Portuguese).

5-BEI-92
1992 Beirão, João Carlos: *Análise Matemática* [Mathematical Analysis], Textbook, Instituto Superior Pedagógico, Maputo (Mozambique), Vol. 1, 194 p.; Vol. 2, 230 p. (in Portuguese).

5-BEI-93
1993 Beirão, João Carlos: *Funções de variável complexa* [Functions of complex variable], Textbook, Instituto Superior Pedagógico, Maputo (Mozambique), 175 p. (in Portuguese).

5-BEI-05
2005 Beirão, João Carlos & Cassy, Bhangy [Mozambique]: *Cálculo diferencial em R^n* [Differential calculus in R^n], Imprensa Universitária, Maputo (Mozambique), 226 p.

5-BELG-97
1997 Belgacem, Fethi [Tunisia]: *Elliptic boundary value problems with indefinite weights: variational formulations of the principal eigenvalue and applications*, Addison Wesley Longman (Pitman research notes in mathematics series, 368), London (UK), 236 p.

5-CAD-99
1999 Cadete, Manuel D. O. [Angola]: *Mathematical models for the management of education in countries with an economy in transition*, doctoral thesis, Tula State Pedagogical University, Tula (Russia), 115 p. (in Russian).

5-CHID-03
2003 Chidami, Mohamed [Morocco] (together with Curto, R., Mbekhta, M., Vasilescu, F.-H., Zemánek, J. (Eds.)): *Operator theory and Banach algebras. Proceedings of the international conference in analysis held in Rabat, Morocco, April 12--14, 1999*, Theta, Bucharest (Hungary), 161 p.

5-CHU-92
1992 Chukwu, Ethelbert Nwakuche [Nigeria]: *Stability and time-optimal control of hereditary systems. With application to the economic dynamics of the U.S.*, Academic Press, Boston MA (USA), 508 p. [2nd edition: Series on Advances in Mathematics for Applied Sciences, Vol. 60. World Scientific Publishing, River Edge NJ (USA), 2001, 495 p.]

5-CHU-01
2001 Chukwu, Ethelbert Nwakuche: *Differential models and neutral systems for controlling the wealth of nations*, Series on

Advances in Mathematics for Applied Sciences, Vol. 54. World Scientific, River Edge NJ (USA), 513 p.

5-CHU-03
2003 Chukwu, Ethelbert Nwakuche: *Optimal control of the growth of wealth of nations. Stability and Control: Theory, Methods and Applications*, Taylor & Francis, London (UK), 384 p.

5-DZI-84
1984 Dzinotyiweyi, Henri A. M. [Zimbabwe]: *The analogue of the group algebra for topological semigroups*, Pitman Advanced Publishing Program, Boston (USA), 196 p.

5-DZI-86
1986 Dzinotyiweyi, Henri A. M.: *A first course in mathematical analysis*, C. B. S. Publishers, Delhi (India).

5-ELY-01a
2001a El Yacoubi, Nouzha [Morocco] (Ed.): *Proceedings of the first AMUPAMO Symposium held in Kairouan, Tunisia, from the 31st of October to the 6th of November 2000 with as Theme: Pan African Mathematics Olympiads, Training and Research*, Presses Universitaires de Yaoundé, Yaoundé (Cameroon), 214 p.

AMUPAMO stands for African Mathematical Union Commission on Pan African Mathematics Olympiads. The proceedings include a report of the symposium and the papers presented in English or French at the plenary sessions:

* Aderemi Kuku: Mathematical sciences and other sciences (107-124);
* Jan Persens: Mathematics development – Striving for a balance between pure and applied mathematics, even at school level (125-136);
* Saliou Touré: Un exemple de coopération dans les pays francophones d'Afrique et de l'Océan Indien [An example of co-operation between the French-speaking countries of Africa and the Indean Ocean] (137-142) (in French);
* Claude Deschamps: Les Olympiades Internationales de Mathématiques [The International Mathematics Olympiads] (143-152) (in French);

* Nouzha El Yacoubi: Olympiades Pan Africaines de Mathématiques de l'Union Mathématique Africaine [The Pan-African Mathematics Olympiads of the African Mathematical Union (155-168) (in French);
* Francisco Bellot Rosado: La compétition mathématique méditerranéenne [The Mediterranean mathematics competition] (169-171) (in French);
* Walter Mientka: The road to the International Mathematical Olympiad (173-177).

5-ELY-01b

2001b El Yacoubi, Nouzha (Ed.), *Actes, 11e Edition des Olympiades Pan Africaines de Mathématiques*, AMUPAMO & La Société Mathématique de Côte d'Ivoire, Abidjan (Côte d'Ivoire), 48 p.

Proceedings of the 11th Pan African Mathematics Olympiad held in Ouagadougou, Burkina Faso (July 15-22, 2001) and organised by the African Mathematical Union Commission for the Pan-African Mathematical Olympiad (AMUPAMO).

5-ELY-02

2002 El Yacoubi, Nouzha [Morocco] & John Webb [South Africa] (Eds.), *Proceedings of the 12th Pan African Mathematics Olympiad of the African Mathematical Union*, AMUPAMO & Foundation for Education, Science and Technology, Pretoria (South Africa), 49 p.

Proceedings of the 12th Pan African Mathematics Olympiad held in Pretoria, South Africa (April 6-14, 2002).

5-ELY-03

2003 El Yacoubi, Nouzha [Morocco]; Ismael Cassamo Nhêze; Luís do Nascimento Paulo & Paulus Gerdes [Mozambique] (Eds.), *Proceedings of the 13th Pan African Mathematics Olympiad of the African Mathematical Union*, AMUPAMO & Ministry of Education, Maputo (Mozambique), 2003, 64 p.

Proceedings of the 13th Pan African Mathematics Olympiad held in Maputo, Mozambique (April 19-27, 2003), organized by the African Mathematical Union Commission for the Pan-African Mathematical Olympiad (AMUPAMO) and hosted by the Ministry of education of Mozambique. Includes the paper "From African 'sona' drawings to

the discovery of new symmetries and matrices" (51-64) by Paulus Gerdes.

5-ESO-82
1982 Esogbue, Augustine O. [Nigeria] (together with R. Bellman & I. Nabeshima): *Mathematical aspects of scheduling and applications*, Pergamon, Oxford (UK), 329 p.

5-ESO-89
1989 Esogbue, Augustine O.: *Dynamic programming for optimal water resources systems analysis*, Prentice Hall, Englewood Cliffs NJ (USA), 435 p.

5-ESO-99
1999 Esogbue, Augustine O. (together with Liu, Baoding): *Decision criteria and optimal inventory processes*, Kluwer, Boston (USA), 210 p.

5-EZI-88
1988 Ezin, Jean-Pierre [Benin] (Ed.): *Fibre bundles: their use in physics*, World Scientific, Singapore, 175 p.

5-FATU-85
1985 Fatunla, Simeon Ola [Nigeria] (Ed.): *Computational mathematics I*, Boole Press Conference Series, Vol. 8, Boole Press, Dún Laoghaire (Ireland), 141 p.

Proceedings of the first international conference on numerical analysis and its applications held in Benin City (Nigeria), November 2-4, 1983.

5-FATU-87
1987 Fatunla, Simeon Ola (Ed.): *Computational mathematics II*, Boole Press Conference Series, Vol. 11, Boole Press, Dún Laoghaire (Ireland), 221 p.

Proceedings of the Second International Conference on Numerical Analysis and its Applications held in Benin City (Nigeria), January 27-31, 1986.

5-FATU-88
1988 Fatunla, Simeon Ola: *Numerical methods for initial value problems in ordinary differential equations*, Computer Science

and Scientific Computing, Academic Press, Boston MA (USA), 295 p.

5-GAT-74
1974 Gattegno, Caleb [Egypt]: *The common sense of teaching mathematics*, Educational Solutions, New York (USA), 129 p.

5-GER-90
1991 Gerdes, Paulus [Mozambique]: *Examples of applied mathematics in agriculture and veterinary science*, National Education Coordinating Commission, Cape Town (South Africa), 54 p.

5-GER-91
1991 Gerdes, Paulus & Cherinda, Marcos [Mozambique]: *Teoremas famosos da geometry* [Famous theorems of Geometry], ISP, Maputo (Mozambique), 112 p.

5-GER-92
1992 Gerdes, Paulus (Ed.): *Who is who in Mathematics and Mathematics Education in Southern Africa*, Southern African Mathematical Sciences Association (SAMSA), Maputo (Mozambique), 64 p. (Supplement 1993, 18 p.; Supplement 1995, 16 p.)

5-GUID-85
1985 Guidy Wandja, Joséphine & Sah Bi, Jess [Côte d'Ivoire]: *Yao crack en math* [Yao crack in mathematics], Nouvelles Editions Africaines, Abidjan (Côte d'Ivoire), 28 p. (in French).

5-HAS-86
1986 Hassan, Mohamed H. A. [Sudan] (together with Farouk El-Baz (Eds.)): *Physics of desertification*, Kluwer, Dordrecht (Netherlands), 473 p.

5-HAS-91
1991 Hassan, Mohamed H. A. et al. (eds.): *The role of women in the development of science and technology in the Third World*, World Scientific, Singapore, 970 p.

5-HAS-93
1993 Hassan, Mohamed H. A.: *Science and technology for the socio-economic development of Africa*, RANDFORUM Press, Nairobi (Kenya), 29 p.

5-HOG-71
1971 Hogbe-Nlend, Henri [Cameroon]: *Théorie des bornologies et applications* [Theory of bornologies and applications], Springer, Berlin (Germany), 168 p. (in French).

5-HOG-73
1973 Hogbe-Nlend, Henri: *Distributions et bornologie* [Distributions and bornology], Universidade de São Paulo, São Paulo (Brazil), 143 p. (in French).

5-HOG-77
1977 Hogbe-Nlend, Henri: *Bornologies and functional analysis: introductory course on the theory of duality topology-bonology and its use in functional analysis*, North-Holland, Amsterdam (Netherlands), 144 p.

5-HOG-81
1981 Hogbe-Nlend, Henri: (together with Vincenzo B. Moscatelli): *Nuclear and conuclear spaces: introductory courses on nuclear and conuclear spaces in the light of the duality "topology-bornology"*, North-Holland, Amsterdam (Netherlands), 275 p.

5-HOUNK-00
2000 Hounkonnou, Mahouton Norbert [Benin] (together with Jan Govaerts & William Lester Jr.) (Eds.): *Contemporary Problems in Mathematical Physics*, World Scientific, River Edge NJ (USA), 377 p.

Proceedings of the 1st International Workshop on Contemporary Problems in Mathematical Physics held in Cotonou (Benin), October 31 – November 5, 1999.

5-JEN-00
2000 Jenda, Overtoun M. G. [Malawi] (with Edgar Enochs): *Relative Homological Algebra*, De Gruyter Expositions in Mathematics, Vol. 30, Walter de Gruyter, Berlin (Germany).

5-KUK-80

1980 Kuku, Aderemi O. [Nigeria]: *Abstract Algebra*, Ibadan University Press, Ibadan (Nigeria).

5-KUK-85

1985 Kuku, Aderemi O. (together with J. Rawnsley & E. Thoma): *Group representation and its applications*, CIMPA, Nice (France), 221 p.

Lectures given at the International Summer School on Group Representation and its Applications, August 17 - September 11, 1981, Ibadan, Nigeria.

5-KUK-86

1986 Kuku, Aderemi O.: *Axiomatic Theory of Induced Representation of Finite Groups*, CIMPA, Nice (France).

5-KUK-94

1994 Kuku, Aderemi O.: *Basic Computative Algebra*, Lecture Notes Series, National Mathematical Centre, Abuja (Nigeria).

5-KUK-97

1997 Kuku, Aderemi O.: *Topics in Algebraic K-Theory*, Lecture Notes Series, National Mathematical Centre, Abuja (Nigeria).

5-KUK-99

1999 Kuku, Aderemi O. (together with H. Bass & C. Pedrini) (Eds.): *Algebraic K-theory and its applications*, World Scientific, Singapore, 607 p.

Proceedings of a workshop and symposium held at the International Centre for Theoretical Physics (ICTP), Trieste, Italy.

5-KUK-06

2006 Kuku, Aderemi O: *Representation Theory and Higher Algebraic K-theory*, CRC Press (USA), 442 p.

5-KWI-04

2004 Kwuida, Leonard [Cameroon]: *Dicomplemented Lattices: A Contextual Generalization of Boolean Algebras*, Shaker (Germany), 132 p.

5-LAB-93
1993 Labuschagne, Willem [South Africa], *A user-friendly introduction to discrete mathematics for computer science*, University of South Africa, Pretoria (South Africa), 304 p.

5-MAK-00
2000 Makinda, Olewole D. [Nigeria] & Sibanda, Precious [Zimbabwe]: *A mathematical introduction to incompressible flow*, Textbook, University of Zimbabwe Press, Harare (Zimbabwe).

5-MAS-97
1997 Masanja, Verdiana G. [Tanzania] (Ed.): *Conference Proceedings. XI SAMSA Symposium on the Potential of Mathematical Modeling of Problems from the SAMSA Region*, Mathematics Department, University of Dar Es Salaam, Dar Es Salaam (Tanzania), 395 p.

Conference proceedings of the XI Southern Africa Mathematical Sciences Association (SAMSA) Symposium, 18-23 August 1997, Arusha (Tanzania).

5-MASE-74
1974 Masenge, Ralph W. P. [Tanzania] (together with: W. F. Reichert): *Fundamentals of numerical methods*, Dar es-Salaam University Press, Dar es-Salaam (Tanzania), 97 p.

5-MASE-88
1988 Masenge, Ralph W. P.: *Basic Numerical Methods*, Dar es-Salaam University Press, Dar es-Salaam (Tanzania), 232 p.

5-MSH-90
1990 Mshimba, Ali Seif [Tanzania] (together with W. Tutschke (Eds.)): *Functional analytic methods in complex analysis*, World Scientific, Singapore.

5-MSH-92
1992 Mshimba, Ali Seif: *Basic complex analysis*, Dar es-Salaam University Press, Dar es-Salaam (Tanzania).

5-NGUER-01

2001 N'Guérékata, Gaston Mandata [Central African Republic]: *Almost Automorphic and Almost periodic Functions in Abstract Spaces*, Kluwer, New York (USA).

5-NGUER-04

2004 N'Guérékata, Gaston Mandata: *Topics in almost automorphy*, Springer, New York (USA).

5-NGU-90

1990 Nguiffo Boyom, Michel [Cameroon] (edited together with J.-M. Morvan & L. Verstraelen): *Geometry and topology of submanifolds*, World Scientific, Teaneck NJ (USA), 412 p.

Proceedings of the International Meeting held in Avignon (France), May 30 - June 3, 1988.

5-NJO-99a

1999a Njock, Georges Edward [Cameroon]: *Théorie de Galois et Applications*, Presses Universitaires de Yaoundé, Yaoundé (Cameroon), 142 p. (in French).

Lecture notes of a course on Galois Theory and its Applications given to students of the 'Maitrise' program at the Mathematics Department of the University of Yaoundé.

5-NJO-99b

1999b Njock, Georges Edward: *Introduction à la Géométrie Projective*, Presses Universitaires de Yaoundé, Yaoundé (Cameroon), 157 p.

Lecture notes of a course on Projective Geometry given at the University of Yaoundé, principally to students of the 'Licence' program for future mathematics teachers.

5-NKE-05

1997 Nkemzi, Boniface [Cameroon]: *Numerische Analysis der Fourier-Finite-Elemente-Methode für die Gleichungen der Elastizitätstheorie* [Numerical analysis of the Fourier finite elements method for equations from elasticity theory], Tectum Verlag, Marburg (Germany), 109 p.

5-OKI-71
1971 Okikiolu, George Olatokunbo [Nigeria]: *Aspects of the theory of bounded integral operators in L^p-spaces*, Academic Press, London (UK), 522 p.

5-OKI-80
1980 Okikiolu, George Olatokunbo: *Special integral operators. Vol. I. Weierstrass operators and related integrals*, Okikiolu Scientific and Industrial Organization, London (UK), 306 p.

5-OKI-81
1981 Okikiolu, George Olatokunbo: *Special integral operators. Vol. II. Poisson operators, conjugate operators, and related integrals*, Okikiolu Scientific and Industrial Organization, London (UK), 507 p.

5-OLAY-00
2000 Olayi, Gabriel Atah [Nigeria]: *Introductory Numerical Methods*, ABU Press, Zaria (Nigeria), 185 p.

5-OLAY-01
2001 Olayi, Gabriel Atah: *Mathematical Methods*, Bachudo Publ., Calabar (Nigeria), 135 p.

5-OLAY-02
2001 Olayi, Gabriel Atah: *Complex Analysis, A Computational Approach*, Bachudo Science Comp., Calabar (Nigeria), 164 p.

5-ROH-05
2005 Rohwer, Carl [South Africa]: *Nonlinear Smoothing and Multiresolution Analysis*, Birkhauser International Series of Numerical Mathematics, Basel (Switzerland).

5-RUN-81
1981 Rund, Hanno [South Africa]: *Generalized connections and gauge fields on fibre bundles*, University of South Africa, Pretoria (South Africa), 120 p.

5-SAL-74
1974 Salbany, Sergio [South Africa]: *Bitopological spaces, compactifications and completions*, Math Monographs of the

University of Cape Town, Cape Town (South Africa), Volume 1.

5-SEY-72
1972 Seydi, Hamet [Senegal]: *La théorie des anneaux japonais* [The Theory of Japanese Rings], Colloque d'Algèbre Commutative, Université de Rennes, Rennes (France), Vol. 12, 82 p. (in French).

5-SHO-00
2000 Shonhiwa, Temba [Zimbabwe]: *Introduction to Vector Analysis*, University of Zimbabwe Publications, Harare (Zimbabwe).

5-TCHU-91
1991 Tchuenté, Maurice [Cameroon]: *Parallel Computation on Regular Arrays*, Halsted Press, New York (USA).

5-UKO-00
2000 Uko, Livinus Ugochukwu [Nigeria]: *Matematicas Amenas* [Mathematics for Leisure], Editorial Universidad de Antioquia, Medellin (Colombia) (in Spanish).

5-VITH-03
2003 Vithal, Renuka [South Africa]: *In Search of a Pedagogy of Conflict and Dialogue for Mathematics Education*, Kluwer, Dordrecht (Netherlands), 416 p.

Study based on an analysis of the South African context. Published version of Vithal's doctoral thesis (2000, Aalborg University, Denmark).

5-VITH-04
2004 Vithal, Renuka & Adler, Jill [South Africa] (together with Keitel, Christine (Eds.)): *Mathematics Education Research in South Africa: Perspectives, Practices and Possibilities*, Human Sciences Research Council, Pretoria (South Africa).

Appendix 6
Board Games in Africa

See also BEA-55; CEN-63; DOU; KLEP-72; KRA-83; MIZ-71; PAU-71; RAT-91; ZAS; 1-HER-29, 32.

6-ANN-38
1938 Anna, M.: The mweso game among the Basoga, *Primitive Man*, Vol. 11 [Uganda].

6-AVED-71
1971 Avedon, Elliot M. & Sutton-Smith, Brian (Eds.), *The study of games*, Wiley, New York (USA), 530 p. (New edition: Krieger Pub., Huntington NY, 1979).

Includes a reproduction of 6-CUL-94.

6-AVE-06
1906 Avelot, R.: Le jeu des godets [The game of jars], *Bulletin de la Société d'Anthropologie de Paris*, Paris (France), Vol. VII [Gabon / Ghana].

6-AVE-08
1908 Avelot, R.: Le ouri, *Bulletin de la Société d'Anthropologie de Paris*, Paris (France), Vol. IX (in French).

6-BALLO-78
1978 Ballou, Kanga: *Règles et stratégies du jeu l'áwélé* [Rules and strategies of the *awelé* game], Nouvelles Éditions Africaines, Abidjan (Côte d'Ivoire), 55 p. (in French).

6-BALLO-84
1984 Ballou, Kanga: *A guide for playing the game of woaley, a fascinating ancient African game of strategy*, Nouvelles Éditions Africaines, Abidjan (Côte d'Ivoire), 36 p.

6-BEAT-39
1939 Beaton, A. C.: A Bari game – Soro, *Sudan Notes and Records*, Khartoum (Sudan), Vol. 22.

6-BELLR-88
1988 Bell, Robie & Cornelius, Michael: *Board games round the world. A resource book for mathematical investigations*, Cambridge University Press, Cambridge (UK), 124 p.

Contains sections on several games from Africa: *achi* (Ghana, p. 7), *dara* (Nigeria, p. 10), *seega* (Egypt, p. 10), *mankalah* (Ancient and modern Egypt, 24-27), *wari* (West Africa, 28-29), *gabata* (a three-row *mancala* game, Ethiopia, p. 30), *kiarabu* (a four-row *mancala* game, Zanzibar [Tanzania], 31-32), *ise-ozin-egbe* (a solitaire game, Nigeria, p. 32).

6-BENN-28
1928 Bennett, G.: Wari, in: Robert Rattray (Ed.), *Religion and art in Ashanti*, Oxford University Press [Ghana].

Reproduced in 6-KOV-95.

6-BIN-96
1996 Binsbergen, Wim van: Time, space and history in African divination and board-games, in: D. Tiemersma and H. Oosterling (Eds.), *Time and Temporality in intercultural Perspective*, Studies in Intercultural Philosophy, No. 4, Rodopi, Amsterdam (Netherlands), 105-125.

6-BIN-97
1997 Binsbergen, Wim van: Rethinking Africa's contribution to global cultural history: lessons from a comparative historical analysis of mankala board-games and geomantic divination, *Talanta, Special issue 'Black Athena: Ten Years After'*, No. 28-29, 221-254.

6-BRA-31
1931 Braunholtz, H. J.: The game of mweso in Uganda, *Man*, London (UK), Vol. 31.

6-BRIE-86
1986 Briere, B. & Briere, J.: *Awele*, Université de Paris VIII, Vincennes (France) (in French) [Ivory Coast].

6-BROL-95
1995 Broline, Duane & Loeb, Daniel: The Combinatorics of Mancala-Type Games: Ayo, Tchoukaillon, and 1/π, *UMAP Journal*, Cambridge MA (USA), Vol. 16, 21-36.

"Certain endgame considerations in the two-player Nigerian Mancala-type game Ayo can be identified with the problem, of finding winning positions in the solitaire game Tchoukaillon. The periodicity of the pit occupancies in s stone winning positions is determined. Given n pits, the number of stones in a winning position is found to be asymptotically bounded by n^2/π."

6-CHA-56
1956 Chaplain, J. H.: A note on mancala games in Northern Rhodesia, *Man*, London (UK), Vol. 56 [Zambia].

6-COL-10
1910 Collins, G. N.: Kboo, a Liberian game, *National Geographic Magazine*, Washington DC (USA).

6-COU-63
1963 Coupez, A. & Benda, V.: Terminologie du jeu d'igisoro en Rwanda [Terminology of the *igisoro* game in Rwanda], *Africa-Tervuren*, Tervuren (Belgium), Vol. 9, No. 2, 37-41 (in French).

6-COUR-43
1943 Courlander, H.: The Ethiopian game of gobeta, *The Negro History Bulletin*, October.

6-CRA-82
1982 Crane, Louise: *African games of strategy, a teaching manual*, African Outreach Series, No.2, University of Illinois, Urbana-Champaign IL (USA), 53 p.

Informs about and gives detailed playing instructions for some of the most common types of African games involving strategy and mathematical principles: 1. games of alignment [*Shisima* (Kenya), *Achi* (Ghana), *Murabaraba* (Lesotho)]; 2. 'Struggle for territory' games [*Sega* (Egypt), *Kei* (Sierra Leone)]; 3. *Mankala* games: a. two-row versions [*Oware* (Ghana) and variations *Adi* (Ghana), *Awele*

(Ivory Coast), *Ayo* (Nigeria), *Okwe* (Nigeria)], b. four-row versions [*Omweso* (Uganda), *Tshisolo* (Congo / Zaire)].

6-CRO-87
1987 Crowe, Donald: Review of Russ' *Mancala games* (RUS-84), *The Mathematical Intelligencer*, New York (USA), Vol. 9, No. 2, 68-70.

6-CUL-94
1894 Cullin, Stewart: *Mancala, the national game of Africa, Annual Report of the U.S. National Museum for 1894*, Smithsonian Institution, Washington DC (USA), 579-607.

Reproduced in: 6-AVE-71, 6-KOV-95.

6-DAN-09
1909 Dandouau, A.: Jeux malgaches [Malgache games], *Bulletin de l'Academie Malgache*, Vol. 6 (in French) [Madagascar].

6-DELE-77
1977 Deledicq, André & Popova, A.: *Wari et solo. Le jeu de calcul africain* [Wari and solo, the African calculation game], Cedic, Paris (France), 206 p. (in French).

6-DELE-81
1981 Deledicq, André: Le jeu de toute l'Afrique [The game of the whole of Africa], *Jeux et Stratégie*, Paris (France), Vol. 7, 15-19 (in French).

6-DRI-27
1927 Driberg, J. H.: The game of choro or pereauni, *Man*, London (UK) [Uganda].

6-DRIE-72
1972 Driedger, Walter: The game of Bao or Mankala in East Africa, *Mila* (Institute of African Studies, University of Nairobi), Nairobi (Kenya), Vol. 3, No. 1, 7-17.

6-GAM-80
1980 Gama Amaral, Manuel: Libao, in: M. Gama Amaral, *O povo Yao, Subsídios para o estudo de um povo do noroeste de*

Moçambique [The Yao people, a contribution to the study of a people from the Northwest of Mozambique], Instituto de Investigação Científica Tropical, Lisbon (Portugal), 327-332 (in Portuguese).

Describes two variants of *libao* or *mvombwa*, a four-row version of the *mancal*a game.

6-ISM-96

1996 Ismael, Abdulcarimo: A study of the *N'tchuva* game: An ethnomathematical approach, in: F. Mira (Ed.), *Educação, empresas e desenvolvimento em Moçambique*, Pendor, Évora (Portugal), 87-103.

Discusses some mathematical aspects of *n'tchuva*, a four-row 'mancala' game played in the South of Mozambique.

6-ISM-02

2002 Ismael, Abdulcarimo: *An ethnomathematical study of Tchadji – about a Mancala type board game played in Mozambique and possibilities for its use in Mathematics Education*, Ph.D. thesis, University of Witwatersrand, Johannesburg (South Africa), 478 p.

The first part of the thesis includes an analysis of the mathematical considerations (e.g. mental calculation, geometrical pattern recognition, probability) of *tchadji* players belonging to the Makhuwa people in the North-East of Mozambique. It compares this four-row version of 'mancala' with other versions and a section is dedicated to its history. The second part of the thesis discusses the author's experience in using the game in teaching probability theory at the university level, and in teaching some elements of probability in upper secondary schools in the north of the country.

6-JAM-00

2000 Jama, Jama Musse: *Shax: The preferred game of our camel herders and other traditional African entertainments*, Sunmoonlake, Roma (Italy), 40 p.

Presents an introduction to the 'shax' three-in-a-row game from Somalia and two other Somali board games 'Layli Goobaley' and 'Korkabood'. It introduces also a computer program for 'shax' (see also: www.redsea-online.com/games/shax.html).

6-KAI-84
1984 Kaiser, Hans-Ruedi; Keller, Beat & Loesch-Berger, Marie-Cécile: *Awélé: Un programme jouant à partir de plans*, Université de Fribourg, Institut de Psychologie, Bulletin de Recherche No. 46. (in French).

6-KLA-11
1911 Klamroth, H.: Afrikanische Brettspiele [African board games], *Archiv für Anthropologie*, Vieweg (Germany), 253 ff. (in German).

6-KOV-95
1995 Kovach, Roger: *Oware! 'The National game of Africa'. A Winning Numbers Game*, Sapient Software, Bolinas CA (USA), 186 p.

Book and diskette with information on how to play West African versions of the mancala game. In the second part it contains a reproduction of earlier texts and information about its authors: 6-SAW-49 (59-63), 6-MUR-52 (64-90), 6-CUL-94 (91-110), 1-HER-29 (111-120), 1-HER-32 (121-147), 6-BENN-28 (148-159).

6-MAR-31
1931 Marin, G.: Somali games, *Journal of the Royal Anthropological Institute*, London (UK), Vol. 61, 499-512.

6-MAT-64
1964 Matthews, J. B.: Notes on some African games, *NADA, the Rhodesian Ministry of Internal Affairs Annual*, Vol. IX, No. 1.

6-MER-53
1953 Merriam, Allan P.: The game of the Kubuguza among the Abatutsi of North-East Ruanda, *Man*, London (UK), Vol. 53, Nov.

6-MON-50
1950 Monod, Th.: Sur quelques jeux africains à quadrillage [On some African board games], *Notes Africaines*, Paris (France), Vol. 45, 11-13.

Describes briefly some board games from the Sahara and Sahel region.

6-MULL-30
1930 Muller, H. R.: Warri: A West African game of skill, *Journal of American Folklore*, Vol. 43, 169.

6-MUR-52
1952 Murray, H.: Introduction to Mancala games, in: H. Murray, *History of board games other than chess*, Oxford University Press, Clarendon (UK) [partially reproduced in 6-KOV-95].

6-MVE-90
1990 Mve-Ondo, Bonaventure: *L'Owani et le Songa: Deux jeux de calculs africains. Découverts du Gabon* [Owani and Songa: two African calculation games. Discoveries from Gabon], Centre Culturel Français Saint-Exupéry & Sépia Editions, Libreville (Gabon) & Paris (France), 130 p. (in French).

This book on calculation games is structured into five chapters: 1. Rules, 2. Tactics and strategy, 3. Formalisation of *Owani* and *Songa*, 4. Calculation games and traditional social systems, 5. Calculation games and philosophy.

6-NEW-39
1939 Newberry, R. J.: Games and pastimes of Southern Nigeria, *The Nigerian Field*, Vol. VIII.

6-NGU-86
1986 N'Guessan, Assandé G.: *L'apprentissage de l'awélé, étude du processus d'acquisition des tactiques et des stratégies* (Paper presented at the Colloquium "Learning and cultural relativity" at Cerisy-la-Salle, France, mimeo) (in French).

Analyses the process of learning the tactics and strategies of the *awelé* game.

6-NGU-88
1988 N'Guessan, Assandé G.: L'awélé, acquisition des tactiques et des stratégies, in: Bureau, René & Denyse de Saivre (Eds.), *Apprentissage et cultures, les manières d'apprendre*, Éditions Karthala, Paris (France), 230-244.

6-NGU-92

1992 N'Guessan, Assandé Gilbert: *Mécanismes d'appentissage de l'AWELE: l'apprentissage d'un jeu de stratégies typiquement africain (l'AWELE) chez les adolescents et les joueurs d'échecs suisses* [Mechanisms of learning awele : the learning of a typical African strategy game (awele) by Swiss youth and chess players], Ed. Universitaires, Fribourg (Switserland), 311 p.

Published version of a doctoral thesis submitted in 1991.

6-NSI-68

1968 Nsimbi, Michael B.: *Omweso: a game people play in Uganda*, African Studies Center, Occasional Paper, No. 6, Los Angeles CA (USA), 38 p.

6-NSI-69

1969 Nsimbi, Michael B.: *Omweso: a game people play in Uganda*, Uganda Publishing House, Kampala (Uganda).

6-NSI-86

1986 Nsimbi, Michael B.: *Omweso – kyawandiikibwa*, Banana books - Mubaka Printers, Kampala (Uganda) (in Ganda).

6-NSI-68, 6-NSI-69 and 6-NSI-86 escribe the *omweso* game and present its history.

6-ODE-77

1977 Odeleye, A. O.: *Ayo, a popular Yoruba game*, Oxford University Press, Ibadan (Nigeria), 54 p.

Chief Odeleye, a master player, describes *ayo*, a mancala board game among the Yoruba in Nigeria, and analyses several popular strategies for playing it.

6-OWE-38

1938 Owen, T. R. H.: A Bega game – Andot, *Sudan Notes and Records*, Khartoum (Sudan), Vol. XXI.

6-PAN-71

1971 Pankhurst, Richard: Gabata and related board games of Ethiopia and the Horn of Africa, *Ethiopia Observer*, Addis Ababa (Ethiopia), Vol. 14, No. 3, 154-206.

Presents a brief history and rules of several versions of *mancala* type games.

6-PAN-82
1982 Pankhurst, Richard: Gabata and other Board-Games of Ethiopia and the Horn of Africa, *Azania*, Nairobi (Kenya), Vol. 17, 27-41.

6-PIN-95
1995 Pingaud, François & Pascal Reysset: *L'awélé: le jeu des semailles africaines*, Chiron-Algo, Paris (France), 109 p.

Presents the *awele* (*woaley*) game in Ivory Coast.

6-POW-01
2001 Powell, Arthur B. & Oshon L. Temple: Seeding Ethnomathematics with 'Oware': 'Sankofa', *Teaching Children Mathematics*, NCTM, Reston VA (USA), Vol. 7, No. 6 (Focus issue: Mathematics and Culture), 369-375.

Illustrates how *oware*, a *mancala* game from the Akan in Ghana, may be explored in the mathematics classroom.

6-POWE-31
1931 Powell-Cotton, P. H. G.: A mancala board called Songo, *Man*, London (UK), Vol. 31 [Cameroon].

6-PRI-92
1992 Prista, António (Ed.): *Jogos de Moçambique* [Games of Mozambique], Instituto Nacional de Educação Física, Maputo (Mozambique) & Centro de Documentação e Informação Amilcar Cabral, Lisbon (Portugal), 79 p. (in Portuguese).

Includes descriptions of the board games *muravarava* (a three-on-row game, p. 39), and *ntchuva* (a four-row *mancala* game, 52-54).

6-PRO-81
1981 Provenzo, Asterie Baker & Provenzo, Eugene F.: *Play it again: Historic board games you can make and play*, Prentice Hall, Englewood Cliffs NJ (USA).

Contains sections on three African board games: *achi*, a three-on-row game from Central Africa (37-39); *wari*, a *mancala* game from West

Africa (115-118), and *seega*, a modern version of *senat* from Ancient Egypt (162-166).

6-RAA-72

1972 Raabe, Juliette: *Le jeu de l'awélé* [The awele game], Editions de la Courtille, Paris (France), 96 p. [Ivory Coast]

6-RET-84

1988 Retschitzki, Jean; Keller, Beat & Loesch-Berger, Marie-Cécile: L'influence du matériel et du niveau des joueurs sur la rétention de configurations du jeu d'awélé [The influence of the material and the level of players on the retention of configurations of the awele game], *Cahiers de Psychologie Cognitive*, Université d'Aix-Marseille II, Marseille (France), Vol. 4, No. 4, 335-361 [Ivory Coast].

6-RET-88

1988 Retschitzki, Jean: L'apprentissage des stratégies dans le jeu d'awélé [Learning strategies in the awélé game], in: René Bureau & Denyse du Saivre (Eds.), *Apprentissage et cultures, les manières d'apprendre*, Éditions Karthala, Paris (France), 213-229.

6-RET-90

1990 Retschitzki, Jean: *Stratégies des joueurs d'awélé* [Strategies of the players of awélé], L'Harmattan, Paris (France), 240 p.

Studies the learning of the strategies of the *awélé* game in Ivory Coast, including an analysis of the use of calculation and estimation (91-98). Presents strategy simulation computer programmes (201-216).

6-RUS-84

1984 Russ, Laurence: *Mancala games*, Reference Publications, Algonac MI (USA), 111 p.

Mancala is the generic name given by anthropologists to a class of board games played throughout Africa, parts of Asia. Due to the slave trade, the game is also found in the Caribbean and on the eastern coast of South America. The games are played on wooden boards, which have either two, three, or four rows of holes carved into them. When not using boards, the rows of holes may be dug out of the earth. The

book presents the rules, distribution and history of several versions of the *mancala* game.

Review: 6-CRO-87.

6-SAND-13

Sanderson, M. G.: Native games of Central Africa, *Journal of the Royal Anthropological Institute of Great Britain and Ireland*, London (UK), Vol. 43, 726-736.

6-SANT-94

1994 Santos Silva, Elísio: *O "ouri" — Um Jogo Caboverdiano e a sua prática em Portugal* [Ouri — A game from Cape Verde Islands and its practice in Portugal], Associação de Professores de Matemática, Lisbon (Portugal), 85 p. (in Portuguese).

This book published by the Association of Mathematics Teachers in Portugal deals with *ouri* (or *seca* or *ouril*), a game of the *mancala* type, as played on the West African Cape Verde Islands and among immigrants in Portugal. It is also compared with other *mancala* type games from the Cape Verde Islands as *pintôn*, and *pia* or *moura*.

6-SANT-95

1995 Santos Silva, Elísio: *Jogos de quadrícula do tipo mancala com especial incidência nos praticados em Angola* [Board games of the mancala type with special attention for those played in Angola], Instituto de Investigação Científica Tropical, Lisbon (Portugal), 323 p. (in Portuguese).

Edition of text completed in 1970 with recent complementary notes (295-311). Chapter 2 (21-68) describes games of the mancala type in general. Chapter 3 (69-119) describes the mancala games played in Angola: *owela*, *muvalavala*, *tchela*, *lueli*, *mwendo*, *quendo*, *uela*, *gango*, *biri*, *déqui*. Chapter 4 (121-294) discusses the origin of these traditional games in Angola.

Review: 6-TOW-98.

6-SAW-49

1949 Sawyer, Walter W.: The game of oware, *Scripta Mathematica*, New York (USA), Vol. XV, 159-161 [reproduced in 6-KOV-95] [Ghana].

6-SHA-34
Shackel, R. S.: Mweso, *Uganda Journal*, Kampala (Uganda), Vol. II.

6-SHA-35
Shackel, R. S.: More about Mweso, *Uganda Journal*, Kampala (Uganda), Vol. III.

6-SHE-94
1994 Sheppard, Reg & Wilkinson, John: *Strategy Games*, Tarquin, Norfolk (UK), 50 p.

Includes brief descriptions of *wari* (two-row *mancala* game, p. 6), *achi* (three-on-a-row game, p. 7), *yoté* (West Africa, p. 32), and *el-quirkat* (North Africa, p. 35).

6-TOW-76
1976 Townshend, Philip: Autour du jeu de Mankala [About the mancala game], *Zaire-Afrique*, Centre d'études pour l'action sociale, Kinshasa (Congo / Zaire), Vol. 105, 287-297 (in French).

6-TOW-77a
1977a Townshend, Philip: Les yeux de mankala au Zaire, au Rwanda et au Burundi [Mancala games in Zaire, Rwanda and Burundi], *Cahiers du centre d'étude et de documentation africaines* (CEDAF), Brussels (Belgium), Vol. 3, 1-76 (in French).

6-TOW-77b
1977b Townshend, Philip: Mankala Games, *Bulletin of the International Committee on Urgent Anthropological and Ethnological Research*, Vienna (Austria), Vol. 19, 47-54.

6-TOW-77c
1977c Townshend, Philip: The South West African game of Illhus in the wider context of African Mankala, *Journal of the South West African Scientific Society*, Windhoek (Namibia), Vol. 31, 85-98

6-TOW-79a
1979a Townshend, Philip: African Mankala in anthropological perspective, *Current Anthropology*, Chicago IL (USA), Vol. 20, 794-796.

6-TOW-79b
1979b Townshend, Philip: Mankala in Eastern and Southern Africa: a Distributional Analysis, *Azania*, Nairobi (Kenya), Vol. 14, 108-138.

6-TOW-79c
1979c Townshend, Philip: *Anthropological Perspectives on Bao (Mankala) Games*, Institute of African Studies, University of Nairobi, Nairobi (Kenya).

6-TOW-82
1982 Townshend, Philip: Bao (Mankala): The Swahili Ethic in African Idiom, *Paideuma*, *Mitteilungen zur Kulturkunde*, Wiesbaden (Germany), Vol. 28, 175-191.

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Appendix 7
Note on research inspired by the historical reconstruction of mathematical ideas in the 'sona' geometric tradition of Southern-Central Africa

(reproduced from AMUCHMA-Newsletter, No. 27, 2003)

Wolfgang Jaritz of the University of Graz (Austria) may have been the first to do mathematical research inspired by the 'sona' tradition of the Cokwe and related peoples of eastern Angola and neighboring regions of Zambia and Congo. Informed by the anthropological studies of Gerhard Kubik (cf. KUB-86, 87a, 87b, 87c), Jaritz studied the algorithm for drawing a particular class of 'sona' and compared it to the paths of a ball at a billiard table (7-JAR-83). Marcia Ascher of Ithaca College (New York, USA) analyzed several 'sona' as graphs (ASC-88, 91 [Ch. 2]). The book (GER-93d, 94i, 95a, 97a) contributed to the historical reconstruction and analysis of mathematical ideas inherent in the 'sona' tradition. Gerdes has developed further the geometry of the 'sona' introducing the concept of mirror curves and inventing Lunda-designs, presented for the first time in 7-GER-90. Inspired by this research, Slavik Jablan (Belgrade, Serbia) has studied mirror curves and their relationship with mathematical knot theory (7-JAB-95, 01). In the early 1990s Robert Lange (Brandeis University MA, USA) developed 'sona tiles.' Franco Favilli and his student Laura Maffei at the University of Pisa (Italy) have been developing software for the construction of mirror curves and Lunda-designs. Mark Schlatter (Centenary College of Louisiana, USA) is studying mirror curves and permutations (7-SCHL-00, 01; 7-PETER-01). Nils Rossing of the University of Science and Technology (Trondheim, Norway) and Christoph Kirfel of the University of Bergen (Norway) applied methods of 'sona' analysis by mirror curves to the mathematical analysis of a class of traditional Norwegian rope mats (7-ROS-03). Gerdes himself advanced with the study of Lunda-designs (GER-99a [Ch. 4]; 7-GER-96, 97, 99a, 99b, 02a, 02b, 02e, 02i, 05, 06a, 06b, 07) and a sub-class called Liki-designs (7-GER-02c, 02d). He found several interesting classes of matrices, like cyclic (7-GER-02d), helix (7-GER-02f), cylinder (7-GER-02g) and chessboard matrices (7-GER-02h). Several of these papers were published in *Visual Mathematics* (*) and other on-line journals. The book 7-GER-07 gives an introduction to cycle matrices.

Links between Lunda-designs, determinants and magic squares were established (7-GER-00, 02i). The newness and the multiple relationships of mathematical ideas arising from the analysis of the 'sona' tradition with other areas of mathematics reflects the profoundness and the mathematical fertility of the ideas of the Cokwe master drawers.
For a further up-date see the appendix "Mathematical research inspired by the *sona* tradition: the example of mirror curves, Lunda-designs and cycle matrices" in GER-06, 217-232.

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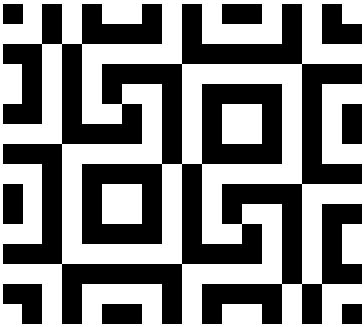

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